

UNIVERSITY OF SANTO TOMAS
ELECTRICAL ENGINEERING DEPARTMENT

**SIMULATION AND ANALYSIS OF ADAPTIVE MPPT (MAXIMUM POWER POINT
TRACKING) ALGORITHMS FOR SOLAR PV SYSTEMS UNDER WEATHER
CONDITIONS IN ARAYAT-MEXICO SOLAR FARM, PAMPANGA**

A Research Project
Submitted to the
Electrical Engineering Department
Faculty of Engineering
University of Santo Tomas

In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Science in Electrical Engineering

by

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MIRASOL, Jonathan O.

May 2026



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APPROVAL SHEET

The Research Project entitled:

SIMULATION AND ANALYSIS OF ADAPTIVE MPPT (MAXIMUM POWER POINT TRACKING) ALGORITHMS FOR SOLAR PV SYSTEMS UNDER WEATHER CONDITIONS IN ARAYAT-MEXICO SOLAR FARM, PAMPANGA

has been adequately prepared and submitted by the following proponents:

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ENDORSEMENT SHEET

This is an endorsement certifying that the Research Proposal/ Project entitled:

**SIMULATION AND ANALYSIS OF ADAPTIVE MPPT (MAXIMUM POWER
POINT TRACKING) ALGORITHMS FOR SOLAR PV SYSTEMS UNDER
WEATHER CONDITIONS IN ARAYAT-MEXICO SOLAR FARM, PAMPANGA**

has been adequately prepared and submitted by the following proponents:

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CERTIFICATE OF ORIGINALITY

We hereby declare that this submission is our work and that, to the best of our knowledge and beliefs, it contains no material previously published or written by another person nor material to which a substantial extent has been accepted for award of any degree or diploma of a university or other institute of higher learning, except where due acknowledgement is made in the text.

We also declare that the intellectual content of this thesis/dissertation is the product of our work, even though we may have assistance from others on style, presentation, and language expression.

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ABSTRACT

The demand for clean energy motivates wider deployment of solar photovoltaic (PV) systems in the Philippines, where tropical weather, frequent cloudiness, rapid irradiance and temperature swings, and typhoons, challenges continuous energy capture. This study evaluates how ten adaptive MPPT algorithms perform under real tropical conditions at the Arayat-Mexico Solar Farm, asking which methods maximize harvested energy, converge fastest, and remain stable during rapid transients.

A MATLAB/Simulink PV model was driven with localized irradiance and temperature records sourced from PAGASA and the Solcast service, plus two benchmark datasets. Ten MPPT approaches (Perturb and Observe, Incremental Conductance, Fuzzy Logic Control, Sliding Mode Control, Ripple Correlation Control, Fibonacci Sequence Based, Artificial Neural Network, Particle Swarm Optimization, Genetic Algorithm, Grey Wolf Optimization) were compared across convergence time, tracking efficiency and accuracy, harvested energy, and cloud-transient recovery. Performance was summarized via an entropy-weighted Algorithm Ranking Stability (ARS) and tested with Friedman and Nemenyi nonparametric analyses.

Classical and control-based algorithms, particularly P&O, IncCond, RCC, and SMC, demonstrated consistently strong performance with tracking efficiencies above 96% and higher harvested energy. SMC achieved the highest overall ranking due to superior convergence time and energy capture. FLC achieved the highest tracking efficiency and stability but slightly lower energy yield during dynamic conditions. Metaheuristic algorithms (PSO, GA, GWO) showed lower performance under the tested implementation parameters.

Classical and control-theoretic MPPT algorithms demonstrated the most consistent and competitive performance for solar PV systems under Philippine tropical irradiance conditions. Among them, SMC achieved the highest Algorithm Ranking Stability (0.8419) due to its superior harvested energy and convergence time, with results statistically validated across both benchmark and real tropical datasets. Metaheuristic algorithms ranked lowest under the tested parameters, though this was mainly due to implementation constraints, suggesting that different parameter settings could produce improved performance in future studies.

Keywords: *Maximum Power Point Tracking (MPPT) algorithms, solar photovoltaic (PV) systems, Perturb & Observe, Incremental Conductance, Fuzzy Logic, Sliding Mode Control, Ripple Correlation Control, Fibonacci Series Based, Artificial Neural Network, Particle Swarm Optimization, Genetic Algorithm, Grey Wolf Optimization, MATLAB/Simulink*

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NOMENCLATURE

A. Acronyms and Abbreviations

Symbol/Unit	Name/Description
ANN	Artificial Neural Network
CTRT	Cloud Transient Recovery Time
DC	Direct Current
EWM	Entropy Weight Method
FLC	Fuzzy Logic Control
FSB	Fibonacci Series Based
GA	Genetic Algorithm
GWO	Grey Wolf Optimization
IncCond	Incremental Conductance
kWh	Kilowatt-hour
MCDM	Multi-Criteria Decision-Making
MPP	Maximum Power Point
MPPT	Maximum Power Point Tracking
LCOE	Levelized Cost of Electricity
PAGASA	Philippine Atmospheric, Geophysical and Astronomical Services Administration
P&O	Perturb & Observe

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PID	Proportional-Integral-Derivative
PSO	Particle Swarm Optimization
PV	Photovoltaic
RCC	Ripple Correlation Control
W	Watts
Wh	Watt-hour
W/m ²	Watts per square meter
s	Second/s
SMC	Sliding Mode Control

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B. Variables and Symbols

Symbol	Description	Unit
G	Solar irradiance	W/m^2
T	Temperature	$^{\circ}\text{C}$
t	Time	s
V	Output voltage of PV system	V
I	Output current of PV system	A
P	Output power of PV system	W
P_{in}	Input power of PV system	W
P_{out}	Output power of converter	W
P_{max}	Maximum power point	W
V_{mp}	Voltage at maximum power point	V
I_{mp}	Current at maximum power point	A
t_{conv}	Convergence time of MPPT algorithm	s
η	Tracking efficiency	%
E	Harvested Energy	Wh
CTRT	Cloud transient recovery time	s
D	Duty cycle of DC-DC converter	–
ΔV	Change in voltage	V
ΔI	Change in current	A
ΔP	Change in power	W

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ΔD	Change in duty cycle	—
$\frac{dP}{dV}$	Power-voltage slope used in MPPT decision	—
N_{hold}	Number of holding iterations before update	—
ω	Entropy weight of a performance metric	—
ARS	Algorithm Ranking Stability index	—
x_{ij}	Performance value of algorithm i under metric j	—
p_{ij}	Normalized probability value in entropy method	—
e_j	Entropy value of metric j	—
d_j	Degree of diversification of metric j	—
k	Number of algorithms evaluated	—
N	Number of datasets or scenarios	—
m	Number of alternatives (algorithms)	—
n	Number of evaluation criteria	—

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CHAPTER 1.0. INTRODUCTION

1.1. Background of the Study

Global demand for renewable energy continues to rise, fueled by population growth, economic development, and technological progress. Solar photovoltaic (PV) systems have become one of the preferred available renewable technologies, due to their sustainability and declining costs. The estimated levelized cost of electricity (LCOE) for utility-scale solar is currently in the range of \$35 and \$72 per megawatt-hour (MWh), making solar more competitive compared to traditional power generation from gas or coal [1].

The Philippines has highly favorable solar resource potential, with average solar irradiance of about 3.68 - 5.73 kWh/m²/day [2], making it highly suitable for PV deployment. Still, the nation has not yet maximized the use of this resource: the Department of Energy's latest power statistics reveal that solar only contributed 3% to national gross electricity generation [3]. Installation of PV systems in tropical climates like the Philippines is plagued by various challenges, including variable weather patterns and high ambient temperatures that reduce array performance. Even though recent developments have yielded cost-effective and efficient PV modules and system components (as of 2025), constraints in the form of finance, grid integration, and site costs still hinder scaling rapidly. The National Renewable Energy Program (NREP) and incentives by the government, seek to deploy more renewables and promote residential and utility-scale solar adoption [4].

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The Maximum Power Point Tracking (MPPT) algorithm is a crucial part of maximizing the power output of the PV system, particularly when the environmental factors are dynamically changing. The MPPT algorithm shifts the operating point in response to the changes in the irradiance and temperature, hence increasing the efficiency of the system and the power output, which is a significant factor in tropical areas.

This study simulated and evaluated selected MPPT algorithms using realistic Philippine meteorological data including irradiance, temperature, and weather variability. Tropical datasets have different features than standard or midlatitude datasets, including a higher variability of short-term irradiance, frequent cloud transients, and high values of ambient temperatures. Because of these unique features, these datasets were crucial in studying MPPT algorithm performance for realistic conditions.

The increasing use of PV systems in the Philippines, mostly utilize typical mid-latitude or generic high-variability benchmark datasets from MPPT studies, which may not fully capture the temporal dynamics and persistence of irradiance fluctuations characteristic of tropical climates. This points to the need for a Philippine-specific benchmarking dataset whose characteristic features can be quantitatively validated and assessed.

This study referenced the Arayat–Mexico Solar Farm, located in the Municipalities of Arayat and Mexico, Pampanga, Central Luzon, Philippines [5]. The farm is one of the largest utility-scale photovoltaic installations in the region, with an operational capacity of 72 MW in Phase 1 and further expansion planned to 116 MW [5], [6]. In addition, it is of strategic relevance to the research objectives, as it provides a realistic context for MPPT algorithms under the tropical conditions of the Philippines.

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The employment of site-specific tropical datasets was necessary because the PV systems in the Philippines have higher short-term variability of irradiance, frequent cloud transients, and elevated ambient temperatures compared to midlatitude or standard datasets [7]. These unique characteristics allowed a more realistic simulation of MPPT algorithm performance under dynamic tropical conditions.

Satellite and reanalysis studies identify Luzon, including Central Luzon and the province of Pampanga, as an area of high solar PV potential but with significant seasonal and short-term variation driven by the monsoon cycles, cloud transients, and atmospheric dynamics. The comparison of remote sensing analyses using AHI-8/9 and FY-4A solar radiation products shows measurable diurnal and seasonal variations in the irradiance of the Philippines, thereby validating the natural variability of Pampanga for MPPT testing [8]. NREL's Competitive Renewable Energy Zones, or CREZ, study identifies portions of Luzon, including those that include Central Luzon, as a high-solar-resource area for utility-scale PV deployment, further confirming the suitability of Central Luzon to host large PV installations [9]. A local study conducted on a grid-tied PV installation in San Fernando, Pampanga, also reports favorable resource availability, which confirms that localized irradiance and temperature data are available for Pampanga and can be used to drive realistic MPPT simulations [10].

Regions such as Tarlac, Nueva Ecija, and Pampanga are considered solar hotspots due to them exhibiting the highest amounts of short-wave radiation. Pampanga, in particular, is a good geographic location for solar energy due to the wide presence of solar PVs. Moreover, the region is the second largest contributor of solar energy systems with an estimated 293,000 Gwh generation of solar power in 2019. Since Pampanga is

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already known for their PV installations, there may already be local familiarity and likely LGU experience with solar projects [11]. While not specific to Pampanga, neighboring regions such as Tarlac was chosen for the Raslag-6 project because the site could be connected to a 69 kV transmission line of the National Grid Corporation of The Philippines (NGCP). This suggests that Central Luzon offers essential grid connectivity for large-scale solar projects. The existence of possible grid connectivity makes sites such as Pampanga a high-priority criterion in site selection [12].

The algorithms investigated in this study are Perturb & Observe (P&O), Incremental Conductance (IncCond), Fuzzy Logic Control (FLC), Sliding Mode Control (SMC), Ripple Correlation Control (RCC), Fibonacci Series Based Algorithm (FSB), Artificial Neural Network (ANN), Particle Swarm Optimization (PSO), Genetic Algorithm (GA) and Grey Wolf Optimization (GWO). Performance was assessed by metrics such as tracking efficiency, power output, and dynamic responsiveness under representative weather conditions. Moreover, the selected farm features monitoring systems and DC/AC telemetry. This enabled data-driven simulation of irradiance-temperature variations and facilitated the duplication of tropical weather conditions and dynamic PV operating scenarios without the need for in-person site visits. With these aspects, the study could closely represent real-world conditions for researching the performance, efficiency, and robustness of the 10 selected MPPT algorithms. This is further supported by ACEN reports and regional studies, which verify the appropriateness of the selected site and confirm the relevance of the research, as well as its local applicability in renewable energy deployment [1],[2],[13].

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Moreover, the application of simulation-based evaluation allows the derivation of advanced performance metrics, such as cumulative tracking response time, rank stability of the algorithm across different datasets, and power oscillation dynamics, which were hard to distinguish using experimental testing alone. Such performance metrics offer more in-depth information on the adaptability and robustness of the MPPT algorithm in rapidly changing tropical environments.

The objective of this study was to determine the MPPT approach that provides better performance for PV systems in the Philippines, which provides a basis towards enhanced solar energy harvesting and the country's renewable energy goals.

1.2.Statement of the Problem

The need for clean and sustainable energy has resulted in the popularity of solar photovoltaic (PV) systems in most countries, including the Philippines. As a tropical archipelagic nation, the Philippines hosts a special combination of high solar potential and quickly alternating weather patterns, such as frequent cloudiness, abrupt changes in temperature, and seasonal typhoons. These climatic variations substantially impact the operation of solar PV systems, especially with the system's efficiency and energy output. The Maximum Power Point Tracking (MPPT) technology is one of the central technologies that evolved to counter these issues. It dynamically changes the system's operating point to realize the maximum achievable power from solar panels. The following questions identified the particular issue this research aimed to solve about the adaptive MPPT algorithm performance within the country's climate:

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1. What were the differences in convergence speed, tracking efficiency, tracking accuracy, harvested energy, and cloud transient recovery time among the Perturb & Observe (P&O), Incremental Conductance (IncCond), Fuzzy Logic Control (FLC), Sliding Mode Control (SMC), Ripple Correlation Control (RCC), Fibonacci Series Based Algorithm (FSB), Artificial Neural Network (ANN), Particle Swarm Optimization (PSO), Genetic Algorithm (GA) and Grey Wolf Optimization (GWO) MPPT algorithms when subjected to rapid irradiance and temperature fluctuations typical of the Pampanga climate?
2. Which MPPT algorithms provided lower convergence speed, tracking efficiency, tracking accuracy, harvested energy, and cloud transient recovery time in various weather scenarios and datasets, including benchmark datasets and a proposed tropical dataset from the Arayat–Mexico Solar Farm?
3. Which of the MPPT algorithms gave the best overall performance ranking according to entropy-derived weights and nonparametric statistical comparisons carried out for several datasets and environmental conditions?
4. How did the performance metrics, dynamic response, and ranking results of the MPPT algorithms differ when tested with the localized tropical irradiance data from the Arayat-Mexico Solar Farm compared to the typical mid-latitude and cloud-transient irradiance datasets?

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1.3.Objectives of the Study

With the increasing need for clean and renewable energy, solar photovoltaic (PV) systems have been identified as one of the primary solutions to meet energy requirements in tropical nations such as the Philippines. Nonetheless, weather variability, such as changing radiation, temperature fluctuations, and occasional cloud coverage, is a major challenge in tapping the full potential of solar panels. In order to get rid of such problems, MPPT algorithms are used to keep changing the operation point of the PV system in such a way that maximum energy can be harvested. This study endeavored to replicate, compare, and analyze the performance of these algorithms when applied under actual tropical weather conditions within Pampanga, with the utilization of localized data. Through this research, the competence of these algorithms in terms of response time, efficiency, and energy output was evaluated to ascertain their viability for actual solar PV applications within the country.

In pursuing these objectives, the research was guided by the following particular objectives:

1. To quantitatively assess the effect of rapid irradiance and temperature variations on the convergence speed, tracking efficiency, tracking accuracy, harvested energy, and cloud transient recovery time of the Perturb & Observe (P&O), Incremental Conductance (IncCond), Fuzzy Logic Control (FLC), Sliding Mode Control (SMC), Ripple Correlation Control (RCC), Fibonacci Series Based Algorithm (FSB), Artificial Neural Network (ANN), Particle Swarm Optimization (PSO), Genetic Algorithm (GA) and Grey Wolf Optimization (GWO) algorithms

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using localized weather data from Arayat, Pampanga and other benchmark datasets.

2. To compare the performance of Perturb & Observe (P&O), Incremental Conductance (IncCond), Fuzzy Logic Control (FLC), Sliding Mode Control (SMC), Ripple Correlation Control (RCC), Fibonacci Series Based Algorithm (FSB), Artificial Neural Network (ANN), Particle Swarm Optimization (PSO), Genetic Algorithm (GA) and Grey Wolf Optimization (GWO) algorithms using measurable indicators including convergence time (s), tracking efficiency (%), tracking accuracy (stability), harvested energy (Wh), and cloud transient recovery time (s) obtained from MATLAB/Simulink simulations, and to rank the algorithms using an entropy-derived weights to compute Algorithm Ranking Stability (ARS).
3. To statistically assess and evaluate significant performance differences across Perturb & Observe (P&O), Incremental Conductance (IncCond), Fuzzy Logic Control (FLC), Sliding Mode Control (SMC), Ripple Correlation Control (RCC), Fibonacci Series Based Algorithm (FSB), Artificial Neural Network (ANN), Particle Swarm Optimization (PSO), Genetic Algorithm (GA) and Grey Wolf Optimization (GWO) algorithms by applying non-parametric methods such as the Friedman test and Nemenyi post-hoc, and integrating the results with entropy-derived weights performance metrics into an overall ranking.
4. To assess and quantify the differences in the performance metrics, dynamic response characteristics, and ranking results of the MPPT algorithm simulations

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results based on the localized tropical irradiance data of the Arayat-Mexico Solar Farm compared to conventional mid-latitude and cloud-transient datasets.

1.4. Significance of the Study

This study addressed the Philippines' rapid adoption of renewable energy, particularly utility-scale solar PV, driven by the country's tropical climate and high solar resource potential. Although PV systems are well-suited to the Philippines, their performance is strongly affected by local weather dynamics, rapid cloud transients, fluctuating irradiance, and thermal variability, which can degrade energy harvesting when MPPT algorithms are not adaptive or robust.

The Arayat–Mexico Solar Farm in Pampanga is an ideal reference site for this study due to the fact that it demonstrates those typical Philippine irradiance patterns: a high amount of solar exposure with frequent and rapid fluctuations.

By simulating and comparing the performance of different MPPT algorithms, mainly Perturb & Observe (P&O), Incremental Conductance (IncCond), Fuzzy Logic Control (FLC), Sliding Mode Control (SMC), Ripple Correlation Control (RCC), Fibonacci Series Based Algorithm (FSB), Artificial Neural Network (ANN), Particle Swarm Optimization (PSO), Genetic Algorithm (GA) and Grey Wolf Optimization (GWO), this research gained important insights into how effective these are under Philippine weather conditions. The results were used to identify which algorithm had the best compromise between convergence speed, tracking efficiency, tracking accuracy, harvested energy, and cloud transient recovery time (CTRT), resulting in better PV system efficiency.

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This research is helpful in a number of stakeholders:

1. **For solar PV designers and engineers**, it gives algorithm selection guidelines based on recommendations provided by data for selecting appropriate MPPT algorithms in local contexts.
2. **For policymakers and energy planners**, the findings can be used to make better-informed decisions on deploying more efficient solar technologies in the national energy mix.
3. **For research and academic communities**, it adds to the literature on adaptive MPPT methods in tropical regions.
4. **For local populations and users of solar energy**, it encourages the construction of more efficient and robust PV systems, which can assist in lowering energy expenses and enhancing access to clean energy.

Ultimately, this study helps maximize solar energy systems in the Philippines, aligning with the overall objective of sustainable and resilient energy development.

1.5.Scope and Limitations

This research focused on the simulation and performance evaluation of Perturb & Observe (P&O), Incremental Conductance (IncCond), Fuzzy Logic Control (FLC), Sliding Mode Control (SMC), Ripple Correlation Control (RCC), Fibonacci Series Based Algorithm (FSB), Artificial Neural Network (ANN), Particle Swarm Optimization (PSO), Genetic Algorithm (GA) and Grey Wolf Optimization (GWO) MPPT algorithms for solar photovoltaic systems under Pampanga climate conditions. MATLAB/Simulink was

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utilized to perform simulations, using localized irradiance and temperature data from SolCast, emphasizing the Arayat-Mexico Solar Farm in Arayat, Pampanga. The suitability of each algorithm for local conditions will be tested to see how well it performs in terms of convergence speed (s), tracking efficiency (%), tracking accuracy (stability), total harvested energy (Wh), and cloud transient recovery time (s).

The reliance on a simulation-based modeling limits the research despite using a realistic irradiance and temperature dataset. The models used are assumed to be ideal or near-ideal behavior for the system, which excludes real-world and non-ideal factors such as switching and conduction losses in power converters. Control-system-related losses are also excluded, such as sensor inaccuracies, measurement noise, and communication delays. Furthermore, PV model parameters were considered constant, which excludes degradation effects or parameter drift present in actual installations.

Additionally, the temporal resolution of the datasets is not fully representative of high-frequency irradiance fluctuations, which are caused by the fast-moving cloud transients. Because of this, the dynamic performance of the MPPT is significantly influenced by means of convergence speed, tracking efficiency, and cloud transient recovery. In actual hardware implementations, these limitations should be considered when interpreting the results and their applicability to real-world PV system deployment.

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1.6. Framework of the Study

Theoretical Framework

This study is grounded on the principles of photovoltaic (PV) power generation and Maximum Power Point Tracking. Photovoltaic panels convert sunlight into power through the photovoltaic effect. When photons hit a silicon semiconductor, electrons are energized and start to flow through an external circuit, producing current. Since power is the product of voltage and current ($P=IV$), the operating point of a PV module determines the amount of energy generated.

The voltage-current characteristics of a PV panel are nonlinear and are affected by external conditions such as irradiance and temperature. At a certain operating point, the product of the voltage and current at its maximum is called the Maximum Power Point (MPP). However, this varies dynamically with weather conditions.

To address this, Maximum Power Point Tracking (MPPT) Algorithms are utilized. These algorithms are designed to continuously monitor and adjust the operating point of the PV system to ensure maximum power harvest. In this study, ten theoretical approaches were considered and categorized:

1. **Classical MPPT Algorithms** - These algorithms rely on a feedback loop based on the P-V curve slope to adjust the operating point.
 - **Perturb & Observe (P&O)**, which operates on an iterative basis of periodically perturbing the array voltage and observing power variations.

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- **Incremental Conductance (IncCond)**, which relies on the principle that at Maximum Power Point (MPP), the derivative of power with respect to voltage is zero.
2. **Control Theoretic / Model Based Algorithms** - These algorithms utilize specific control theories or numerical sequences to optimize tracking of MPP.
- **Sliding Mode Control (SMC)**, which is a non-linear control technique to force the system to reach and remain on a predefined “sliding surface”.
 - **Ripple Correlation Control (RCC)**, which makes use of the switching ripples present in power converters to guide the system toward MPP.
 - **Fibonacci Series Based Algorithm (FSB)**, which is a numerical search technique that utilizes the Fibonacci sequence and is able to narrow the search window of MPP.
3. **Intelligent & AI-Based MPPT Algorithms** - These algorithms utilizes soft computing approaches to handle non-linearities which do not require precise mathematical models of the PV system.
- **Fuzzy Logic Control (FLC)**, which applies fuzzy set theory to model nonlinear system behavior and decision-making, making it effective under rapidly changing weather conditions.
 - **Artificial Neural Network (ANN)**, which mimics the processing capability of the human brain which has input, hidden, and output layers trained to recognize the non-linear relationship of inputs and the MPP.

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4. **Metaheuristic / Optimization-Based MPPT algorithms** - These algorithms are usually inspired by behaviour of animals, biological processes, or humans.

- **Particle Swarm Optimization (PSO)**, which is inspired by bird flocking or fish schooling behaviour that utilizes a population of particles to move through a search space to find MPP.
- **Genetic Algorithm (GA)**, which is based on Darwin's “survival of the fittest” theory wherein a population of solution evolves through numerous selections, crossovers, and mutations to find MPP.
- **Grey Wolf Optimization (GWO)**, which mimics the leadership hierarchy and hunting mechanism of grey wolves to find three fittest solutions to converge and find MPP.

Conceptual Framework

This research adopted the Input-Process-Output Model to illustrate the flow of study:

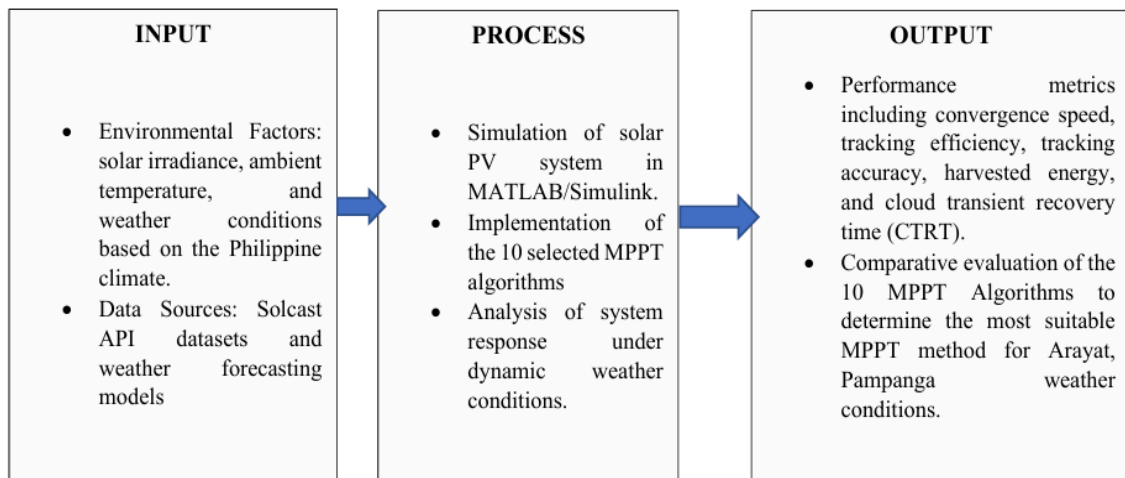


Figure 1.1 The Input-Process-Output Model

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Through this framework, the study systematically evaluated how environmental variables affect solar PV performance and how different MPPT algorithms optimize power generation.

1.7. Definition of Terms

- **Algorithm Rank Stability (ARS):** This is a combined performance index used to evaluate MPPT algorithms. It ranks the performance of MPPT algorithms using multiple performance indicators with entropy weights. It indicates the overall performance of each MPPT algorithm in terms of speed, efficiency, tracking accuracy, harvested power, and cloud transient recovery time.
- **Artificial Neural Network (ANN):** A soft computing MPPT approach based on the operation of the human brain utilizing layers of “neurons” to estimate the optimal duty cycle based on inputs (solar irradiance and temperature). [18]
- **Cloud Transient Recovery Time (CTRT):** This is the time taken by an MPPT algorithm to resume its operation near the MPP after the sudden change in solar irradiance due to the shadowing effect of passing clouds. Low values of cloud transient recovery time are desirable to improve the dynamic response to changing weather conditions.
- **Crisp value:** A precise numerical value taken from measurements or calculations and is used along with fuzzy values in fuzzy logic. [19]

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- **Defuzzification:** Takes a fuzzy value and converts it into a crisp value and is the end of a fuzzy logic process. [20]
- **Fibonacci Series Based Algorithm (FSB):** A numerical search algorithm that utilizes the Fibonacci sequence and can narrow down the search range to converge to the MPP with reduced iterations. [18]
- **Fuzzification:** Takes a crisp value and converts it into a fuzzy value and is the start of a fuzzy logic process. [20]
- **Fuzzy Logic Control:** An algorithm which isn't limited to the values 0 and 1 but can use values between 0 and 1. This reasoning has more resemblance to human decisions as it takes account of vagueness and uncertainty to its reasoning. [21]
- **Fuzzy value:** A vague value that is between 0 and 1 which tells how close an input is to either value. Used along with crisp values in fuzzy logic. [20]
- **Genetic Algorithm (GA):** An algorithm based on Darwin's principle of "survival of the fittest" which evaluates a population of optimal duty cycles through selection, crossover, and mutation to reach MPP. [22]
- **Grey Wolf Optimization (GWO):** An algorithm that mimics the leadership hierarchy and hunting mechanisms of grey wolves to track MPP. [22]
- **Incremental Conductance (IncCond):** An algorithm that refers to the ratio of derivative of conductance with respect to voltage and

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instantaneous conductance to determine needed changes in power systems.
[23]

- **Maximum Power Point (MPP):** This is the operation point on solar power systems wherein maximum power is generated ideal for harvesting energy. [24]
- **Maximum Power Point Tracking (MPPT):** A technique used to maximize the maximum power generated from a solar power system by adjusting the operation point using various algorithms. [25]
- **Partial Shading:** An occurrence wherein a part of the solar panel is blocked due to presence of objects such as buildings and trees, preventing sunlight from hitting the panels and decreasing power output. [26]
- **Particle Swarm Optimization (PSO):** An algorithm inspired by flocking behaviour (bird or fish) that uses a population called “particles” to iteratively find MPP based on best positions of the population. [22]
- **Perturb & Observe (P&O):** Widely used MPPT that operates by periodically perturbing the PV array voltage and observing changes to output power to adjust MPP. [22]
- **Photovoltaic (PV) Systems:** These are power systems that are composed of solar panels which convert sunlight into electrical power through the use of photovoltaic effect. [27]
- **Ripple Correlation Control (RCC):** An algorithm which exploits the naturally occurring ripples in PV power to guide the system to MPP. [28]

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- **Simulation:** This is the process of creating an ideal model of a system to mimic operations of real world components using software. In this experiment, MATLAB/Simulink was the simulation software used. [29]
- **Sliding Mode Control (SMC):** A non-linear control technique that follows a predefined “sliding surface” for PV operations which corresponds to the MPP. [18]
- **Solar irradiance:** Solar irradiance is the electromagnetic radiation emitted by the sun on a surface which can be captured for electrical energy and is expressed in W/m^2 or kWh/m^2 . [27]
- **Steady State Oscillation:** A performance metric which refers to the amplitude of power output around the MPP once the system has reached a steady operating condition, where lower oscillation means higher efficiency. [22]
- **Voltage Regulation:** This is the measurement of a system’s capability of maintaining stable voltage from a sending-end to a receiving-end. [30]

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CHAPTER 2.0 REVIEW OF RELATED LITERATURES

2.1. Overview of the Chapter

This chapter presents the relevant theories, a review of the related literature, studies based on the variables of this study, the conceptual framework, and definitions of essential terminology.

2.2 Related Literature

This section contains various pieces of literature based on the variables used in the study.

2.2.1 Photovoltaic Systems employed with different Maximum Power Point Tracker (MPPT) Algorithms and Techniques.

2.2.1.1 Photovoltaic Systems Types and Components

Photovoltaic (PV) systems have become one of the most preferred renewable energy sources due to their low operating costs, their almost maintenance-free nature, and their environmentally friendly, clean energy approach. The use of PV systems has good long-term economic benefits. However, the main problem with the acceptance of PV systems lies in their initial investment costs. One of the main reasons is the price of the solar panels. Over the past decade, the cost per watt has declined, but it would still cost more than the average cost of traditional electricity. Another issue with PV systems is the Grid-parity issue, in which the PV power cost per unit is lower than the electricity provided by power utility companies [31].

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A PV system is composed of the following main basic components: solar panels, solar charge controller, battery (DC storage), and a direct current (DC) to alternating current (AC) inverter. Solar panels convert light into electrical energy, which is regulated by the solar charge controller. The solar charge controller manages the voltage and current output of the solar panel to prevent overcharging the battery. The battery stores excess energy or provides energy when the solar panel output is too low to supply the load. Solar panel output may be low due to cloudy weather, partial shading, or nighttime, when no solar energy can be captured. Lastly, the inverter converts the DC output of the solar panel into AC electricity, which household appliances and grid systems operate on [32].

PV cells are mainly categorized into monocrystalline, polycrystalline, and thin-film types. Commercial-scale applications commonly use monocrystalline and polycrystalline PV cells. Monocrystalline and polycrystalline are classified based on the number of crystals shared per cell. [33]. The crystal structure of monocrystalline materials is homogeneous, meaning it is a single, continuous crystal structure, and its lattice orientation and electronic properties are uniform throughout the material. This is manufactured using the Czochralski Growth Method, in which cylindrical ingots are grown and sliced into wafers. Phosphorus and boron are doped to produce the p-n junction. Monocrystalline PV cell production is an expensive process, but it has been shown to be highly efficient at converting light energy into electricity, even in low-light conditions. To reduce costs, polycrystalline is used instead. The difference in efficiency between monocrystalline and polycrystalline is ~11%. Polycrystalline is composed of multiple small-grain crystals per cell, with varying crystallographic orientations. It is

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manufactured by casting molten silicon into blocks, which are then sliced into wafers, or by crystallizing amorphous silicon (a-Si) via chemical vapor deposition (CVD). Both monocrystalline and polycrystalline are used for high-power applications. Among the three main types of solar PV cells, thin-film is the cheapest to make but less efficient than polycrystalline. Thin-film PV cells are commonly manufactured by vapor-depositing a thin silicon layer onto a metal or glass substrate. Thin-film is made with A-Si, and it is the non-crystalline form of silicon. Thin-film A-Si PV cells have 3 layers to capture the sunlight. Because they are inexpensive, thin-film PV cells are commonly used in calculators, households, and buildings [34].

2.2.1.2 Maximum Power Point Tracking (MPPT) Algorithms and Techniques.

Numerous works and studies exist to improve PV systems with regard to Grid-parity, which are valuable for enhancing the overall PV system performance [35]. Maximum Power Point Tracking (MPPT) is one of the most popular topics in solar energy. The main goal of MPPT is to track the maximum power output of the PV system under any weather or environmental conditions. The I-V operating point of the PV system matches the load characteristic. To maintain this maximum power, the PV panel's voltage, current, and output power are regulated at Maximum Power Point (MPP) [36].

2.2.2 Utilization of solar energy in the Philippines

2.2.2.1 Solar energy potential in the Philippines

When it comes to solar energy, the Philippines is falling short of its potential to harness it. The Philippines has high levels of solar irradiance, making it well-suited for solar power generation. There have been goals to expand renewable energy in the

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Philippines, with a major aim of doubling capacity, which will reduce the country's reliance on coal. Despite these goals, a 2018 renewable energy symposium reports that installed renewable energy in the country increased by only 3% over 2005-2016. Coal imports increased an average of 12.8% annually from 1989 to 2015, which conflicts with the goal of reducing coal-based power in the country [37].

The Philippines currently derives 77% of its energy from fossil fuels, with the remaining 23% from renewable sources such as hydro, geothermal, and solar power. As statistics show, only 1% of the Philippines' power consumption comes from solar energy, despite its abundant sunlight. This is further supported by a data report from the Philippine Department of Energy (PDOE), which led them to push for more solar projects. The demand and popularity of solar power in the Philippines are high, as households and businesses consider solar energy as an option for their use [38]. Although solar energy is popular and in demand among these populations, the use of hydro and geothermal power is increasingly favored. This leaves the potential for solar energy usage to be better for rural areas.

The solar market in the Philippines is expected to grow at a Compound Annual Growth Rate (CAGR) of 25.2% from 2025 to 2030 [39]. The renewable energy market is expected to grow further, leading to a decline in the importance of fossil fuels and cleaner, more renewable energy options. Renewable energy sources are in increasing demand, and implementation costs are lower, enabling better accessibility in other areas of the country. The goal of renewable energy is to lower demand for fossil fuels and reduce overall greenhouse gas emissions [40].

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2.2.2.2 Solar irradiance distribution and weather variability on PV systems performance

The Philippine weather is tropical, meaning that the average temperature is hot and humid with a highly variable weather pattern. Temperature and solar irradiance greatly affect the performance of PV systems. The following are studies that evaluated PV system output performance and power potential in the Philippines. Principe et al. (2024) conducted a nationwide assessment of solar PV power potential using geospatial data, and the project is named Project SINAG (Solar PV Resource and Installation Assessment using Geospatial Technologies). Project SINAG is a “joint research program between the Philippine Department of Science and Technology (DOST) and China’s Ministry of Science and Technology (MOST)” [42].

By using remotely sensed data, Project SINAG can integrate factors such as solar irradiance, land elevation, slope, and temperature to evaluate the nation’s theoretical PV potential. The webGIS (Web Geographic Information System) of Project SINAG is publicly available but takes around 3-4 weeks to process data requests. Through this project, they found that the western and central regions of the Philippines have the highest potential for solar energy, while the eastern and mountainous regions receive lower irradiance, making them the least suitable for solar farms. These regions experience frequent cloud cover, which blocks sunlight from reaching the solar panels. The study, however, did not specify the Global Horizontal Irradiance (GHI) data for the Philippines and only shows the country's solar irradiance map [41].

A study by Sotto et al. (2024) shows that cloud optical thickness reduces solar irradiation by 19.36% during the cool season (December to January), 16.08% during the

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rainy season (June to November), and 10.54% during the hot dry season (March to May). From this data gathered by Sotto et al.'s study, it can be concluded that the rainy season's clouds and rainfall degrade solar availability in the country the most, and even with the least presence of clouds during dry seasons, the solar potential can still be degraded but shows minimal effect compared to all seasons [7].

Another study conducted by Galvan (2021) reported an average solar insolation of 4.92 kWh/m²/day in Cagayan Province, with 3.94 kWh/m²/day in December and 5.93 kWh/m²/day in April. This data aligns with the study by Principe et al. (2024) and confirms that the Philippines experiences fluctuations in solar energy, underscoring the country's potential to harness abundant solar energy. These studies demonstrate that weather variability and geographical factors significantly affect the potential of Philippine solar energy [42].

2.2.3 Simulation Tools and Techniques

2.2.3.1 Simulation Models and Flowcharts

To simulate photovoltaic (PV) systems, we need to understand how they generate power. When exposed to sunlight, PV cells produce current that flows into a load. The simplest ideal equivalent circuit of a PV cell consists of a PV cell in parallel with a diode and a load. Since it's ideal, there is no series resistance, meaning losses are neglected. When in short-circuit condition, the generated current of the PV cell is the measured short-circuit current. When in open circuit condition (load is removed), the generated current is equal to the difference between the generated current and the current flowing through the diode. A constant fraction of the open-circuit voltage of a PV cell corresponds to the maximum power point of a PV system. A more accurate equivalent

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circuit of a PV cell involves using 2 diodes, a parallel resistance, and a series resistance connected to the load [33]. When simulated in MATLAB/Simulink, a simple model would consist of multiple PV cells mentioned earlier. The voltage output would be multiplied by the number of PV cells connected in series used on the model. Output parameters will also be obtained via MATLAB/Simulink, where the P-V-I characteristics will be displayed on a graph.

The PV model would be used along with a Maximum Power Point Tracking (MPPT) algorithm. MPPT algorithms are important because they help adjust the PV system's impedance to keep it close to or at the peak power point under varying conditions. These conditions include solar irradiance, temperature, partial shading, varying weather conditions in the Philippines, and load demands. MPPT algorithms are typically used as controllers for PV systems due to their ability to adapt to variable factors. MATLAB provides a tutorial on implementing MPPT algorithms to adjust the operating voltages and duty cycles of a PV system [43].

For Perturb and Observer (P&O), conditional statements are used to adjust duty cycles and operating voltages based on conditions. Internal values are updated for every change. The flowchart of the P&O algorithm is shown below:

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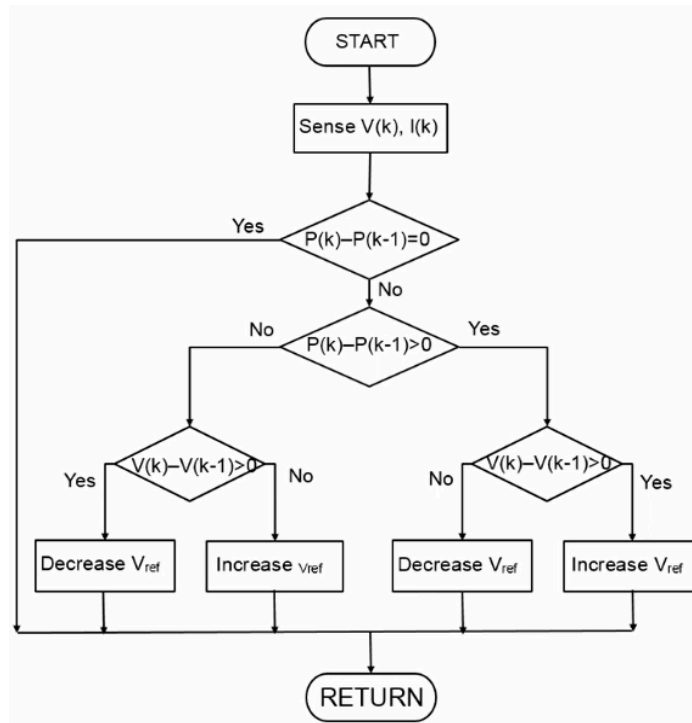


Figure 2.1 P&O Algorithm Flow Chart

Incremental Conductance (IncCond) is more complex than P&O. IncCond's main idea is to compare incremental conductance to instantaneous conductance to determine how the PV system will adjust. To implement logic, MATLAB has a state-flow chart wherein you can implement the flowchart, and the logic flow is visualized in the software. The flowchart for an IncCond algorithm is shown below [43]:

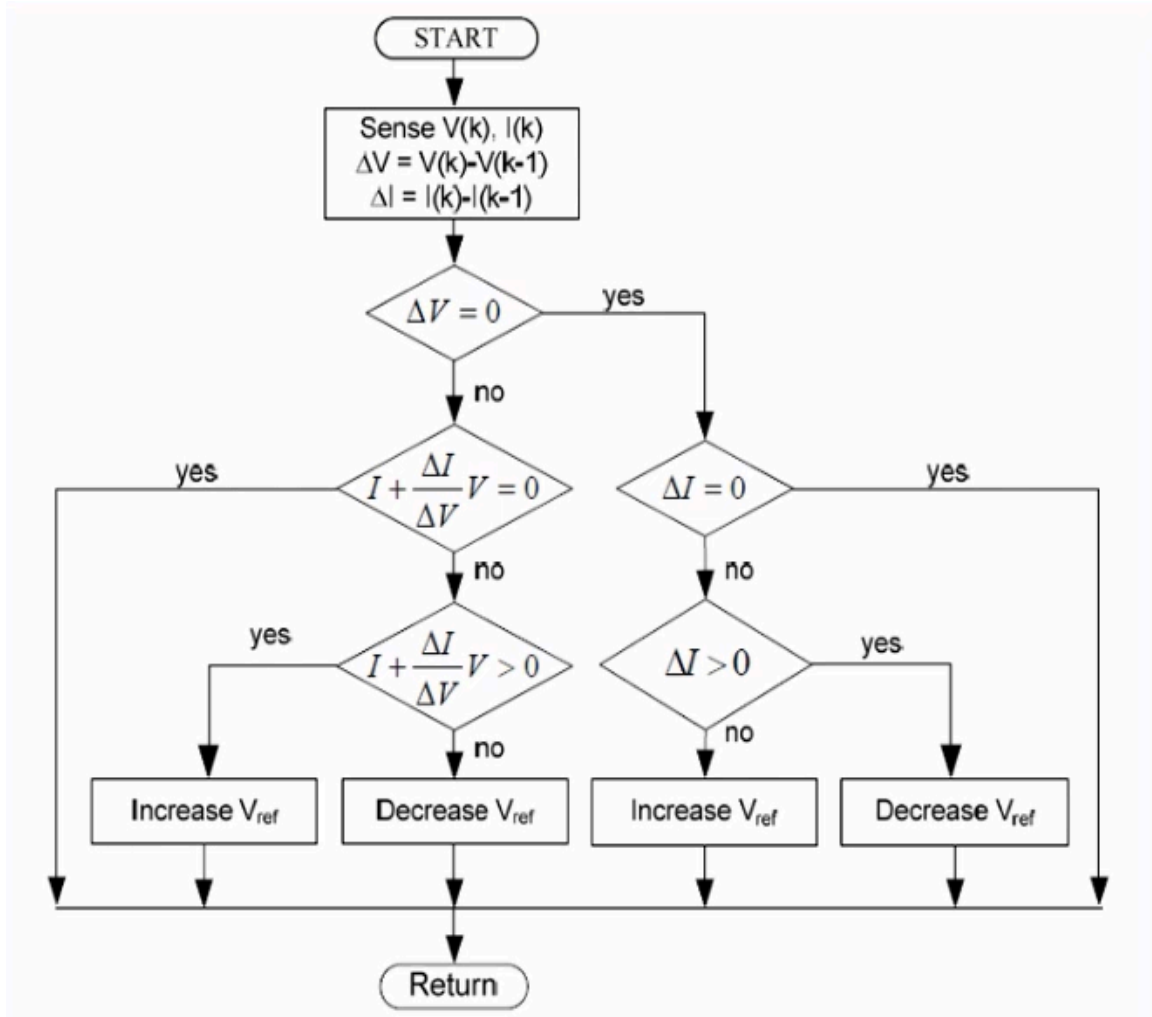


Figure 2.2 *IncCond Algorithm Flowchart*

Sliding Mode Control (SMC) MPPT is a robust, nonlinear control technique that forces the photovoltaic (PV) system's operating point to track a predefined sliding surface corresponding to the maximum power point. It continuously adjusts the converter's duty cycle based on the sign of the switching function, ensuring fast convergence and strong disturbance rejection during rapid changes in irradiance or temperature. The flowchart for SMC MPPT is shown below [44]:

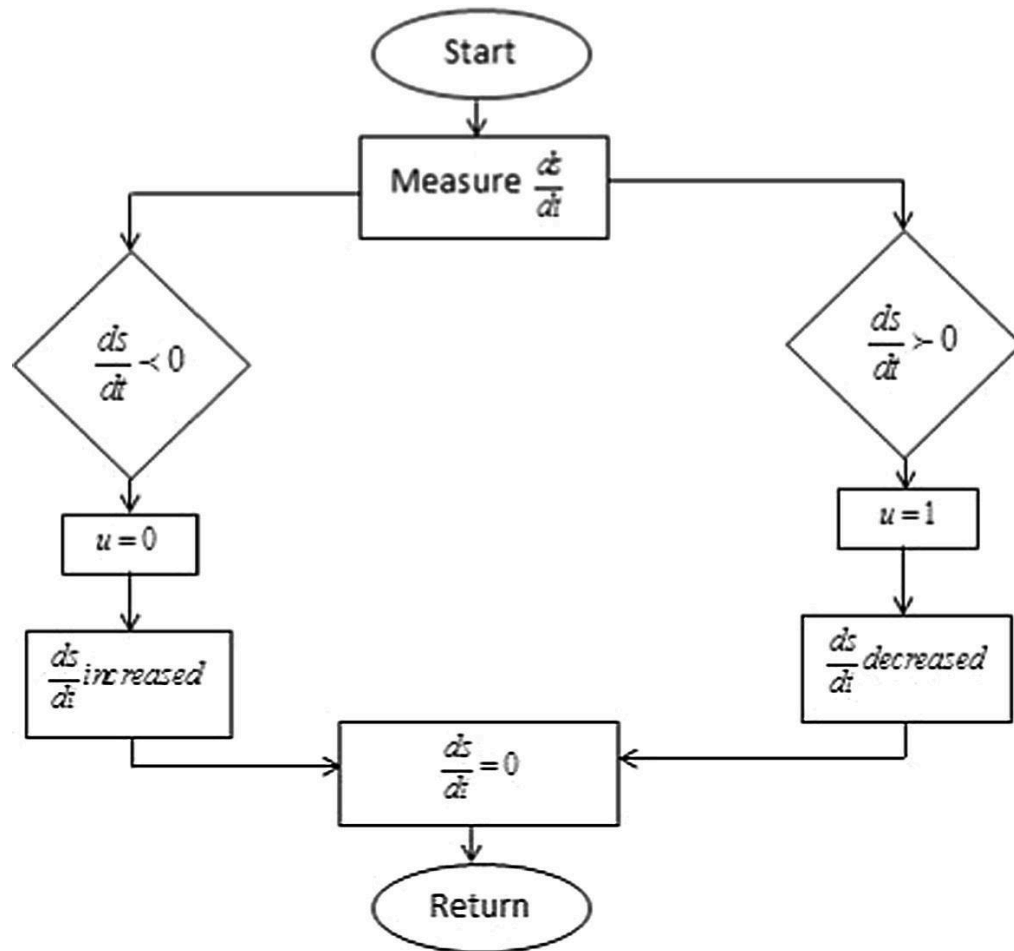


Figure 2.3 Sliding Mode Control Flowchart

Ripple Correlation Control (RCC) MPPT is a technique that exploits the natural switching ripple in PV voltage and current to determine the direction of increasing power without injecting an external perturbation. By correlating the signs of voltage/current ripple with power ripple, the controller adjusts the duty cycle toward the maximum power point. The flowchart for RCC MPPT is shown below [45]:

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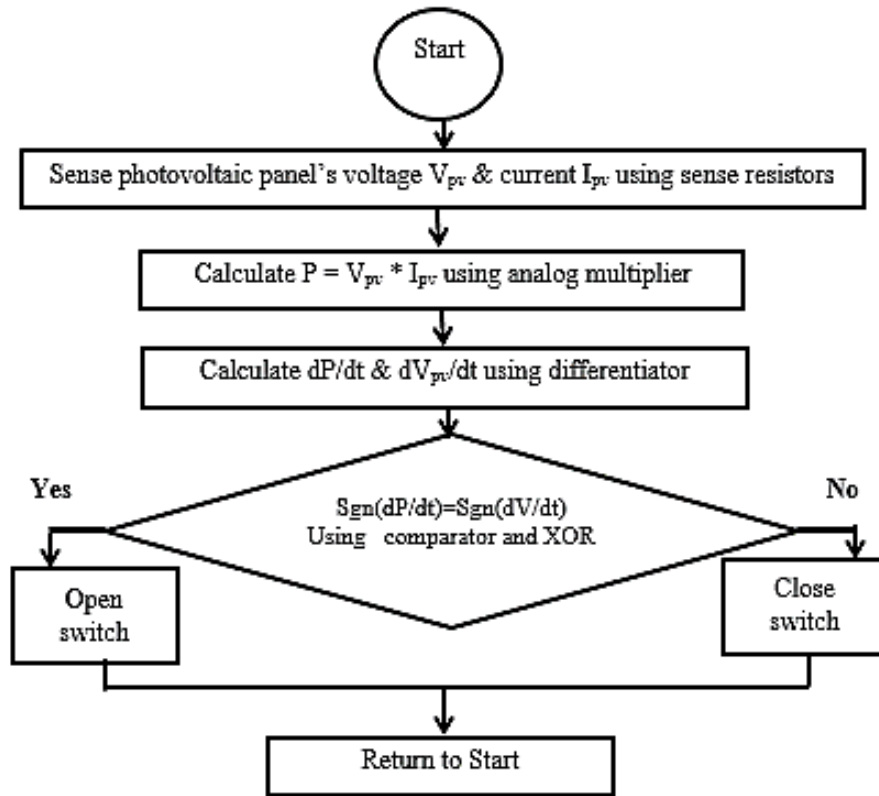


Figure 2.4 Ripple Correlation Control Flowchart

Fibonacci Series-Based (FSB) MPPT uses the numerical pattern of the Fibonacci sequence to generate adaptive step sizes when searching for the maximum power point. By increasing or decreasing the step size according to Fibonacci progression, the algorithm achieves faster convergence during large power deviations and finer adjustment near the peak. The flowchart for FSB MPPT is shown below [46]:

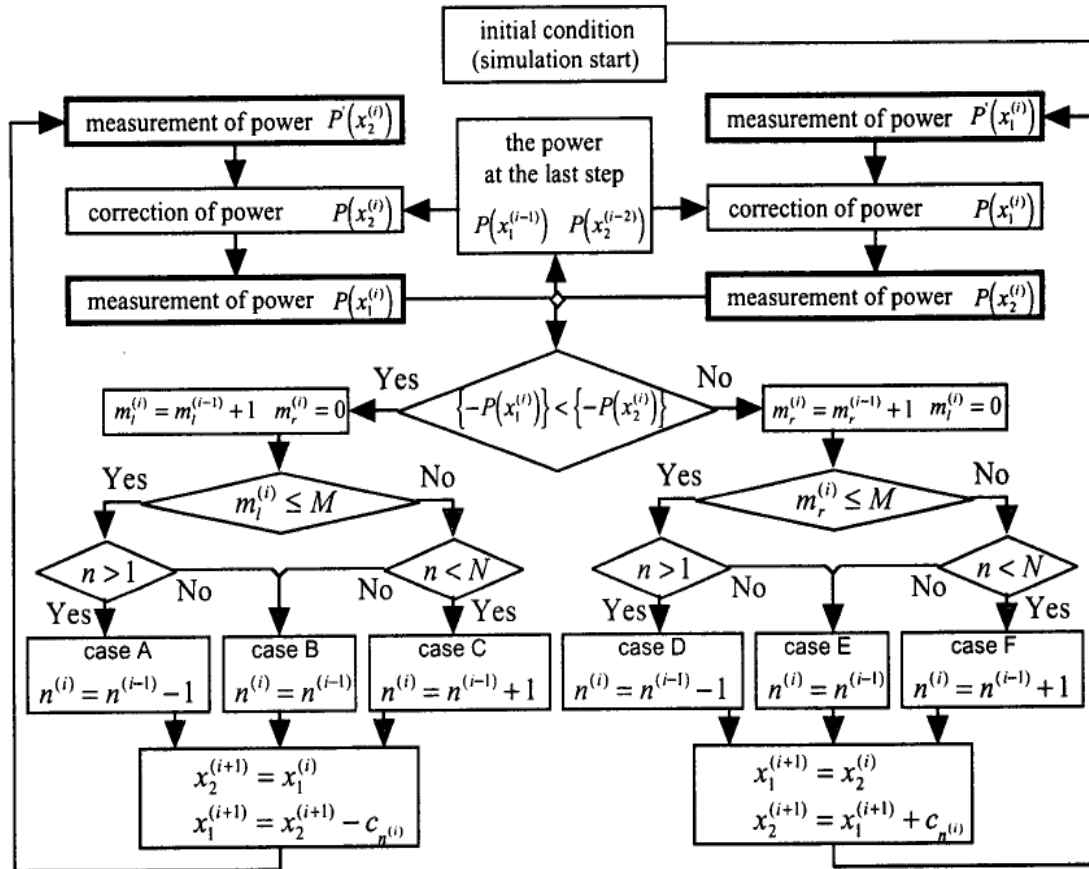


Figure 2.5 Fibonacci Series-Based MPPT Flowchart

Fuzzy Logic uses values between 1 and 0, unlike conventional binary systems. Using this algorithm for MPPT allows a more adaptive approach to decisions based on the changes of power and voltages. A sample flowchart for a Fuzzy Logic MPPT algorithm is shown below [47]:

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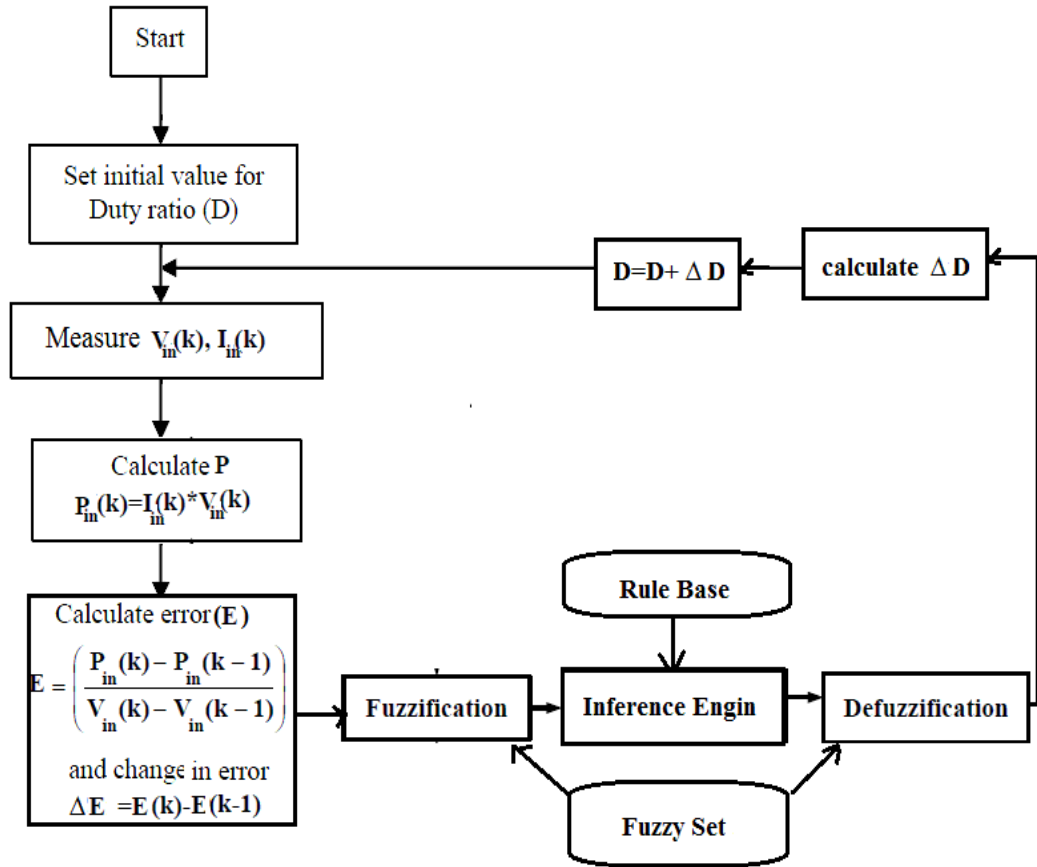


Figure 2.6 Fuzzy Logic Algorithm Flowchart

An Artificial Neural Network (ANN) MPPT uses a trained neural network to estimate the optimal operating voltage or duty cycle that maximizes PV power, based on inputs such as voltage, current, irradiance, and temperature. The flowchart for ANN MPPT is shown below [48]:

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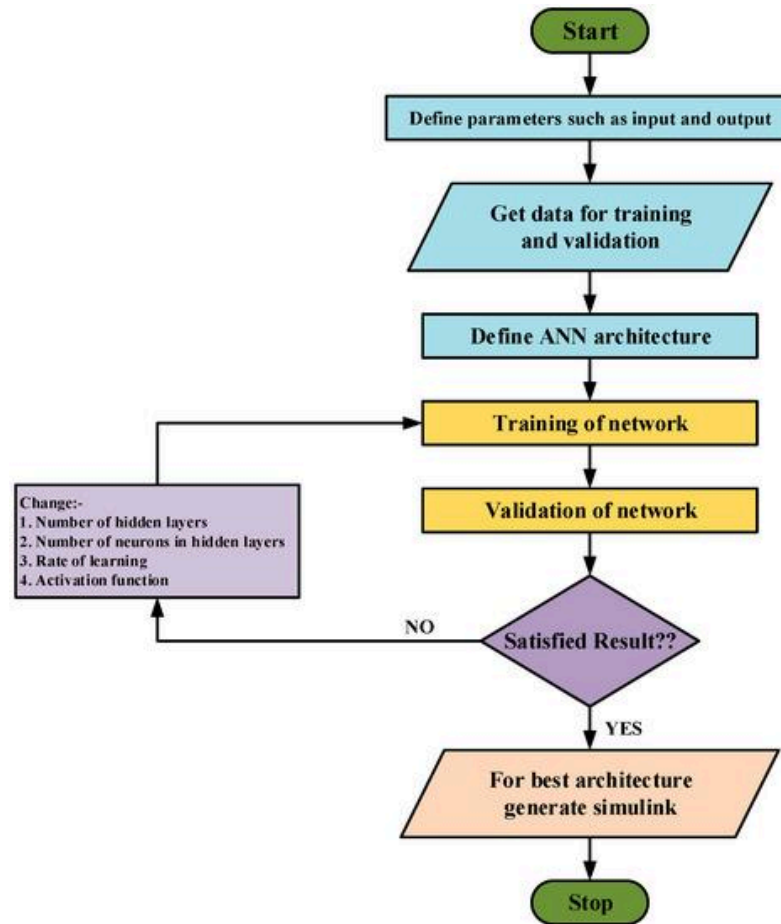


Figure 2.7 Artificial Neural Network MPPT Flowchart

Particle Swarm Optimization (PSO) MPPT uses a population of candidate solutions (“particles”) that move within the search space to find the operating point that yields maximum PV power. Each particle updates its position based on its own best performance and the group's best performance, enabling fast global convergence even under partial shading. The flowchart for PSO MPPT is shown below [49]:

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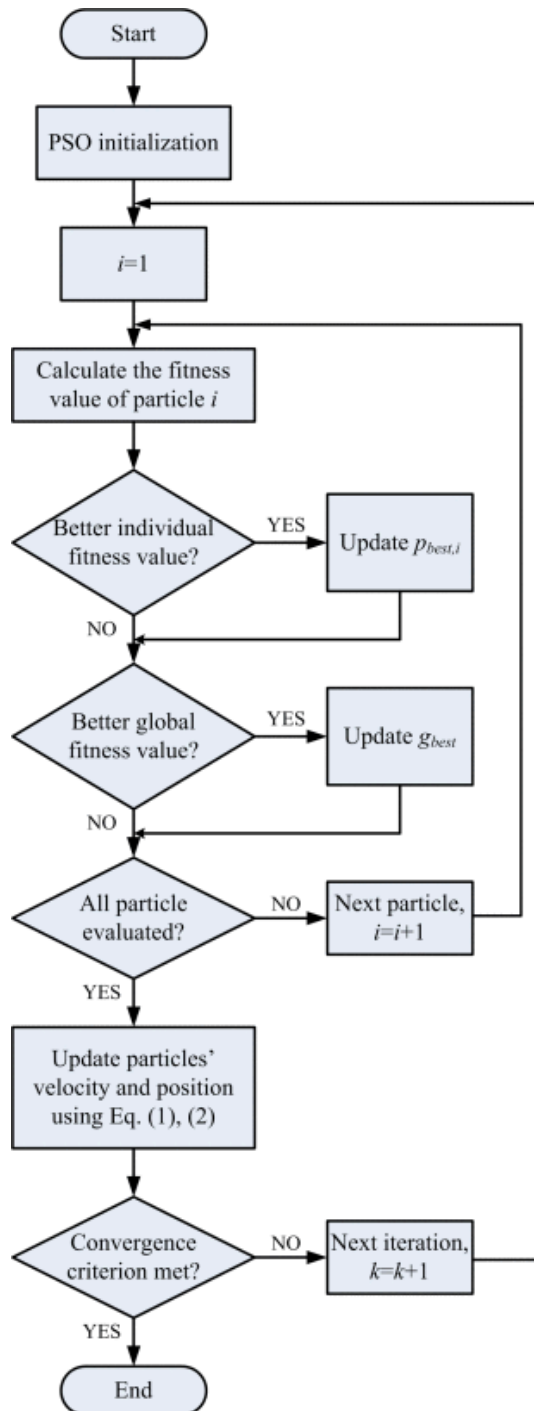


Figure 2.8 Particle Swarm Optimization MPPT Flowchart

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Genetic Algorithm (GA) MPPT uses evolutionary optimization through selection, crossover, and mutation to find the operating point that gives the maximum PV power. It evaluates a population of candidate voltages or duty cycles and gradually improves them until the best solution emerges. The flowchart for GA MPPT is shown below [22]:

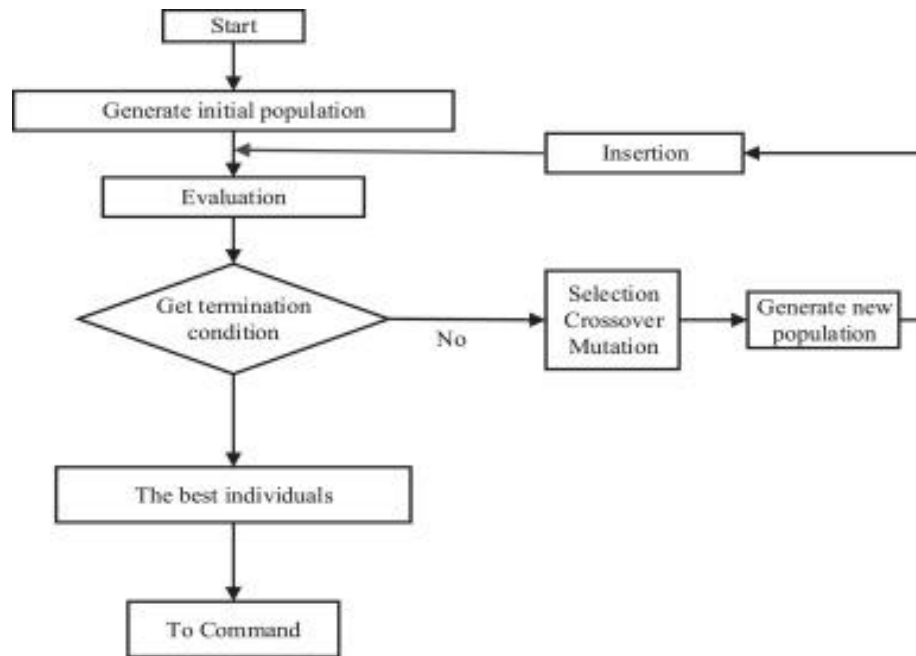


Figure 2.9 Genetic Algorithm MPPT Flowchart

Grey Wolf Optimization (GWO) MPPT mimics the hunting and leadership hierarchy of grey wolves to locate the operating point that maximizes PV power. The algorithm updates candidate solutions by modeling encircling, tracking, and attacking behaviors, enabling it to search globally while still converging efficiently. The flowchart for GWO MPPT is shown below [22]:

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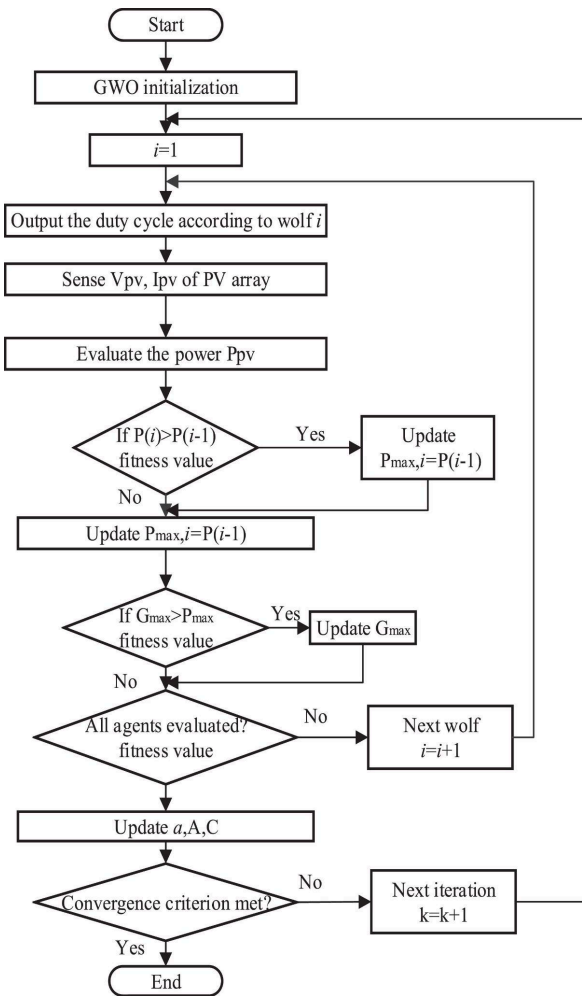


Figure 2.10 Grey Wolf Optimization MPPT Flowchart

To start a simulation run, the irradiance (measured in W/m^2) is input into the system before clicking the run button. Other factors can also be included to simulate varying conditions for the MPPT algorithm to adjust to. Voltage, current, and power graphs will be shown in another window after running the simulation.

2.2.3.2 Simulation Tools and Platforms used in PV systems

Simulating MPPT requires simulation software that models, analyzes, and optimizes PV system performance, which can be extremely useful before hardware

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implementation. Commonly used simulation software includes MATLAB/Simulink, PSIM, and PVSyst, due to the flexibility and features they offer. MATLAB/Simulink has been the most dominant among these three because of its ability to display detailed block diagram models and to model converters dynamically. One downside of MATLAB/Simulink is its slower simulation speed, but it is the most robust of the 3 simulation software [50]. PVSyst focuses more on system-level design and long-term energy harvest of PV systems under varying environmental conditions. Both MATLAB/Simulink and PVSyst can be used to complement each other since MATLAB/Simulink focuses more on being able to implement and test MPPT algorithms under rapidly changing weather conditions, while PVSyst can provide a more comprehensive evaluation of the PV System by analyzing system losses, long-term energy yield, and overall performance [51]. PSIM software excels in its simulation speed, transient waveform analysis, and converter efficiency calculation. Compared to MATLAB/Simulink, PSIM is simpler but less feature-rich. Nonetheless, all 3 simulation software packages can complement each other. MATLAB/Simulink can be used to simulate MPPT algorithms and assess transient response, tracking efficiency, and algorithmic oscillations. PSIM can be used for its faster transient simulations to evaluate waveforms, and PVSyst can analyze the controller's long-term yield and system losses [52].

2.2.3.3 Data gathering sources for simulation

Data needed for the simulation requires the solar irradiance of the area and weather conditions, such as temperature and partial shading. Sites that offer this data in the Philippines are Solcast and PAGASA. PAGASA can provide data on solar irradiance

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(W/m²), minimum and maximum ambient temperatures (°C), relative humidity, and wind speed. To request raw data on PAGASA, parameters must be specified, including the station location, date range, and data parameters such as solar irradiance, humidity, and temperature. Previous data is also available on their site, but recent data is very likely not to be included. Data requests may take 7-15 working days (3-4 weeks of processing time) to be fulfilled, depending on how difficult it is to obtain the data, which is determined by the type of data and the length of the time frame [53]. Solcast offers a wider range of parameters for requests, and its data comes from satellites. Solcast can be accessed for free with an account, but a free account has a limited number of requests available. For MPPT purposes, Solcast can provide Global Horizontal Irradiance (GHI), Direct Normal Irradiance (DNI), Diffuse Horizontal Irradiance (DHI), Air temperature (AIR_TEMP), Relative humidity (RELATIVE_HUMIDITY), and cloud opacity (CLOUD_OPACITY) [54].

2.3 Relevant Theories

This section presents the theoretical foundations of the study, highlighting key principles and concepts in maximum power point tracking (MPPT), photovoltaic (PV) energy conversion, and adaptive MPPT control algorithms.

2.3.1 Photovoltaic (PV) Conversion Theory

2.3.1.1 PV Systems Principles and Formula

Photovoltaic (PV) systems convert sunlight into electrical energy. It consists of PV cells arranged in a solar panel. These PV cells are semiconductor devices that use the

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photovoltaic effect to generate electric current when exposed to sunlight, operating as a p-n junction diode. As light strikes the semiconductor, photons will excite the electrons across the bandgap, which will generate free charge carriers that will flow through the circuit to generate current. Electrons will go toward the n-side and the holes on the p-side. The generation rate of electric current depends on numerous factors, including solar irradiance, semiconductor bandgap, surface reflectance, carrier mobility, temperature, and recombination rate. The relationship between voltage and current in a PV cell is expressed by the I-V and P-V characteristics, which vary with the factors mentioned earlier. An Ideal PV cell is represented by the formula:

$$I = I_{pv,cell} - I_{0,cell} \left[\exp \left(\frac{qV}{akT} \right) - 1 \right]$$

where $I_{pv,cell}$ is the current generated by the incident light (it is directly proportional to the Sun irradiation), $I_{0,cell}$ is the reverse saturation or leakage current of the diode, q is the electron charge ($1.60217646 \times 10^{-19}$ C), k is the Boltzmann constant ($1.3806503 \times 10^{-23}$ J/K), T (in Kelvin) is the temperature of the p-n junction, and a is the diode ideality constant.” [33].

A solar cell’s simplest equivalent circuit consists of a current source in parallel with a diode. A series resistance is also included, representing the losses within the solar cell and its connections. This can shape the I-V curve between the maximum power point and the open-circuit voltage to provide a more realistic representation of PV behavior. The diode here defines the I-V characteristics of the solar cell. The diode quality factor, n, can also affect the I-V curve and be adjusted to account for real-world operating variations [55].

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2.3.1.2 Performance Characteristics of PV Systems

The performance of PV systems is affected by the electrical characteristics of their solar cells and by the environmental conditions under which they operate. Performance evaluation is determined by both the I-V and P-V curves, which describe the relationships among current (I), power (P), and voltage (V). The I-V curve describes the operating point of the cell under given conditions, and the P-V curve describes the maximum power point (MPP), where the PV system delivers the highest power. Open-circuit voltage (V_{oc}) and short-circuit current (I_{sc}) are key parameters in a PV system and are used to plot the P-V and I-V curves. V_{oc} refers to the maximum voltage of the system, while I_{sc} is the maximum current. As solar irradiance increases, the I-V and P-V curves shift upward, resulting in higher I_{sc} . This translates to higher overall power generation in the PV system, with only a slight increase in open-circuit voltage. As the temperature rises, V_{oc} decreases significantly, while I_{sc} increases slightly. This would result in a lower overall MPP and decreased PV system efficiency. With both effects combined, the MPP would continue to shift and dynamically adjust the operating points of the PV system. To minimize the shifting of MPP, Maximum Power Point Tracking (MPPT) techniques are used [55].

2.3.2 Maximum Power Point Tracking (MPPT)

2.3.2.1 MPPT Importance in PV Systems

PV system output is represented by the I-V curve. The highest point of the I-V curve is called the Maximum Power Point (MPP), which can change depending mainly on solar irradiance, temperature, partial shading, and other factors. When this value

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changes, the total output power of a PV system changes, since the MPP is its operating point. To track the change of MPP, Maximum Power Point Tracking (MPPT) algorithms are implemented in PV systems. These algorithms maximize energy harvest to ensure that the change in the P-V characteristic curve is as close to zero as possible. MPPT algorithms can change the duty cycle of the converters. The output current and voltage are measured to match the source impedance to the load to track MPP. This results in a better efficiency and output power in PV systems [18].

Commonly used methods of MPPT algorithms are Constant Voltage (CV) Method, Open Circuit Voltage (OCV) Method, Short Circuit Current (SCC) Method, Perturb & Observe (P&O) Method, Incremental Conductance (IncCond) Method, and Fuzzy Logic Method. These techniques can generally be arranged from least to most complex in terms of control precision and computational requirements, with the first three relying on fixed reference parameters and the latter using adaptive control strategies that yield higher-accuracy outputs under varying environmental conditions [56].

2.3.2.2 Classification of MPPT Techniques

There are four classifications of algorithms used for MPPT: Classical MPPT control techniques, Intelligent MPPT control techniques, Optimization techniques, and hybrid techniques. Classical MPPT control techniques include, but are not limited to, Perturb and Observe (P&O), Incremental Conductance (IncCond), and Constant Voltage (CV) or Fractional Open Circuit Voltage Method. Classical MPPT control algorithms are typically less computationally intensive, less accurate during rapid irradiance and weather changes, and more prone to oscillations upon implementation. In Classical MPPT, the effects of partial shading are neglected. Using Classical MPPT control techniques tends

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to result in poorer performance. The P&O method monitors power changes and adjusts the duty cycle accordingly. This process is repeated until the change in power over the change in voltage is equal to zero. The CV method assumes a fixed MPP voltage, which adjusts the PV output by changing the duty cycles relative to this value. Incremental Conductance [18].

Intelligent MPPT control techniques include Fuzzy Logic Control (FLC), Artificial Neural Networks (ANN), Sliding Mode Control (SMC), Fibonacci Series-Based MPPT (FSB MPPT), and Gauss Newton Technique (GNT). These are more advanced than classical MPPT control and use machine learning to optimize the MPP. These techniques yield higher accuracy and faster tracking, even under rapid weather and irradiance changes, compared to Classical MPPT control techniques [18].

The last 2 classifications, which are optimization techniques and hybrid techniques, are more advanced compared to the first 2 classifications. Hybrid techniques combine 2 algorithms to address the weaknesses of each other in a single new algorithm. Due to their complexity, these algorithms will not be used in the study, but they are a great idea for future optimizations, as they offer the best results when used for MPPT [18].

Optimization techniques are metaheuristic and typically based on and designed around animals. Examples of these include the Artificial Bee Colony (ABC) and Particle Swarm Optimization (PSO). ABC is based on how bees find their food: the maximum power is generated at the food, and the duty cycle is the food's position. PSO is based on the behavior of a flock of birds or schooling fish, which is referenced to designing the algorithm based on the amount of food (this is the power output for MPPT), the position

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of the swarm (duty cycle), and varying food availability (variables such as partial shading and weather in MPPT) [18].

2.3.3 Classical MPPT Algorithms

2.3.3.1 Perturb and Observe Method (P&O)

P&O is widely used due to its simplicity and low cost of implementation. P&O works by perturbing the PV array terminal voltage and observing its changes in output power. If the power increases, the perturbation continues to adjust in the same direction; otherwise, it reverses the perturbation's direction. Although this method is popular, it suffers from low efficiency and is slower than other algorithms. At steady state, P&O oscillates around the maximum power point (MPP), resulting in power losses. P&O also lacks the ability to account for partial shading conditions when tracking the MPP. [22]

P&O calculates with the following formulas:

$$P(t) = U(t) \times I(t); \Delta = P(t) - P(t - 1)$$

Wherein:

$P(t)$ is the PV power, $P(t-1)$ is the previous maximum power, $U(t)$ is the control signal, and Δ is the difference between $P(t)$ and $P(t-1)$. When $\Delta = 0$, the MPP is already tracked. When $\Delta \neq 0$, the control signal, $U(t)$, is adjusted. When the difference is positive, if $U(t) > U(t-1)$, $U(t)$ is incremented, it is decremented. The logic reverses if the difference is negative. [57]. Given its status as the commercial standard for many existing inverters, it serves as the benchmark to quantify the performance gains of the proposed MPPT algorithms. Moreover, P&O is one of the best options when budget is limited, and the efficiency loss is negligible.

2.3.3.2 Incremental Conductance (IncCond)

IncCond's main idea is to "compare the ratio of derivative of conductance with the instantaneous conductance". This is derived from the formula: $\frac{dP}{dV} = \frac{d(VI)}{dV} = I + V \frac{dI}{dV} = 0$ where at MPP, the derivative of power with respect to voltage ($\frac{dP}{dV}$) is zero. Rearranging the formula gives us a better look at the IncCond method: $-\frac{I}{V} = \frac{dI}{dV} \approx \frac{\Delta I}{\Delta V}$. From this formula, the increments of PV current (ΔI) over increments of PV voltage (ΔV) shows the conditions of IncCond when compared to the derivative of current with respect to voltage. When $\frac{dI}{dV} = -\frac{I}{V}$, the system is at MPP. When $\frac{dI}{dV} > -\frac{I}{V}$, the system is at the left side of MPP. When $\frac{dI}{dV} < -\frac{I}{V}$, the system is at the right side of MPP. Using this relationship, we can get the marginal error (ε) with the formula: $\frac{dI}{dV} + \frac{I}{V} = \varepsilon$ wherein $\frac{I}{V}$ is the instantaneous conductance and $\frac{dI}{dV}$ is the incremental conductance [23].

Compared to Perturb and Observe (P&O), which is a method with simpler and lower computational complexity, the IncCond method enables the system to move precisely toward the MPP without oscillating around it. P&O continuously perturbs the operating voltage which causes the oscillations in the algorithm while IncCond precisely moves toward the MPP and will hold the value without oscillations once $\frac{dI}{dV} = -\frac{I}{V}$ is true [58]. IncCond is used as a benchmark along with P&O for multiple MPPT studies. In comparison to other classical algorithms, IncCond shows the most remarkable results and can perform under rapidly changing atmospheric conditions. By comparing the more advanced algorithms to IncCond, the study can determine if the added complexity

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provides a statistically significant improvement particularly under the rapidly changing atmospheric conditions typical of Pampanga [28].

2.3.4 Control-Theoretic / Model-Based Algorithms

2.3.4.1 Sliding Mode Control (SMC) MPPT

The SMC MPPT is mainly used for nonlinear systems. For SMC, the following must be fulfilled:

$$P = VI = I^2 R; \frac{dP}{dI} = 0 \text{ which means } 2R + I \frac{dR}{dI} = \sigma = 0$$

Wherein σ is the sliding surface, R is the resistance, V is the voltage, I is the current and P is power. The duty ratio denoted as δ controls the converter to track MPP will be updated according to the equation below:

$$\delta(i + 1) = \begin{cases} \delta_i + \Delta\delta \text{ for } \sigma > 0 \\ \delta_i - \Delta\delta \text{ for } \sigma < 0 \end{cases}$$

$i+1$ indicates the i th iteration of changing the duty ratio and $\Delta\delta$ is the change in duty ratio. [18].

SMC is known to be very stable due to its steady state oscillations being very minimal. This can be seen in a study by Mostafa et al. (2020) wherein the steady state of voltage, power, and current graphs is almost like a straight line compared to the oscillations of using an algorithm such as P&O. It also has a very short transient time of 9ms compared to the 220ms of using P&O. The study aims to verify if SMC's rapid tracking capability can minimize the energy lost during the transition periods of weather compared to other algorithms to smooth the power fed into the grid [44].

2.3.4.2 Ripple Correlation Control (RCC) MPPT

Power converters in a PV array impose voltage and current ripples during operation on the PV array. This adds ripples into the system, causing power losses into the system. Ripples induce stress into hardware components wherein high levels of ripples can cause equipment to wear down quickly. Despite being a power loss, RCC MPPT takes advantage of these ripples by using ripples as information to track the maximum power point (MPP) [28]. RCC relies on the fact that $\frac{dP}{dV}$, the derivative of power with respect to the voltage, is zero at the MPP, positive to the left, and negative to the right. When the voltage or current rises and power rises, the operating point is below the MPP. On the other hand, when the voltage or current rises but power decreases, the operating point rises above the MPP. RCC uses the equation below:

$$\frac{dP}{dt} \times \frac{dV}{dt} = 0 \text{ or } \frac{dP}{dt} \times \frac{dI}{dt} = 0$$

Where $\frac{dP}{dt}$ is the derivative of power with respect to the time, $\frac{dV}{dt}$ is the derivative of voltage with respect to the time, $\frac{dI}{dt}$ is the derivative of current with respect to the time [18].

Due to RCC using the ripples rather than adjusting against the ripples, this MPPT shows very low steady state oscillations and has been proven to be better than the benchmarks P&O and IncCond. Its behaviour of using the ripples also minimizes the hardware stress the system experiences, making it a great choice for long-term operations. This allows for fast tracking speed even under rapid irradiance while maintaining minimal steady-state power losses [59].

2.3.4.3 Fibonacci Series Based (FSB) MPPT

Tracking the maximum power point (MPP) across the entire search region can take a lot of time to calculate. Unlike other algorithms, FSB MPPT focuses on a certain range of operations to track MPP. FSB MPPT makes use of the minimum and maximum voltage of the PV system as test points to check which direction the change of operation must take place. It utilizes the fibonacci sequence and uses the following equation to track MPP:

$$R_{(n+2)} = R_{(n+1)} + R_n ; (n = 1, 2, 3.... \text{ and } R_1 = R_2 = 1)$$

In this equation, R represents the points of the PV curve used to track MPP and n is the iteration count of steps used in determining the step size of adjustment [18].

This algorithm is selected for the study for its search efficiency. For large arrays, scanning the entire voltage range is time-consuming. A study by Meng et al. (2025) demonstrates the use of this MPPT for their study. It shows that FSB MPPT has a 110ms response speed, medium complexity in implementation, has a $\pm 0.8\%$ tracking accuracy error, and is suitable for rapid irradiance scenarios [60].

2.3.5 Intelligent & AI-Based MPPT Algorithms

2.3.5.1 Fuzzy Logic Control (FLC)

Fuzzy sets are elements with values ranging from 0 to 1, rather than just 0 and 1. This allows representation of uncertainties and approximation in values. This is an extension of binary logic wherein we now introduce levels to system states such as “low”, “medium”, and “high”, allowing reasoning on complex and imprecise environments [21].

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The process of how a Fuzzy Logic works involves a nonlinear mapping from an input feature vector to a scalar output. The vector output of this system will be represented as a collection of multi and single-output systems. Fuzzy Logic systems consider a huge amount of possibilities which lead to a large amount of different mappings. Because of this, Fuzzy Logic can be incorporated in a lot of specific applications. Although there are a lot of possibilities in the usage of the Fuzzy Logic system, it does not require thorough understanding. All that's important is to choose the proper method or technique for which algorithm it is to be used for [21].

Fuzzy Logic has been proven to reduce oscillations, shorten settling time, and reduce power loss in a PV system under varying irradiance, temperature, and shading conditions. This makes this algorithm generally more effective in weather conditions that would fluctuate rapidly throughout one day [61].

2.3.5.2 Artificial Neural Network (ANN) MPPT

ANN MPPT is a soft computing approach inspired by how the human brains work in processing information. A network is created using “neurons” which functions as the decision makers of the network. There are layers used in the ANN approach: the input layer, the hidden layer, and the output layer. The input layer receives the atmospheric data (solar irradiance and temperature) and PV parameters (I_{sc} and V_{oc}), the hidden layer processes these signals, and the output signal generates the control signal (duty cycle) for the converter. A diagram is shown below to easily visualize the ANN layers along with the “neurons” [18]:

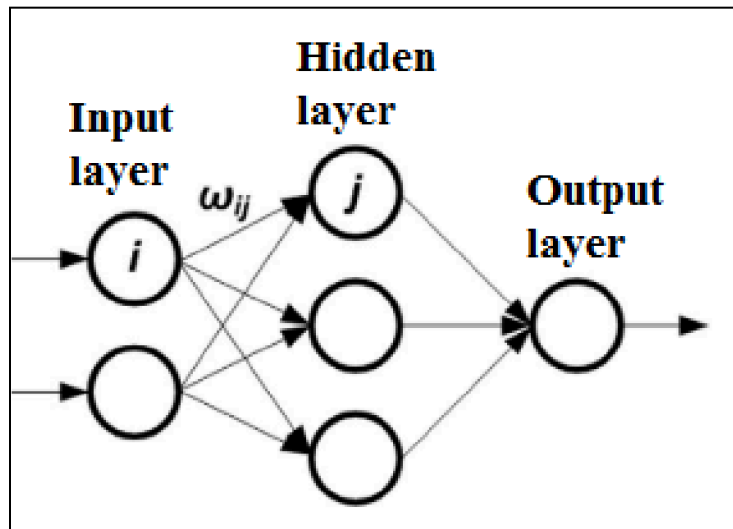


Figure 2.11 ANN Layer Visualization

Although ANN is a very smart algorithm, the downside of using this is that these algorithms must be specifically trained for each panel. This means that changing the solar panel hardware specifications will render the existing ANN algorithm used for the current solar panel obsolete. This also applies to if the solar panel hardware degrades over time wherein ANN cannot optimize to track MPP properly [18].

ANN is used for the study since it is a highly intelligent algorithm that offers stable steady state oscillations and high efficiency compared to conventional methods. It is an algorithm that properly represents the use of machine learning in tracking the MPP. ANN offers a search-free type of tracking once trained properly meaning it can easily jump or search for the peak. This makes it a great algorithm for rapid irradiance under tropical climate [22].

2.3.6 Metaheuristic / Optimization-Based MPPT Algorithms

2.3.6.1 Particle Swarm Optimization (PSO) MPPT

The core idea of PSO is inspired by the social flocking behaviour of birds or schooling of fish to solve non-linear optimization problems. In MPPT, PSO finds the best solution to track MPP by utilizing a population of search agents called “particles” that represents a solution such as the correct duty cycle for adjusting power towards MPP. The particle then dynamically adjusts its position based on its own search experience and the collective success of the entire swarm. The general equation for the new position of a particle in PSO is [22]:

$$x_i^{k+1} = x_i^k + v_i^{k+1}$$

Where i represents the specific particle, k is the k th iteration, x_i represents the position, v_i represents the velocity factor. Each particle moves at a specific velocity wherein the velocity factor for each particle is calculated by the following equation [22]:

$$v_i^{k+1} = wv_i^k + c_1r_1(P_{besti} - x_i^k) + c_2r_2(G_{besti} - x_i^k)$$

The c_1 and c_2 are the acceleration constants, w represents the inertia weight, r_1 and r_2 are random values between $[0, 1]$, P_{besti} is the best power value found by a specific particle so far, and G_{besti} is the best power value found by any particle in the entire swarm. The equation can be broken down into the following parts: wv_i^k , $c_1r_1(P_{besti} - x_i^k)$, and $c_2r_2(G_{besti} - x_i^k)$. wv_i^k is responsible for creating a smooth search path for the particles. The $c_1r_1(P_{besti} - x_i^k)$ represents the personal experience of each

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particle. Lastly, $c_2 r_2 (G_{besti} - x_i^k)$ is the distance of the particle from the entire swarm attracting the particle toward the global leader. Adding these altogether gives us the resultant velocity factor [22].

PSO is known for its easy implementation and simple structure. Once the particles find MPP, they settle and don't have to move much when they are on the right spot. However, the initial computation speed of this algorithm suffers from slow tracking speed due to the calculation demand of the entire population of particles. This is usually the case if the initialization values aren't carefully selected making it the crucial step in creating a successful PSO algorithm [22]. Despite the slow tracking speed, it has been seen that PSO MPPTs perform excellently under varying weather and partial shading conditions ideal for a tropical region such as the Philippines. The study shall also aim to see if the algorithm's immunity to drift and steady state oscillation results in a higher net energy yield despite the losses due to slow tracking speed [18].

2.3.6.2 Genetic Algorithm (GA) MPPT

GA MPPT is popular for using Darwin's principle of "survival of the fittest" as its computing technique. Unlike conventional derivative based optimization methods such as IncCond wherein the objective functions must be differentiable and continuous, GA can handle non-linear functions with discontinuities or piecewise segments. The algorithm operates on a "population" composed of "individuals" which are referred to as the potential solutions with each individual representing as a candidate duty cycle. The individuals are optimized using a "fitness function". There are three fundamental steps in the GA MPPT: selection, crossover and mutation [22].

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The selection process chooses individuals to act as the parents for the next generation. The selection process is governed by the fitness function which in the context of MPPT is the PV output power. Individuals which yield higher power will be more likely to be chosen to ensure beneficial traits (optimal duty cycles) are preserved and propagated.

The crossover process is the primary mechanism to exchange genetic information between two selected parent chromosomes. In the context of MPPT, this involves the combination of duty cycle values of two parents to form an offspring which inherits the best attributes of both parents to land closer to the MPP.

Mutation is a random and low probability alteration to the genes of offspring. This random encounter ensures that the genetic pool of the population is diverse. In the context of MPPT, the population could be stuck below the MPP until this mutation occurs by randomly altering a duty cycle.

For better visualization, the flowchart of a GA MPPT is shown below [22]:

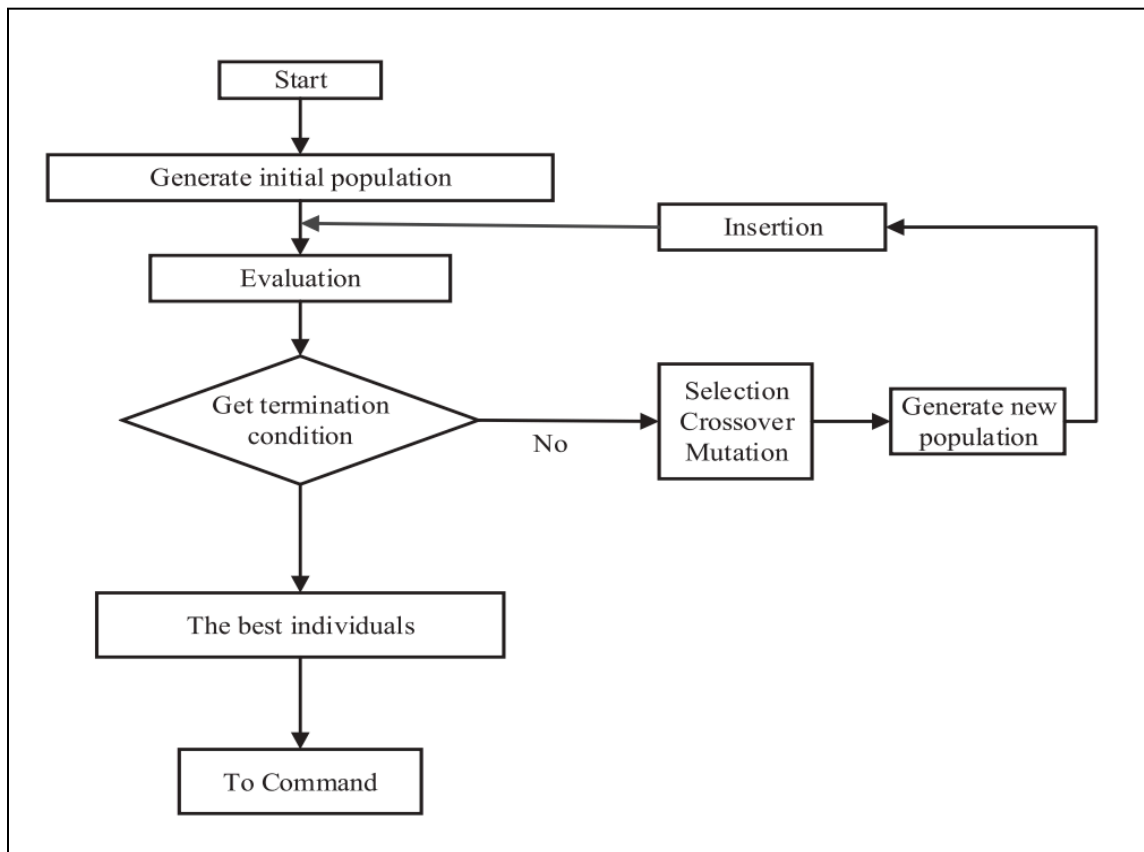


Figure 2.9 Genetic Adaptation MPPT Flowchart

The GA algorithm is also used to train other MPP algorithms such as ANN and can optimize them to track quickly and accurately. Moreover, one of the advantages of using GA is its capabilities to optimize the specific setup that gives the lowest cost for the highest energy [18]. With the algorithm being cost efficient, has a robust search mechanism, and stability, it is a good choice for MPPT usage on tropical weather conditions in the Philippines. The study also uses GA to determine if a more randomized, evolutionary algorithm provides better reliability in finding the MPP during the most complex shading scenarios observed at the site.

2.3.6.3 Grey Wolf Optimization (GWO) MPPT

GWO MPPT is a bio-inspired meta-heuristic algorithm inspired by the leadership hierarchy and hunting mechanism of grey wolves in nature. In the context of MPPT, the position of the wolf is treated as the duty cycle and the PV output power is the prey. A social hierarchy is established in the GWO MPPT composed of the alpha (α), beta (β), delta (δ), and omega (ω). The alpha represents the leader with the solution for the best power found so far, the beta represents the second-best solution, the delta represents the third-best solution and the omega is the rest of the population or the followers of the pack. The GWO updates positions based on the top three leaders alpha beta and delta. Their average position is triangulated to avoid reliance on a single data point for a smoother search trajectory. The general model of the GWO is shown as the equation as follows: [22]

$$D = |C \cdot X_p(t) - X(t)|; X(t + 1) = X(t) - A \cdot D$$

Where D is the distance, X_p is the position of the prey, X is the wolf's positions, and A and C are coefficient factors that decrease linearly over iteration to transition the pack to search for MPP.

GWO is proven to be more efficient than conventional algorithms in terms of accuracy, robustness and convergence time. This demonstrated a high energy yield and tracking accuracy. GWO was selected for the study as it is often compared to the chosen PSO which is already an efficient algorithm. Compared to PSO, GWO has fewer control variables. The study can observe if faster convergence speeds and reduced computational burden can achieve better or same performance as well [62].

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2.3.7 Performance summary of algorithms

To summarize the algorithm's performances based on the previous literatures, a table is formed below:

Algorithm	Efficiency	Stability	Speed	Complexity	Description
Perturb and Observe (P&O)	Low	Not Stable	Slow	Low	Widely used as a cheap and simple option. Often used by studies as a benchmark [22].
Incremental Conductance (IncCond)	Average or High	Stable	Slow or Average	Average	Offers better yield compared to P&O, also used as a better benchmark [28].
Sliding Mode Control (SMC)	High	Very Stable	Very Fast	High	SMC can reach steady state very quickly with minimal steady state oscillations [45].
Ripple Correlation Control (RCC)	High	Very Stable	Fast	Low	RCC utilizes the PV's ripples, thus introducing no external interference to the system [60].
Fibonacci Series Based	High	Very Stable	Very Fast	Average	Limits the search to a

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(FSB)					certain range, which is great for large scale operations allowing for fast convergence [62].
Fuzzy Logic Control (FLC)	High	Very Stable	Average or High	High	Possess strong robustness and applicability to nonlinear systems, which excels under varying atmospheric conditions [58].
Artificial Neural Network (ANN)	High	Very Stable	Very Fast or Medium	Extremely High	Works similarly to the human brain and has a fast response and high steady-state accuracy [58].
Particle Swarm Optimization (PSO)	High	Very Stable	Fast	Average or High	Simulates the behaviour of swarms that can efficiently solve complex multi-peak P-V curves [58].
Genetic Algorithm (GA)	High	Very Stable	Fast	High	Uses Darwin's "survival of the fittest principle", searching for

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					the optimum using a population of individuals to find MPP [22].
Grey Wolf Optimization (GWO)	High	Very Stable	High or Very Fast	Low or Medium	Highly efficient algorithm based on mimicking the hunting hierarchy of wolves, offering remarkable convergence speed and tracking capability [63].

Table 2.1 MPPT Algorithms Performance Summary

2.3.8 Adaptive MPPT Techniques

2.3.8.1 Concept and Mechanism of Adaptive MPPT Techniques

MPPT Techniques can also be classified as adaptive which are conventional MPPT techniques but with modifications made for the system to self-adjust depending on the situation or location. These adaptive MPPT techniques use adaptive control theory as its foundation. Compared to the fixed parameters of conventional MPPT algorithms, adaptive control allows for continuous updates of the duty cycle depending on system feedback. For instance, a study by Ibrahim et al. (2023) used IncCond for their study but with a modification wherein a fuzzy self-tuning controller was added into the algorithm to adjust on varying weather conditions. The fuzzy self-tuning controller enabled the PV

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system to adjust the duty cycle based on varying weather condition compared to a standalone IncCond algorithm that adjusts until $\frac{dI}{dV} = -\frac{I}{V}$ is true. This adaptive MPPT was tested under varying climate scenarios which shows efficiency results of 99.80%, 99.76%, and 99.73% [63].

2.3.8.2 Adaptive P&O, IncCond and Fuzzy-based adaptive systems

Adaptive variants of P&O and IncCond are based on modifying fixed control parameters to improve the response time and reduce oscillations in the PV system. Adaptive P&O modifies the step-size by making it variable to increase the perturbation magnitude if the system is far from achieving MPP and decrease perturbation magnitude the closer it approaches MPP value. Adaptive IncCond is similar to how an adaptive P&O is modified by adjusting its incremental step size and implementing self-tuning elements such as the fuzzy self-tuning controller mentioned earlier [63]. These modifications retain the simplicity of P&O and IncCond but have significantly increased the performance of these algorithms, especially under varying weather conditions [64].

Fuzzy logic-based adaptive systems are often in hybrid with other MPPT techniques such as IncCond, PSO, and P&O as they are very effective in working as an auto-tune system. This is because fuzzy logic follows human reasoning, which means it can follow a more varied range making it very good at transitioning to MPP smoothly [65]. Adaptive MPPTs can get very complex and are not considered for the study but it is a great addition for future implementations.

2.4.Related Studies

The following presents international and local studies containing related materials for this study.

2.4.1 Perturb & Observe

Several studies have analyzed the performance, strengths, and limitations of the Perturb and Observe (P&O) Maximum Power Point Tracking (MPPT) method in photovoltaic (PV) systems, particularly under conditions relevant to tropical climates. P&O remains one of the most widely implemented MPPT techniques due to its structural simplicity and low computational requirements. Femia et al. [66] demonstrated that the algorithm effectively drives PV systems toward the maximum power point under steady atmospheric conditions, making it suitable for low-cost and small-scale applications. Similarly, Sera et al. [67] emphasized that P&O exhibits reliable tracking when irradiance is constant but becomes less effective under rapidly changing environmental conditions.

Studies concerning dynamic weather behavior, such as those typical of tropical regions, highlight several limitations of P&O. Wasynczuk [68] reported that fast irradiance fluctuations, often caused by rapidly moving cloud formations, can lead the algorithm to misinterpret natural power variations as responses to perturbations, causing incorrect tracking directions. Patel and Agarwal [69] further showed that P&O can become trapped in local maxima under partial shading, a common occurrence in regions with dense vegetation or irregular urban shading patterns. These findings are especially relevant for tropical climates where shading and weather transitions occur frequently.

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Temperature effects also play a critical role in P&O performance. According to Esham and Chapman [70], rising module temperatures significantly alter the PV power-voltage curve, yet the P&O algorithm lacks explicit temperature compensation. This results in moments where the algorithm incorrectly interprets temperature-induced power changes as irradiance variations, temporarily lowering tracking efficiency. Such temperature sensitivity is important in tropical regions, where high ambient temperatures and intense solar exposure coexist.

2.4.2 Incremental Conductance

The incremental conductance (IncCond) algorithm follows the MPP through the comparison of instantaneous and incremental conductance (dI/dV) of the PV array [23]. In its simplest configuration, IncCond employs a fixed voltage or duty-cycle increment, which again forces a trade-off between convergence rate and steady-state accuracy. Adaptive or variable-step IncCond algorithms have since been designed to overcome this. For instance, Li et al. (2016) proposed an adaptive IncCond algorithm in which the step size is automatically controlled through an adjustment coefficient. The method adjusted initial step sizes with fractional open-circuit voltage and then power slope-modulated step size to provide fast dynamic response and enhanced steady-state performance under large changes in irradiation [71]. Their experiments and simulations verified that this adaptive IncCond generated more energy than previous fixed-step systems, being effective and easy to implement.

Ali et al. (2021) further developed a fuzzy-logic-based variable-step IncCond MPPT. In their IEEE Access paper, they established fuzzy areas around the MPP and

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applied fuzzy rules for adjusting the duty-cycle step. Emulating a grid-integrated PV array under different sunlight, the fuzzy-IncCond method presented significant improvements: "the simulation results depict a worthy improvement in static and dynamic behaviors from that of the conventional IncCond method with the change of the environmental conditions. Moreover, it improves the output DC power and decreases the convergence time to steady state" [72]. Practically, this implies more power being delivered in a shorter time. Similarly, Ibrahim et al. (2023) designed an adaptive IncCond with a fuzzy-tuned PID controller (INC-FST). Under PLOS ONE simulations, they compared INC-FST to conventional IncCond, P&O, and an IncCond-fuzzy approach, for various climate scenarios. The INC-FST demonstrated outstanding tracking efficiency ($\approx 99.8\%$ of the optimum MPPT) in all scenarios, and it generated minimal overshoot and very rapid rise time, effectively exploiting PV power output [63].

Dandoussou et al. (2016) present additional evidence that adaptive IncCond performs well in practice. In their Simulink simulation of a 60 W crystalline silicon module with parameters updating every few seconds, the variable step-size IncCond algorithm achieved the highest reliability. It raised the daily peak power to 50.76 W (compared to 34.40 W for a fixed-parameter model) and lifted the daily energy harvested to 263 Wh (compared to 195 Wh) [73]. This indicates that under realistic tropical variability, a properly optimized adaptive IncCond can harvest much more energy than a non-adaptive scheme.

2.4.3 Sliding Mode Control

Recent studies highlight Sliding Mode Control (SMC) as a robust MPPT technique capable of handling the rapid irradiance and temperature fluctuations common in tropical regions such as the Philippines. Gursoy *et al.* demonstrate that both classical and higher-order SMC variants provide fast dynamic response and strong disturbance rejection, with super-twisting SMC showing reduced chattering and improved efficiency during rapid irradiance transitions [74]. Jamshidi *et al.* likewise report that improved SMC schemes, such as fuzzy-tuned and backstepping-based SMC, achieve faster settling time and lower steady-state oscillation than conventional controllers, indicating suitability for climates with frequent cloud transients [75].

Studies focusing on shading conditions, which are prevalent in tropical urban environments, further support the advantages of SMC. Hadj Salah *et al.* integrate a Cuckoo Search global optimizer with super-twisting SMC to reliably reach the global MPP under partial shading, achieving faster and more stable tracking compared to standalone MPPT approaches [76]. Similarly, recent advancements such as enhanced super-twisting SMC for dual-MPPT inverter systems and adaptive sliding-mode MPPT show strong performance under fast-changing irradiance and temperature profiles, conditions typical in equatorial climates [77], [78]. Collectively, the literature.

2.4.4 Ripple Correlation Control

The seminal experimental and theoretical treatment by Esham *et al.* demonstrated RCC's very fast dynamic response and parameter insensitivity, showing asymptotic convergence to the MPP and practical circuit implementations suitable for converters

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with inherent ripple [79]. Recent reviews and benchmarking studies confirm RCC as a strong candidate for environments with rapidly changing irradiance because it uses the converter's high-frequency information to react faster than many conventional low-frequency methods (e.g., fixed-step P&O) [80], and RCC has been successfully re-implemented in modern converter technologies (GaN, SiC) for dynamic applications [85]. Work done in Philippine and regional institutions also demonstrates RCC's practical viability: an analog 5 V RCC MPPT IC and related implementations were developed and tested for low-voltage PV/battery applications in the Philippines, indicating the method's real-world applicability in tropical deployment scenarios [45].

RCC's advantages for tropical climates, where cloud transients and temperature swings cause fast MPP shifts, include faster tracking and lower steady-state losses compared with many classic algorithms; some experimental and simulation studies report tracking efficiencies $\geq 95\%$ and convergence times on the order of 10–100 ms in fast-changing test conditions (exact figures depend on converter topology, ripple amplitude and filtering). [79], [82]. However, RCC also has limitations that are especially relevant in tropical field deployments: it requires a usable ripple source (so converters/filters must be designed to provide measurable ripple), can be sensitive to measurement noise, and may need tuning if the converter switching/frequency or PV capacitance changes substantially [79], [80]. Given the Philippines' combination of high irradiance, frequent cloud transients, and common partial shading in urban/vegetated sites, RCC (especially modern discrete-time or IC implementations) is well suited for

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rapid MPPT, provided the converter and sensing hardware are designed to preserve and measure the required ripple signals [79], [80], [81], [45].

2.4.5 Fibonacci Series Based

Improved/modified Fibonacci approaches have been proposed to cope with partial shading and fast irradiance changes by (a) splitting the P–V domain into subintervals, (b) re-initializing search intervals on abrupt illumination changes, and (c) applying a local Fibonacci search inside each interval to find local maxima before comparing for the global peak [83], [60]. These improvements reduce computational load versus brute-force global search, and several recent simulation/experimental studies report fast tracking: for example, a 2025 implementation using an improved Fibonacci linear search tracked the MPP in the test case within about 0.20–0.32 s after a large irradiance step and located the correct global peak under localized shading (simulated array power ≈ 37.25 kW in that scenario). Other benchmarking reviews indicate that modern Fibonacci-based and other interval-search MPPTs can achieve tracking accuracies often reported ≥ 95 –99% and typical convergence times on the order of 0.1–0.3 s in well-tuned implementations (exact figures depend on converter ripple, sampling, and array topology) [60], [84].

For tropical deployments (high irradiance, frequent cloud transients, and common partial shading in urban/vegetated sites like the Philippines), FSB MPPT has clear strengths and trade-offs. Its structured interval narrowing and multi-interval handling reduce the chance of settling on a local maximum under non-uniform illumination, and the algorithm's small-area search reduces computation and switching activity compared with exhaustive search methods [85]. Limitations noted in the literature include

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sensitivity to (a) accurate detection of array topology/illumination pattern (required to set subintervals), (b) measurement noise and converter filtering (which affect evaluation of candidate points), and (c) potential time penalty if many subintervals must be scanned, although improved variants mitigate this by adaptive reinitialization and hybridization with faster local methods [85], [84]. Overall, recent studies and reviews conclude that properly tuned/modified Fibonacci search MPPTs are practical and effective candidates for PV systems in tropical climates, often achieving sub-second convergence to the true global MPP while maintaining high tracking efficiency when implemented with suitable hardware and sampling rates [84], [60], [85].

2.4.6 Fuzzy Logic

Fuzzy-logic MPPT controllers use linguistic rules and membership functions to decide voltage or duty-cycle adjustments. They naturally manage the PV's nonlinear I–V curve without requiring a precise mathematical model [86]. Fuzzy MPPT always outperforms fixed-step algorithms in simulations and modeling research in stability and energy harvested. For instance, Robles-Algarín et al. (2017) compared a fuzzy controller with P&O on a 65 W PV system model and concluded that the fuzzy controller had faster settling time and much lower power loss and oscillation at steady state [61]. Similarly, Giurgi et al. (2022) modeled a fuzzy-logic MPPT and an IncCond MPPT during abrupt irradiance transitions; they presented that the fuzzy controller enabled the inverter to capture more than 99% of available PV power, while IncCond took only roughly 80%, with greatly minimized power oscillations using fuzzy control [87].

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It is supported by other research that fuzzy MPPT's advantages apply to tropical or highly changing climates. For example, Siddula (2020) used a hybrid fuzzy-IncCond algorithm and demonstrated through MATLAB simulations to track the MPP reliably under varying irradiance and significantly minimize output oscillations over conventional approaches [59]. Based on tabulated data from various papers, fuzzy controllers tend to demonstrate near-zero overshoot and low oscillation about the MPP. Zhani et al. (2024) compared fuzzy, IncCond, and P&O controllers under changing irradiance/temperature and concluded that "all obtained results confirm the superiority of the FLC method in terms of oscillation dynamics and response time...at start-up and to trace the MPP" [88]. In summary, fuzzy MPPT methods have the tendency to increase tracking accuracy and stability: they treat nonlinear variations smoothly with high tracking efficiency and low steady-state oscillations as opposed to conventional algorithms [89].

2.4.7 Artificial Neural Network

Recent experimental work demonstrated ANN-MPPT achieving tracking efficiencies $\approx 98.1\text{--}98.3\%$ and substantial reductions in tracking time versus classical methods; for example, a controlled experimental validation reported an ANN tracker with 98.16% MPPT efficiency and a shortest tracking time of 1.3 s (measured on a low-cost microcontroller setup under realistic transients). These results show ANNs can reduce oscillation around the MPP and maintain high extraction under dynamic conditions[90].

Contemporary ANN-MPPT studies also emphasize practical deployment issues important for tropical field use: (a) input selection (e.g., using PV current + temperature rather than pyranometer radiation to reduce sensor noise and latency), (b) lightweight

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network topologies and quantized models for microcontroller implementation, and (c) hybrid schemes that combine ANN prediction with a local fine-tuner (e.g., P&O or INC) to guarantee final convergence. Comparative simulation and field tests reported in recent reviews and papers indicate ANN and other ML-based MPPTs commonly achieve tracking accuracies in the mid-90s to high-90s % range ($\approx 95\text{--}99\%$) and improved robustness under partial shading and rapid cloud transients when properly trained and validated on representative datasets [90], [91].

2.4.8 Particle Swarm Optimization

Experimental work on real PV hardware shows PSO can substantially reduce energy losses during transient irradiance events: del Rio et al. implemented a PSO-based MPPT on a commercial PV panel and found PSO cut settling time from 31 s to 12 s and reduced transient energy loss from 31.96% to 9.72% compared with a P&O baseline in their test protocol [92]. Simulation and lab studies under static and dynamic partial shading also report PSO achieving much higher tracking accuracy than P&O/IC, e.g., a 2023 comparative study reported PSO tracking error $\approx 3.04\%$ vs P&O $\approx 21.79\%$ (static shading) and an overall PV-system efficiency gain of $\approx 4.66\%$ over P&O in dynamic shading tests; uniform-insolation tracking times reported in that work were on the order of 0.01–0.0125 s for common algorithms, with PSO slightly slower in raw iteration time but far more accurate at finding the GMPP [93].

Recent work focuses on reducing PSO's computational delay and improving convergence under fast cloud transients typical of tropical climates. Advanced and hybrid PSO variants (improved PSO / IQPSO / adaptive-PSO / swarm-size-reduction PSO)

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consistently report faster convergence and higher GMPP success rates than vanilla PSO. For example, IQPSO and other quantum/variant PSOs show markedly faster convergence and higher accuracy than classical PSO in multi-peak and irradiance-varying tests (benchmarked on ~2 kW test rigs), with the authors reporting conventional PSO as the slowest and least accurate among the compared biomimetic methods in their experiments [94]. Practical adaptive strategies, such as dynamically reducing swarm size or adaptive inertia/acceleration, have been shown (in simulations and preprints) to cut PSO convergence time by up to ~50–75% while keeping failure rates low, which makes these approaches promising for tropical PV installations where sub-second convergence is desirable to reduce energy loss from rapid cloud cover [95], [96].

2.4.9 Genetic Algorithm

Recent simulation and experimental studies show GA-based MPPT outperforms classic methods (P&O, INC) under partial shading: for example, a 2024 MDPI study comparing GA vs P&O in several shading cases reports that P&O produced only ~51.22% steady-state extraction in one heavily shaded case while the GA successfully located the GMPP for that case (MDPI case results). The same paper demonstrates GA methods recover substantially more power under nonuniform illumination and recommends GA for scenarios with frequent cloud transients and heterogeneous shading typical of tropical climates [96]. Experimental work that uses GA to optimize fuzzy-logic MPPTs also confirms practical gains: test-bench results validate improved accuracy and robustness under fast varying conditions compared with baseline controllers [97].

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Work within the last five years has focused on reducing GA computational cost and convergence time so the method is practical for field use in fast-changing tropical weather. Comparative benchmarking and metaheuristic surveys report that modern GA variants and GA-hybrids typically reach tracking efficiencies in the mid-to-high 90% range and can converge to the GMPP within sub-second to a few-second windows depending on population size, mutation rate and hardware (several studies report convergence targets within $\sim 0.5\text{--}2.0$ s for small arrays or test rigs) [98], [99]. Practical implementations for tropical deployments therefore recommend (a) using GA as a global search initializer (or periodically) and combining it with a faster local fine-tuner (P&O/INC) for steady-state maintenance, and (b) tuning GA parameters (population, generations, mutation) to trade off accuracy vs computational delay based on the expected rate of irradiance change for the site (e.g., Manila's frequent cloud transients).

2.4.10 Grey Wolf Optimization

Grey Wolf Optimization (GWO) and its improved/hybrid variants have repeatedly demonstrated very high MPPT performance in recent experimental and simulation studies. For example, the GWO-VINC hybrid reported steady-state tracking efficiencies of 99.80% (static shading simulation), 98.82% (dynamic shading simulation), 99.43% (experimental static shading) and 98.51% (experimental dynamic shading), while reducing average tracking time by $\sim 50.0\%$ (46.49% faster in simulation vs. plain GWO and 89.34% faster in experiment vs. VINC) and improving tracking accuracy by $\approx 1.9\%$ over the baseline algorithms [100]. An IGWO-VINC variant attained a 99.76% tracking efficiency in reported tests [101], and an Enhanced GWO (EGWO) variant reached

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99.65% tracking efficiency with reported convergence/steady-state times as low as ~ 3 s (EGWO), i.e., reductions in tracking time of up to $\sim 45\%$ compared with original GWO variants in the same study [102]. Hybridizations that combine GWO with PSO (GWO–PSO) or with local fine-search methods likewise show systematically faster convergence to the global MPP and higher steady-state MPPT efficiencies (typically $>98\text{--}99\%$) than classical P&O or INC under partial-shading / rapidly changing irradiance scenarios [103].

These algorithmic results are especially relevant for tropical countries like the Philippines, where high mean irradiance (national averages commonly reported in the $\sim 4.5\text{--}5.5$ kWh/m²/day range, with typical site values $\sim 4.9\text{--}5.1$ kWh/m²/day) coexists with rapid cloud transients, monsoonal variability and widespread partial-shading risks from vegetation and built obstructions [104], [105]. Philippine field studies (e.g., Muntinlupa City and regional assessments) emphasize that weather parameters (irradiance, temperature, humidity) produce measurable hourly and daily swings in PV output and that optimal tilt/operation choices vary seasonally, conditions under which the fast convergence and $>98\%$ steady-state efficiencies reported for GWO hybrids could significantly reduce real energy losses compared with conventional MPPTs [105], [106]. Nevertheless, while laboratory and co-simulation studies report sub-second to a few-second convergence and very high tracking efficiencies, there is a clear gap in field-validated GWO hybrid MPPT trials using Philippine site irradiance traces; local on-site testing (using measured GHI/ambient temperature traces and typical PV strings) is therefore recommended to confirm the lab/simulation gains (e.g., the cited $\approx 99\text{--}99.8\%$

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efficiencies and multi-tens-percent tracking-time reductions) under true tropical variability [100], [101], [102], [103], [105].

2.4.11 Energy Yield Improvement: With MPPT vs Without MPPT

All the research concurs that employing any MPPT significantly enhances energy harvesting over direct (fixed) coupling. In a case study with tropical irradiation, a solar system without MPPT (direct-coupled) yielded much lower power than when using MPPT. The MPPT-equipped system achieved 100% load demand and even enabled grid export, whereas the no-MPPT case was short of it. Precisely, the application of MPPT "resulted in a 62% increase in power harvesting" from the same array [107]. In practical applications, this is to say that the inclusion of MPPT can augment annual yield by tens of percent in the tropics. For instance, one study indicated that the application of MPPT obviated a 2550 kWh yearly deficit in end-of-season production [107]. Shortly stated, any MPPT algorithm significantly outperforms none: gains of the order of 50–60% increase in energy harvested are reported under variable tropical conditions [108]. This highlights the importance of MPPT in the Philippines.

- With MPPT: The controller keeps adapting to varying conditions, and the PV is always in or close to peak power. In reported studies, this enabled the system to meet the energy requirement fully and even supply surplus to the grid [108].
- Without MPPT (direct coupling): Without MPPT (direct coupling): The panel is held at a sub-optimum position, losing considerable potential. In the cited case, , the MPPT eliminated a 2–3% yearly energy deficit and boosted total harvest by ~62% compared to direct coupling [108].

2.4.12 Impact of Weather Conditions on MPPT Performance

Research on MPPT under tropical weather conditions shows emphasis on adaptive, hybrid and predictive strategies to cope with the rapidly changing operating landscapes caused by partial shading and cloud transients. Deep and memetic reinforcement-learning approaches have been proposed to learn robust tracking policies under partial shading and ANN based controllers have been explored to improve transient response [109], [110]. The principal challenge that was identified across literature is maintaining tracking accuracy during very fast irradiance ramps and irregular partial shading. Controllers must balance fast reaction against stability and avoidance of local-peak trapping, while operating within the computation, sampling and fixed-point constraints of low-cost embedded hardware commonly used in the Philippines.

The Philippine climate presents frequent cloud cover, high humidity and temperature fluctuations, all of which significantly impacts the performance of adaptive MPPT algorithms. Cloud movement leads to irradiance variation and partial shading, creating multiple local peaks in the power-voltage curve that can trap conventional MPPT methods in suboptimal points [91]. High Humidity and dew formation reduce effective irradiance and modify the PV output characteristics, introducing measurement noise and transient power losses [111]. Meanwhile, temperature swings shift the maximum power point voltage due to the negative temperature coefficient of solar cells, requiring real-time thermal compensation in control logic [112].

2.4.13 AI and machine learning-based adaptive MPPT approaches

A study by Phan et al. (2020) used deep reinforcement learning (DRL) based MPPT control which uses AI to learn an optimal tracking policy for its development. The DRL MPPT control interacts with the PV system to learn about its voltage and current inputs through a deep neural network and makes decisions based upon this information to maximize the energy yield of the PV system. DRL does not rely on mathematical models for its evaluation and instead adapts the PV system dynamically depending on the irradiance and temperature variations in the environment which results in fast and stable operation to reach the MPP. This study proves that DRL achieved a higher tracking efficiency, lesser oscillations, and it was able to adapt fast on new conditions despite these conditions being untrained on machine learning [109].

2.4.14 Global Maximum Power Point Tracking (GMPPT)

Devireddy et al. (2022) proposed a novel Global Maximum Power Point Tracking (GMPPT) algorithm designed to address the limitations of classical MPPT techniques such as P&O and IncCond which often fail to detect the global peak under partial shading conditions. Under non-uniform irradiation, PV systems exhibit complex characteristics with multiple peaks in their P-V and P-I curves, often causing traditional algorithms to get trapped at local peaks, thereby reducing system efficiency [113].

The study resolved this issue by introducing an algorithm that can track the global peak using the P-I curve rather than the usual P-V curve. The authors stated that using the P-I curve is advantageous since the search region varies with irradiation and the PV

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current is proportional to irradiation meaning that by using the current as reference, the tracking time is significantly reduced [114].

2.5 Research Gaps

The research for MPPT algorithms is transitioning from conventional algorithms (P&O and InCond) towards intelligent, model-based, adaptive and AI-based MPPT to handle fast-varying irradiance and partial shading. In current times, the trends for MPPT studies is the application of more artificial intelligence-based approaches (ex. Fuzzy Logic Control) instead of more conventional algorithms (ex. Incremental Conductance). Intelligent MPPT algorithms are observed to show higher tracking efficiency and less steady-state oscillations for rapidly changing weather conditions [114]. Moreover, there is also a rise in studies of model-based and predictive approaches such as Model Predictive Control (MPC) and Finite-Control-Set MPC, which is used to improve transient response and satisfy operational constraints [115]. Last is the hybridization of algorithms which combines intelligent controllers with conventional approaches. This approach uses the fast local tracking of simple algorithms under normal operating conditions and the high tracking efficiency and low steady-state oscillations from intelligent algorithms when under rapid change from weather conditions [110].

Most existing studies evaluate MPPT algorithms on standard test irradiance profiles or on datasets from temperate climates, which results in a lack of studies that are applicable for the Philippine climate. There are existing studies in the Philippines that explore the use of MPPT algorithms however there is a lack of studies that use Philippine

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climate data. This matters as Philippine specific weather conditions such as rapid cloud transients, humidity effects and geographical location changes the statistics of irradiance compared to standard test irradiance profiles. Moreover, there is also a lack of literature that studies the optimal MPPT algorithm to be used for a specific local area.

As stated before, most studies use standard test irradiance profiles which do not exactly represent the climate of a specific area. Furthermore, there is a lack of comprehensive studies that identify the optimal MPPT algorithms in specific Philippine locations. Most of these evaluations rely on a generalized profile which does not represent the distinct behavior of solar energy in geographically unique areas.

Additionally, the topic of evaluating algorithm performance is already widely documented, the study has a different approach on the matter, mainly filling significant gaps regarding the dynamic robustness of these algorithms across different datasets through simulation. Most existing algorithm evaluations rely on standardized test conditions such as the EN 50530 standard. They do not explore further into investigating the Algorithm Rank Stability (ARS) which refers to the algorithm's performance consistency when transferred into different climate zones. It is yet unknown whether mid-latitude benchmarks or standardized test conditions maintain their dominance in comparison to the tropical climate fluctuations of the Philippines. On the other hand, the Cloud-Transient Recovery Time (CTRT), a performance metric which measures the time required to recover from sudden irradiance changes caused by cloud-transients also remains unquantified by existing studies. This leaves a gap in understanding transient energy losses.

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MPPT studies focused in the Philippines do not really tackle the evaluation of multiple different MPPT algorithms. For instance, a study by Salvador et al. (2024) focuses on creating an adaptive IncCond algorithm and it is compared to the conventional IncCond algorithm. They were able to prove that the adaptive approach performs better than conventional but recommended the use of other algorithms such as AI based algorithms for further evaluation of their created adaptive IncCond algorithm [116]. Another study by Villapana et al. (2023) focused on evaluating different solar charge controllers. These controllers already have their built-in MPPT algorithms which would optimize the solar PV cell output. Comparing this to the study, Villapana et al. evaluated different solar charge controllers in the Philippines rather than evaluating different MPPT algorithms in the Philippines [117].

With the stated research gaps, there is a clear case for a simulation approach that combines realistic tropical irradiance and temperature inputs with baseline and adaptive MPPT controllers. Under tropical irradiance and temperature, the simulation study shall compare the performance of both baseline algorithms (P&O and IncCond) to other algorithms categorized as control-theoretic/model-based algorithms, intelligent and AI-based algorithms, and metaheuristic/optimization-based algorithms. As the trends with the recent studies moving from conventional MPPT algorithms (P&O and IncCond) to AI-based algorithms (Fuzzy Logic, ANN and PSO), this paper aims to test that trend to the Philippine tropical climate, specifically in Arayat-Mexico Solar Farm in Pampanga for solar panel optimization.

CHAPTER 3.0. METHODOLOGY

3.1. Introduction

This chapter presents the overall methodology being implemented in the study, defining the systematic approach used to model, simulate, and analyze multiple MPPT algorithms under the tropical weather conditions of Arayat-Mexico, Pampanga. This chapter provides a clear explanation of how the research evaluates the performance of MPPT algorithms in solar photovoltaic systems by integrating local meteorological data, benchmark datasets, simulation modeling, and algorithmic comparison.

The chapter is organized into key stages. Firstly, data were acquired from Solcast and benchmark PV datasets to get irradiance and temperature profiles, including periods with extreme weather variations. The next stage was modeling the PV system, in which the PV array, DC load, and all other components were simulated in MATLAB/Simulink under realistic conditions at the Arayat-Mexico Solar Farm. The next stage was the development and implementation of MPPT algorithms into the model. It was subjected to various simulation conditions emulating clear-sky, cloudy, ramp-change, and fluctuating irradiance scenarios representative of tropical environments to test the system. Conclusively, the performance evaluation and comparative analysis are done through quantitative metrics such as convergence speed (s), tracking efficiency (%), tracking accuracy (stability), harvested energy (Wh), and cloud-transient recovery time (s). The methodology aimed to produce realistic, comprehensive analysis results that would enable the selection and optimization of MPPT algorithms well-suited to Pampanga's tropical weather.

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3.2. Research Design

3.2.1 Type of Research

This study employed a quantitative, simulation-based experimental design to evaluate and compare the performance of multiple MPPT algorithms in a solar photovoltaic (PV) system. Simulation experiments were done in MATLAB/Simulink because this platform is widely utilized for modeling and analysis of dynamic renewable energy systems.

The design focused on creating a controlled, consistent simulation environment that accurately reflects the weather conditions in Arayat-Mexico, Pampanga, along with two benchmark PV datasets representing stable and highly variable irradiance conditions. Solar irradiance and ambient temperature data from Solcast were integrated to ensure the reliability and validity of results under realistic operational conditions.

A comparative analysis approach was utilized to assess the performance differences among classical, control-based, intelligent, and metaheuristic MPPT techniques. Each algorithm was subjected to variable irradiance and temperature profiles, which included clear-sky, cloudy, ramp-change, and rapidly fluctuating scenarios representative of tropical conditions in Pampanga. Quantitative assessments of convergence speed (s), tracking efficiency (%), tracking accuracy (stability), harvested energy (Wh), and cloud-transient recovery time (s) were conducted for each algorithm.

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This research design helps in identifying the strengths and limitations of each MPPT approach, providing key insights for optimizing solar PV systems in tropical regions such as Pampanga.

3.2.2 Research Procedure

The overall research process follows a sequence that ensures the accuracy of comparisons for each MPPT algorithm. The step-by-step process is represented in Figure 3.1 and presents the research flow from data collection to results evaluation.

The study obtained meteorological data, including solar irradiance and ambient temperature, from Solcast, along with corresponding benchmark datasets for stable and highly variable irradiance conditions. These were further processed in a simulated environment to reflect realistic local weather patterns, including cloud-induced irradiance fluctuations typical of tropical and typhoon-affected conditions.

Next, a PV system model was developed in MATLAB/Simulink to establish all necessary parameters, including module specifications, array configuration, converter details, and environmental inputs. Once the base system was completed, the set of MPPT algorithms under evaluation was integrated into the model to regulate the PV array's power output.

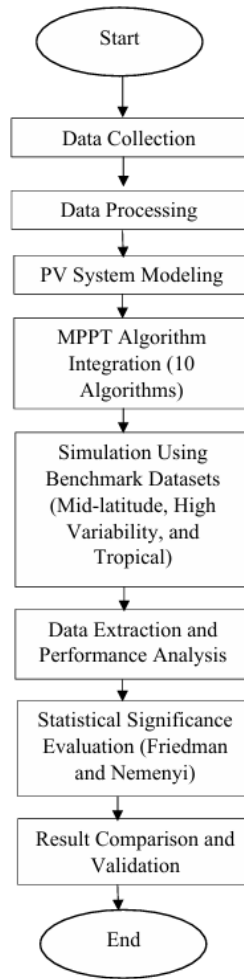
Simulation tests were then conducted under identical environmental conditions, including constant irradiance, gradual ramps, and rapidly changing irradiance and temperature profiles, to ensure a fair and consistent comparison across all algorithms. The resulting simulation outputs, such as power, voltage, current, and energy curves, were collected and analyzed for performance assessment.

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In the context of the quantitative analysis, descriptive statistical measures, comparative performance analysis, and non-parametric statistical methods were used to assess the performance of the MPPT algorithms under varying conditions. The performance assessment considered the performance indicators such as tracking efficiency, convergence time, tracking stability, energy harvesting, and cloud transient recovery time (CTRT). This analysis provided a comprehensive comparison of MPPT algorithm performance. Additionally, the analysis provided a comprehensive insight into the performance of each MPPT algorithm.

The results were then validated and compared to identify which MPPT algorithms have demonstrated stronger performance and adaptability in tropical weather in Pampanga. The results were used to suggest improvements to MPPT control strategies in Philippine solar PV systems.

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***Figure 3.1** Research Procedures*

3.3 System Architecture and Simulation Environment

3.3.1 Mathematical model (single-diode vs double-diode)

The photovoltaic (PV) cell may be modeled using electrical equivalent circuits that account for its nonlinear current–voltage behavior under varying irradiance and temperature. Of all the methods proposed, the single-diode and double-diode models are the most popular in solar PV simulation and performance analysis. The single-diode model reduces the physical behavior of the PV cell to a single exponential diode term with good enough accuracy, but at low computational complexity [34]. At the same time,

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the double-diode model adds an extra diode to account for recombination losses in the depletion region, enhancing precision at low irradiance but at the cost of additional parameterization and simulation effort [118], [119].

In this study, the single-diode model was used due to its good accuracy-to-computational complexity trade-off, which supports dynamic MPPT algorithm comparison under MATLAB/Simulink. The equivalent circuit of the single-diode includes a current source for the photocurrent, a diode, a series resistance R_s , and a shunt resistance R_{sh} , as shown in Figure 3.2.

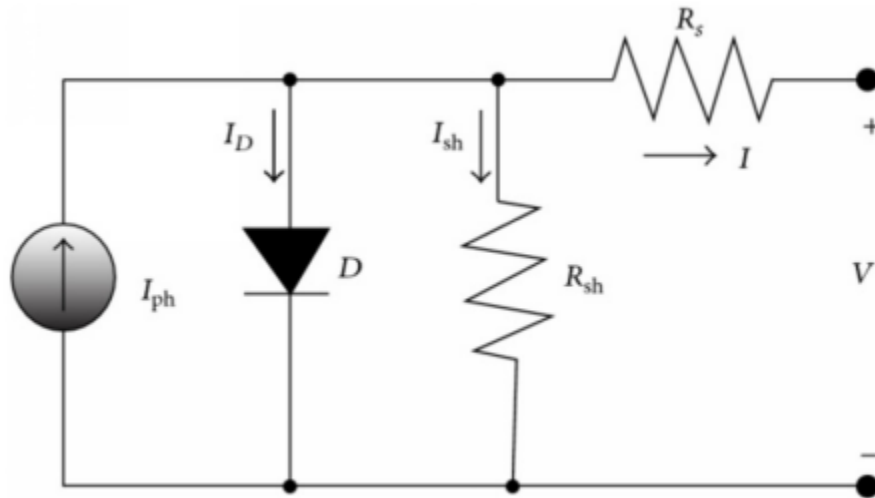


Figure 3.2 Single Diode Model Equivalent Circuit of a PV cell [120]

The standard single-diode model formulation is given with the current-voltage relationship of a PV cell:

$$I = I_{ph} - I_o \left(e^{\frac{q(V+IR_s)}{nkT}} - 1 \right) - \left(\frac{V+IR_s}{R_{sh}} \right)$$

Where:

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I_{ph} = photocurrent (A)

I_o = diode reverse saturation current (A)

q = electron charge (1.602×10^{-19} C)

k = Boltzmann constant (1.381×10^{-23} J/K)

T = cell temperature (K)

n = diode ideality factor

R_s = series resistance (Ω)

R_{sh} = shunt resistance (Ω)

V = output voltage (V)

I = output current (A)

The photocurrent I_{ph} depends linearly on solar irradiance and temperature, and can be modeled as:

$$I_{ph} = [I_{sc, ref} + \alpha(T - T_{ref})]\left(\frac{G}{G_{ref}}\right)$$

where $I_{sc, ref}$ is the short-circuit current at standard test conditions (STC), α is the temperature coefficient of current (A/°C), G is the actual irradiance (W/m²), and G_{ref} is the reference irradiance (1000 W/m²).

This single-diode model provided a sufficiently accurate description of PV behavior to support system-level simulations, MPPT tracking, and performance analysis under varying tropical conditions, such as those at the Arayat-Mexico Solar Farm.

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3.3.2 PV System Parameters and Reference Panel Details

To produce realistic, credible simulation outputs, this research used a commercially purchased photovoltaic (PV) module as the basis for the system model. The module used was the *Seraphim SRP-540W series (PERC Mono, 144 cells)*, one of the most commonly used modules in large-scale solar farms throughout the Philippines due to its cost-effectiveness, tropic hardiness, and compatibility with local grid conditions. The parameters of this PV module are presented in *Table 3.1* and were used as the basis to define the simulation constants and input data in MATLAB/Simulink.

MODULE TYPE	SRP-540-BMA-BG		
	Front STC	Front NOTC	Back STC
Maximum Power - $P_{mp}(W)$	540	406	378
Open Circuit Voltage - $V_{oc}(V)$	49.50	46.18	49.48
Short Circuit Current - $I_{sc}(A)$	13.81	11.16	9.74
Maximum Power Voltage - $V_{mp}(V)$	41.55	38.39	41.61
Maximum Power Current - $I_{mp}(A)$	13	10.59	9.09
Module Efficiency STC - $\eta(\%)$	20.90		
Power Tolerance (W)	(0, +4.99)		
Pmax Temperature Coefficient	-0.34%/°C		
Voc Temperature Coefficient	-0.26%/°C		
Isc Temperature Coefficient	+0.05%/°C		

Table 3.1 PV Module Characteristics [121]

These conditions represent Standard Test Conditions (STC), an irradiance of 1000 W/m², a cell temperature of 25 °C, and an air mass (AM) of 1.5, as defined by the

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International Electrotechnical Commission (IEC 61215) and indicated in the Seraphim SRP-540-BMA-HV datasheet.

The temperature-dependent variation of the PV output voltage and current was accounted for based on the following equations [34]:

$$I_{sc}(T) = I_{sc,ref} [1 + \alpha I (T - T_{ref})]$$

$$V_{oc}(T) = V_{oc,ref} [1 + \beta v (T - T_{ref})]$$

$$P_{max}(T) = P_{max,ref} [1 + \beta p (T - T_{ref})]$$

These relationships ensured that the PV array model responded to temperature variations common in tropical environments like those in Arayat-Mexico, Pampanga, where module temperatures can exceed 60°C at peak irradiance.

To simulate the photovoltaic system, the MATLAB/Simulink PV Array block was used. This block internally includes a photovoltaic array that comprises a series of modules connected in series and parallel. In a photovoltaic system, the voltage and current of the photovoltaic array are theoretically determined by the modules connected in series and parallel, respectively. The photovoltaic array output can be expressed as:

$$V_{array} = N_s \times V_{mpp}, \quad I_{array} = N_p \times I_{mpp}$$

where N_s and N_p represent the number of modules connected in series and parallel, respectively. On the other hand, V_{mpp} and I_{mpp} refer to the module voltage and current at the MPP. This equation demonstrates how the PV arrays are connected to attain the desired voltage and current levels for the DC-DC converter.

In this paper, the built-in PV Array model in MATLAB/Simulink is used to simulate the photovoltaic system. In this way, the simulation accurately emulates the

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electrical characteristics of the PV array under varying irradiance and temperature conditions.

3.3.3 Boost Converter Design and Justification

A DC–DC boost converter was utilized as the power conditioning stage within this research. It acts as a link between the photovoltaic (PV) array and the resistive DC load, enabling the PV module to operate at its Maximum Power Point (MPP) through the control of the duty cycle using chosen MPPT algorithms. The topological choice is the boost converter because it can step up the PV array's input voltage to a higher output voltage, making it particularly suitable for low-voltage PV applications. In addition, it offers simpler control, fewer components, and greater efficiency than buck–boost or Cuk converters, making it a popular choice for simulation-oriented performance investigation under different irradiance and temperature conditions [122], [123].

The MPPT algorithm provides a Pulse-Width Modulation (PWM) duty cycle signal that controls the converter switch (often a MOSFET) to manage the output voltage and maintain continuous operation around the MPP. The converter is in continuous conduction mode (CCM) for reliable energy transfer, with the inductor current never reaching zero.

The mathematical relations governing the converted design are as follows:

- Duty Cycle (D):

$$D = 1 - \frac{V_{in}}{V_{out}}$$

where V_{in} is the input voltage from the PV array, and V_{out} is the regulated output voltage.

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- Inductor Design (L):

$$L = \frac{V_{in} \times D}{\Delta I_L \times f}$$

where ΔI_L is the peak-to-peak inductor current ripple, and f_s is the switching frequency.

- Output Capacitor (C):

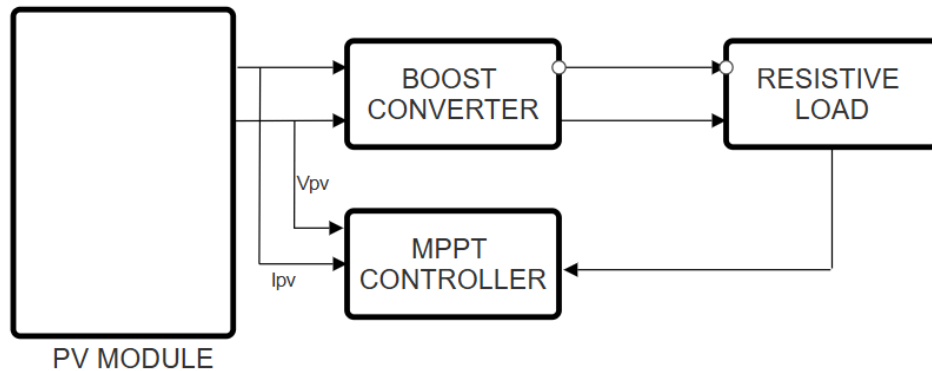
$$C = \frac{I_o \times D}{\Delta V_o \times f_s}$$

where I_o is the output current, and ΔV_o is the allowable output voltage ripple.

Here, the switching frequency (f_s) was set to 10 kHz to balance switching losses and transient response efficiency in the simulation. The selected component ratings ensure stable converter performance and low output ripple, even under rapidly changing irradiance characteristics of tropical Philippine weather conditions [124].

The boost converter dynamically responded to real-time duty-cycle adjustments provided by the MPPT algorithms, in order for each technique, whether classical, control-based, intelligent, or metaheuristic, to effectively regulate power extraction under variable operating conditions.

As seen in Figure 3.3, the overall system architecture and simulation environment consists of the PV module blocks, MPPT controller, boost converter, and resistive load for performance testing.



***Figure 3.3** System Architecture*

3.3.4 Power Output Analysis and Load Configuration

This research utilized a DC resistive load as the system output element. Using a resistive load is preferred over a DC bus because it makes it easier to assess MPPT algorithm performance by eliminating inverter-related losses and grid interaction complexities. In such a case, analysis could be performed solely on how efficiently MPPT algorithms extract maximum power from a PV array. The resistive-load setup ensured accurate measurement of the real-time tracking efficiency, convergence speed, and steady-state stability of both MPPT algorithms, free of AC-side disturbances.

The load was maintained as a fixed resistive load to provide a stable and controlled operating condition during the simulations. Using a constant load allows the study to isolate the behavior of the MPPT algorithms and evaluate their tracking capability under changing irradiance and temperature without additional variability introduced by load fluctuations. This approach improves the repeatability and

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comparability of results, ensuring that any changes in PV output are primarily due to environmental variations and the algorithm's control actions.

This research used a DC resistive load to provide a controlled, analytically transparent evaluation of MPPT performance. Compared with DC bus and grid-connected configurations, a resistive load provides a direct, deterministic relationship between voltage, current, and power: $P = V^2/R$. This allows straightforward power measurement validation without the influence of power conversion stages.

The use of DC bus or grid-connected configurations introduces converter-switching losses, control-loop interactions, and grid-synchronization effects. These factors mask the intrinsic performance of MPPT algorithms, making it difficult to determine whether performance is mainly due to MPPT algorithms or downstream system components. Using a DC resistive load ensures that all performance comes from the PV array and the implemented MPPT control strategy.

Furthermore, maintaining a fixed resistive load ensures operating conditions are consistent. This allows fair benchmarking between the MPPT algorithms. A constant load prevents stochastic fluctuations and allows the study to focus on the algorithms' adaptability in changing irradiances and temperature, which are the primary disturbances in PV systems. This implementation allows for repeatability, comparability, and statistical reliability.

From a control systems perspective, a resistive load prevents the introduction of additional feedback loops and dynamic states, thereby preventing unintended interactions that could affect metrics such as convergence speed and tracking accuracy. This makes a

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resistive load particularly suitable for transient response and steady-state oscillation evaluation, which are key metrics in evaluating MPPT performance.

However, it is acknowledged that this configuration does not fully represent real-world PV systems where power converters and grid interactions are present. This simplified setup is intentionally chosen to isolate MPPT algorithm performance and provide a baseline performance evaluation. This ensures that the study reflects the true performance of the MPPT algorithm before integration into more complex systems.

3.4. MPPT Algorithm Development

The MATLAB/Simulink code used in this study was developed through a combination of the research team's own implementation work and commissioned external coding assistance. The research team was responsible for designing the simulation framework, defining the metrics to evaluate algorithms, which algorithms to use, the dataset inputs, and validating all outputs compared to expected theoretical behavior. A working base code of the algorithms, metrics calculation, and data compilation, as well as a Simulink PV environment, was established with the assistance of a commissioned external MATLAB expert. The researchers reviewed, debugged, and refined the code to further meet the specific requirements of the study, such as tweaking the algorithms to fit closer to what's needed for the study and removing any bugs in the code.

All algorithm implementations follow the canonical mathematical formulations documented in the literature cited in Chapter 2. For classical algorithms (P&O, IncCond), the duty-cycle update rules follow the standard gradient-sign logic. For control-theoretic

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algorithms (SMC, RCC, FSB), the sliding surface, correlation-based, and interval reduction formulations are implemented as described in section 2.3.4. For intelligent algorithms (FLC, ANN), the fuzzy inference rules and neural network architecture follow the configurations described in section 2.3.5. For metaheuristic algorithms (PSO, GA, GWO), the population initialization, update equations, and termination conditions follow the standard formulations with parameter values based on section 2.3.6.

Code validation was carried out by testing each algorithm under a standard constant irradiance scenario ($G = 1000 \text{ W/m}^2$, $T = 25^\circ\text{C}$) to verify that it converges to the expected theoretical maximum power point (MPP). Results are then cross-checked with literature to verify if it functions as expected from published simulation results from related studies. Once the output of the validation is reviewed and tested by the research team, it is integrated into the simulation framework.

The complete source code, including all algorithm scripts, the unified Simulink model (.slx), dataset inputs, and the metrics computation script, is publicly available at Appendix A.

3.4.1 Overview of MPPT Techniques

MPPT algorithms are control methods that allow a PV system to operate at its maximum power point even under varying environmental conditions. MPPT algorithms solve this by continuously adjusting the duty cycle of the DC–DC converter so that the PV operating voltage remains at or near the instantaneous MPP [70], [66].

In MPPT control, the DC–DC converter acts as the interface that controls the PV array voltage. The MPPT algorithm calculates an updated duty cycle based on measured

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PV voltage, current, or estimated gradients and applies this duty cycle to the converter's switching device in order to shift the operating point toward the MPP. Under fast-varying irradiance conditions, such as cloud transients and tropical weather fluctuations, the MPPT methods must respond quickly while maintaining stability with minimum oscillations around the MPP.

Different MPPT methods have been developed, which balance the trade-off among speed, accuracy, stability, and energy harvesting performance. The classical methods usually calculate the slope of the PV power curve or compare instantaneous values to decide on the direction of adjustment. Control-based approaches are essentially based on system dynamics or sliding surfaces. On the other hand, intelligent and metaheuristic approaches incorporate fuzzy rules, neural networks, swarm behavior, or population-based optimization to estimate or search for the global MPP under rapidly changing or nonlinear conditions [125].

In all approaches, MPPT algorithms are developed for the same aim: to continuously track the MPP in dynamic environmental conditions with the purpose of optimizing energy harvesting.

3.4.2 Classical Algorithms (P&O, IncCond)

In this work, the classical MPPT algorithms were implemented in MATLAB/Simulink using a standard duty-cycle adjustment logic. The P&O method was modeled here by applying a periodic perturbation to the operating voltage and observing the resulting change in power to determine the sign of the next update in the duty cycle. For the implementation of the IncCond algorithm, incremental and instantaneous

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conductance values are computed, and changes in the duty cycle are determined by the sign of the slope of the power-voltage characteristic. Limitations may arise in P&O oscillations after convergence, a well-known problem in PV MPPT systems. Similar to P&O, IncCond noise sensitivity issues cause the algorithm to hover around the MPP, reducing efficiency.

Both classical algorithms operated on real-time measurements of PV voltage and current and updated the duty cycle at the predefined controller sampling interval. A single smooth and accurate MPP curve is produced and output. Their implementation serves as the baseline reference against which the other MPPT categories were compared.

3.4.3 Control-Based Algorithms (SMC, RCC, FSB)

In this work, control-based MPPT algorithms were implemented based on the modeling of the dynamic behavior of the PV system and using various control-law formulations in order to adjust the converter duty cycle. The SMC or Sliding Mode Control method applies a predefined switching surface in order to drive the PV voltage towards the MPP. It is assumed for SMC that PV dynamics are sufficiently known and that the switching surface is properly defined to ensure convergence to the MPP. System states are also assumed to be measurable. RCC utilized intrinsic ripple components of the PV signal in order to estimate the direction of power change without additional perturbation. Measurable ripple components are assumed to exist in the PV voltage and current due to converter switching. Ripple signals are not significantly attenuated due to filtering to allow meaningful correlation between ripple components and MPP direction. In contrast, the Fibonacci Search-Based (FSB) algorithm performed operating voltage

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range narrowing using interval reduction logic, which is derived from the Fibonacci sequence. It is assumed for FSB that the MPP lies within a predefined voltage search range and that the PV power-voltage characteristic is continuous and unimodal, enabling interval reduction to convergence.

Each control-based algorithm comes with its limitations. SMC is prone to chattering, which causes high-frequency oscillations at the sliding surface. This leads to increased switching losses and reduced system efficiency. Performance is also sensitive to switching surface and control gain design. The RCC algorithm is highly dependent upon the presence and quality of ripple signals. Under conditions with weak ripple signals or heavy filtering, algorithm tracking may become unreliable. This method is also sensitive to measurement noise, which can distort ripple estimation. FSB may exhibit slower convergence if the initial search estimate is wide. The performance of FSB is dependent upon the selection of the search interval, which may be less suitable for rapidly changing irradiance due to its iterative computation process.

All the control-based methods were implemented in the Simulink model using the same sampling interval and converter interface as the classical MPPT algorithms to unify the implementation and ensure a totally fair comparison among scenarios.

3.4.4 Intelligent Algorithms (FLC, ANN)

The rule-based and data-driven control structures were considered to implement intelligent MPPT algorithms. In the Fuzzy Logic Control (FLC) model, the membership functions were defined for both error and change-in-error signals, while fuzzy inference rules were applied to obtain a computed duty-cycle adjustment. It is assumed that the

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defined membership functions are sufficient to represent the PV system behavior across its operating range. It is also assumed for FLC that input variables capture system dynamics for effective control. For the Artificial Neural Network (ANN) method, an estimation of optimum operating voltage was performed in this study with a neural model trained on irradiance-and temperature-related inputs. It is assumed for ANN that the training dataset for irradiance and temperature is representative of the actual operating conditions of the PV system. Further assumptions include that the trained model can generalize well with unseen conditions and that input measurements are accurate.

Performance limitations are present for FLC depending on the membership function design and rule base. Improper tuning will lead to suboptimal tracking performance and increased steady-state oscillations. ANN is limited by the amount of available training data, which is problematic when operations differ from the training data range. This will ultimately lead to poor generalization and inaccurate MPP estimation. Additionally, ANN requires a training phase, which introduces higher computational complexity compared to classical and control-based algorithms.

Both intelligent controllers were implemented in the Simulink environment with the same interface as classical and control-based algorithms in order to guarantee consistent sampling, duty-cycle output, and interaction with the converter for comparative evaluation.

3.4.5 Metaheuristic Algorithms (PSO, GA, GWO)

The metaheuristic MPPT algorithms were implemented by using population-based optimization routines to iteratively search for the operating voltage

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corresponding to the maximum power point. Particle Swarm Optimization updates candidate solutions depending on the individual and global best positions. while the Genetic Algorithm applied selection, crossover, and mutation operators to evolve a population toward the optimal voltage. In PSO, it is assumed that the PV voltage search space is properly defined and contains the global maximum power point. PV power-voltage characteristic is assumed to remain stable for each iteration process of the algorithm. The Grey Wolf Optimizer modeled the leadership hierarchy and hunting behavior to drive the search process towards the MPP. For both PSO and GWO, population size and convergence are able to reach convergence within the given simulation timeframe. It is also assumed that output power is accurate and measurement signals are reliable. Lastly, variations in irradiance are assumed to be slower than the convergence capability of the algorithm to ensure a valid search operation.

Despite global search capability, metaheuristic algorithms have several limitations. The first limitation is their computational complexity. For each iteration, multiple candidate solutions are evaluated, which increases processing requirements when compared to the previously discussed methods. The second limitation is their slow convergence in real-time applications when the population size and search space are large. When under rapidly changing irradiance, the algorithms may not converge quickly enough to track the true MPP. Lastly, both algorithms' performance relies heavily on parameter selection. Improper tuning of population size, number of iterations, and algorithm coefficients may lead to premature convergence, trapping of local optima, and longer computational time.

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All three metaheuristic algorithms ran at the same controller sampling rate and were set to provide an updated duty cycle at every iteration. The implementation of these into the Simulink environment also followed the same converter interface and measurement structure as the other MPPT categories for a consistent and fair comparison of all test scenarios.

3.4.6 Unified Implementation in MATLAB/Simulink

All MPPT algorithms were implemented within a unified MATLAB/Simulink environment to ensure that all scenarios are consistently evaluated. Each algorithm input real-time measurements of PV voltage and current, and provided an updated duty-cycle command to the DC-DC boost converter. In all methods, there was a common sampling interval, flow of signal structure, and measurement interface. This ensured direct and fair comparison of algorithm performance by eliminating variations caused by different controller configurations.

3.4.7 Duty-Cycle Update Interface

Each MPPT algorithm is interfaced with the DC-DC boost converter via a standard duty cycle update interface. Each algorithm calculated a new duty cycle value, together with real-time measurements of PV voltage and current. The new duty cycle was then fed directly to the PWM generator that drove the converter's switching device. By implementing a standard interface, all algorithms drove the converter in the same manner under identical timing constraints and, thus, provided a fair comparison of MPPT performance for all test scenarios.

3.5. Weather Data Acquisition and Preprocessing

3.5.1 Data Sources and Selected Study Locations

To achieve a realistic and location-based simulation environment, this research used historical and real-time weather data applicable to the solar farms of Arayat-Mexico, Pampanga. The researchers obtained high-resolution, time-specific weather data, such as solar irradiance decreases and temperature changes, from Solcast, which provides solar irradiance and forecasting data services [54].

Two benchmark datasets were added in order to allow cross-dataset comparison of the MPPT algorithms in addition to the local tropical data. Benchmark Dataset 1 represents stable and standard midlatitude PV operating conditions, while Benchmark Dataset 2 contains high-variability cloud-transient profiles suitable for testing dynamic MPPT performance.

These datasets collectively enabled the simulation to model a wide range of weather conditions, abrupt irradiance drops, temperature fluctuations, and rapid cloud-induced transitions typical of tropical climates. All collected data were preprocessed to meet input format requirements for MATLAB/Simulink and imported as time-series signals into the PV array block. The study, however, did not consider long-term PV module degradation because the limited scope was restricted to short-term, simulation-based performance analysis only under varying environmental conditions.

3.5.2 Data Processing

Preprocessing of weather datasets was done to ensure that they could be used effectively within MATLAB/Simulink. Preprocessing consisted of cleaning, the removal of invalid or missing data, outliers, and replacing missing values in between through interpolation for continuity of the signals. All data were then resampled at a single time step equivalent to the MPPT control loop sampling rate.

Test profiles representing key operating conditions relevant to MPPT performance evaluation were extracted directly from the tropical and benchmark datasets. These included:

- Clear-Sky Scenario
- Ramp-Change Scenario
- Step-Change Scenario
- Mixed-Weather Scenario

These processed profiles provided a set of standardized test cases for the comparison of MPPT algorithms on various datasets. They allowed consistent and realistic testing for a wide range of environmental conditions.

3.6 Simulation Setup

3.6.1 Hardware Specifications

All MATLAB/Simulink simulations were performed on a personal computer with the following hardware specifications:

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- Processor: 11th Gen Intel(R) i9-11900H @ 2.50GHz
- RAM: 32 GB DDR4-3200 SODIMM
- Operating System: Windows 11 Home Single Language, 64-bit
- MATLAB Version: MATLAB R2024b
- Plugins required: Simscape Electrical and Simulink

No GPU acceleration or parallel computing toolboxes were used, and all simulations were executed on a single CPU core. The simulation time step was set to 1×10^{-5} , and each simulation consisted of a 60-second scenario. Simulation time across all 120 runs was approximately 1-1.2 hours long.

3.6.2 Simulation Environment

The simulation was done in MATLAB/Simulink, which allowed an integrated modeling and analysis environment for the given PV system. The model had three main subsystems: the PV array block, the DC-DC boost converter, and the MPPT control block. Based on its manufacturer's datasheet [121], the Seraphim SECO high-efficiency 540 W bifacial module was used to model the PV array. A boost converter regulated the output voltage via pulse-width modulation controlled by the MPPT algorithm block.

A fixed-step discrete solver was utilized with a step size of 1×10^{-5} to accurately resolve the switching behavior at a standard 10 kHz PWM frequency applied in the converter model. For longer runs where energy yield and steady-state behavior were of primary interest, a slower supervisory sampling interval was employed, such as 1×10^{-3} s, to lighten computational load without a loss of accuracy in the results [126], [127].

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3.6.3 Simulation Scenarios

A variety of realistic solar conditions observed in the Philippines—such as clear-sky, partial-cloud, typhoon-induced, overcast, rapid-transient events, as well as dry, cool, and rainy seasonal variations- were represented in the irradiance and temperature profiles used in the simulation experiments, as defined in Section 3.5.2 (Weather Data Acquisition and Preprocessing). There were 60 seconds for each scenario to capture transient and steady-state responses related to changes in the irradiance. To provide a fair comparison across all MPPT algorithms, the same PV array model, converter configuration, and control settings were used in every test run. Performance indicators such as convergence speed (s), tracking efficiency (%), tracking accuracy (stability), harvested energy (Wh), and cloud transient recovery time (s) were computed from the resulting simulation outputs.

3.6.4 Parameter Settings

The irradiance profile for each scenario was sourced from a satellite-derived irradiance dataset from Solcast to capture the rapid weather fluctuations and extreme events not fully represented in the local measurements, as discussed in detail in Section 3.5.2. All irradiance and temperature profiles were standardized to the controller sampling interval and the simulation time step used in the MATLAB/Simulink model. The system used a DC–DC boost converter with a switching frequency of 10 kHz, which offered an appropriate balance between dynamic response and switching losses, and follows standard PV power-electronics practice. Fixed PV module parameters and

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identical converter settings were used for all simulation runs, ensuring that MPPT algorithms were compared fairly and under identical conditions.

3.7 Entropy Weight Method for Performance Metric Weighting

To objectively assess the relative significance of the performance metrics used for the evaluation of the MPPT algorithms, the Entropy Weight Method (EWM) was used. The entropy weight method is a popular MCDM technique, which uses the concept of entropy to determine the weights of the criteria for the evaluation purpose. The criteria with higher variation in the results are considered to have more information content, hence higher weights are assigned to them during the evaluation procedure [151], [152].

In the present study, the entropy weight method was used to determine the weights for the results obtained from the simulation of the MPPT algorithms using the MATLAB/Simulink environment. The results were used to determine the Algorithm Ranking Stability (ARS), which is a composite measure of the overall performance of the MPPT algorithms. The ARS is used as a single figure of merit for the comparison of the MPPT algorithms, which are tested under different weather conditions.

The entropy weight method was used to determine the ranking of the MPPT algorithms, as follows:

3.7.1 Construction of the Decision Matrix

In this step, a decision matrix is formed with the performance results of each MPPT algorithm for a set of chosen performance parameters.

$$X = [x_{ij}]$$

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where:

- x_{ij} represents the value of algorithm i under criterion j
- $i = 1, 2, \dots, m$ represents the MPPT algorithms
- $j = 1, 2, \dots, n$ represents the performance metrics

In this study:

- $m = 10$ MPPT algorithms
- $n = 5$ evaluation metrics

The five performance metrics extracted from MATLAB/Simulink simulations are:

1. Convergence Time (s)
2. Tracking Efficiency (%)
3. Tracking Accuracy (stability)
4. Harvested Energy (Wh)
5. Cloud Transient Recovery Time (s)

Each row in the decision matrix represents one MPPT algorithm, and each column represents a performance metric.

3.7.2 Identification of Benefit and Cost Criteria

Before the normalization step, the metrics have to be classified as either a benefit criterion or a cost criterion.

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Cost criteria are metrics for which lower values are better, whereas benefit criteria are metrics for which higher values are better.

For the purpose of the current study, the metrics are classified as follows:

Cost Criteria:

- Convergence Time
- Cloud Transient Recovery Time

Benefit Criteria:

- Tracking Efficiency
- Tracking Accuracy
- Harvested Energy

This classification ensures that the normalization process correctly reflects the desired optimization direction for each metric.

3.7.3 Normalization of the Decision Matrix

Due to the different units of the evaluation metrics, the decision matrix is normalized to enable the comparison.

For benefit criteria, normalization is done as follows:

$$r_{ij} = \frac{x_{ij}}{\max(x_j)}$$

For cost criteria, normalization is done as follows:

$$r_{ij} = \frac{\min(x_j)}{x_{ij}}$$

This step normalizes all the parameters into dimensionless numbers between 0 and 1.

3.7.4 Computation of the Probability Matrix

After normalization, the normalized performance values are converted to probabilities using the following formula:

$$P_{ij} = \frac{r_{ij}}{\sum_{i=1}^m r_{ij}}$$

where P_{ij} is the probability value corresponding to the normalized performance value r_{ij} .

The probability matrix is needed to compute the entropy of each performance metric.

3.7.5 Entropy Calculation

The entropy value for each performance metric is computed using:

$$E_j = -k \sum_{i=1}^m P_{ij} \ln(P_{ij})$$

Where

$$k = \frac{1}{\ln(m)}$$

And

- E_j represents the entropy value of criterion j
- M represents the number of alternatives (MPPT algorithms)

A higher entropy value implies that there is less variability between the alternatives; hence, the criterion is providing less discriminative information [151], [153].

3.7.6 Diversification Degree

The degree of diversification is then calculated to determine the amount of useful information provided by each criterion:

$$d_j = 1 - E_j$$

where d is the diversification degree of criterion j . The criteria with higher diversification values are more significant to the decision-making process.

3.7.7 Determination of Entropy Weights

The entropy weights of each performance metric are calculated by normalizing the values of diversification as follows:

$$w_j = \frac{d_j}{\sum d_j}$$

where:

- w_j represents the weight assigned to criterion j

These weights reflect the relative importance of each performance metric on the basis of the variability of the results obtained through the simulation.

3.7.8 Computation of Algorithm Ranking Stability (ARS)

The last step is calculating the Algorithm Ranking Stability (ARS) for all MPPT algorithms. The ARS is calculated by the weighted sum of the normalized performance values as follows:

$$ARS_i = \sum_{j=1}^n w_j r_{ij}$$

where:

- ARS_i represents the overall performance score of algorithm i

The algorithms are ranked according to their ARS values. The algorithm with a high ARS value has good performance.

3.8 Performance Evaluation Metrics

To test the performance of the proposed MPPT algorithms under different irradiance levels, five performance parameters are considered in this study: convergence speed, tracking efficiency, tracking accuracy, harvested energy, and cloud transient recovery time. These parameters are obtained based on the output of MATLAB/Simulink simulation of each algorithm.

Convergence speed (s) is a performance metric that measures the time required by the MPPT algorithm to track the maximum power point after changes in environmental conditions. This performance metric is related to the ability of the controller to adapt quickly to changes in solar irradiance levels or temperature changes. The faster convergence of the MPPT algorithm means that it is able to quickly identify the optimal operating point of the photovoltaic system.

Tracking efficiency (%) is a performance metric that measures the ability of the proposed MPPT algorithm to maximize the extracted power from the photovoltaic system during normal operation. The higher tracking efficiency of the proposed MPPT algorithm means that it is able to track the actual MPP of the photovoltaic system despite changes in irradiance and system conditions.

Tracking accuracy (stability) indicates the degree of fluctuation in the PV system's power output with respect to its maximum power point when steady-state is achieved. It is observed that algorithms with lower levels of fluctuation around the maximum power point are more stable and result in lower power losses when steady-state is achieved.

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Harvested Energy (Wh) indicates the total electrical energy generated by the solar power system throughout the entire simulation time period. This parameter indicates a cumulative measure of electrical energy harvested by the MPPT control algorithm, reflecting its ability to maximize solar power harvesting throughout a given time period.

Cloud transient recovery time (s) assess the ability of the MPPT control algorithm in recovering from a sudden change in irradiance, which is usually caused by cloud passage. This measure indicates how long it takes for an MPPT control system to return to its maximum power point following a sudden change in irradiance levels. The lower the recovery time, the better a system will perform in changing weather conditions.

The values obtained from the above five performance metrics were utilized to obtain the decision matrix for the MPPT algorithms under evaluation. The values of the performance metrics were further processed to obtain the relative importance of the decision criteria using the entropy weighting technique as explained in Section 3.7. Finally, the entropy weights obtained were used to calculate the Algorithm Ranking Stability (ARS) for the MPPT algorithms. The ARS represents the overall performance index of the MPPT algorithms derived from the weighted values of the normalized performance metrics.

3.9 Data Analysis Procedure

3.9.1 Computation of Performance Metrics

The performance metrics for each of the 120 simulation cases were computed from the solar PV simulated voltage (V_{pv}), current (I_{pv}), Power (P_{pv}), and output voltage (V_{out}) values, which were logged by the MATLAB/Simulink model. Output

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current and output power were derived from the logged output voltage and the fixed load resistance of $20\ \Omega$ as follows:

$$I_{out} = V_{out}/R_{load}$$

$$P_{out} = V_{out}/I_{out}$$

All metric computations were applied consistently across all algorithms, datasets, and scenario combinations using the metric computation MATLAB script.

Tracking efficiency was computed as the ratio of the total output energy delivered to the load ($Energy_out$) over the total energy harvested by the PV array across the entire 60-second simulation window ($Energy_in$). The formula is as follows:

$$\eta_t = (Energy_out / Energy_in) \times 100\%$$

Where:

- $Energy_in = \sum P_{pv} \times \Delta t$
- $Energy_out = \sum P_{out} \times \Delta t$
- with Δt representing the uniform simulation timestep

The tracking efficiency metric reflects the converter-level energy transfer efficiency of the MPPT controller over the 60-second simulation of each simulation case, including all transient and steady-state periods.

Convergence time was defined as the first time instant at which the PV output power has reached 95% of the steady-state power value and was computed as:

$$t_{conv} = \min\{t: P_{pv}(t) \geq 0.95 \times P_{ss}\}$$

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Where P_{ss} was estimated as the mean of the final 2000 samples of the simulation, and $P_{pv}(t)$ is the PV output power. This formula is in the dedicated metrics MATLAB computation script, which calculates the time did the PV array power to first reach 95% on its own steady-state value.

Tracking accuracy (dynamic stability) was assessed using a normalized stability index computed from the steady-state PV output power. The final 5000 samples of each simulation were designed as the steady-state window, and the stability index was computed as the coefficient of variation of output power within that window:

$$Stability = \sigma(P_{ss}) / \mu(P_{ss})$$

where $\sigma(P_{ss})$ and $\mu(P_{ss})$ are the standard deviation and mean of the PV output power in the steady-state window, respectively. A lower stability calculated with the formula indicates reduced steady-state oscillation and more precise tracking of the MPP. This metric is equivalent to the normalized ripple ratio reported in the results.

Harvested energy was computed as the total electrical energy delivered to the fixed load resistance of 20 Ω over the 60-second simulation duration:

$$E = \sum P_{out}(t) \times \Delta t / 3600$$

Energy (E) is in watt-hours units (Wh). This formula multiplies the instantaneous output power (watts) by the duration of each time step (seconds). This metric directly captures the practical energy yield of each algorithm under the simulated irradiance profile and scenario. Based on the entropy-weight calculation, this metric serves as the

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primary basis for the entropy-weighted Algorithm Ranking Stability (ARS), given its highest entropy weight of 0.4350.

The Cloud Transient Recovery Time (CTRT) was defined as the total cumulative duration over which the PV output power remained below 95% of the rolling maximum power throughout the 60-second simulation period. The rolling maximum serves as a dynamic reference and follows the irradiance as it changes throughout the simulation. A rolling maximum with a window of 500 samples was applied to Ppv to obtain the reference power trajectory, and the CTRT was computed as:

$$CTRT = \sum \Delta t, \text{ for all } t \text{ where } Ppv(t) < 0.95 \times P_{ref}(t)$$

Where $P_{ref}(t)$ is the rolling maximum of Ppv at time t. This formulation accumulates all periods during which the algorithm has not yet recovered to near-maximum power following an irradiance event. This provided a continuous measure of cumulative tracking lag across the full simulation horizon rather than the response time of any single transition.

The computed metrics were aggregated into a master results table containing one row per algorithm-dataset-scenario, for a total of 120 rows. Per-algorithm averages across all 12 scenario-dataset combinations are used as the basis for the entropy weight method and ARS ranking. Comparative bar charts and plots were generated to visualize multi-metric performance differences across algorithms.

3.9.2 Statistical and Comparative Analysis

This section includes the statistical and comparative analyses to accurately evaluate the performance of the MPPT algorithms. These analyses aim to guarantee that

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the differences among the algorithms are statistically significant and scientifically reliable under various and fluctuating conditions.

3.9.2.1 Descriptive Statistical Analysis

Simulation raw data were subjected to descriptive statistical analysis first. The mean (μ), standard deviation (σ), and variance (σ^2) of the tracking efficiency, convergence time, steady-state oscillation, total energy yield, CTRT, and computation time were computed using the formulas:

$$\mu = \frac{1}{N} \sum_{i=1}^N Xi, \quad \sigma^2 = \frac{1}{N-1} \sum_{i=1}^N (Xi - \mu)^2 \quad \sigma = \sqrt{\sigma^2}$$

Here, N represents the number of algorithms evaluated (N = 10), and X_i is the average performance metric value of the i-th algorithm across all benchmark datasets and irradiance scenarios. Sample standard deviation and variance (with N-1 in the denominator) were employed here since the ten algorithms represent a sample of possible MPPT approaches and not the whole population of all possible control strategies. These formulas represented the central tendency and dispersion of algorithm performance across the evaluated set. A smaller standard deviation indicated that algorithms performed more consistently and closely to one another on a given metric, while a larger standard deviation indicated greater performance spread and higher discriminatory value of that metric in differentiating algorithm behavior [71], [133]. Descriptive statistical analysis also served as the basis for the succeeding inferential testing analyses [110].

3.9.2.2 Comparative Performance Analysis

After computing all the performance metrics and statistical parameters, a comparative performance analysis was performed. This analysis highlighted which MPPT algorithms behaved better under tropical and benchmark conditions. It considered a few metrics for evaluation, namely: tracking efficiency, convergence time, steady-state oscillation, total harvested energy, CTRT, and computational effort, as defined in Section 3.8.

The percent improvement of one algorithm over another for pairwise comparisons was determined using:

$$Improvement\ (\%) = \frac{X_A - X_B}{X_B} \times 100$$

Where X here referred to the performance metric being evaluated, metrics for which higher values are desirable, such as tracking efficiency and total energy yield, and those for which lower values are desirable, such as oscillation amplitude, CTRT, and convergence time, were interpreted accordingly to deduce better performance. Variance and standard deviation were also assessed in observing the stability and robustness of algorithms under fast-changing irradiance conditions.

Algorithm Rank Stability (ARS) was then calculated by observing the differences in rankings using entropy-based weighting. ARS helped determine the stability of rankings of various algorithms with respect to varying levels of irradiance. Finally, to ensure that any differences found in rankings are statistically significant, non-parametric tests such as the Friedman test and Nemenyi post-hoc test, as discussed in Section 3.10.7.

3.9.2.3 Step-by-Step Statistical and Comparative Analysis

The full statistical and comparative analysis followed the step-by-step procedure below. Figure 3.4 presents the corresponding flowchart.

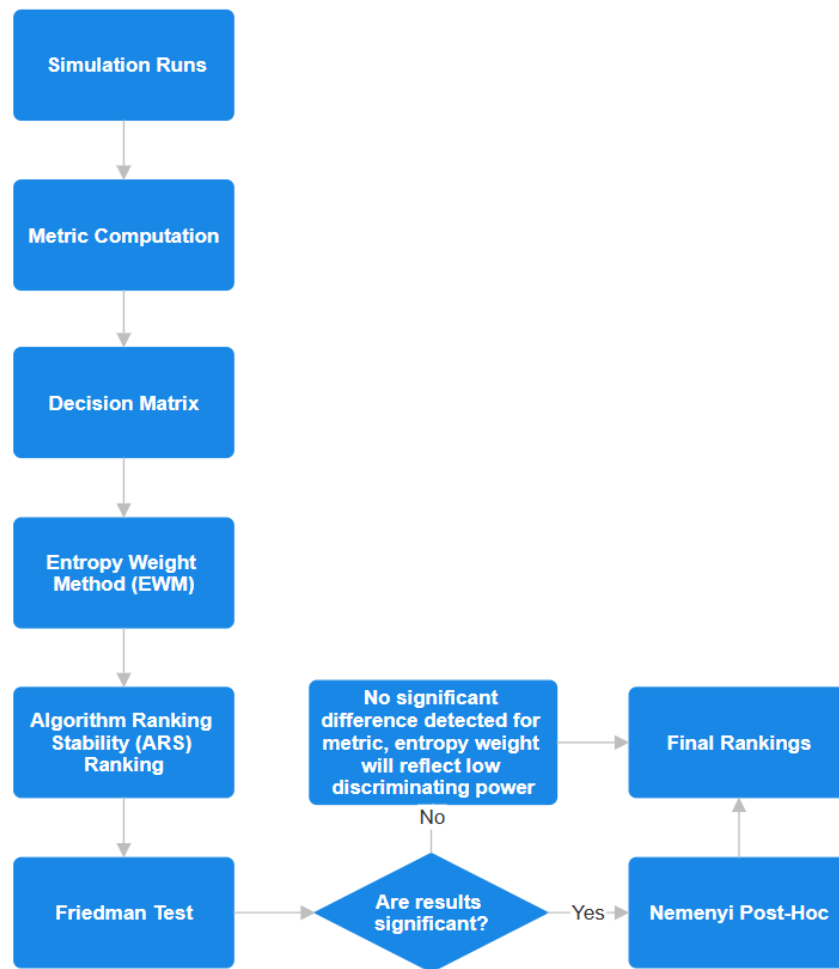


Figure 3.4 *Flowchart of Statistical and Comparative Analysis*

Step 1 is to collect simulation outputs. For each of the 10 MPPT algorithms, 12 simulation runs were executed (3 datasets x 4 weather scenarios). Each run produces a time-series of PV output power ($P_{pv}(t)$) sampled at a uniform timestep (Δt) over a

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60-second simulation window. These time series are stored as individual .mat output files and are loaded into the metrics computation MATLAB script.

Step 2 computes the five metrics (Convergence Time, Tracking Efficiency, Tracking Accuracy, Harvested Energy, Cloud Transient Recovery Time), wherein each metric is computed as follows:

1. Convergence Time (s): $t_{\text{conv}} = \min\{t: P_{\text{pv}}(t) \geq 0.95 \times P_{\text{ss}}\}$, where P_{ss} is estimated as the mean of the final 2000 samples. Input is $P_{\text{pv}}(t)$, and Output is scalar t_{conv} per run.
2. Tracking Efficiency (%): $\eta_t = (\text{Energy}_{\text{out}} / \text{Energy}_{\text{in}}) \times 100$, where $\text{Energy}_{\text{in}} = \sum P_{\text{pv}} \times \Delta t$ and $\text{Energy}_{\text{out}} = \sum P_{\text{out}} \times \Delta t$ over the full 60 s window. Inputs include $P_{\text{pv}}(t)$ and $P_{\text{out}}(t)$, and the output is a scalar η_t per run.
3. Tracking Accuracy (Stability): $\text{Stability} = \sigma(P_{\text{ss}}) / \mu(P_{\text{ss}})$, computed over the final 5,000 samples of the simulation window. Input is $P_{\text{pv}}(t)$, and Output is the scalar ripple ratio per run.
4. Harvested Energy (Wh): $E = \sum P_{\text{out}}(t) \times \Delta t / 3600$ over the full 60 s window. Input is $P_{\text{out}}(t)$ while Output is scalar E per run.
5. Cloud Transient Recovery Time (s): $\text{CTRT} = \sum \Delta t$ for all t where $P_{\text{pv}}(t) < 0.95 \times P_{\text{ref}}(t)$, where $P_{\text{ref}}(t)$ is a rolling maximum of P_{pv} using a 500-sample window. Input includes $P_{\text{pv}}(t)$, and the output is scalar CTRT per run.

Step 3 is to aggregate into a decision matrix. Per-algorithm averages are computed across all 12 scenario-dataset combinations to produce a 10×5 decision matrix of $X = [x_{ij}]$, where i = algorithm index (1 to 10) and j = metric index (1 to 5). This matrix serves as the input to the Entropy Weight Method (EWM).

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Step 4 is the application of the EWM, where the decision matrix is normalized, entropy values are computed per metric, diversification degrees are derived, and entropy weights (w_j) are calculated as described in 3.7.

Step 5 computes the Algorithm Ranking Stability (ARS) wherein $ARS_i = \sum w_j \times r_{ij}$, where r_{ij} is the normalized rank of algorithm i on metric j . Higher ARS indicates better overall performance. Inputs include X and W , while the output of this step is the ARS score per algorithm.

Step 6 applies the Friedman test independently to each of the five metrics using the 12 scenario-dataset combination per algorithm values as observations. The test statistic χ^2_F follows a chi-squared distribution with $k-1 = 9$ degrees of freedom. Input is the raw metric matrix, and the outputs are χ^2_F and p-value per metric.

Step 7 uses the Nemenyi Post-Hoc test, where the Friedman test gives $p < 0.05$ to identify which specific algorithm pairs differ significantly. The critical difference is computed as $CD = q_\alpha \times \sqrt{\frac{k(k+1)}{6N}}$ where q_α is the critical value for $\alpha = 0.05$ and $N = 12$ observations. Algorithm pairs with mean rank differences exceeding CD are considered statistically significantly different. Input is the Friedman test results, and the output is a pairwise significance matrix.

3.9.3 Data Interpretation

Metric interpretation was utilized for understanding the interpretation of each performance metric with respect to the MPPT algorithm performance. Tracking efficiency and total energy indicated the capability of extracting maximum power, where better performance was denoted by higher values of efficiency or energy. Smaller

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convergence time and low values of CTRT indicated quicker adjustments for irradiation, as well as reduced accumulated error ranges for maximum performance. Amplitude of oscillations for the steady-state oscillations denotes stability at MPP, with lower oscillations indicating reduced power losses [135], [136].

Statistical interpretation was a complement to metric analysis in that it not only focuses on average performance but also on consistency. A low value of sensitivity coefficients was preferable as it is less sensitive to variations in weather conditions [137].

Additionally, graphical interpretation helped support the analysis with time-domain graph illustrations of voltage, current, and power for the PV system. These provided an effective means of showing convergence, stability, and tracking rates.

Finally, a framework of comparison was established by integrating the normalized metrics, weights, and stability of ranking. The ARS offered a solid basis for choosing MPPT algorithms with outstanding energy generation, dynamic performance, stability, computational complexity, and consistency for both benchmark and tropical datasets. An integrated analysis assisted in forming well-informed decisions on the appropriateness of the MPPT algorithm to be used in the Pampanga PV scenario [138].

3.10 Algorithm Comparison Across Multiple Datasets

3.10.1 Overview of Cross-Dataset Evaluation

All MPPT algorithms were tested with various irradiance and temperature profiles using the cross-dataset test framework. There were three datasets: two benchmark datasets that represent standard and high variability PV systems, respectively, and one candidate tropical dataset for Arayat-Mexico Solar Farm. Testing the MPPT algorithms

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with these datasets made it possible to evaluate their consistency, flexibility, and robustness for various weather patterns.

Within this context, standard performance measures were calculated for each dataset individually, such as convergence time, tracking efficiency, tracking accuracy (stability), harvested energy, and cloud transient recovery time. These measures quantified both instantaneous and accumulated performances of an algorithm for a given set of environmental conditions.

To generalize performance analysis on datasets, the paper also explored the stability of the rankings of the algorithms on datasets using Algorithm Rank Stability (ARS). ARS was a more generalized metric of the stability of relative rankings of MPPT algorithms with respect to various irradiance conditions.

Simulation setup and controller configurations remained the same for all datasets to ensure that differences in observed performance are due to the environments simulated and not driven by differences in system or algorithm models.

3.10.2 Description of Datasets Used

3.10.2.1 Benchmark Dataset 1: Standard Mid-Latitude PV Dataset

Benchmark Dataset 1 represents a standard mid-latitude PV operating environment that encompasses stable irradiance, moderate temperature variations, and relatively smooth transitions throughout the day. This dataset is based on the publicly available irradiance and meteorological data provided by the National Renewable Energy Laboratory (NREL) through the National Solar Radiation Database (NSRDB). This is usually used as a baseline in MPPT studies because the operating conditions are less

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volatile compared to tropical climates. It provides a controlled reference profile for evaluating how each MPPT algorithm performs under stable weather conditions. Processing and formatting time-series irradiance and temperature signals from this dataset will be applied to the PV system model with identical simulation settings to those adopted in processing the tropical dataset [154].

3.10.2.2 Benchmark Dataset 2: High-Variability Cloud-Transient Dataset

Benchmark Dataset 2 is representative of a high-variability irradiance profile, including rapid cloud transients with steep irradiance ramps and frequent short-term fluctuations. This dataset is based on the irradiance test sequences defined in the European Standard EN 50530, which is one of the most used benchmarks in MPPT efficiency tests. The EN 50530 has irradiance ranges which are low to medium irradiance version (100-500 W/m²) and medium to high irradiance version (300-1000 W/m²). The EN 50530 test profiles include different ramp rates to simulate varying irradiance at fast or slow rates measured in 10-100 W/m²/s. Equivalent time-series implementations of EN 50530 dynamic irradiance profiles are widely used in simulation-based MPPT studies and are supported by reference materials from Sandia National Laboratories [155].

3.10.2.3 Proposed Tropical Dataset: Arayat–Mexico Solar Farm

The proposed tropical dataset represented local operating conditions of the Arayat-Mexico Solar Farm in Pampanga, having high-resolution irradiance and temperature measurements. Elevated temperatures, rapid cloud movements, and sharp irradiance drops during typhoon periods are included in this dataset. Data was sourced from Solcast to give full coverage of all seasonal and short-term variations. After

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preprocessing, the dataset was then converted into time-series inputs for MATLAB/Simulink and applied under the same simulation configuration utilized for the benchmark datasets. This dataset served as the primary basis for evaluating MPPT algorithm performance under Philippine tropical conditions.

3.10.3 Data Preprocessing and Standardization

3.10.3.1 Data Cleaning

All data is passed through an automated data cleaning process before being utilized in the simulation environment. Missing or incomplete irradiance and temperature data values were identified, either removed or replaced via short-gap interpolation in order to keep the continuity of the signal intact. Sudden spikes, negative values, or physically unfeasible measurements were filtered out in order not to distort MPPT performance evaluation results. Outliers caused by sensor fault, logging errors, or transmission noise were also found and corrected to ensure that every dataset represents real operating conditions. The cleaned dataset was, therefore, be prepared for time alignment and normalization steps in the subsequent preprocessing stages.

3.10.3.2 Time Alignment and Resampling

All time series data were aligned to a uniform time base in order to compare values consistently between sources. Because the benchmark and tropical datasets may be on different sampling intervals, each dataset was resampled to have the equivalent controller sampling rate as that used within the simulation model. Linear interpolation was performed when down-sampling lower-resolution data is converted to higher-resolution time steps. On the other hand, averaging was performed when

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higher-frequency datasets are being down-sampled. This ensured that every algorithm had synchronized irradiance and temperature inputs at each simulation step, thereby eliminating timing inconsistencies that could impact MPPT performance evaluation.

3.10.3.3 Normalization of Input Variables

All the inputs related to irradiance and temperature were normalized and then fed into the simulation model to maintain numerical scaling consistency among datasets. Normalization transformed every dataset into a standardized range that MPPT algorithms, especially ANN- and metaheuristic-based techniques, can process without any disturbance in terms of level of magnitude variation. This avoids bias caused by dataset-specific scaling and allows algorithm behavior to be driven by differences in real performance rather than variations in input units or ranges. After normalization, the processed data were ready to use in all simulation scenarios.

3.10.4 Experimental Scenarios

3.10.4.1 Clear-Sky Scenario

The clear-sky profile corresponded to a stable irradiance situation, having low fluctuations, a smooth curve of irradiance, and a consistent temperature. This scenario was used to determine how well each MPPT algorithm tracks the maximum power point during stable environmental conditions. This provides a basis for tracking efficiency, steady-state oscillations, and convergence behavior without the interference of fast weather changes. All algorithms were run under the same clear-sky profile for uniformity and fairness in baseline comparison performance.

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3.10.4.2 Ramp-Change Scenario

The ramp change simulated a range of slow, progressive increases or decreases in irradiance over time, indicating situations such as those presented by slow-moving clouds and during morning and afternoon transitions. This scenario investigated how well each MPPT algorithm may adapt smoothly to continuous environmental changes and if it can maintain high tracking efficiency during an extended irradiance ramp. Responsiveness, stability, and capability for tracking a shifting maximum power point without excessive oscillations or delay were measured by applying the same ramp profile to all datasets and algorithms.

3.10.4.3 Step-Change Scenario

The step-change scenario modeled sudden and abrupt changes in irradiance, such as those caused by fast-moving clouds or short events of shadowing. Within this scenario, the transient response for every MPPT algorithm was assessed, specifically in terms of how well it can promptly detect and track the new maximum power point after a rapid change in the environmental conditions. Key performance indicators in this scenario included convergence time, overshoot behavior, and stability during the immediate period following the event of a step change in irradiance. All algorithms run through the same step-change profile to ensure consistent and comparable assessment.

3.10.4.4 Mixed Weather Scenario

A mixed-weather scenario was a combination of clear-sky, ramp changes, and fast step variations. This simulation gave a realistic, day-long tropical weather behavior. It evaluated the general adaptability and robustness of each algorithm under constantly

varying conditions. It assessed tracking efficiency, stability, and responsiveness because of the changing weather irradiance within one run. These mixed characteristics of weather ensured a holistic performance evaluation across diverse and unpredictable operating conditions.

3.10.5 Algorithm Execution Protocol

3.10.5.1 Initialization Conditions

The simulation environment initialized all algorithms with identical system and environmental conditions. Each algorithm had identical PV voltage (V_{pv}), PV current (I_{pv}), and STC irradiance and temperature that corresponded with the starting dataset. The boost converter operated in Continuous Conduction Mode (CCM), with the capacitor voltage and inductor current pre-charged to their steady-state conditions, to avoid artificial transients that may bias the results [139], [140].

Algorithm-specific internal parameters, such as previous voltage, previous current, estimated derivatives, membership function outputs, swarm positions, or population values, were reset at the start of each simulation [141], [142]. Resetting prevented learning-based and metaheuristic algorithms from carrying over residual information from previous simulations. As defined earlier in Section 3.6, we set the solver settings, sampling time, and switching frequency to maintain uniform behavior across all MPPT controllers.

These standardized initializations ensured that algorithm performance is solely due to controller design, and not influenced by unequal starting conditions or residual simulation information.

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3.10.5.2 Algorithm Run Procedure

All MPPT algorithms followed a uniform execution procedure to maintain consistency and fairness across all simulations. For every simulation run, the irradiance and temperature dataset was loaded into the PV Array block. The PV array, boost converter, and algorithm were initialized at $t = 0$ in accordance with sections 3.11.5.1 and 3.11.5.2. During the simulation, each algorithm read the PV voltage (V_{pv}) and PV current (I_{pv}) to compute the appropriate control action, which outputted a corresponding duty cycle for the boost converter. The boost converter then updated the PV operating point in response, after which the process repeated each sampling interval as time advanced. Execution of the algorithm simulation remained identical for all. A 60-second run time was used and expired for each scenario per dataset. Ensuring that every algorithm followed the same procedural steps attributed any observed performance difference to algorithm behavior rather than unequal simulation handling [149], [145].

3.10.6 Performance Metrics per Dataset

In Section 3.8, various performance parameters are discussed. These performance parameters for each MPPT technique were calculated for various datasets analyzed in this research work. Additionally, the benchmark datasets as well as the tropical datasets provided independent sources for extracting respective performance parameters for each of the datasets. These performance parameters included efficiency of tracking, convergence time, amplitude of oscillation in stationary points, total energy output, complexity, and Cumulative Tracking Response Time.

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These simulated PV voltage, current, and power outputs provided inputs for the analysis of all the metrics on the dataset level. All metrics were calculated separately for each data set to reveal the algorithms' performance corresponding to their own irradiance and temperature conditions. CTRT was also calculated for each data set since it is responsible for a number of dynamic tracking transitions between the irradiances.

After completing the computations at the dataset level, the obtained performance metrics were normalized and weighted using the Entropy Weight Method. The resulting weighted values were used to establish the relative performance rankings of the algorithms for each dataset and served as the basis for the subsequent statistical analyses described in Section 3.10.7.

Algorithm Rank Stability (ARS) was not measured at the dataset level. It was actually calculated in the next stage of aggregation through the analysis of the variability of rankings of algorithms using entropy-based weighting. This was important to ensure that ARS is a measure of stability among rankings of datasets, not their performance.

3.10.7 Statistical Comparison Across Datasets

Determination of MPPT algorithm performance required statistical significance across the three datasets. Non-parametric tests were conducted as the study had a repeated-measure structure, while the MPPT performance metrics have a non-normal distribution. These tests were carried out on performance metrics related to the data set (tracking efficiency, convergence time, tracking accuracy, energy harvested, and CTRT)

First, the Friedman test was used to compare all ten algorithms simultaneously across each performance metric. Each dataset was processed using the Entropy Weight

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Method, after which it was evaluated using the Friedman test to determine if there are statistically significant differences that exist among algorithms based on the obtained performance metric values.

Second, if the Friedman test identified significant differences, the Nemenyi post hoc test was conducted to identify which algorithm pairs differ in performance. This allowed for pairwise comparisons without assuming normality.

Third, to aid pairwise comparisons, the Friedman and Nemenyi post hoc test was used for pairwise comparisons between datasets to verify consistency across different environmental conditions. This test was very helpful for checking the validity of differences for both CTRT and convergence metrics.

Statistical significance was evaluated using p-values with a significance level of $p < 0.05$. These non-parametric methods allowed robust statistical validation of algorithm performance differences and ensured that comparative conclusions from the metric-based analysis are reliable and not a result of random variations [150].

Algorithm Rank Stability (ARS) did not undergo inferential statistical analysis, as it was calculated from the result of rankings on various datasets instead of the distribution of the metrics. ARS was employed as a descriptive statistic to determine the stability of rankings by the algorithm after a valid performance analysis.

Lastly, results from the test were integrated into an expert-weighted performance index to produce final algorithm rankings.

3.10.8 Result Aggregation and Comparative Evaluation

The results obtained from these non-parametric tests were then aggregated such that a comparative evaluation of these in a unified manner may be performed. The mean ranks were developed for each metric to outline both stability and consistency of the behavior obtained by each algorithm. It incorporated tracking efficiency, convergence time, steady-state oscillation amplitude, total energy yield, computational complexity, and CTRT at the dataset level.

After the metric values were normalized to a common scale, entropy-based weighting was applied to each metric to objectively determine its relative importance based on the discriminatory information it carried across all algorithms. Metrics that exhibited greater variation across algorithms received higher entropy weights, reflecting their greater contribution to differentiating algorithm performance.

Along with this, statistical validation using the Friedman test and Nemenyi post-hoc specifies algorithm pairs that differ across the three datasets. Significance of those differences was determined using p-values below the $\alpha = 0.05$ threshold.

The ARS was calculated by studying how many algorithm rankings have changed between the benchmark and tropical datasets. ARS was merely a descriptive indicator of ranking consistency and provided one with the names of algorithms that maintain stable relative performance across a diversity of irradiance profiles.

The overall comparative performance assessment, which identifies the algorithms that perform best under both benchmark and tropical conditions, was derived by combining the algorithm rankings obtained from the weighted performance metrics and

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the results of the statistical analyses. This approach ensures that the final conclusions reflect not only statistically validated performance differences but also consistent algorithm rankings and practical engineering considerations.

3.10.9 Justification for Incorporating a Tropical Dataset

The tropical dataset for the Arayat-Mexico solar farm was incorporated to ensure that the comparative analysis for all MPPT algorithms reflected real-world operating conditions of the PV site. Compared to standard benchmark datasets, tropical datasets exhibited higher irradiance fluctuations, higher temperature, higher humidity levels, prevalent typhoons and monsoons, and frequent cloud transients. These characteristics affected PV performance significantly in terms of tracking response and converter stability.

Operating conditions such as these were important in the context of CTRT, since transitions in irradiance tended to enhance the cumulative dynamics of MPPT algorithms. This made tropical data critical for assessing algorithm robustness in places with greater variability compared to mid-latitude regions.

Also, a local tropical dataset captured unique environmental stressors present in Philippine solar PV sites. Incorporating this dataset further enabled ARS analysis in the context of testing whether algorithm rankings derived from benchmark datasets remained consistent under realistic tropical operating conditions. Therefore, the algorithm performance assessment using the tropical dataset was directly applicable to Philippine PV operations.

3.11 Methodology for Ensuring Research Novelty

This research ensured novelty through the extraction of new, robust, and dynamics-oriented insights by re-simulating all ten algorithms under a single environment. Each MPPT algorithm had already been well-documented in existing literature, prior works, and review papers, where results were obtained from heterogeneous models, irradiance profiles, and evaluation protocols, which may have limited the comparability and transferability of conclusions.

The study performed a controlled comparative simulation of ten algorithms, which included conventional, control-based, intelligent, and metaheuristic approaches. The PV model, PV parameters, control parameters, sampling rates, and execution procedures were identical for all algorithms tested.

A multi-dataset testing framework was used with three distinct irradiance datasets: the standard mid-latitude benchmark dataset, the high-variability cloud transient benchmark dataset, and a proposed localized tropical dataset derived from the Arayat-Mexico Solar Farm. Beyond simple cross-validation, this framework aimed to investigate whether conclusions about MPPT algorithm performance would remain consistent across different datasets. In this study, this effect was quantified by introducing Algorithm Rank Stability (ARS) across datasets, which measured the consistency of algorithm performance across different irradiance conditions. The introduction of ARS enabled the evaluation of whether MPPT algorithms maintained their relative performance when applied under realistic tropical conditions.

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The study also incorporated Cloud-Transient Recovery Time (CTRT), a dynamic performance metric that captured the time required for MPPT algorithms to recover after sudden changes or spikes in irradiance caused by cloud transients. CTRT was distinct from convergence and steady-state efficiency in the sense that it analyzed the performance of MPPT algorithms in real-world disturbance scenarios, which were not possible to analyze in steady-state conditions.

Finally, statistical analysis tests such as the Friedman and Nemenyi post-hoc test ensured that the results were statistically significant and not just a form of numerical valuation, particularly in the case of the newly proposed ARS and CTRT metrics. All these factors enabled the proposed study to provide a comprehensive and data-aware benchmark, which enabled the formulation of replicable and statistically valid conclusions regarding the performance of MPPT algorithms in tropical PV systems.

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CHAPTER 4.0. Results and Discussion

4.1 Overview

This chapter presents the results obtained from the simulation and analysis of the chosen Maximum Power Point Tracking (MPPT) algorithms when subjected to different weather conditions. The simulation was carried out using MATLAB/Simulink to analyze the efficiency of the chosen MPPT algorithms when applied to a PV system using environmental data that corresponds to the Arayat-Mexico Solar Farm in Pampanga.

The chosen MPPT algorithms' performances were analyzed using five different performance parameters: convergence time, tracking efficiency, tracking accuracy (stability), harvested energy, and cloud transient recovery. These parameters were chosen to comprehensively analyze the efficiency of the chosen MPPT algorithms when applied to a PV system.

The results obtained from the simulation for the different performance parameters will first be discussed individually. Then, a comparison will be made to analyze the different performances of the chosen MPPT algorithms. To comprehensively analyze the overall efficiency of the chosen MPPT algorithms, the Entropy Weight Method will be applied to determine the weights for the chosen performance parameters. The weights will then be used to determine the Algorithm Ranking Stability(ARS).

The results will be compared using different weather datasets in order to compare the relative consistency of the results of the respective algorithms. Statistical analysis will be conducted in order to investigate the relationship between the results of the respective algorithms and environmental factors. Finally, the chapter will be concluded by

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highlighting the major findings and observations from the results obtained, which will be the basis of the conclusions presented in the next chapter.

4.2 Simulation Setup Summary

This section gives a brief overview of the simulation setup that was performed in order to assess the performance of the selected Maximum Power Point Tracking (MPPT) algorithms. The PV system and the MPPT control algorithm were simulated using the MATLAB/Simulink environment. The simulations were performed in order to assess the performance of the selected algorithms under various weather conditions that are characteristic of the weather patterns experienced at the Arayat-Mexico Solar Farm in Pampanga.

Ten MPPT algorithms were implemented and evaluated using the same PV system setup in order to assess their performance under various weather conditions. The performance of the algorithms was evaluated using various weather scenarios in order to assess their performance in tracking the maximum power point under various weather conditions.

Five performance metrics were used in evaluating the performance of each algorithm. They include convergence time, tracking efficiency, tracking accuracy/stability, harvested energy, and cloud transient recovery. These metrics were selected to capture different aspects of MPPT performance, including speed of response, tracking precision, energy capture capability, and robustness during sudden irradiance fluctuations.

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Simulation results obtained from the scenarios provided form the basis of the performance evaluation, comparative analysis, and ranking of the various MPPT algorithms discussed in the succeeding sections of this chapter.

4.3 Performance Metric Results

4.3.1 Convergence Time

The term "convergence time" is used to define the period of time it takes for the MPPT algorithm to achieve the maximum power point (MPP) after a change in conditions of irradiance. Therefore, it is desirable for the algorithm to have less convergence time; in other words, it should be able to adapt quickly to changing conditions in order to maximize power extraction from the photovoltaic (PV) array. This metric is particularly relevant in dynamic conditions where the rate of change of irradiance is high.

Benchmark Dataset 1: Standard Mid-Latitude PV Dataset

Algorithm	Clear	Mixed	Ramp	Step
P&O	1.80599999427 795	1.76900005340 576	1.76900005340 576	1.28499996662 14
IncCond	1.79900002479 553	1.76900005340 576	1.76900005340 576	1.29200005531 311
SMC	1.78100001811 981	1.75800001621 246	1.75800001621 246	1.28900003433 228
RCC	1.79700005054 474	1.75899994373 322	1.75899994373 322	1.29600000381 47
FSB	1.65400004386 902	1.55299997329 712	1.55299997329 712	1.15400004386 902

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FLC	1.90699994564 056	1.83899998664 856	1.83899998664 856	1.35300004482 269
ANN	1.81799995899 2	1.97200000286 102	1.94500005245 209	1.61300003528 595
PSO	1.01100003719 33	0.77999997138 9771	0.44499999284 7443	0.29399999976 1581
GA	1.04499995708 466	1.125	1.06500005722 046	1.125
GWO	2.26699995994 568	1.30200004577 637	1.09599995613 098	1.37300002574 921

Table 4.1 Convergence Time of MPPT Algorithms under Standard Mid-Latitude PV Dataset

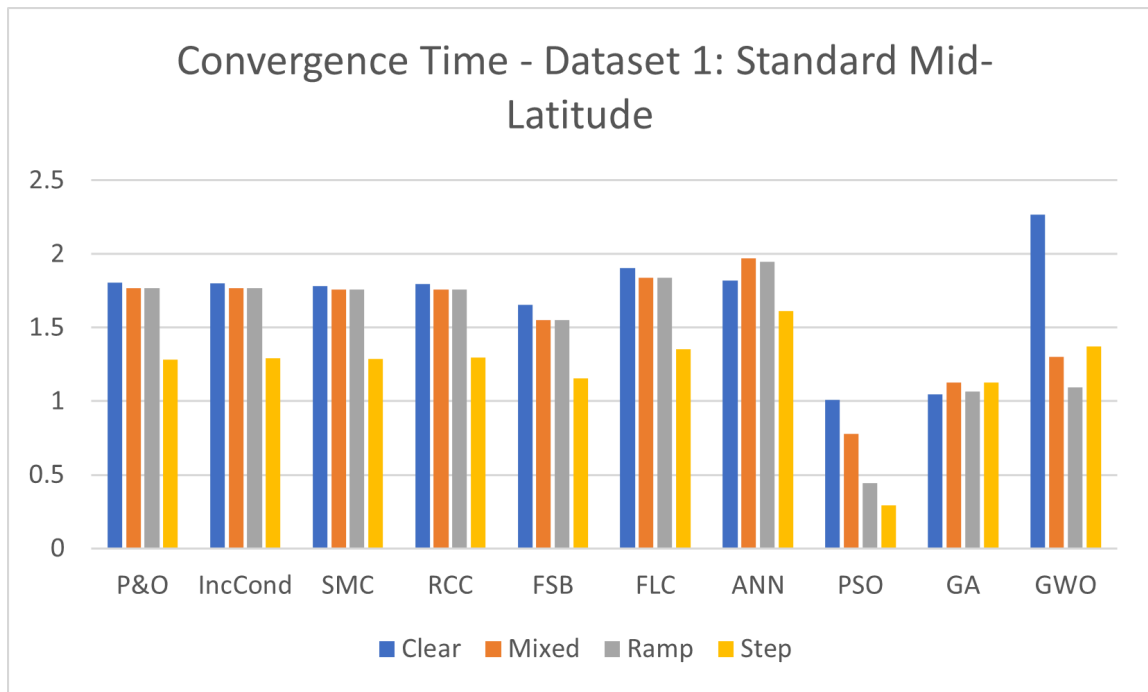


Figure 4.1 Convergence Time of MPPT Algorithms under Standard Mid-Latitude PV Dataset

Table 4.1 and Figure 4.1 shows the convergence time of all MPPT algorithms under clear, mixed, ramp, and step conditions using standard Mid-Latitude data.

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Convergence time is defined as the time taken by an MPPT algorithm to track the MPP from an initial point after a change in irradiance conditions. Lower convergence time is desirable, and thus an algorithm with lower convergence time is said to have better dynamic performance.

From the table and figure, it can be observed that the Particle Swarm Optimization (PSO) algorithm has the fastest convergence time under all conditions. It has recorded convergence times of 1.011 s, 0.780 s, 0.445 s, and 0.294 s under clear, mixed, ramp, and step conditions, respectively. The Genetic Algorithm (GA) has also recorded fast convergence times under all conditions, particularly under mixed and ramp conditions.

Among all Classical MPPT algorithms, P&O and Incremental Conductance (IncCond) algorithms have recorded almost similar convergence times under all conditions, ranging from 1.7 s to 1.8 s. This is due to similar search mechanisms employed in these algorithms. Among Control-Theoretic algorithms, SMC and RCC have shown improved convergence times compared to Classical algorithms. FSB has recorded lower convergence times, indicating a faster response in locating the maximum power point.

However, Fuzzy Logic Control (FLC) and Artificial Neural Network (ANN) show relatively higher convergence times for various irradiance cases. Notably, ANN shows higher convergence times under mixed and ramp irradiance conditions, which means that it takes longer to adapt to changes in irradiance. The Grey Wolf Optimizer (GWO) shows the lowest convergence time under clear sky conditions at 2.267 seconds, which is a result of the nature of metaheuristic optimization algorithms that require iterative search.

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Overall, it is evident from the results that metaheuristic algorithm methods like PSO and GA have better convergence abilities, while classical and control-theoretic methods maintain moderate but consistent response times under the standard mid-latitude PV dataset.

Benchmark Dataset 2: High-Variability Cloud-Transient Dataset

Algorithm	Clear	Mixed	Ramp	Step
P&O	1.56200003623 962	0.90200001001 358	0.88499999046 3257	0.21099999547 0047
IncCond	1.57500004768 372	1.06900000572 205	1.06900000572 205	0.33599999547 0047
SMC	0.98799997568 1305	0.44699999690 0559	0.44699999690 0559	0.21099999547 0047
RCC	1.57700002193 451	0.90200001001 358	0.90200001001 358	0.21099999547 0047
FSB	1.35899996757 507	0.79299998283 3862	0.79299998283 3862	0.05999999865 88955
FLC	1.71300005912 781	1.00399994850 159	1.00399994850 159	0.21099999547 0047
ANN	1.58800005912 781	0.44699999690 0559	1.06500005722 046	0.20399999618 5303
PSO	1.40600001811 981	0.20100000500 679	0.20100000500 679	0.20100000500 679
GA	1.125	0.20100000500 679	0.20100000500 679	0.01799999922 51396
GWO	3.39000010490 418	0.20399999618 5303	0.44699999690 0559	0.20399999618 5303

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Table 4.2 *Convergence Time of MPPT Algorithms under High-Variability
Cloud-Transient Dataset*

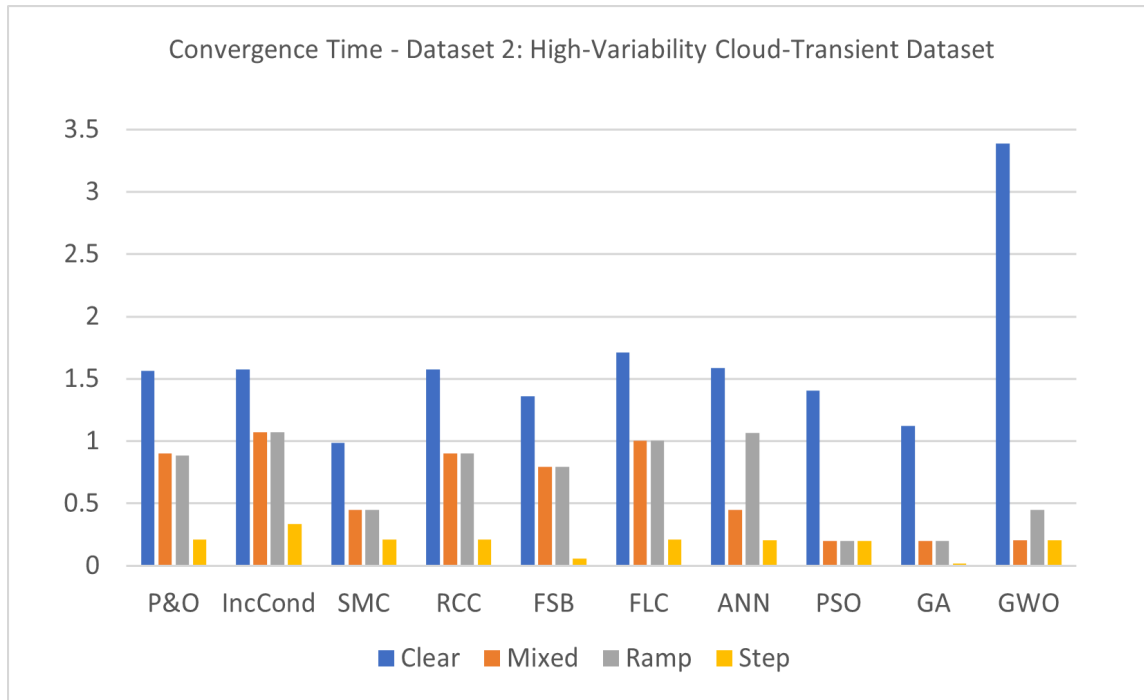


Figure 4.2 *Convergence Time of MPPT Algorithms under High-Variability
Cloud-Transient Dataset*

Table 4.2 and Figure 4.2 shows the convergence time of all MPPT algorithms under clear, mixed, ramp, and step irradiance conditions using high variability cloud transient data. As shown in the table, lower convergence times indicate faster stabilization to the maximum power point after sudden changes in irradiance levels.

From Table 4.2 and Figure 4.2, it is clear that the GA has the fastest convergence time under step irradiance conditions with a value of approximately 0.018 seconds. In contrast, the FSB algorithm has fast convergence times under the same conditions with a

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value of approximately 0.060 seconds. Additionally, the PSO algorithm has fast responses to dynamic conditions such as mixed, ramp, and step irradiance changes with a constant convergence time value of approximately 0.201 seconds.

From the Classical MPPT algorithms, P&O and IncCond have moderate convergence times under all irradiance conditions, indicating fast responses to irradiance changes. From the Control-Theoretic MPPT algorithms, SMC demonstrates improved convergence relative to classical methods under mixed and ramp scenarios.

In contrast, the Grey Wolf Optimizer (GWO) has the slowest convergence speed under clear conditions, which is around 3.39 seconds. Therefore, it was concluded that the exploration-based search mechanism of some of the metaheuristic algorithms might cause some delay before the optimal operating point is achieved.

From the above analysis and the simulation results, it was concluded that the metaheuristic algorithms have better convergence properties even under fluctuating irradiance conditions. On the other hand, the classical methods provide stable but comparatively slower response times.

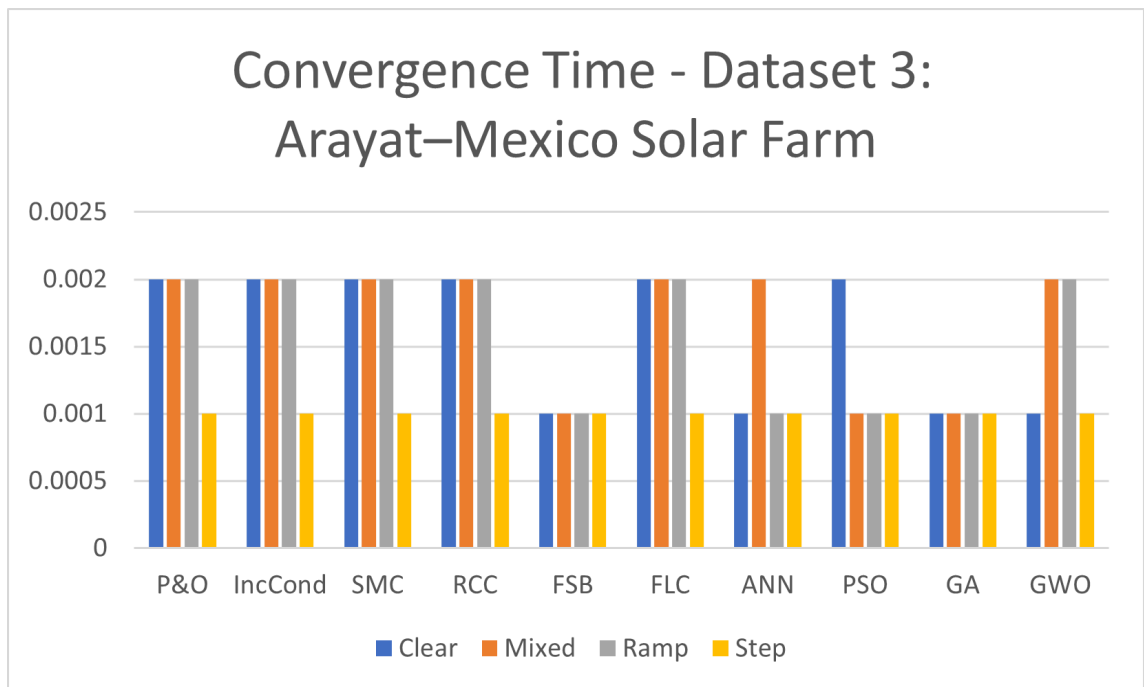
Benchmark Tropical Dataset 3: Arayat–Mexico Solar Farm

Algorithm	Clear	Mixed	Ramp	Step
P&O	0.00200000009 49949	0.00200000009 49949	0.00200000009 49949	0.00100000004 749745
IncCond	0.00200000009 49949	0.00200000009 49949	0.00200000009 49949	0.00100000004 749745
SMC	0.00200000009 49949	0.00200000009 49949	0.00200000009 49949	0.00100000004 749745

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RCC	0.00200000009 49949	0.00200000009 49949	0.00200000009 49949	0.00100000004 749745
FSB	0.00100000004 749745	0.00100000004 749745	0.00100000004 749745	0.00100000004 749745
FLC	0.00200000009 49949	0.00200000009 49949	0.00200000009 49949	0.00100000004 749745
ANN	0.00100000004 749745	0.00200000009 49949	0.00100000004 749745	0.00100000004 749745
PSO	0.00200000009 49949	0.00100000004 749745	0.00100000004 749745	0.00100000004 749745
GA	0.00100000004 749745	0.00100000004 749745	0.00100000004 749745	0.00100000004 749745
GWO	0.00100000004 749745	0.00200000009 49949	0.00200000009 49949	0.00100000004 749745

Table 4.3 Convergence Time of MPPT Algorithms under Arayat-Mexico Solar Farm Tropical Dataset



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***Figure 4.3 Convergence Time of MPPT Algorithms under Arayat-Mexico Solar Farm
Tropical Dataset***

The results for the convergence time of the tropical data, which represent the irradiance conditions of the Arayat-Mexico Solar Farm, are presented in Table 4.3 and Figure 4.3.

The results indicate that the convergence times of this data are much smaller compared to the previous benchmark data. All the algorithms were able to converge to the MPP within 0.001 to 0.002 seconds, indicating the rapid stabilization of the MPPT controllers under the tropical irradiance conditions.

Among the tested algorithms, the FSB, GA, and GWO methods had the shortest convergence times for several irradiance conditions, particularly for clear, step, and other conditions, where the convergence times were close to 0.001 seconds. This indicates the faster response of these methods in adjusting the operating point of the photovoltaic system under the particular irradiance conditions.

On the other hand, algorithms such as P&O, Incremental Conductance (IncCond), SMC, and FLC generally exhibited convergence times close to 0.002 seconds, which are slightly higher than those of the fastest algorithms but still represent rapid convergence.

The fast convergence speed for MPPT is an advantage, but it does not guarantee adaptability to changing environmental conditions. There is a compromise in choosing whether to prioritize fast convergence or adaptive tracking.

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Despite these small differences, the results demonstrate that all evaluated MPPT algorithms exhibit very fast convergence under the tropical dataset, indicating that the controllers are able to quickly stabilize near the maximum power point under the irradiance conditions represented in the Arayat–Mexico Solar Farm dataset.

4.3.2 Tracking Efficiency

Tracking efficiency is defined as the ability of an MPPT algorithm to extract power close to the theoretical maximum power point (MPP) of the photovoltaic system. Higher tracking efficiency means that an MPPT algorithm is able to operate close to the optimum power point with minimal loss of energy. It is an important measure of the performance of an MPPT algorithm in maximizing the energy available from solar irradiance under various environmental conditions.

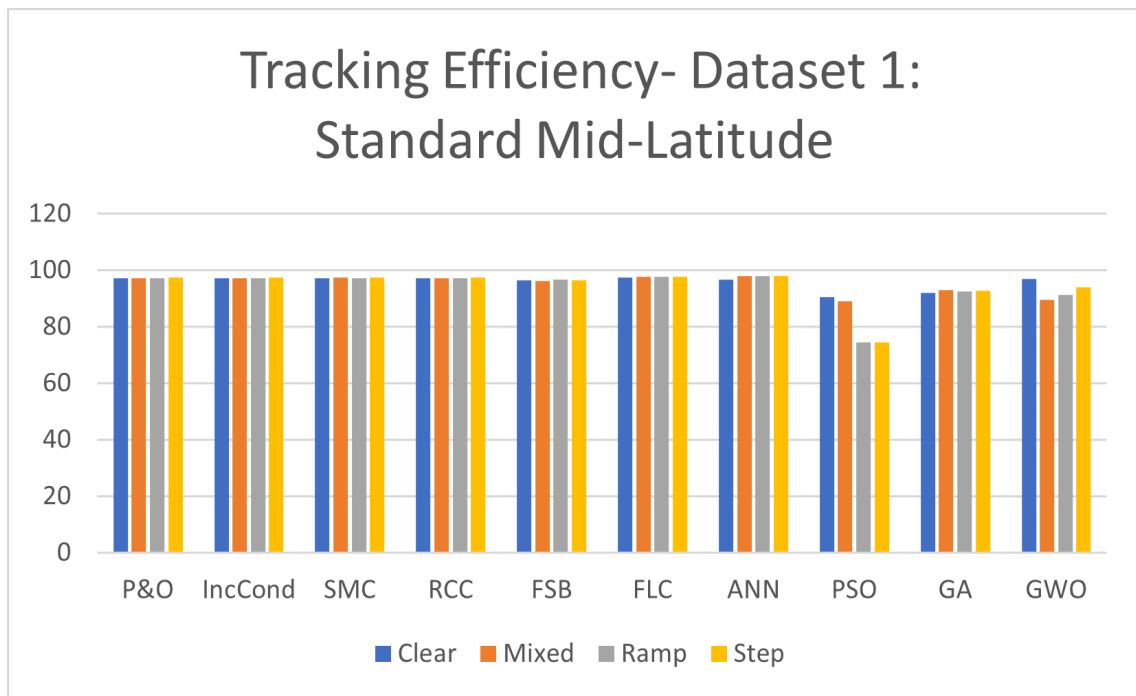
Benchmark Dataset 1: Standard Mid-Latitude PV Dataset

Algorithm	Clear	Mixed	Ramp	Step
P&O	97.1356178340 313	97.2544490478 087	97.2463219879 861	97.3576191560 385
IncCond	97.1341591731 568	97.2549553642 545	97.2469899037 715	97.3577471811 207
SMC	97.1330413644 606	97.2629856639 784	97.2453135268 595	97.3618458509 821
RCC	97.1353524237 914	97.2555892807 531	97.2465693579 956	97.3570977325 177
FSB	96.5078324140 138	96.1698170571 028	96.5400457446 619	96.3778797669 507
FLC	97.4865601975	97.5453503004	97.6029831046	97.6500759947

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	345	241	019	933
ANN	96.5829896500 02	97.8107595808 249	97.8299318333 325	97.7892556488 507
PSO	90.4355277132 505	89.0869281417 268	74.4452004163 555	74.3117226961 624
GA	92.0271402289 811	93.0533062921 21	92.3220323194 342	92.7410975700 018
GWO	96.8047676381 346	89.4530754448 712	91.0968224018 762	93.8370342503 828

Table 4.4 Tracking Efficiency of MPPT Algorithms under Standard Mid-Latitude PV Dataset



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***Figure 4.4 Tracking Efficiency of MPPT Algorithms under Standard Mid-Latitude PV
Dataset***

The results obtained for the tracking efficiency of the MPPT algorithms, as shown in Table 4.4 and Figure 4.4, are based on the clear, mixed, ramp, and step irradiance conditions for the standard mid-latitude PV dataset.

The results indicate that the majority of the MPPT algorithms obtained high tracking efficiency, greater than 97%, which demonstrates the stable performance of the MPPT algorithms under the standard mid-latitude irradiance conditions. Among the MPPT algorithms, the Artificial Neural Network (ANN) obtained the highest tracking efficiency under mixed and ramp irradiance conditions, reaching approximately 97.81% and 97.83%, respectively. Similarly, the Fuzzy Logic Control (FLC) obtained high tracking efficiency for the majority of the irradiance conditions, having results between 97.48% and 97.65%.

Among the Classical MPPT methods, P&O and Incremental Conductance methods show stable efficiency between 97.13% and 97.36%, thus ensuring reliable tracking in all conditions. Among the Control-Theoretic methods, both SMC and RCC show efficiency comparable to the Classical methods, thus ensuring effective tracking for the standard dataset.

On the other hand, the efficiency of the Particle Swarm Optimization (PSO) algorithm was significantly lower compared to other algorithms. For instance, in ramp and step conditions, the efficiency of the PSO algorithm was reduced to around 74.45% and 74.31%, respectively.

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Similarly, the efficiency of the Genetic Algorithm (GA) was lower compared to most classical algorithms and was in the range of 92-93%.

From the above analysis, it is observed that intelligent control algorithms like ANN and FLC have better tracking efficiency, while the efficiency of certain metaheuristic algorithms is reduced for standard mid-latitude data.

Benchmark Dataset 2: High-Variability Cloud-Transient Dataset

Algorithm	Clear	Mixed	Ramp	Step
P&O	97.1384159585 306	97.2918290983 853	97.2827132789 094	97.4596172140 651
IncCond	97.1390938765 606	97.2907332727 065	97.2803732425 806	97.4593259515 064
SMC	97.1490827903 284	97.2997468230 348	97.2826919156 354	97.4603219796 526
RCC	97.1382472426 871	97.2918222079 281	97.2832676270 655	97.4593856991 306
FSB	96.4528237242 92	96.1972230033 434	96.2432776628 619	96.1425374358 036
FLC	97.5120641351 854	97.5803659268 269	97.5901906157 094	97.7316569641 355
ANN	96.4882393194 225	97.7257167390 62	98.1228605573 721	97.8856115796 755
PSO	96.8152637133 825	96.4240869801 715	95.2566455711 557	97.5739018834 793
GA	92.0761912229 555	94.4712126462 419	94.4815065332 133	94.4192555842 445
GWO	89.8850071217 904	96.1790955103 601	96.4165614153 498	95.9625942438 69

Table 4.5 *Convergence Time of MPPT Algorithms under High-Variability*

Cloud-Transient Dataset

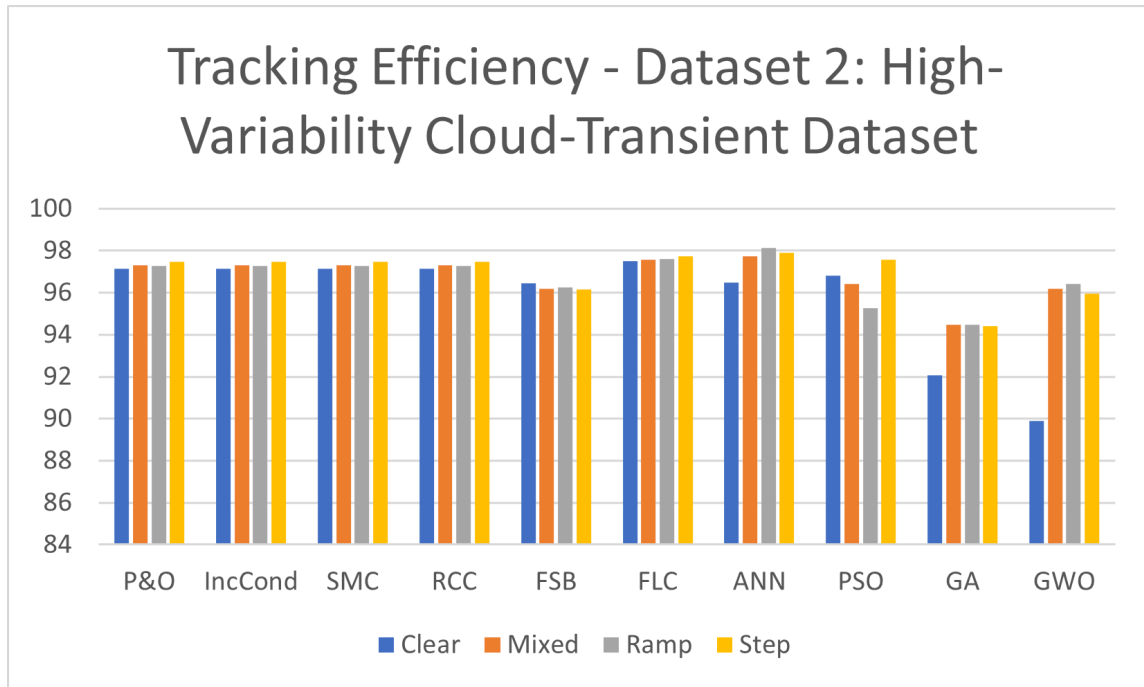


Figure 4.5 *Convergence Time of MPPT Algorithms under High-Variability*

Cloud-Transient Dataset

The results obtained for the high variability cloud transient dataset, which simulates rapidly changing irradiance levels, are shown in Table 4.5 and Figure 4.5.

The results show that the majority of the algorithms maintained high tracking efficiency levels, with the majority being over 97%. This indicates the ability of the algorithms to adapt to rapidly changing irradiance levels. Out of the tested algorithms, the ANN achieved the highest efficiency under ramp conditions, with an efficiency of approximately 98.12%.

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The Fuzzy Logic Control (FLC) also achieved high efficiency levels in all scenarios, ranging from 97.51% to 97.73%, which indicates the robustness of the FLC in cloud-transient environments. Similarly, the classical algorithms such as P&O, IncCond, SMC, and RCC maintained their efficiency levels between 97.13% and 97.46%, which indicates their reliability, even in rapidly changing irradiance levels.

The efficiency of the Particle Swarm Optimization (PSO) algorithm was moderate under this dataset, with values ranging between 95.26% and 97.57%. This is a considerable improvement compared to its performance in Dataset 1.

The Grey Wolf Optimizer (GWO) algorithm had the lowest efficiency under clear conditions at 89.89%. This implies that it performs poorly in terms of tracking under this particular scenario. The efficiency of the Genetic Algorithm (GA) was relatively low compared to all the algorithms. Its efficiency values were between 92% and 94%.

From the analysis presented above, it is clear that all MPPT algorithms have high efficiency in tracking, and particularly, ANN and FLC performed very well. Overall, the results presented show that both classical and intelligent control-based algorithms have high tracking efficiency under rapidly varying conditions, and some metaheuristic algorithms show comparatively lower performance.

Benchmark Tropical Dataset 3: Arayat–Mexico Solar Farm

Algorithm	Clear	Mixed	Ramp	Step
P&O	96.0625280462 922	95.9640306676 603	96.0848917691 889	96.4484095687 932
IncCond	96.0611475616 994	95.9616414618 369	96.0833797954 72	96.4471894854 636

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SMC	96.0617441357 958	95.9512155033 722	96.0868602892 153	96.4568706022 125
RCC	96.0617120804 394	95.9624067668 588	96.0842866832 2	96.4477534540 597
FSB	95.9692345851 266	95.6570781820 584	95.4313965016 798	96.3237635483 542
FLC	96.4426446761 812	96.2810471948 897	96.3810300983 185	96.7623227912 436
ANN	96.0313597750 812	95.7306432107 005	95.9607723316 583	96.3621837663 315
PSO	96.1758485823 518	92.3158025643 079	92.3684460137 496	92.2892951997 896
GA	90.7546215529 69	90.0315902761 371	90.2426742805 898	91.1879066445 83
GWO	91.5205931376 721	88.8297519696 949	92.3625054693 947	89.2588279060 596

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Table 4.6 Tracking Efficiency of MPPT Algorithms under Arayat-Mexico Solar Farm

Tropical Dataset

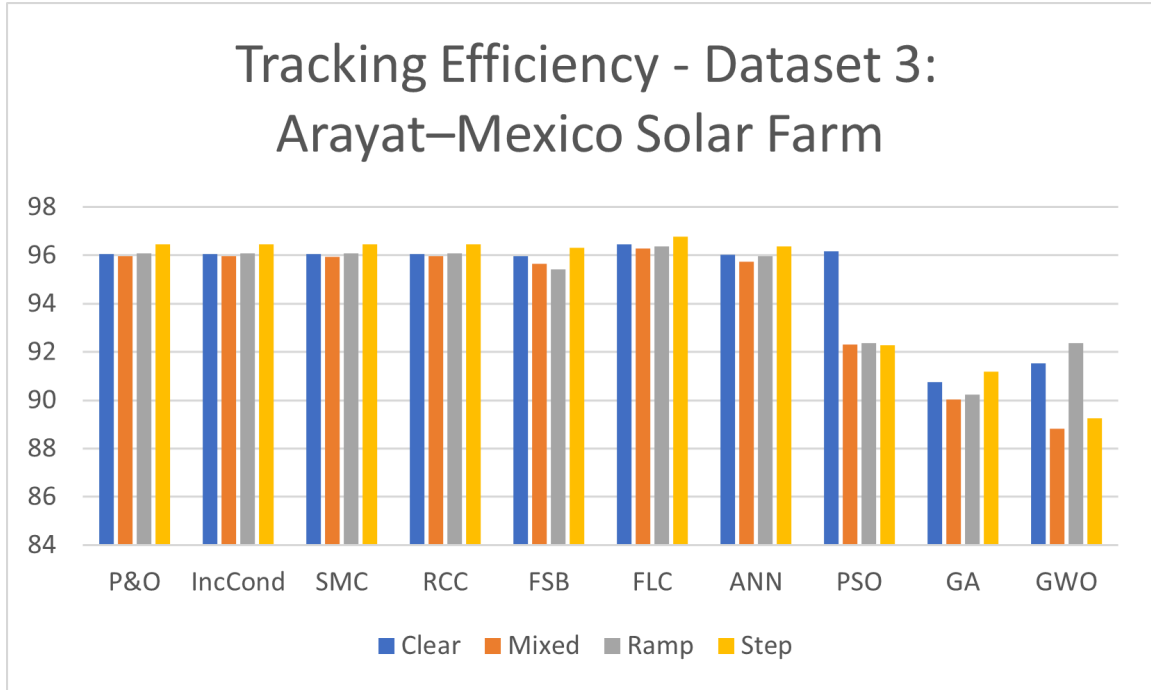


Figure 4.6 Tracking Efficiency of MPPT Algorithms under Arayat-Mexico Solar Farm

Tropical Dataset

Table 4.6 and Figure 4.6 show the performance of the tracking efficiency of the tropical dataset representing the irradiance conditions of the Arayat-Mexico Solar Farm.

From the results, it is seen that all algorithms were able to maintain tracking efficiency at around 95% to 96%. This indicates stable MPPT operation under tropical irradiance conditions. Among all the algorithms used in the simulation, it is seen that the Fuzzy Logic Control (FLC) was able to achieve maximum efficiency under step change conditions at around 96.76%. This shows that the FLC is able to effectively track under rapid change conditions.

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It is also seen that all classical algorithms like P&O, IncCond, SMC, and RCC were able to achieve maximum efficiency at around 96% for all conditions. This indicates consistent tracking capability of classical algorithms under tropical conditions.

The Artificial Neural Network (ANN) also showed a stable efficiency level, ranging from 95.73% to 96.36%. This shows a high level of reliability in terms of efficiency under different irradiance levels.

On the other hand, the Genetic Algorithm (GA) and Grey Wolf Optimizer (GWO) showed the lowest level of efficiency in the dataset, ranging from 88-91%. The Particle Swarm Optimization (PSO) method also showed a low level of efficiency in mixed and ramp conditions, at around 92%. This shows a low level of tracking efficiency in comparison to classical and intelligent control-based methods.

Even though the tracking efficiency is quite high for several algorithms, tracking efficiency cannot be considered the sole indicator of performance tradeoffs in real-time applications. For example, the tracking efficiency of algorithms like the FLC and ANN can be high, yet some of them suffer from reduced energy yield during real-time operations.

In conclusion, the results show that classical and intelligent control-based methods have a high level of tracking efficiency under tropical irradiance conditions, and some of the metaheuristic methods have comparatively low levels of efficiency.

4.3.3 Tracking Accuracy

Tracking Accuracy The term "tracking accuracy (stability)" is used to define the degree to which an MPPT algorithm maintains its operating point at or near the true

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maximum power point (MPP) during steady-state operation, quantified as the ratio of the standard deviation of the output power over a terminal sampling window to its mean value. A lower stability value indicates that the output power fluctuates minimally around its mean, reflecting tight, consistent MPP tracking with low oscillation amplitude and small steady-state error. Conversely, a higher stability value signifies greater power dispersion, implying that the algorithm's operating point drifts or oscillates more significantly away from the true MPP during sustained operation.

Therefore, it is desirable for the algorithm to produce the lowest possible stability value; unlike metrics where proximity to unity is optimal, here a value approaching zero is ideal, as it corresponds to near-perfect suppression of steady-state oscillations and minimal continuous power loss due to MPP deviation.

Benchmark Dataset 1: Standard Mid-Latitude PV Dataset

Algorithm	Clear	Mixed	Ramp	Step
P&O	1.019678879	1.021171517	1.021375475	1.026021918
IncCond	1.020200275	1.021918877	1.02191888	1.02738875
SMC	1.026678728	1.027344026	1.027557636	1.029007987
RCC	1.019778595	1.019097302	1.021943921	1.027424645
FSB	1.074125787	1.052282395	1.054700238	1.041027243
FLC	1.013998221	1.015036148	1.01503644	1.023590114
ANN	1.02327374	0.8966791709	1.028625446	1.021540215
PSO	1.013317834	1.016750643	1.042155363	1.026301933

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GA	1.560545592	1.532099018	1.193210113	1.1817711
GWO	1.024376533	0.8907915683	1.400512594	1.032459335

***Table 4.7 Tracking Accuracy of MPPT Algorithms under Standard Mid-Latitude PV
Dataset***

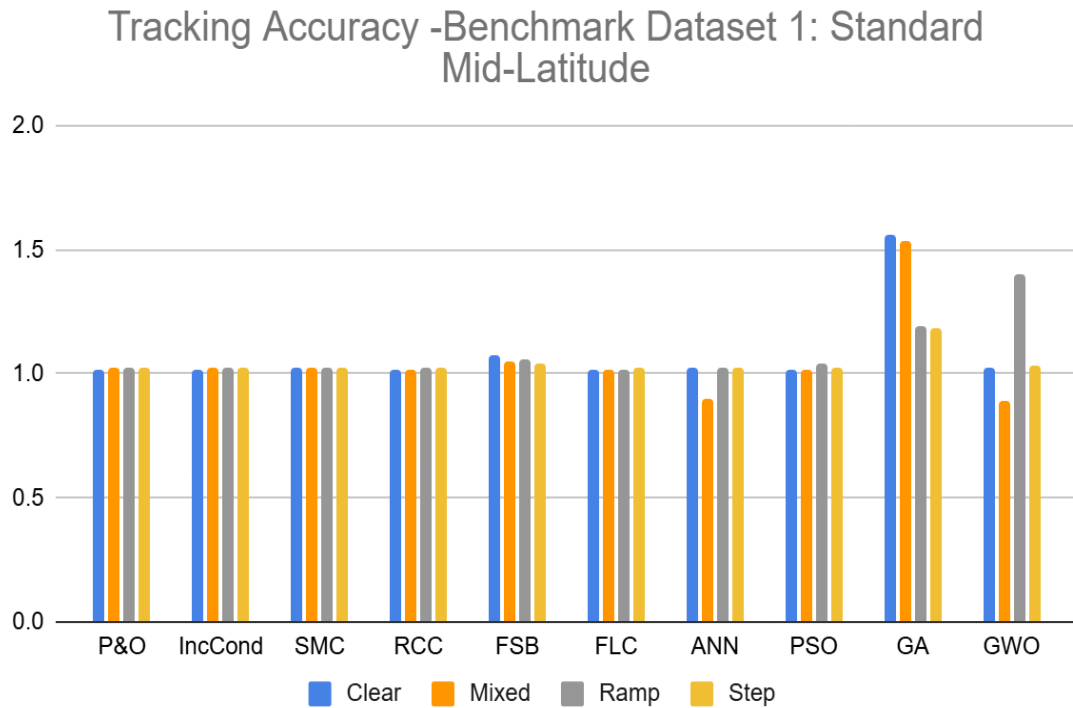


Figure 4.7 Tracking Accuracy of MPPT Algorithms under Standard Mid-Latitude PV Dataset

Table 4.7 illustrates the tracking accuracy (stability) of all tested MPPT algorithms under clear, mixed, ramp, and step irradiance conditions using the Standard Mid-Latitude PV Dataset. Among all algorithms tested, FLC and PSO consistently delivered the best tracking accuracy across all irradiance conditions, posting some of the lowest stability values in the intelligent and metaheuristic categories respectively. The Classical Algorithms (P&O and IncCond) performed reliably and uniformly, with stability values hovering narrowly across Clear, Mixed, Ramp, and Step conditions, indicating consistent but not exceptional accuracy. Control-Based Algorithms showed more spread, SMC and RCC performed comparably to classical methods, while FSB was

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notably the worst performer within its category, recording the highest values across nearly all conditions (e.g., 1.074 under Clear, 1.052 under Mixed).

The Metaheuristic category exhibited the widest performance variance of any group. GA was the worst overall performer across the entire benchmark, with strikingly high stability values in Clear (1.561) and Mixed (1.532) conditions, suggesting poor convergence and highly unstable tracking. GWO also underperformed significantly in Ramp (1.401) and Mixed (0.891, deviating far from the cluster), reinforcing that metaheuristic algorithms are generally the least reliable for dynamic tracking accuracy in standard mid-latitude conditions. In contrast, Intelligent Algorithms (FLC and ANN) trended toward lower, more stable values on average, making them the strongest-performing category overall, though ANN showed some inconsistency under Mixed conditions.

Benchmark Dataset 2: High-Variability Cloud-Transient Dataset

Algorithm	Clear	Mixed	Ramp	Step
P&O	1.004403201	0.987990657	0.9816955854	0.9989267881
IncCond	1.004341376	0.9868822517	0.9868822517	0.9938228799
SMC	1.012465242	1.107869017	1.199484446	1.237467052
RCC	1.002248317	0.9903636381	0.9903636238	1.009441425
FSB	1.035895175	0.9928347737	0.9912797231	0.977726353
FLC	0.9974131341	1.216906902	1.216906906	1.035524102
ANN	1.006291607	1.02941712	1.002719642	1.026367402

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PSO	1.012958829	1.166083457	1.41415903	1.046701427
GA	1.141410277	1.293218444	1.461238639	1.247902956
GWO	1.009283016	2.840360522	1.023393652	1.039821397

Table 4.8 Tracking Accuracy of MPPT Algorithms under High-Variability

Cloud-Transient Dataset

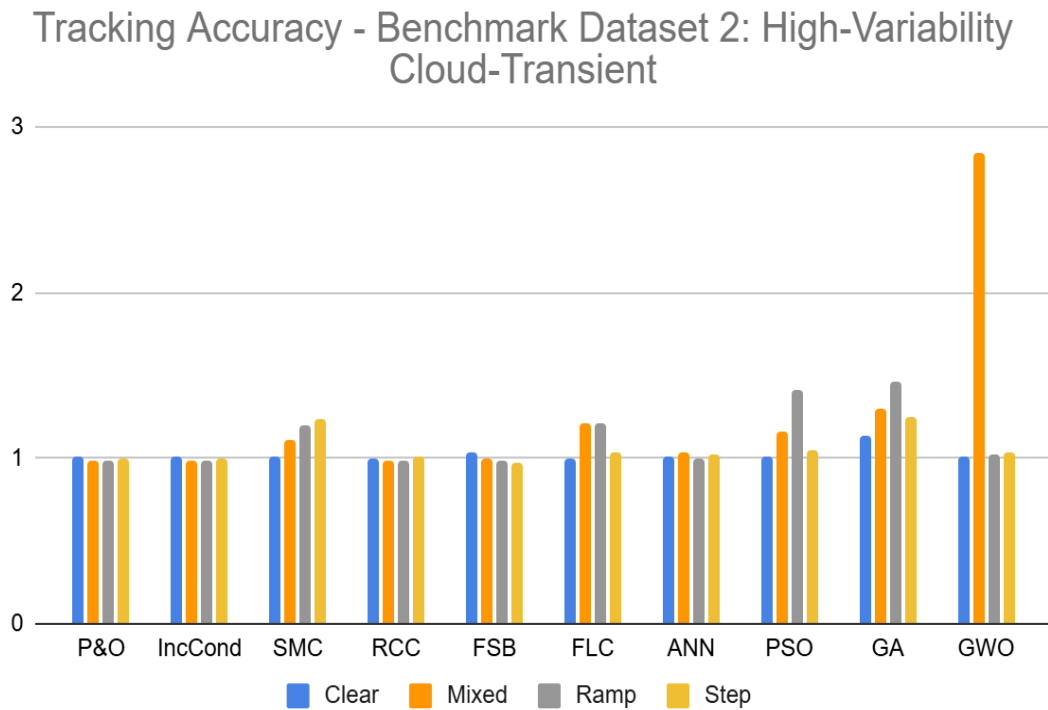


Figure 4.8 Tracking Accuracy of MPPT Algorithms under High-Variability

Cloud-Transient Dataset

Table 4.8 illustrates the tracking accuracy (stability) of all tested MPPT methods under clear, mixed, ramp, and step irradiance conditions using the high-variability cloud-transient dataset. The Classical Algorithms (P&O and IncCond) emerged as the best-performing category overall, consistently recording the lowest stability values across

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all irradiance profiles, with several values dipping below 1.0, indicating exceptionally tight steady-state tracking. Within the Control-Based category, RCC and FSB also performed competitively, posting low values particularly in Mixed and Ramp conditions, while SMC was the clear outlier, degrading significantly under dynamic conditions and reaching values as high as 1.237 under Step, making it one of the worst individual performers in this dataset. The Intelligent Algorithms showed diverging behavior: ANN maintained relatively low and consistent values, whereas FLC spiked heavily in Mixed and Ramp conditions (1.217), undermining its category average.

The Metaheuristic category again proved to be the most unstable group under this more demanding dataset. GWO recorded the single worst value in the entire benchmark at 2.840 under Mixed conditions, representing a dramatic failure in tracking stability during rapid cloud transients. GA continued its poor showing with high values across all conditions, reinforcing its status as the worst overall metaheuristic performer. PSO also degraded notably under Ramp (1.414), further confirming that metaheuristic algorithms as a category struggle most under high-variability irradiance, likely due to their search behavior conflicting with the demands of fast-changing conditions.

Benchmark Tropical Dataset 3: Arayat–Mexico Solar Farm

Algorithm	Clear	Mixed	Ramp	Step
P&O	0.8025693012	0.8048388799	0.804839025	0.9304576736
IncCond	0.8030877488	0.8062858927	0.8050925601	0.9299471384
SMC	0.8293241491	0.8321279462	0.8321278871	0.9734510879

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RCC	0.8027960918	0.8054083415	0.8053745997	0.9312052671
FSB	0.9918224255	0.9961345351	0.9961345255	0.9896777459
FLC	0.7770756965	0.7791426029	0.779141117	0.9357081091
ANN	0.9922182628	0.8105268846	0.9987536089	0.9876372999
PSO	0.809350155	0.9773698732	0.977369869	0.9625455578
GA	0.9136152604	0.9636076439	1.25489425	0.9384610626
GWO	1.139340423	0.8550110442	0.9871172939	1.405258479

Table 4.9 Tracking Accuracy of MPPT Algorithms under Arayat-Mexico Solar Farm Tropical Dataset

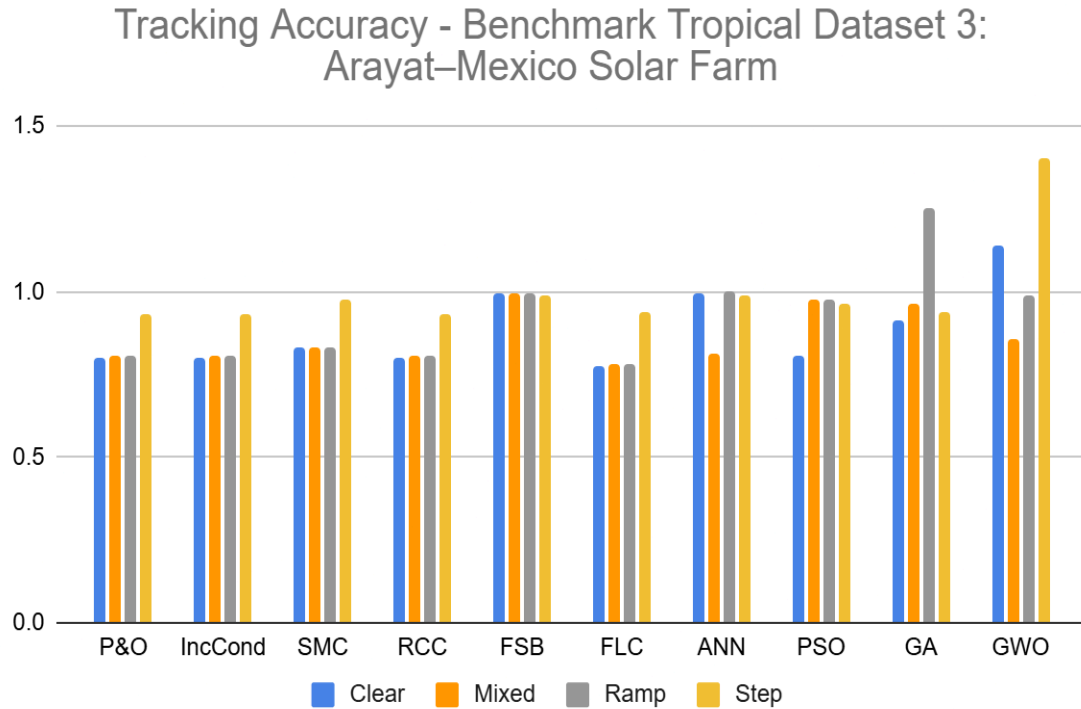


Figure 4.9 Tracking Accuracy of MPPT Algorithms under Arayat-Mexico Solar Farm Tropical Dataset

Table 4.9 illustrates the tracking accuracy (stability) of all tested MPPT methods under clear, mixed, ramp, and step irradiance conditions using the Benchmark Tropical

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Dataset from the Arayat–Mexico Solar Farm. FLC stood out as the best individual performer across the entire benchmark, consistently recording the lowest stability values in Clear, Mixed, and Ramp conditions (averaging around 0.779), indicating the tightest and most stable steady-state tracking of all algorithms tested. The Classical Algorithms (P&O and IncCond) also performed strongly, with low and uniform values across all profiles, reaffirming their reliability as a category even under tropical conditions. Within the Control-Based category, SMC and RCC tracked closely to the classical algorithms and performed well, while FSB lagged noticeably, posting values approaching 0.99 across most conditions, the highest within its category and reflecting comparatively looser tracking stability.

The Metaheuristic category again showed the most inconsistency as a group. GWO was the worst individual performer in this dataset, spiking to 1.139 under Clear and 1.405 under Step conditions, the highest single value recorded across the entire table, suggesting severe instability under tropical step irradiance. GA also underperformed, particularly under Ramp conditions where it reached 1.255, making it the second worst overall. PSO similarly trended higher than the classical and intelligent algorithm categories. Notably, the Intelligent Algorithms diverged considerably: while FLC was the top performer overall, ANN posted values close to 0.99–1.0, dragging the category average up and highlighting that FLC's advantage does not extend uniformly across the intelligent algorithm group.

While lower ripple ratios indicate higher tracking accuracy, they do not indicate better performance across all scenarios. Reduced oscillations may cause continuous shifts

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from the MPP rather than precise tracking. This is present in highly variable irradiance conditions where tracking accuracy tradeoffs with CTRT.

4.3.4 Harvested Energy

The term “harvested energy” is used to define the total amount of electrical energy extracted from the photovoltaic (PV) system by the MPPT algorithm over a given period of operation. It represents the cumulative energy obtained from the PV array while the algorithm continuously tracks the maximum power point (MPP) under varying irradiance and temperature conditions. This metric reflects the overall effectiveness of the MPPT method in maximizing energy production throughout the entire operating interval.

Therefore, it is desirable for the algorithm to maximize harvested energy; in other words, it should consistently track the MPP during both transient and steady-state conditions in order to capture as much available solar energy as possible. High harvested energy indicates that the MPPT algorithm is able to respond efficiently to environmental changes while maintaining stable operation, thereby minimizing energy losses due to slow convergence or oscillations around the MPP.

Benchmark Dataset 1: Standard Mid-Latitude PV Dataset

Algorithm	Clear	Mixed	Ramp	Step
P&O	4.303094894	3.886244958	3.936534445	3.421657708
IncCond	4.302614572	3.887418413	3.93852253	3.422974498
SMC	4.293615458	3.870897623	3.925194135	3.404337301

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RCC	4.304487946	3.886241966	3.937704935	3.422124025
FSB	3.654538623	3.016774908	3.197114595	2.548783785
FLC	3.970669409	3.577420144	3.608509191	3.130465882
ANN	3.592836927	2.853819477	2.926023284	2.622038087
PSO	1.179467113	0.8737072659	0.308818651	0.2505134849
GA	1.426421604	1.404451236	1.290650003	1.116570492
GWO	1.705925378	0.9019032722	1.086606715	1.217383462

Table 4.10 Harvested Energy of MPPT Algorithms under Standard Mid-Latitude PV

Dataset

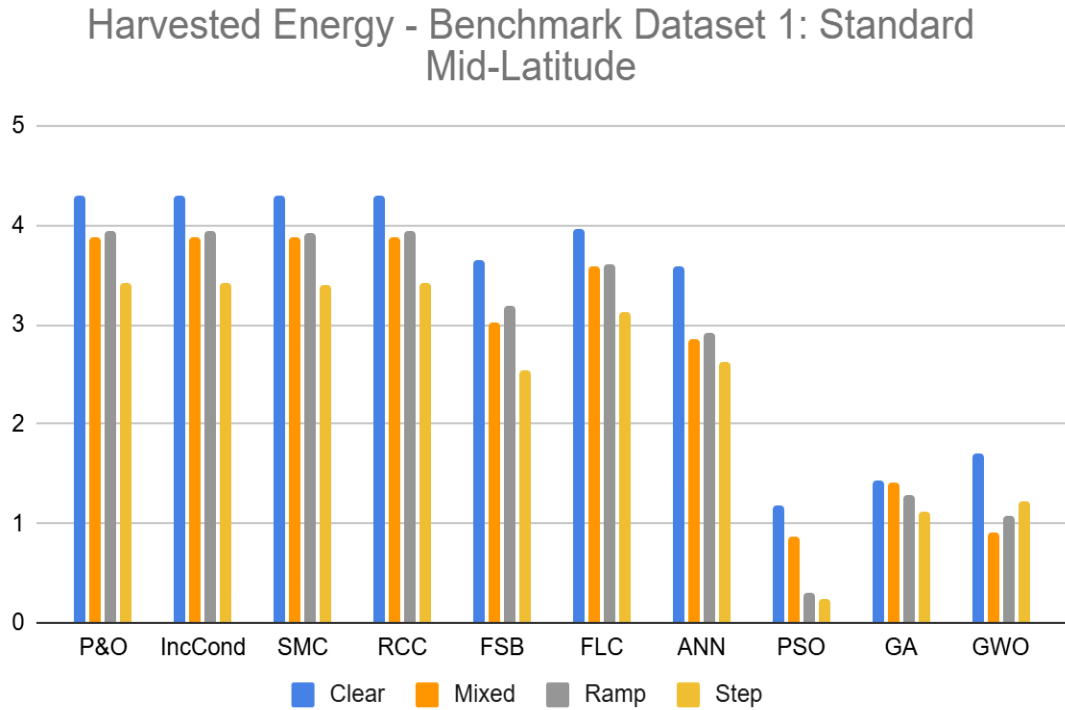


Figure 4.10 Harvested Energy of MPPT Algorithms under Standard Mid-Latitude PV
Dataset

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Table 4.10 illustrates the harvested energy (Wh) of all tested MPPT methods under clear, mixed, ramp, and step irradiance conditions using the standard mid-latitude PV dataset. Harvested energy reflects the total usable energy extracted by each algorithm, where higher values indicate more effective power extraction from the PV system. From the table, it is evident that P&O, IncCond, RCC, and SMC consistently harvest the highest energy across all irradiance conditions, with values clustering between approximately 3.40 and 4.30 Wh throughout all scenarios. Among these, RCC records the highest harvested energy under clear sky conditions at 4.3045 Wh, while P&O and IncCond perform comparably across all conditions. FLC also demonstrates competitive harvested energy, recording 3.9707 Wh under clear sky and maintaining values above 3.13 Wh across all conditions, placing it among the better-performing algorithms in this dataset. ANN, while slightly lower than the top performers, still achieves reasonable energy harvest ranging from 2.622 to 3.593 Wh across conditions. In contrast, PSO exhibits the lowest harvested energy across all irradiance conditions, recording particularly poor values of 0.3088 Wh under ramp and 0.2505 Wh under step conditions, a dramatic drop that suggests PSO's optimization behavior leads to significant energy losses during dynamic irradiance transitions. GA and GWO similarly yield considerably lower harvested energy, with GA ranging from 1.117 to 1.426 Wh and GWO from 0.902 to 1.706 Wh across all conditions. Overall, the results clearly demonstrate that Classical MPPT and Control-Theoretic algorithms consistently extract the highest energy, followed competitively by Intelligent and AI-Based methods, while Metaheuristic algorithms perform significantly worse, with PSO recording the poorest harvested energy despite its noted convergence speed advantages.

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Benchmark Dataset 2: High-Variability Cloud-Transient Dataset

Algorithm	Clear	Mixed	Ramp	Step
P&O	3.83194183	3.428857438	3.536330564	2.997740212
IncCond	3.83327064	3.429018458	3.534528388	2.998403409
SMC	3.824071561	3.412864064	3.520540588	2.972764988
RCC	3.833508786	3.4284669	3.538099405	2.998357657
FSB	3.228430942	2.508412654	2.646923527	1.988493117
FLC	3.522362086	3.138602737	3.226747795	2.707307709
ANN	3.089283589	2.861051832	1.67898443	2.225383293
PSO	2.672988409	2.594326168	2.038744372	2.318429182
GA	1.280445817	1.548121751	1.476458716	1.231663421
GWO	0.9258312412	1.60724833	1.231738664	1.252456759

***Table 4.11** Harvested Energy of MPPT Algorithms under High-Variability
Cloud-Transient Dataset*

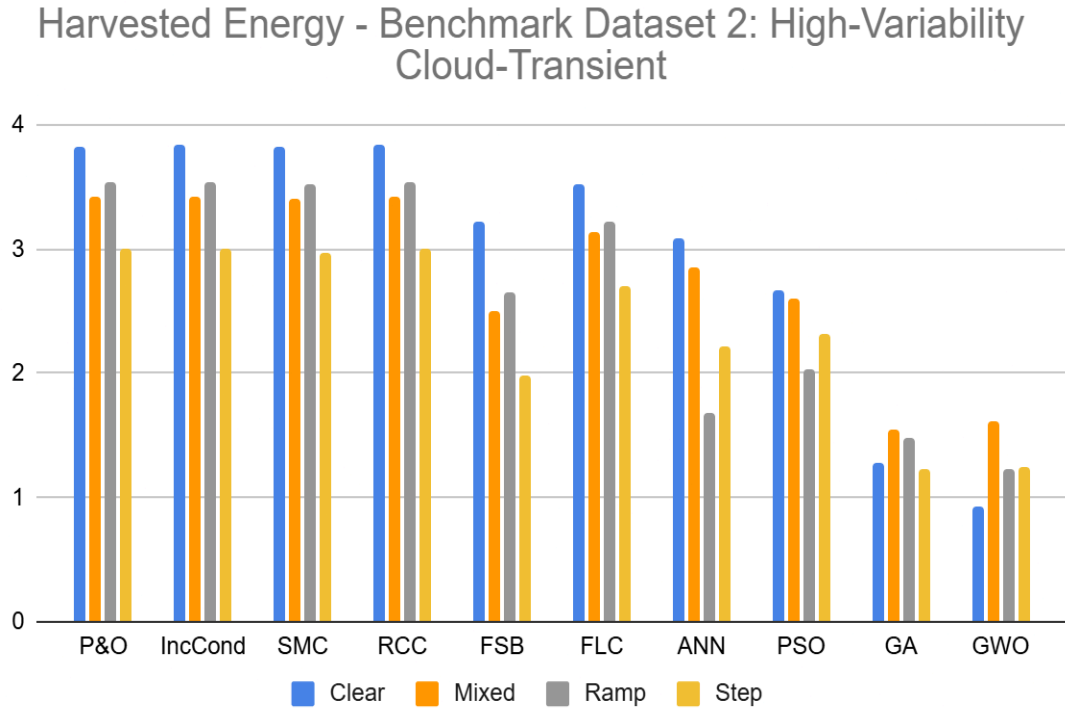


Figure 4.11 Harvested Energy of MPPT Algorithms under High-Variability
Cloud-Transient Dataset

Table 4.11 illustrates the harvested energy (Wh) of all tested MPPT methods under clear, mixed, ramp, and step irradiance conditions using the high-variability cloud-transient dataset. From the table, it is evident that P&O, IncCond, RCC, and SMC again dominate in terms of energy harvesting performance, consistently recording the highest values across all irradiance conditions. RCC achieves the highest harvested energy under clear sky at 3.8335 Wh, and all four algorithms remain tightly clustered between approximately 2.97 and 3.83 Wh throughout all scenarios, demonstrating robust and reliable energy extraction even under the high-variability conditions characteristic of this dataset.

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FLC maintains a competitive position, recording 3.5224 Wh under clear sky and sustaining values above 2.71 Wh across all conditions, while ANN performs reasonably well under clear and mixed conditions but drops notably to 1.6790 Wh under ramp irradiance, suggesting sensitivity to rapid irradiance transitions. FSB records comparatively lower harvested energy, particularly under mixed and step conditions at 2.5084 and 1.9885 Wh, respectively, indicating reduced effectiveness under highly variable cloud-transient irradiance. In contrast, GA and GWO yield the lowest harvested energy in this dataset, with GWO recording the poorest clear sky performance at just 0.9258 Wh and GA ranging between 1.231 and 1.548 Wh across all conditions. PSO, while performing better than GA and GWO, still falls considerably short, with values ranging from 2.038 to 2.673 Wh. Overall, the results consistently reinforce that Classical MPPT and Control-Theoretic algorithms, with the exception of FSB, outperform other categories in harvested energy under the high-variability cloud-transient dataset, with Intelligent and AI-Based methods remaining competitive, and Metaheuristic algorithms, particularly GWO and GA, demonstrating the weakest energy extraction performance.

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Benchmark Tropical Dataset 3: Arayat–Mexico Solar Farm

Algorithm	Clear	Mixed	Ramp	Step
P&O	2.157639481	1.908947099	1.932058691	2.107616261
IncCond	2.15640543	1.907845341	1.931297057	2.106496443
SMC	2.153058178	1.899237645	1.923220849	2.100914455
RCC	2.157597031	1.908660632	1.931915081	2.107581065
FSB	1.867053822	1.511557513	1.411098399	1.848509427
FLC	1.993682716	1.769902365	1.770136333	1.948747909
ANN	1.795084132	1.699169668	1.652547477	1.77870667
PSO	0.9975870782	0.6816485349	0.6691650509	0.7531485651
GA	0.6564776267	0.5287113911	0.5633310964	0.6547641629
GWO	0.6795252535	0.576508617	0.8641678097	0.4978278601

***Table 4.12 Harvested Energy of MPPT Algorithms under Arayat-Mexico Solar Farm
Tropical Dataset***

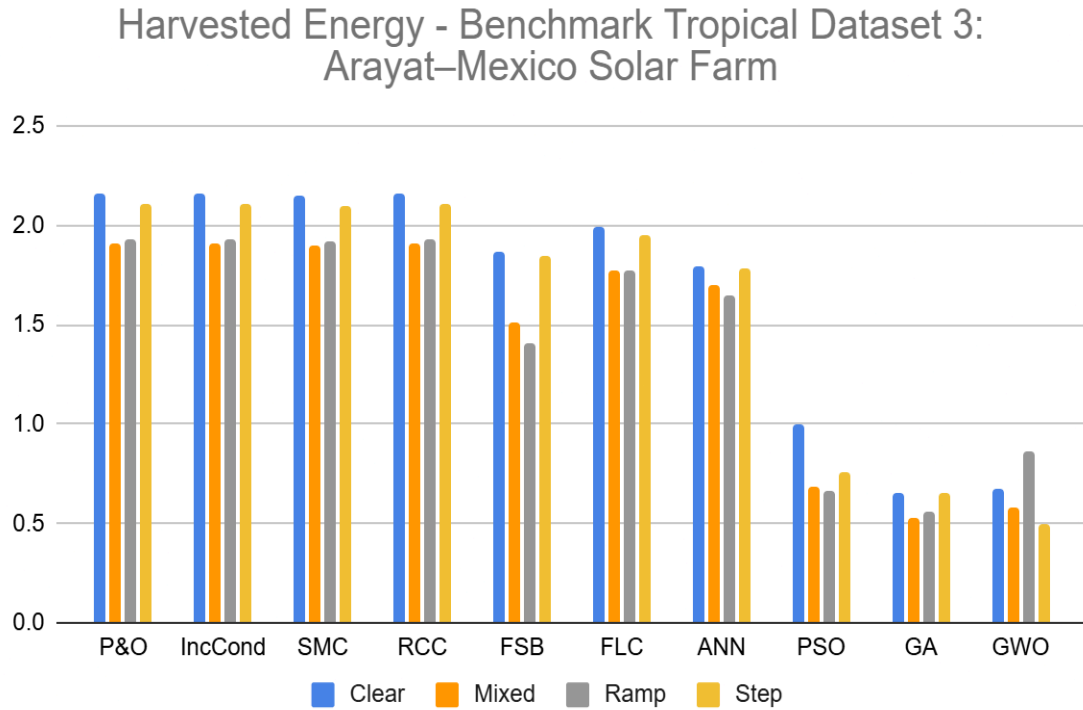


Figure 4.12 Harvested Energy of MPPT Algorithms under Arayat-Mexico Solar Farm
Tropical Dataset

Table 4.12 illustrates the harvested energy (Wh) of all tested MPPT methods under clear, mixed, ramp, and step irradiance conditions using the Benchmark Tropical Dataset from the Arayat–Mexico Solar Farm. From the table, it is evident that P&O, IncCond, RCC, and SMC once again record the highest harvested energy across all irradiance conditions, with values ranging between approximately 1.899 and 2.158 Wh throughout all scenarios. P&O and RCC achieve the highest clear sky performance at 2.1576 Wh respectively, and all four algorithms remain closely clustered across all conditions, demonstrating consistent and reliable energy extraction under tropical irradiance profiles. Notably, the overall harvested energy values in this dataset are

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considerably lower than those observed in the mid-latitude and cloud-transient datasets, which is likely attributable to the distinct high-temperature and diffuse irradiance characteristics of the tropical environment that reduce overall PV output.

FLC performs competitively, recording 1.9937 Wh under clear sky and maintaining values above 1.769 Wh across all conditions. ANN follows closely, ranging from 1.653 to 1.795 Wh. FSB records moderate performance with a notable dip to 1.4111 Wh under ramp conditions, underperforming relative to RCC and SMC within the same Control-Theoretic category. In stark contrast, GA and GWO exhibit the weakest energy harvesting performance in this dataset, with GA recording values as low as 0.5287 Wh under mixed conditions and GWO dropping to just 0.4978 Wh under step irradiance, the lowest single value recorded across the entire dataset. PSO also performs poorly, ranging between 0.669 and 0.998 Wh across all conditions. Overall, the results consistently demonstrate that Classical MPPT and Control-Theoretic algorithms vastly outperform other categories in harvested energy under tropical conditions, with Intelligent and AI-Based methods remaining moderately competitive, and Metaheuristic algorithms, particularly GWO and GA, showing the poorest energy extraction performance, highlighting their unsuitability for tropical PV environments.

The effects of convergence speed, tracking accuracy, and tracking efficiency culminate in possible harvested energy. This makes harvested energy a comprehensive performance metric. However, algorithms optimized for fast convergence or tracking speed do not always produce the best energy yield. This demonstrates a trade-off between instantaneous tracking and harvested energy of the MPPT algorithm.

4.3.5 Cloud Transient Recovery

The Cloud Transient Recovery Time (CTRT) is a novel metric proposed in this study to quantify the total duration during which a PV output power ($P_{pv}(t)$) remains below an acceptable recovery threshold relative to a dynamic maximum power reference. It is desirable for an MPPT algorithm to have a low CTRT value as it indicates a quick restoration of operation near the maximum power point after an irradiance disturbance. Conversely, a high CTRT value indicates a prolonged tracking underperformance resulting in energy losses during and after transient events.

The recovery threshold used in CTRT is defined at 95% of the reference power, corresponding to the 5% settling time criterion defined in classical control systems theory, which defines system recovery as reaching 95% of the final value [156, 157]. This criterion has been independently applied in published engineering literature by Barraza et al. [158], who explicitly state that settling time is calculated using the criterion of 95% of the final value, and in MPPT benchmarking by Nguyen [159], who formally defines convergence time t_{95} as the first time PV output sustains at or above $0.95 \times G$ which is similar to the threshold formulation used in the study. The 5% criterion was selected over the alternative 2% criterion, as Ogata [156] notes that the choice depends on design objectives, with the 5% criterion being standard for dynamic transient response evaluation, while the 2% criterion is more appropriate for steady-state precision analysis.

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Based on this criterion, the CTRT metric is defined as the total cumulative duration in seconds during which the $P_{pv}(t)$ remains below 95% of the rolling maximum reference power $P_{ref}(t)$:

$$CTRT = \sum \Delta t, \text{ for all } t \text{ where } P_{pv}(t) < 0.95 \times P_{ref}(t)$$

$$0.5s \text{ (Reference window)} = 0.001 \text{ (data time step)} \times 500 \text{ (samples)}$$

This summation is evaluated at each time step across the 60-second simulation period, where $P_{ref}(t)$ is the time-varying rolling maximum of $P_{pv}(t)$, recomputed at every time step using MATLAB's `movmax` function over a local 500-sample centered window applied to $P_{pv}(t)$. This produces a dynamic reference point that captures the local peak power envelope across the simulation without requiring a fixed irradiance-based reference. The 500-sample window was determined through expert consultation documented in Chapter 3. The 0.5-second reference window is calculated by multiplying the 500 samples with the simulation data timestep of 0.001 seconds.

The cloud transient is not represented explicitly in the formula. Rather, it enters through the irradiance input data that drives both $P_{pv}(t)$ and $P_{ref}(t)$ throughout the simulation. When a cloud-induced irradiance drop occurs, $P_{pv}(t)$ falls, while $P_{ref}(t)$ temporarily retains the pre-cloud peak power level, creating the gap that the CTRT metric measures. As shown in the irradiance and power output graphs, the Cloud Transient dataset exhibits sharp irradiance drops, while the tropical dataset shows cloud-induced irradiance variability based on Solcast satellite-derived measurements [54]. These input disturbances produce corresponding drops in $P_{pv}(t)$ that fall below the 95% threshold, as indicated by the vertical markers on the power graphs.

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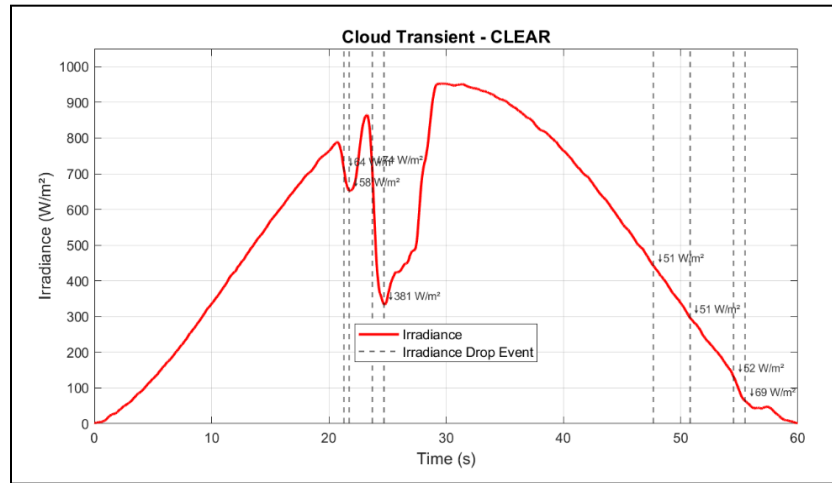


Figure 4.13 Irradiance vs. Time graph for Cloud Transient Clear Dataset

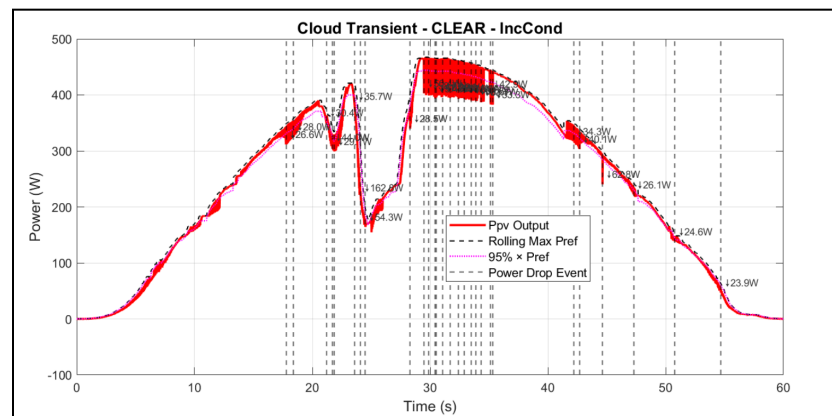


Figure 4.14 Sample Power Output vs. Time Graph - Cloud Transient Clear - IncCond

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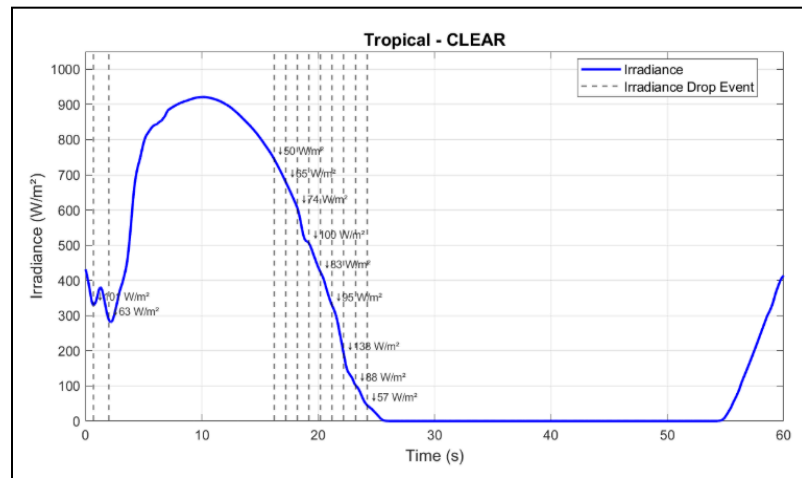


Figure 4.15 Irradiance vs. Time graph for Tropical Clear Dataset

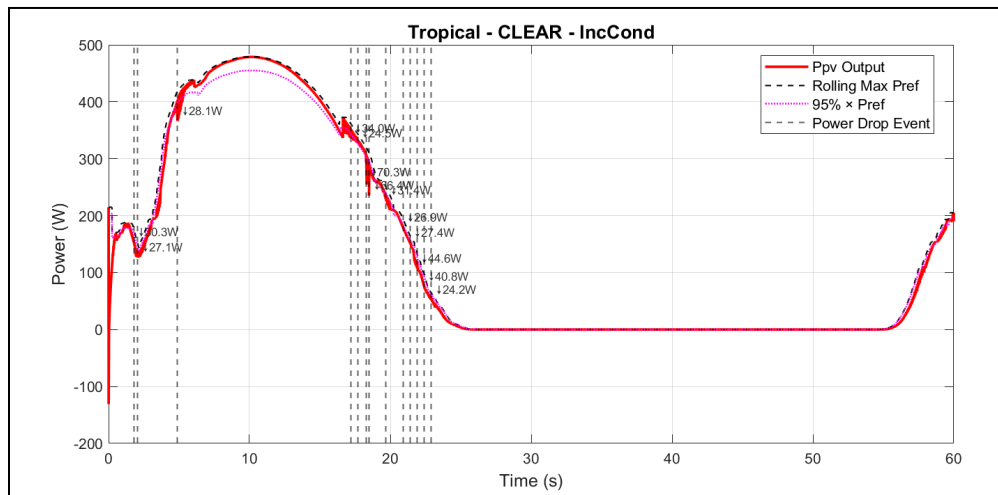


Figure 4.16 Sample Power Output vs. Time Graph - Tropical Clear - IncCond

The rolling maximum reference replaces a fixed static reference because MPP shifts continuously under dynamic irradiance, making fixed-reference evaluation unreliable [23, 67]. This dynamic reference ensures that CTRT reflects actual tracking underperformance relative to what the algorithm was recently capable of achieving, rather than against an idealized static baseline.

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CTRT synthesizes three concepts from existing literature: the 5% settling criterion for the recovery threshold [156, 157, 158, 159], the dynamic rolling reference approach, motivated by findings that MPP shifts continuously under variable irradiance making fixed-reference evaluation unreliable [23, 67], and the cumulative duration measurement across the full simulation window, extending Nguyen’s per-event t_{95} metric [159] into a single aggregate performance indicator.

Benchmark Dataset 1: Standard Mid-Latitude PV Dataset

Algorithm	Clear	Mixed	Ramp	Step
P&O	20.152	22.089	21.807	22.96
IncCond	20.086	22.077	21.797	22.703
SMC	20.824	23.045	23.264	23.783
RCC	19.876	21.929	21.889	22.797
FSB	33.572	37.874	36.522	39.396
FLC	28.488	26.431	25.83	23.676
ANN	24.516	23.024	21.967	23.919
PSO	56.493	52.367	38.022	58.638
GA	58.201	58.279	58.137	58.961
GWO	50.605	59.332	57.912	52.352

Table 4.13 *Cloud Transient Recovery of MPPT Algorithms under Standard Mid-Latitude PV Dataset*

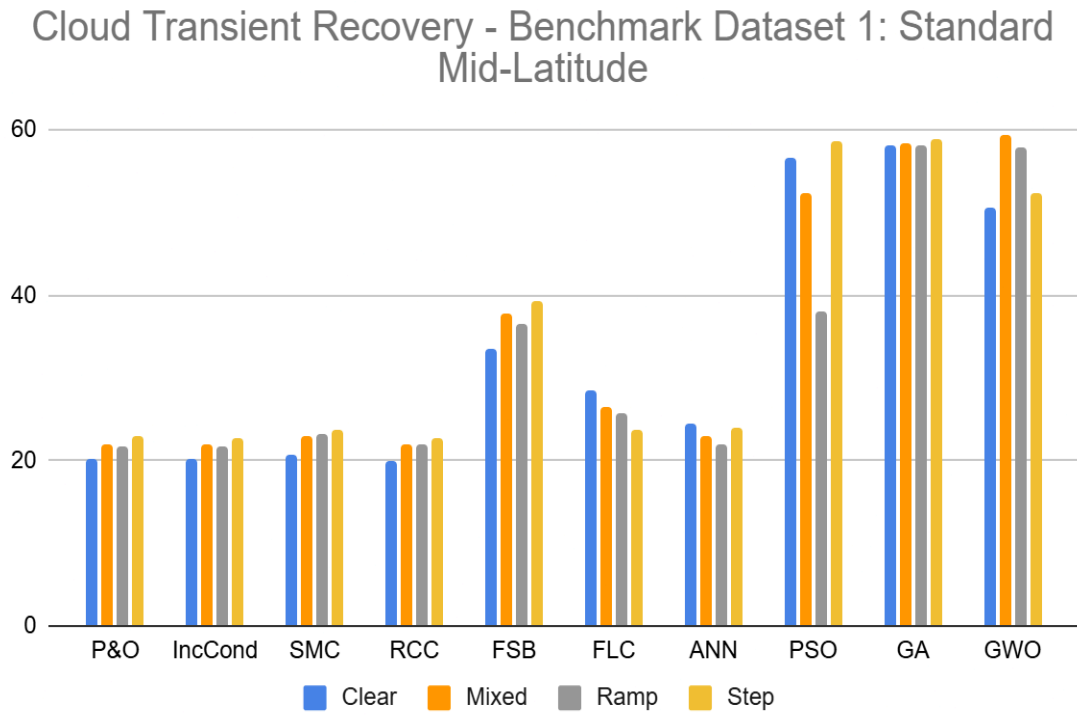


Figure 4.17 Cloud Transient Recovery of MPPT Algorithms under Standard Mid-Latitude PV Dataset

Table 4.13 illustrates the cloud transient recovery time (s) of all tested MPPT methods under clear, mixed, ramp, and step irradiance conditions using the standard mid-latitude PV dataset. Cloud transient recovery time measures how quickly each algorithm restores operation to the maximum power point following a cloud transient event, where lower values indicate faster and more responsive recovery. From the table, it is evident that P&O, IncCond, RCC, and SMC consistently achieve the fastest recovery times across all irradiance conditions, with values ranging between approximately 19.9 and 23.8 seconds throughout all scenarios. Among these, RCC records the shortest recovery time under clear sky conditions at 19.876 s, while P&O and IncCond remain

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closely matched across all conditions, reflecting their similar control mechanisms and rapid response characteristics.

ANN also demonstrates competitive recovery performance, recording 24.516 s under clear sky and improving notably to 21.967 s under ramp conditions, placing it reasonably close to the top performers in overall responsiveness. FLC shows moderate recovery times that improve progressively from 28.488 s under clear sky down to 23.676 s under step conditions, suggesting that FLC adapts more effectively under abrupt irradiance transitions. FSB, despite belonging to the Control-Theoretic category alongside RCC and SMC, records considerably slower recovery times ranging from 33.572 s under clear sky to 39.396 s under step conditions, indicating that its variable-step search mechanism hinders rapid re-convergence following cloud transient events. In stark contrast, GA records the slowest recovery times across all irradiance conditions, with values consistently clustering around 58 seconds regardless of irradiance type, ranging from 58.137 to 58.961 s, indicating a near-complete inability to recover swiftly from cloud transient events. PSO similarly performs poorly, with recovery times reaching 56.493 s under clear sky and spiking to 58.638 s under step conditions, while GWO also exhibits significantly delayed recovery ranging from 50.605 to 59.332 s. Overall, the results demonstrate that Classical MPPT and Control-Theoretic algorithms, with the exception of FSB, are substantially superior in cloud transient recovery speed under the standard mid-latitude dataset, with Intelligent and AI-Based methods also performing competitively, while Metaheuristic algorithms, particularly GA, PSO, and

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GWO, exhibit critically slow recovery times that would be highly detrimental in real-world PV applications.

Benchmark Dataset 2: High-Variability Cloud-Transient Dataset

Algorithm	Clear	Mixed	Ramp	Step
P&O	25.277	32.52	31.107	32.148
IncCond	25.561	32.625	31.245	32.176
SMC	26.028	34.501	32.84	33.959
RCC	25.397	32.611	30.975	31.863
FSB	36.326	47.384	46.043	54.091
FLC	29.038	34.042	36.385	30.036
ANN	29.107	24.867	27.572	26.159
PSO	33.336	46.818	57.686	32.607
GA	58.265	57.623	58.145	58.771
GWO	51.216	38.196	52.437	50.929

Table 4.14 Cloud Transient Recovery of MPPT Algorithms under High-Variability Cloud-Transient Dataset

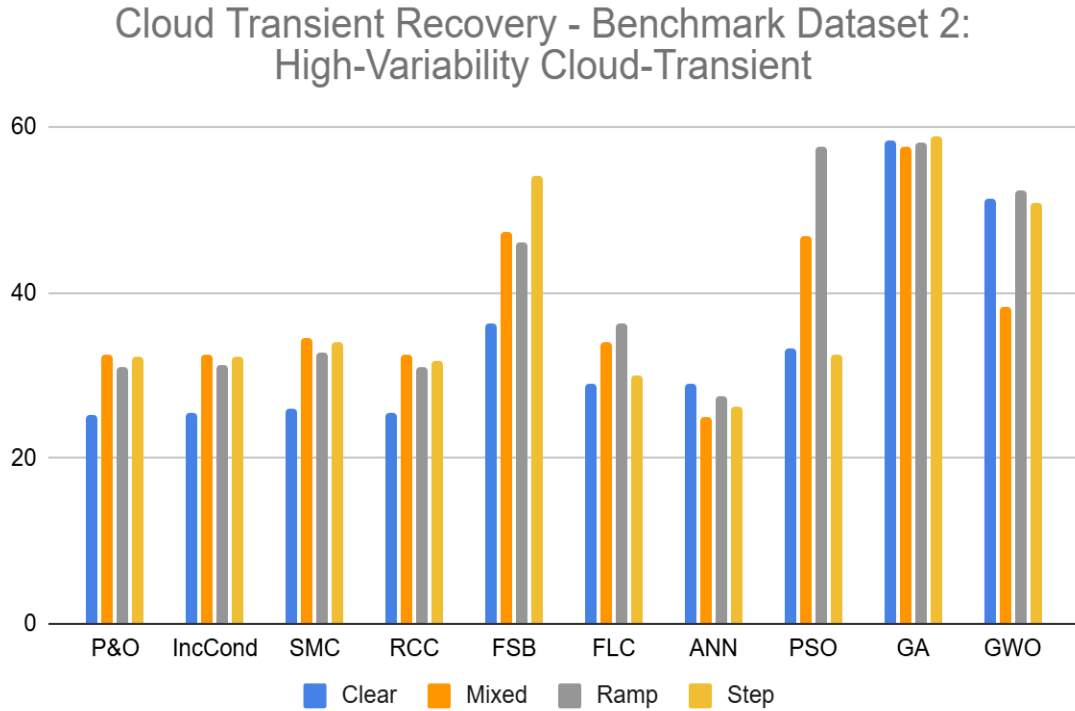


Figure 4.18 Cloud Transient Recovery of MPPT Algorithms under High-Variability Cloud-Transient Dataset

Table 4.14 illustrates the cloud transient recovery time (s) of all tested MPPT methods under clear, mixed, ramp, and step irradiance conditions using the high-variability cloud-transient dataset. From the table, it is evident that P&O, IncCond, RCC, and SMC again demonstrate the fastest recovery times across all irradiance conditions, with values ranging between approximately 25.3 and 34.5 seconds throughout all scenarios. Notably, recovery times in this dataset are generally higher than those observed in the mid-latitude dataset, which is expected given the increased irradiance variability characteristic of the cloud-transient environment. Among the top performers, RCC records the shortest clear sky recovery at 25.397 s, and P&O and IncCond remain closely matched across all conditions, once again reflecting their consistent and reliable recovery behavior.

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ANN stands out as a notable performer in this dataset, recording among the lowest recovery times under mixed, ramp, and step conditions at 24.867, 27.572, and 26.159 s respectively, demonstrating strong adaptability under high-variability irradiance. FLC also performs reasonably well, with its best recovery recorded under step conditions at 30.036 s, though it exhibits a notable spike under ramp conditions at 36.385 s. FSB shows considerably slower recovery, with values rising to 47.384 s and 46.043 s under mixed and ramp conditions respectively, and reaching 54.091 s under step irradiance, indicating that its variable-step search mechanism is particularly disadvantaged under high-variability cloud-transient conditions. GA again records the worst overall performance, with recovery times consistently hovering near 58 seconds across all conditions, ranging from 57.623 to 58.771 s. GWO also performs poorly overall, though it records a comparatively lower value of 38.196 s under mixed conditions. PSO shows highly variable recovery, ranging from a competitive 32.607 s under step conditions to a poor 57.686 s under ramp irradiance. Overall, the results reinforce that Classical MPPT algorithms and the stronger Control-Theoretic methods, particularly RCC and SMC, maintain significantly superior cloud transient recovery under high-variability conditions, with Intelligent and AI-Based methods also performing competitively, while Metaheuristic algorithms, with GA as the worst performer, and FSB exhibit the slowest and least reliable recovery times in this dataset.

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Benchmark Tropical Dataset 3: Arayat–Mexico Solar Farm

Algorithm	Clear	Mixed	Ramp	Step
P&O	23.649	24.1	23.167	23.688
IncCond	23.548	23.447	23.005	23.671
SMC	24.157	24.595	23.866	24.21
RCC	23.673	24.512	23.148	39.849
FSB	24.919	28.372	29.829	26.505
FLC	29.404	28.901	26.23	29.729
ANN	25.904	24.186	25.273	25.458
PSO	26.856	31.033	29.13	28.659
GA	58.416	58.442	58.563	58.37
GWO	58.645	56.358	56.526	40.561

Table 4.15 Cloud Transient Recovery of MPPT Algorithms under Arayat-Mexico Solar Farm Tropical Dataset

Cloud Transient Recovery - Benchmark Tropical Dataset 3:
Arayat–Mexico Solar Farm

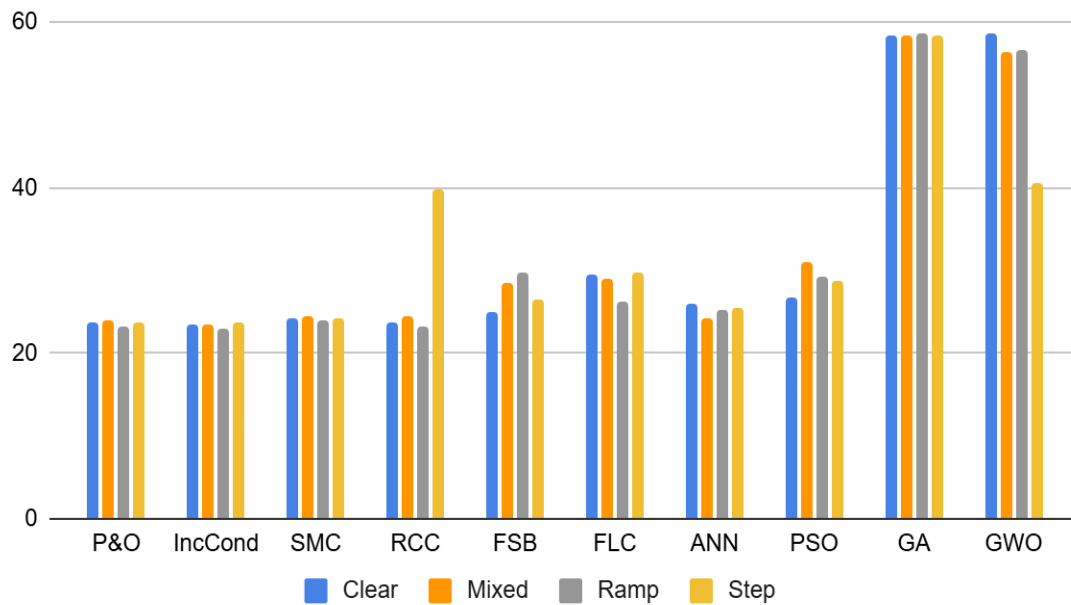


Figure 4.19 Cloud Transient Recovery of MPPT Algorithms under Arayat-Mexico Solar Farm Tropical Dataset

Table 4.15 illustrates the cloud transient recovery time (s) of all tested MPPT methods under clear, mixed, ramp, and step irradiance conditions using the Benchmark Tropical Dataset from the Arayat–Mexico Solar Farm. From the table, it is evident that P&O, IncCond, and SMC achieve the fastest and most consistent recovery times across all irradiance conditions, with values tightly clustered between approximately 23.0 and 24.6 seconds throughout all scenarios, the narrowest performance spread observed across all three datasets. IncCond records the best overall recovery, achieving its lowest value of 23.005 s under ramp conditions, while P&O and SMC remain comparably close across all conditions, demonstrating exceptional recovery consistency under tropical irradiance profiles.

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ANN again performs competitively, recording recovery times of 25.904, 24.186, 25.273, and 25.458 s across all conditions, placing it very close to the top performers and reinforcing its strong adaptability across diverse irradiance environments. RCC exhibits a notable anomaly under step irradiance at 39.849 s, a significant spike compared to its otherwise competitive values under clear, mixed, and ramp conditions, suggesting a specific vulnerability to abrupt step changes in the tropical environment. FSB demonstrates only moderate recovery performance, with values generally ranging between 24.919 and 29.829 s, consistent with its variable-step search behavior. FLC demonstrates similarly moderate recovery, with values ranging between 26.230 and 29.729 s. PSO shows slightly elevated but relatively stable recovery times between 26.856 and 31.033 s. In stark contrast, GA and GWO record dramatically slower recovery times, with GA consistently clustering near 58 seconds across all conditions and GWO ranging from 40.561 to 58.645 s. Overall, the tropical dataset results confirm that Classical MPPT algorithms, particularly P&O and IncCond, alongside SMC from the Control-Theoretic category, deliver the fastest and most reliable cloud transient recovery, while Intelligent and AI-Based methods remain competitive, and Metaheuristic algorithms, particularly GA and GWO, remain critically slow regardless of the irradiance environment tested.

While shorter recovery times indicate faster dynamic response to irradiance changes, they do not necessarily guarantee higher overall performance. Algorithms with fast convergence may still experience steady-state oscillations or suboptimal tracking during transient periods, illustrating a tradeoff between CTRT and tracking efficiency.

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4.4 Descriptive and Comparative Analysis

Table 4.16 presents the average performance of all ten tested MPPT algorithms across all benchmark datasets and irradiance scenarios, serving as the basis for the descriptive statistical analysis and comparative performance discussion that follows.

Algorithm	Average Tc (Convergence Time)	Average Efficiency	Average Accuracy	Average Energy	Average Recovery
ANN	0.8880833466	97.02669367	0.9838	2.397910739	25.16266667
FLC	0.9064166596	97.21385767	0.9893	2.863712856	29.01583333
FSB	0.7435833305	96.16774247	1.0168	2.452307609	36.73608333
GA	0.4924166686	92.31737793	1.1806	1.098172276	58.34775
GWO	0.8574166735	92.63388638	1.1032	1.045593613	52.08908333
IncCond	0.8904166868	96.89306136	0.9574	3.120732932	25.16175
P&O	0.8496666675	96.8938703	0.9582	3.120721965	25.222
PSO	0.3786666695	90.62488912	1.0418	1.27821199	40.97041667
RCC	0.8508333316	96.89362421	0.9680	3.121228786	26.54325
SMC	0.7238333375	96.8959767	1.0280	3.10839307	26.256

Table 4.16 Algorithm Average Performance Across All Benchmarks and Scenarios

4.4.1 Descriptive Statistical Analysis

Convergence Speed. The mean convergence time across all algorithms is $\mu = 0.758$ s, with a standard deviation of $\sigma = 0.182$ s and variance of $\sigma^2 = 0.033$. The wide spread, ranging from a minimum of 0.379 s (PSO) to a maximum of 0.906 s (FLC), reflects the fundamental differences in search and tracking mechanisms between metaheuristic and intelligent algorithm categories. The Classical group exhibits the tightest internal consistency ($\sigma = 0.029$), indicating that P&O and IncCond converge with predictable, stable timing across conditions, while the Metaheuristic group shows the

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highest internal variance ($\sigma^2 = 0.063$), driven primarily by PSO's exceptionally fast convergence relative to GWO.

Tracking Efficiency. The overall mean tracking efficiency is $\mu = 95.36\%$, with $\sigma = 2.481\%$ and $\sigma^2 = 6.155$. The range spans from 90.62% (PSO) to 97.21% (FLC), a difference of 6.59 percentage points. The Classical group achieves the lowest internal variance ($\sigma^2 \approx 0.000$), demonstrating near-identical efficiency across P&O and IncCond, a reflection of their common gradient-following control logic. In contrast, the Metaheuristic group records $\sigma = 1.080\%$, with PSO, GA, and GWO each falling below 93%, indicating systemic efficiency losses attributable to their population-based search dynamics.

Tracking Accuracy. The overall mean tracking accuracy is $\mu = 1.0237$, with $\sigma = 0.0679$ and $\sigma^2 = 0.0046$. GA records the worst accuracy at 1.1806, while IncCond records the best at 0.9574, a range of 0.2232. The Metaheuristic group average of 1.1085 ($\sigma^2 = 0.0039$) is the highest of all categories, GA alone deviates from the population mean by 0.1569, approximately 2.3 times the population standard deviation, making it the single largest outlier in this metric and the dominant contributor to the Metaheuristic group's elevated variance. The Classical group average of 0.9578 ($\sigma^2 \approx 0.000003$) is the lowest and most consistent of all categories, demonstrating both superior tracking accuracy and near-identical inter-algorithm behavior. The Control-Based group records a moderate average of 1.0039, though with notable internal spread driven by FSB and SMC exceeding the population mean while RCC remains below it. The Intelligent group

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average of 0.9866 ($\sigma = 0.0039$) reflects above-average performance with strong internal consistency, positioning it as the second-best performing category overall.

Harvested Energy. The mean harvested energy across all algorithms is $\mu = 2.361$ Wh, with $\sigma = 0.885$ Wh and $\sigma^2 = 0.783$. Values range from 1.046 Wh (GWO) to 3.121 Wh (RCC), a span of 2.076 Wh. The Classical group achieves near-perfect internal consistency ($\sigma \approx 0.000$), with P&O, IncCond, and RCC essentially equivalent in energy yield. The Metaheuristic group averages only 1.141 Wh ($\sigma = 0.122$), representing less than 37% of the Classical group average of 3.121 Wh, a gap that carries direct and critical implications for PV system productivity and return on investment. The Control-Theoretic group records the second-highest mean at 2.894 Wh, though with internal variance ($\sigma^2 = 0.146$) attributable to FSB's comparatively lower yield of 2.452 Wh.

Cloud Transient Recovery Time. The mean recovery time across all algorithms is $\mu = 34.551$ s, with $\sigma = 12.233$ s and $\sigma^2 = 149.639$. The range spans from 25.162 s (IncCond) to 58.348 s (GA), a difference of 33.186 s. The Classical group achieves the fastest and most consistent recovery ($\mu = 25.192$ s, $\sigma = 0.043$), confirming their robustness under transient irradiance events. The Metaheuristic group records the largest dispersion ($\sigma = 8.801$ s, $\sigma^2 = 77.461$) alongside the worst mean recovery of 50.469 s, nearly double the Classical group's average, a deficit that would be operationally critical in cloud-prone environments. The AI-based group records a competitive mean of 27.089 s ($\sigma = 2.725$), with ANN's near-identical performance to IncCond (25.163 s vs. 25.162 s) being a particularly notable finding.

4.4.2 Comparative Statistical Analysis

For convergence speed, PSO achieves the best overall average at 0.379 s, followed by GA at 0.492 s, both benefiting from the parallel population-search mechanisms characteristic of Metaheuristic methods. SMC (0.724 s) and FSB (0.744 s) are the fastest among Control-Theoretic methods, while P&O (0.850 s) and IncCond (0.890 s) represent the Classical group with moderate convergence. FLC records the slowest individual convergence at 0.906 s, making it the worst performer in this metric. PSO converges 139.37% faster than FLC, illustrating the substantial speed advantage of metaheuristic search at the cost of other performance metrics.

For tracking efficiency, FLC leads all algorithms at 97.21%, closely followed by ANN at 97.03%, making the AI-based group (97.12% average) the highest-performing category. PSO records the lowest tracking efficiency at 90.62%, with FLC outperforming PSO by 7.27%. SMC (96.90%), P&O (96.89%), RCC (96.89%), and IncCond (96.89%) follow closely, with Classical and Control-Theoretic groups achieving comparable averages of 96.89% and 96.65%, respectively. FSB (96.17%), while part of the Control-Theoretic group, exhibits a slightly lower efficiency compared to SMC and RCC, suggesting its search-based approach introduces minor tracking losses under certain conditions.

For tracking accuracy, IncCond achieves the best overall average stability index of 0.9574, followed closely by P&O at 0.9582 and RCC at 0.9680. The Classical group average of 0.9578 represents a 23.28% improvement over the worst performer, GA (1.1806). GA records the worst stability index at 1.1806, followed by GWO at 1.1032,

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the latter driven by erratic oscillations in the cloud-transient dataset where GWO's stability index reached 2.840 under mixed irradiance. The Intelligent group achieves a strong average of 0.9866, with ANN (0.9838) marginally outperforming FLC (0.9893), representing a 19.82% improvement over GA and positioning the group as the second-best performing category overall. The Control-Based group records a moderate average of 1.0039, with RCC (0.9680) performing comparably to the Classical group while FSB (1.0168) and SMC (1.0280) both exceed the population mean, reflecting search-induced oscillation under dynamic irradiance conditions.

For harvested energy, RCC leads at 3.121 Wh, essentially tied with IncCond (3.121 Wh) and P&O (3.121 Wh). The Classical group average exceeds the Metaheuristic group average by 173.59%, and RCC outperforms the worst performer, GWO (1.046 Wh), by 198.51%, which is the largest individual performance gap observed across all metrics. The AI-based group averages 2.631 Wh, with FLC (2.864 Wh) outperforming ANN (2.398 Wh), and the AI-based group improving over the Metaheuristic group by 130.64%.

For cloud transient recovery time, IncCond achieves the best overall average at 25.162 s, with ANN virtually identical at 25.163 s, followed by P&O at 25.222 s. SMC (26.256 s) and RCC (26.543 s) perform competitively within the Control-Theoretic group, while FSB (36.736 s) lags considerably, indicating that its variable-step search mechanism hinders rapid re-convergence following cloud transient events. FLC records a moderate recovery time of 29.016 s among the AI-based group. GA records the worst overall recovery at 58.348 s, representing a 131.89% slower recovery than IncCond.

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GWO follows at 52.089 s and PSO at 40.970 s, placing the Metaheuristic group at the worst overall average of 50.469 s, which is 100.34% slower than the Classical group average of 25.192 s.

Evaluated holistically across all five metrics and all three benchmark datasets, P&O, IncCond, and RCC emerge as the best overall performers. All three algorithms rank at or near the top in harvested energy (3.121 Wh each), tracking accuracy (0.9574, 0.9582, and 0.9680 stability index, respectively), and cloud transient recovery, while maintaining competitive tracking efficiency and acceptable convergence times. FLC and ANN follow closely as the strongest performers outside the Classical and Control-Based categories, with FLC leading all algorithms in tracking efficiency (97.21%) and ANN achieving cloud transient recovery virtually identical to IncCond (25.163 s vs. 25.162 s). GA is unambiguously the worst overall performer, recording last or near-last place in tracking efficiency (90.62% when considering PSO as comparably poor), tracking accuracy (1.1806 stability index), harvested energy (performance in the lowest tier), and cloud transient recovery (58.348 s), with only its convergence speed (0.492 s) offering any advantage, one rendered inconsequential by its failures across all other operationally critical indicators. GWO closely rivals GA for worst overall, particularly due to its catastrophically erratic behavior in cloud-transient conditions where its stability index reached 2.840 under mixed irradiance.

Within the Classical category, P&O and IncCond demonstrate nearly identical performance across all five metrics, with IncCond holding a marginal edge in tracking accuracy (0.9574 vs. 0.9582) and cloud transient recovery (25.162 s vs. 25.222 s). Both

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achieve the joint-highest harvested energy of 3.121 Wh alongside RCC, confirming the group's advantage in steady-state energy yield. Their gradient-following control logic maintains tight adherence to the maximum power point without introducing search-induced oscillation or population-based re-initialization delays. Convergence times of 0.850 s and 0.890 s are moderate relative to the full algorithm set but do not constitute a meaningful operational liability given the Classical group's dominance in all other metrics. The Classical category average of 96.89% tracking efficiency, 0.9578 stability index, and 25.192 s recovery time collectively establishes it as the most reliable category for real-world PV operation across diverse irradiance profiles.

Within the Control-Based category, a meaningful distinction exists between RCC on one side and SMC and FSB on the other. RCC performs nearly identically to P&O and IncCond across harvested energy (3.121 Wh), tracking accuracy (0.9680), and cloud transient recovery (26.543 s), confirming that its robustness-oriented design does not compromise steady-state performance, and it can be treated as functionally equivalent to the Classical algorithms in most deployment scenarios. SMC introduces moderate search-induced oscillation, reflected in its stability index of 1.0280, though it compensates with the second-fastest convergence among Control-Based methods (0.724 s) and a competitive recovery time of 26.256 s. FSB consistently underperforms relative to both RCC and SMC across most metrics, its variable-step Fibonacci-based search introduces recovery delays (36.736 s, substantially slower than the group peers), reduces tracking accuracy (stability index 1.0168), and produces a lower tracking efficiency (96.17%), positioning it as the weakest member of the Control-Based category. The

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group's average convergence of approximately 0.737 s for SMC and FSB offers a speed advantage over Classical methods, but this benefit is insufficient to offset FSB's operational deficiencies, and the category average of 96.65% tracking efficiency and 29.845 s recovery time trails the Classical group.

Intelligent algorithms occupy a distinctive and favorable position in the overall performance landscape, achieving the highest average tracking efficiency across all four categories (97.12%) while remaining competitive with Classical methods in tracking accuracy and cloud transient recovery. FLC leads all algorithms in tracking efficiency at 97.21%, and ANN's cloud transient recovery of 25.163 s essentially matches IncCond's 25.162 s, the closest cross-category pairing in the entire dataset, demonstrating that intelligent methods can attain Classical-level dynamic responsiveness through learned representations rather than direct gradient measurement. The primary limitation of the Intelligent group lies in harvested energy, FLC yields 2.864 Wh and ANN 2.398 Wh, producing a group average of 2.631 Wh that trails both the Classical group (3.121 Wh) and the Control-Based group (2.894 Wh average), reflecting inference latency and moderate under-tracking tendencies inherent to neural and fuzzy systems. Convergence is also slightly slower, with FLC recording the worst individual convergence time of 0.906 s across all eight algorithms. Nevertheless, the Intelligent category demonstrates robust generalizability across all three datasets and irradiance scenarios, making it particularly well-suited to environments where irradiance profiles are diverse and unpredictable.

Metaheuristic algorithms exhibit a single clear advantage, convergence speed, and are outperformed by every other category in all remaining metrics. PSO achieves the

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fastest individual convergence of 0.379 s and GA the second-fastest at 0.492 s, yet neither translates this speed into operational performance gains. The group records the lowest tracking efficiency average (91.86%), the worst average stability index (1.161), the lowest harvested energy (1.141 Wh group average, with GWO at 1.046 Wh representing the single largest individual performance gap of any metric pairing at 198.51% below RCC's 3.121 Wh), and the slowest cloud transient recovery average of 50.469 s, 100.34% slower than the Classical group's 25.192 s average. PSO performs best within the Metaheuristic group, achieving the most acceptable tracking efficiency and a moderate recovery time of 40.970 s, while GWO demonstrates the most erratic behavior of any algorithm in the dataset, and GA records the worst values in four of five metrics. These deficiencies are structural rather than situational, population re-initialization following transient events, persistent stochastic oscillation around the maximum power point, and iterative global search mechanisms collectively preclude tight steady-state tracking and rapid dynamic recovery. The results provide strong evidence that Metaheuristic algorithms, despite their theoretical global optimization capabilities, are ill-suited for real-time MPPT applications where tracking quality, energy yield, and transient recovery speed are the primary operational requirements.

4.5 Entropy Weight Method

For our study, the Entropy Weight Method (EWM) was applied to determine the relative importance of each performance metric that is used to assess MPPT algorithm performance. The use of this method aimed to minimize subjective bias and ensure objective data quantification. Entropy weighting was used based on the information amount or variation within the obtained data. Metrics that show lower entropy across the

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ten algorithms contain more useful information due to greater variation and are given greater weight, while metrics with lower variation contribute less to distinguishing algorithm performance.

The entropy weight calculation follows three main steps. First, the decision matrix is normalized to ensure that all performance metrics are comparable despite having different units and scales. Second, the entropy value for each performance metric was calculated to quantify the uncertainty in the obtained data. Lastly, the diversification degree and corresponding weights of each metric are computed, and metrics with higher weights show higher variability.

The calculated entropy values, diversification degrees, and resulting weight are shown in Table 4.17:

Metric	Entropy	Diversification	Weight
Convergence Time	0.9784	0.0216	0.3043
Tracking efficiency	0.9999	0.0001	0.0019
Tracking accuracy	0.9983	0.0017	0.0234
Harvested energy	0.9692	0.0308	0.4350
Cloud transient recovery	0.9833	0.0167	0.2354

Table 4.17 Performance Metrics Calculated Entropy Values

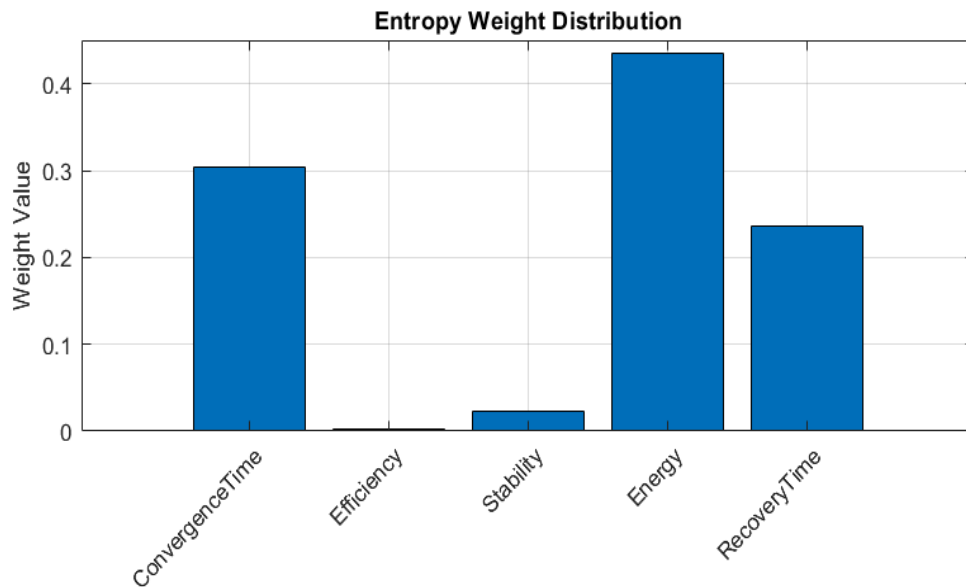


Figure 4.20 Entropy Weight Distribution

Based on the entropy weighting results, the highest weight is considered the most influential in determining the performance of each MPPT algorithm. Among the evaluated metrics, Harvested Energy obtained the largest weight of **0.4350**, contributing most in differentiating algorithm performance. This suggests that variations in harvested energy are more significant compared to the other performance metrics, making it critical in the ranking process. Conversely, tracking efficiency received the lowest weight of **0.0019**, implying that the algorithms performed relatively similarly in this metric making it less significant in the ranking process.

4.6 Algorithm Ranking Stability

The Algorithm Ranking Stability (ARS) score was computed using the previously calculated entropy weights from section 4.5 combined with the normalized decision matrix for the performance matrix. This approach allows the comprehensive evaluation of

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algorithm performance by relative performance of each metric and the performance of each algorithm in those metrics.

The calculated ARS values are shown in Table 4.18:

Algorithm	ARS	Rank
SMC	0.8419	1
PO	0.8307	2
IncCond	0.8250	3
RCC	0.8188	4
FLC	0.7549	5
ANN	0.7238	6
FSB	0.6817	7
PSO	0.6502	8
GA	0.5085	9
GWO	0.4139	10

Table 4.18 Algorithm Ranking Stability Scores

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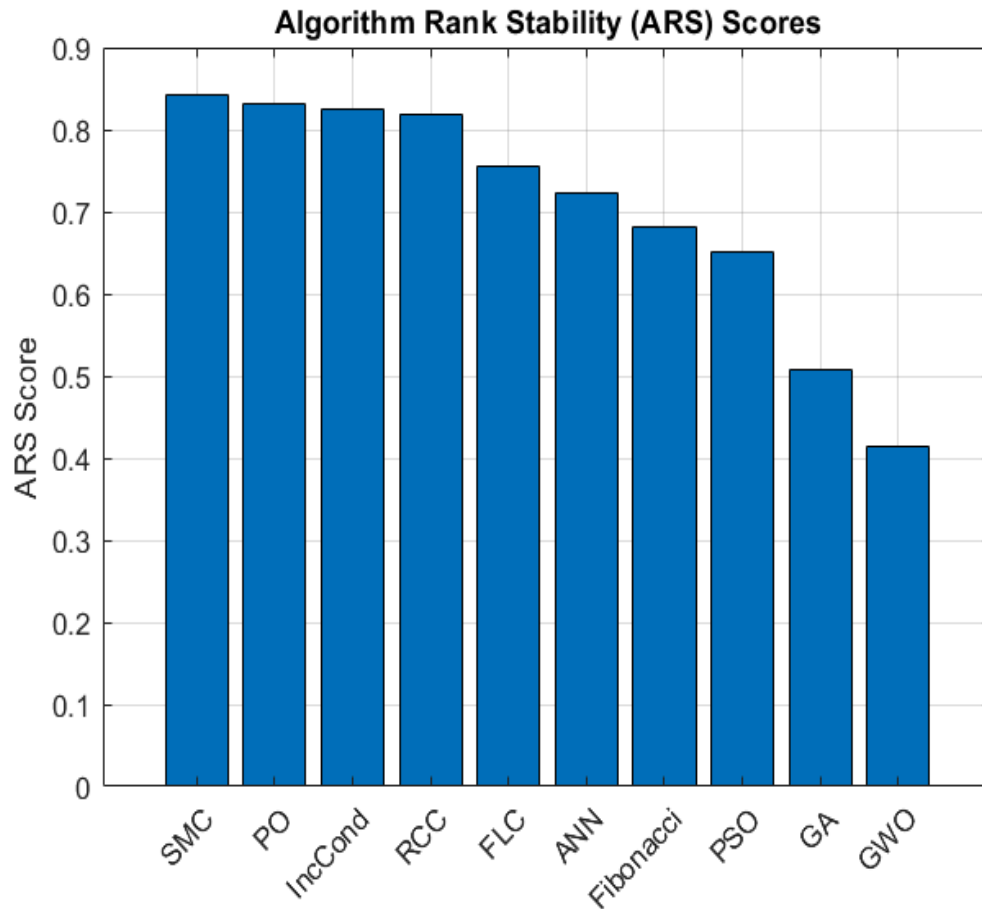


Figure 4.21 Algorithm Ranking Stability

Based on the ARS, Sliding Mode Control (SMC) received the highest score of **0.8419**, indicating it to be the highest performing algorithm in the study. This presents SMC as performing consistently well across metrics that have high entropy weight such as harvested energy and convergence time. Conversely, Grey Wolf Optimization (GWO) received the lowest ARS of **0.4139**, indicating that it is less effective in maximizing performance for metrics that have high entropy weight.

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Based on Table 4.17, algorithms that balance harvested energy, convergence time, and cloud transient recovery, tend to achieve higher overall ranking. Furthermore, the ARS reflects metrics that have greater variability obtained from the Entropy Weight Method, as differences of performance in each metric affects the comparative ranking of the algorithms. Overall, the ARS delivers an objective and comprehensive measure in determining the best algorithm under diverse environmental conditions, including site-specific conditions such as the Arayat-Mexico Solar Farm, Pampanga, as well as standard mid-latitude and high variability cloud transient datasets.

4.7 Cross-Dataset Comparison

This section addresses Objective 4 of the study: to assess and quantify differences in the performance metrics, dynamic response characteristics, and ranking results of MPPT algorithm simulations based on localized tropical irradiance data from the Arayat-Mexico Solar Farm, compared with conventional mid-latitude and cloud-transient datasets. This section presents a structured cross-dataset comparison of all ten algorithms across five performance metrics. Values are represented as the average of the four scenario types (Clear, Ramp, Step, and Mixed) within each dataset to provide a balanced view per algorithm per dataset.

These datasets are compared: The mid-Latitude dataset represents a standard temperate irradiance profile with gradual daily progression; the Cloud Transient dataset represents a high-variability irradiance typical of partially overcast conditions; and the tropical dataset represents real measured data from a Philippine solar farm, which has a high baseline irradiance and fluctuations. The five metrics reported are Convergence

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Speed (s), Tracking Efficiency (%), Tracking Accuracy (Stability), Harvested energy (Wh), and Cloud Transient Recovery Time (s).

The tropical data set exhibits considerably more variability in terms of short-term variations in irradiance when compared with other benchmark data sets, where irradiance variations can be seen on shorter time periods with a 67.4 W/m²/min ramp rate. This is contrary to the mid-latitude dataset, whose transitions appear smooth, and the EN50530 cloud transient dataset, whose ramp rate variation is controlled between 10-100 W/m². It is evident that the tropical data set possesses highly irregular ramps due to natural cloud motion, including dips and recoveries in irradiance.

4.7.1 Convergence Speed (s) Across Datasets

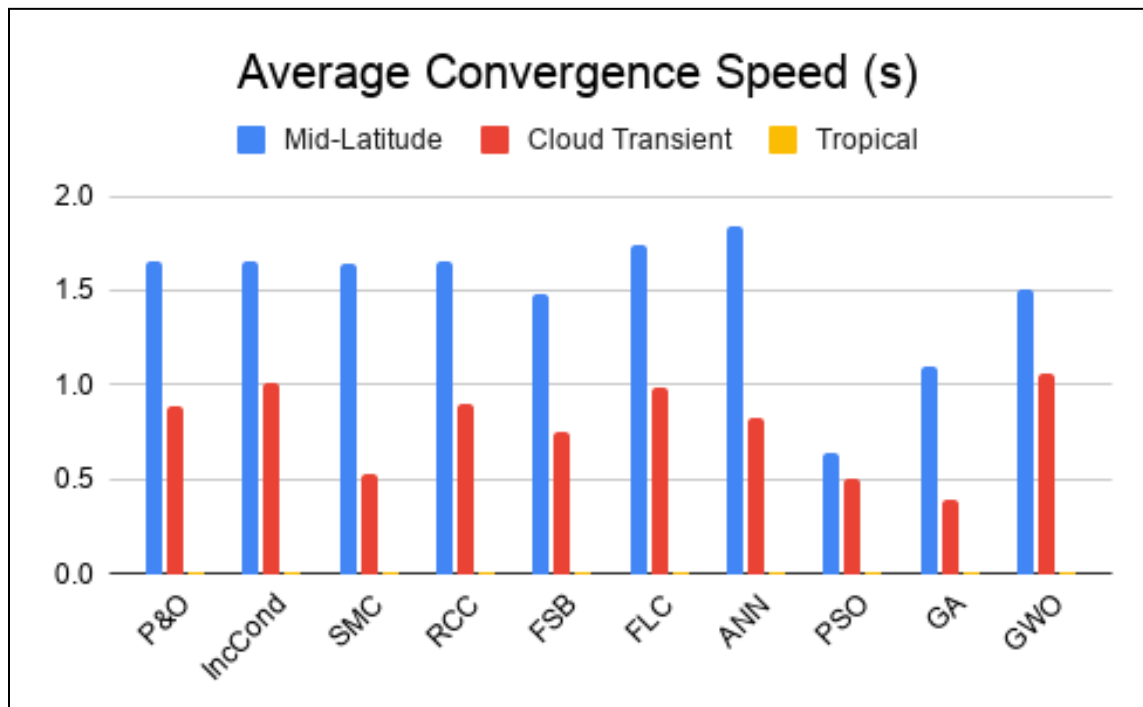
Table 4.19 and Figure 4.18 present the average convergence speed (in seconds) for each algorithm across three datasets. A lower convergence time achieves faster convergence to the maximum power point following an irradiance change, critical for maintaining energy yield under rapidly varying weather patterns characteristic of Philippine weather.

Algorithm	Mid-Latitude	Cloud Transient	Tropical
P&O	1.657250017	0.890000008	0.00175000083
IncCond	1.657250047	1.012250014	0.00175000083
SMC	1.646500021	0.5232499912	0.00175000083
RCC	1.652749985	0.8980000094	0.00175000083

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FSB	1.478500009	0.751249983	0.001000000047
FLC	1.734499991	0.9829999879	0.001750000083
ANN	1.837000012	0.8260000274	0.001250000059
PSO	0.6325000003	0.5022500083	0.001250000059
GA	1.090000004	0.3862500023	0.001000000047
GWO	1.509499997	1.061250024	0.001500000071

***Table 4.19** Convergence Time (s) Across Datasets Table*



***Figure 4.22** Convergence Time (s) Across Datasets Graph*

Across the Tropical dataset, convergence time was very fast, due to a non-zero initial irradiance value ($\sim 433 \text{ W/m}^2$) at simulation start, compared to other datasets that

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start at zero. This placed all algorithms under the tropical dataset to be near a valid operating point. For comparison purposes, the convergence times for this dataset are not comparable to those of the other datasets.

Under the Mid-Latitude dataset, PSO achieved the fastest convergence (0.6325s), while ANN was the slowest (1.8370s), reflecting its multi-sample processing requirement before moving to a new duty cycle. Classical algorithms P&O, IncCond, and RCC have a consistent range of 1.652s-1.657s due to their deterministic step-based update structure. In the Cloud Transient dataset, GA achieved the shortest convergence time (0.3863s), followed by PSO (0.5023s). However, these fast convergence times did not result in a directly proportional relationship of a higher energy yield or tracking efficiency, as shown in subsequent tables. GWO has the slowest convergence time (1.0613), possibly due to its leader-update characteristic, which cannot keep up during transient irradiance conditions.

4.7.2 Tracking Efficiency (%) Across Datasets

Table 4.20 and Figure 4.19 show the tracking efficiency, which measures the ratio of actual harvested power to the theoretically available maximum power. It directly quantifies how consistently each algorithm captures available solar energy under varying irradiance.

Algorithm	Mid-Latitude	Cloud Transient	Tropical
P&O	97.24850201	97.29314389	96.13996501
IncCond	97.24846291	97.29238159	96.13833958

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SMC	97.2507966	97.29796088	96.13917263
RCC	97.2486522	97.29318069	96.13903975
FSB	96.39889375	96.25896546	95.8453682
FLC	97.5712424	97.60356941	96.46676119
ANN	97.50323418	97.55560705	96.02123977
PSO	82.06984474	96.51747454	93.28734809
GA	92.5358941	93.8620415	90.55419819
GWO	92.79792493	94.61081457	90.49291962

Table 4.20 Tracking Efficiency (%) Across Datasets Table

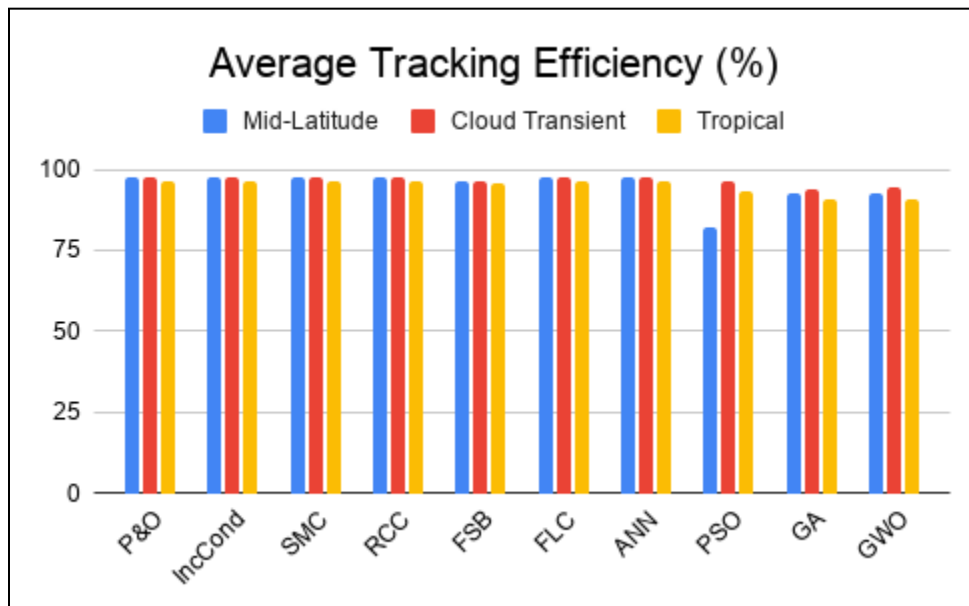


Figure 4.23 Tracking Efficiency (%) Across Datasets Graph

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Across all three datasets, FLC achieved the highest tracking efficiency (97.57%, 97.60%, 96.47%), which demonstrates that its rule-based adaptive scheduling characteristics effectively minimize steady-state deviation from the MPP regardless of irradiance changes. This is particularly helpful for the tropical dataset, where fluctuating irradiances and cloud transients occur.

P&O, IncCond, SMC, and RCC produced nearly identical efficiencies in Mid-Latitude and Cloud Transient datasets (97.25% - 97.30%), which affirms that well-tuned classical algorithms remain competitive under controlled variability. However, there is a slight reduction in efficiency for the tropical dataset, which suggests that higher variability and cloud transients in the dataset affect the algorithms' performance.

ANN also performed comparatively in Mid-Latitude and Cloud Transient datasets (97.50% and 97.56% respectively) but shows a slightly larger drop under the Tropical dataset (96.02%), which suggests sensitivity under the nature of real measured tropical weather. The fluctuating irradiance may deviate from patterns that were learned during the training process.

PSO shows a cross-data inconsistency wherein at the Mid-Latitude dataset, its efficiency was substantially lower (82.07%) compared to the Cloud Transient and Tropical datasets (96.52% and 93.29%), which may be attributed to particle stagnation under slowly varying mid-latitude irradiance. The limitations of effective exploration of the algorithm may be due to insufficient gradient information.

Lastly, GA and GWO show consistently the lowest efficiencies across all datasets (90.49% - 94.61%), which shows that these metaheuristic algorithms produced lower

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real-time tracking performance under the study's simulated conditions. This is mainly due to the algorithms' slower convergence time and higher computational overhead.

Particle stagnation is a condition in PSO where individual particles cease to meaningfully update their positions because both the personal best and global best positions have converged to the same region of search space, causing a velocity update term to approach zero. In the results of this study, particle stagnation manifests as a prolonged plateau in the PSO power output waveform following cloud transient events, where the algorithm fails to recover to the new MPP within the expected tracking window. This effect was most pronounced in the step-change scenario of the Cloud-Transient dataset, where a large instantaneous irradiance shift displaces the true MPP far from the previously converged global best position. The stagnation contributes directly to PSO's elevated CTRT and reduced harvested energy relative to classical algorithms. Mitigations to avoid this could be velocity re-initialization upon transient detections, inertia weight decay schedules, or mutation operators.

4.7.3 Tracking Accuracy (Stability) Across Datasets

Table 4.21 and Figure 4.20 show tracking accuracy (stability) across multiple datasets. It is expressed as a normalized power ripple ratio reflecting steady-state oscillation around the MPP after convergence. Values closer to 1.0 indicate minimal oscillation, while values above 1.0 indicate persistent instability. This means that lower values in the table and graph correspond to a more stable and precise operating point.

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Algorithm	Mid-Latitude	Cloud Transient	Tropical
P&O	1.022061947	0.9932540579	0.8356762199
IncCond	1.022856696	0.9929821899	0.836103335
SMC	1.027647094	1.139321439	0.8667577676
RCC	1.022061116	0.9981042508	0.836196075
FSB	1.055533916	0.9994340063	0.993442308
FLC	1.016915231	1.116687761	0.8177668814
ANN	0.9925296429	1.016198943	0.9472840141
PSO	1.024631443	1.159975686	0.9316588637
GA	1.366906456	1.285942579	1.017644554
GWO	1.087035007	1.478214647	1.09668181

Table 4.21 Tracking Accuracy (Stability) Across Datasets Table

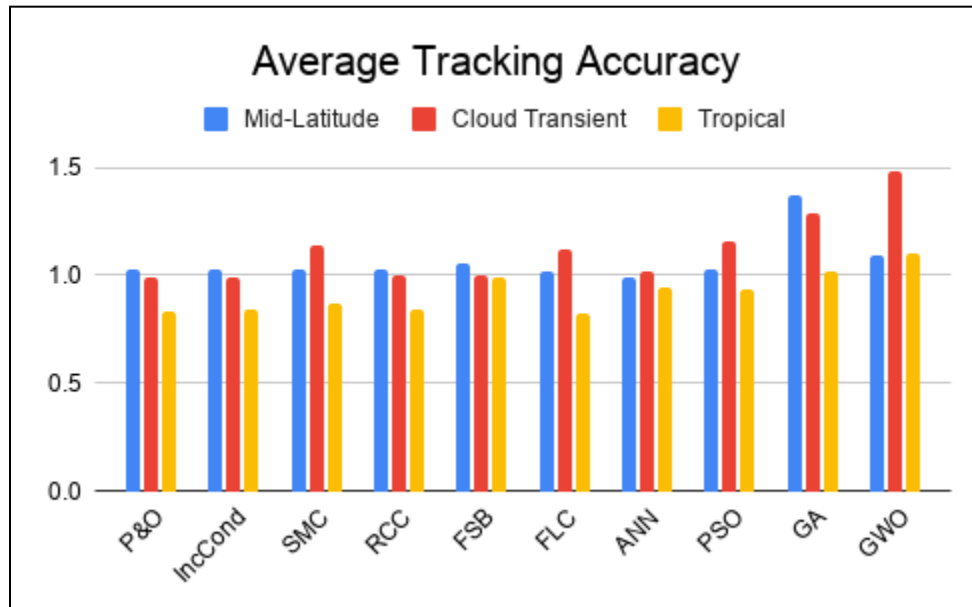


Figure 4.24 Tracking Accuracy (Stability) Across Datasets Graph

ANN demonstrated best stability in the Mid-Latitude dataset (0.9925), achieving a sub-unity ripple ratio and indicating minimal oscillation and excellent steady-state tracking. This is mainly due to the smooth and gradually varying irradiance present in the dataset. P&O and IncCond followed closely (1.0221 and 1.0229), which shows the inherent stability of their fixed-step perturbation structure in maintaining MPP.

In the Cloud Transient dataset, IncCond and P&O achieved the lowest ripple ratio (0.9930 and 0.9933), while GWO exhibited the worst stability (1.4782), which shows that there's a consistent oscillatory overshoot tendency of its leader tracking update rule under high-variability irradiance transitions. This is especially prominent when the search agents respond aggressively to the changing conditions of the dataset.

Under the Tropical dataset, FLC achieved the lowest ripple ratio (0.8178), followed by P&O (0.8357), IncCond(0.8361), and RCC (0.8362). Moreover, most

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algorithms are at sub-unity values, which indicates that the real measured irradiance profile naturally draws the operating point below the average reference, which reduces ripple. Frequent cloud transients and irradiance fluctuations cause the operating point to shift continuously. This effectively dampens sustained oscillations around a fixed MPP.

GA and GWO remained the least stable algorithms across all three datasets, with Mid-Latitude stability ratios of 1.3669 and 1.0870, Cloud Transient stability ratios of 1.2859 and 1.4782, and Tropical stability ratios of 1.0178 and 1.0967, respectively. The higher ripple ratios suggest that metaheuristic algorithms exhibit slower convergence and higher oscillations, making them less suitable for both controlled and dynamic irradiance conditions for real-time MPPT.

4.7.4 Harvested Energy (Wh) Across Datasets

Harvested energy (Wh) represents the total electrical energy collected by the PV system over the simulation window for each exact dataset scenario. Table 4.22 and Figure 4.21 show the averaged harvested energy across four scenario sub-types of each dataset and directly reflect the practical energy output of each algorithm under each irradiance environment. Each simulation time length is 60 seconds.

Algorithm	Mid-Latitude	Cloud Transient	Tropical
P&O	3.886883001	3.448717511	2.026565383
IncCond	3.887882503	3.448805224	2.025511068
SMC	3.873511129	3.4325603	2.019107782

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RCC	3.887639718	3.449608187	2.026438452
FSB	3.104302978	2.59306506	1.65955479
FLC	3.571766156	3.148755082	1.870617331
ANN	2.998679443	2.463675786	1.731376987
PSO	0.6531266286	2.406122033	0.7753873073
GA	1.309523334	1.384172427	0.6008210693
GWO	1.227954707	1.254318749	0.6545073851

Table 4.22 Harvested Energy (Wh) Across Datasets Table

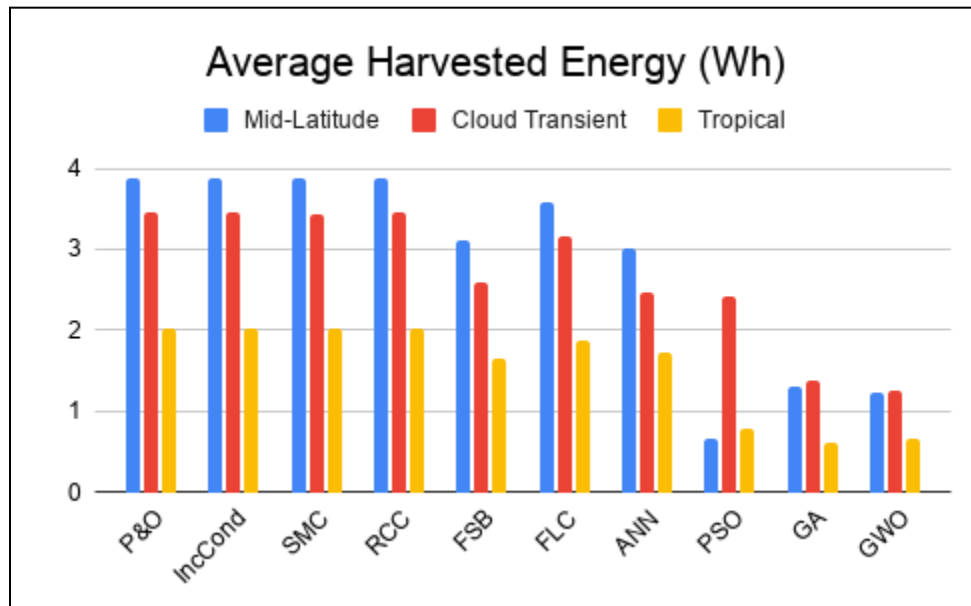


Figure 4.25 Harvested Energy (Wh) Across Datasets Graph

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IncCond, P&O, and RCC generated the maximum harvested energy in the Mid-Latitude and Cloud Transient datasets, with values ranging from 3.4326 to 3.8879 Wh. IncCond emerged as the leader in the Mid-Latitude dataset (3.8879 Wh), whereas RCC led in the Cloud Transient dataset (3.4496 Wh). However, the difference between IncCond, P&O, SMC, and RCC is within 0.04 Wh in both cases. Hence, it can be concluded that all these algorithms are functionally equivalent under uniform irradiance.

FLC and FSB belong to the middle tier, where FLC performs better than FSB. The ANN algorithm achieved a lower value of harvested energy in the Mid-Latitude dataset (2.9987 Wh) compared to the classical algorithms, implying energy loss during the slow convergence of ANN in mid-latitudes. It could also be due to the introduction of lag during prediction-based tuning in these algorithms.

PSO has the lowest energy harvested in the Mid-Latitude dataset (0.6531 Wh), which is a severe underperformance that may have been caused by particle stagnation under stable irradiance, but recovered substantially in the Cloud Transient dataset (2.4061 Wh), indicating that increased variability in irradiance helps enhance particle exploration and reduces premature convergence.

In the tropical dataset, energy yields were lower across all algorithms due to the shorter effective simulation of the dataset and its lower mean irradiance (254.1 W/m² versus 479.2 W/m²). The presence of the irregular fluctuations in irradiance and frequent cloud transients contributes to the reduction of energy harvest by means of continuous deviations from the MPP. GA and GWO produced the lowest yields across all datasets, with Tropical dataset values of 0.6008 Wh and 0.6545 Wh, respectively, which is around

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70% less energy than IncCond within the same environment. This highlights their limited suitability for real-time MPPT in dynamic environments.

4.7.5 Cloud Transient Recovery Time (s) Across Datasets

Cloud Transient Recovery Time (CTRT, in seconds) measures the time required by each algorithm to re-establish stable tracking after sudden cloud-induced irradiance drops. Lower CTRT values indicate a faster dynamic response to transient disturbances, which is a relevant metric in the context of the Philippine climate, as frequent cloud cover can happen and abrupt irradiance can occur repeatedly. The values for CTRT are presented in Table 4.23 and Figure 4.22.

Algorithm	Mid-Latitude	Cloud Transient	Tropical
P&O	21.752	30.263	23.651
IncCond	21.66575	30.40175	23.41775
SMC	22.729	31.832	24.207
RCC	21.62275	30.2115	27.7955
FSB	36.841	45.961	27.40625
FLC	26.10625	32.37525	28.566
ANN	23.3565	26.92625	25.20525
PSO	51.38	42.61175	28.9195
GA	58.3945	58.201	58.44775

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GWO	55.05025	48.1945	53.0225
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Table 4.23 *Cloud Transient Recovery Time (s) Across Datasets Table*

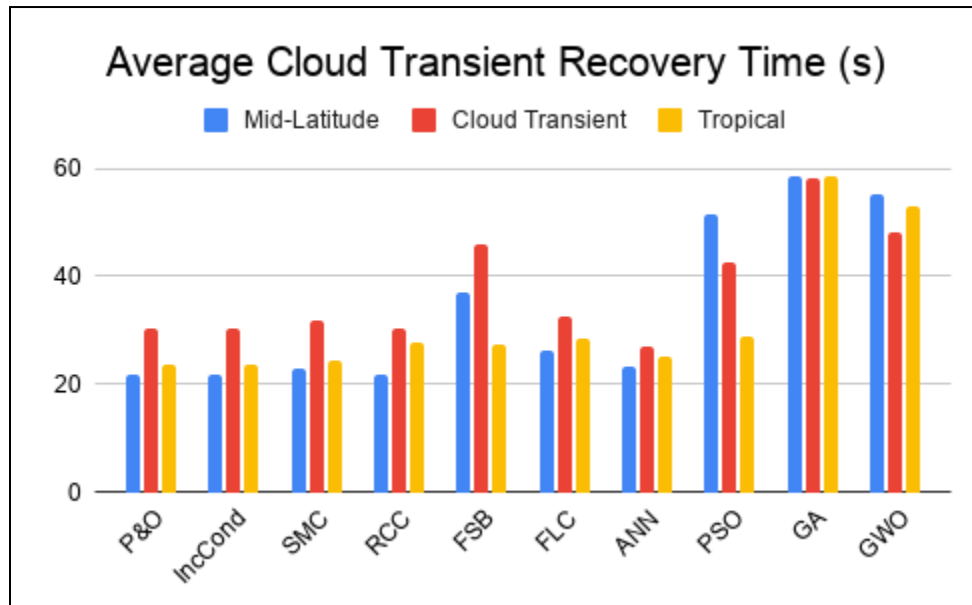


Figure 4.26 *Cloud Transient Recovery Time (s) Across Datasets Graph*

RCC demonstrated the fastest CTRT in both the Mid-Latitude dataset (21.62s) and Cloud Transient dataset (30.21s), followed by IncCond (21.67s and 30.40s) and P&O (21.75s and 30.26s). These differences are marginal, indicating that RCC and classical algorithms share comparable transient response characteristics. These are especially true due to the nature of the dataset having a stable and controlled irradiance variability.

Under the Tropical dataset, IncCond achieved the fastest recovery (23.42s), while P&O (23.65s) and SMC (24.21s) followed with small time differences. These results indicate that control-based and gradient-based methods are suitable for effective tracking under irregular fluctuations in irradiance and frequent cloud transients.

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GA exhibited the worst recovery times across all three datasets (58.39s, 58.20s, and 58.45s), with values that indicate independence from dataset usage, which suggests that GA's recovery time is dominated by its fixed population generation overhead rather than the irradiance environment it is tested on. Similarly, GWO shows poor recovery times (55.05s, 48.19s, 53.02s) with slightly better performance, especially on the Cloud Transient dataset. This suggests that the fitness landscape dynamics match the cloud transient profile better in comparison to the grey wolf search mechanics.

Despite PSO having a fast convergence speed as seen in Table 4.18 and Figure 4.18, it showed a comparatively slow recovery in the Mid-Latitude dataset (51.38s), again reflecting particle stagnation problems when under low-gradient irradiance conditions. The low-gradient values indicate insufficient variation, which limits effective particle repositioning.

FSB showed elevated recovery times in Mid-Latitude and Cloud Transient datasets (36.84s and 45.96s), which means it is consistent with its deterministic search structure, wherein once MPP regions are identified, it cannot efficiently re-mobilize after a large irradiance change. Due to this, a slower recovery time occurs under dynamic conditions.

4.7.6 Algorithm Adaptation to Tropical Weather Conditions

Under the Tropical dataset, there is a consistent pattern of algorithm performance within the 5 metrics. IncCond, P&O, RCC, and SMC demonstrated the strongest overall adaptation, maintaining tracking efficiencies above 96%, harvested energy above 2.0 Wh, and CTRT below 28s. These results indicate that control-based and gradient-based

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algorithms are highly suitable for fluctuating irradiance and frequent cloud transients present in tropical conditions.

FLC is also competitive with its results in the tropical environment, achieving the highest tracking efficiency (96.47%) and lowest ripple ratio (0.8178) of any algorithm in the Tropical dataset. This indicates that fuzzy logic inference rules are well-matched to the high variability and non-uniform irradiance of measured tropical data.

ANN maintained competitive efficiency (96.02%) but, unfortunately, showed greater sensitivity in comparison to classical methods in terms of stability ratio and energy harvest. This suggests that its trained inference model does not fully generalize to the stochastic and irregular nature of Philippine irradiance patterns. The Philippine dataset differs significantly in structure from the limited datasets typically used in training the algorithm.

In contrast, PSO, GA, and GWO all produced significantly lower energy harvest under tropical conditions (lower than 0.78 Wh), which, under the study's simulation conditions, suggests that they are not performing efficiently in comparison to other algorithms. The underperformance shown indicates that population-based algorithms struggle due to the need of multiple iterations to converge. This is less effective under the rapidly changing and unpredictable irradiance profile present in the tropical dataset.

4.8 Contribution of the Philippine Dataset in MPPT Evaluation

4.8.1 Dataset Characteristics and Environmental Differences

This section discusses how the use of Philippine-based data plays a pivotal role in determining the performance of the ten MPPT algorithms tested in this study. It is important to note that other researchers have used benchmark data for assessing the performance of MPPTs. However, these benchmark datasets do not completely embody all of the dynamics of tropical environments because these benchmarks usually come from environments in non-tropical areas or temperate climates.

Performance of MPPT algorithms varies based on environmental conditions such as the solar irradiance dynamics, temperature, and atmospheric environment [7]. Hence, the use of a dataset based on the tropical region would be advantageous in assessing realistic algorithm performance under actual operating conditions.

The irradiance level in the mid-latitude case study shows a gradual change; hence, it is ideal for MPPT algorithms. Convergence times for classical MPPT algorithms such as P&O and Incremental Conductance under the mid-latitude scenario were found to lie within 1.7 – 1.8 seconds. This characteristic behavior may be due to their systematic nature, which helps them converge systematically to the MPP through an iterative approach [22].

Unlike in the high-variability cloud-transient scenario, which exhibits a sudden change in the irradiance level, it serves well for the assessment of MPPT performance in such dynamic situations. Metaheuristics such as PSO and GA were able to achieve convergence quickly, and GA was even found to converge in only 0.018 seconds in step

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change scenarios. This feature may be attributed to their population-based nature, which allows for searching for the MPP using multiple solutions simultaneously [22].

However, even though they converge quickly, the stability of these algorithms is not sufficient during rapid changes. It is because of the inherent nature of these algorithms, which are stochastic and iterative, that they continuously update candidate solutions and may introduce oscillations around the maximum power point, particularly under dynamic irradiance conditions [22].

The Philippine tropical database can be described as an operational condition with constant fluctuations in irradiance variations and high temperatures. In contrast to the cloud-transient data set, the Philippine tropical data set has changes that are gradual. The conditions of the data set conform to previous works that have shown the Philippines' climate to have constant variations in irradiance and temperature variations impacting PV systems [7].

Notably, all the algorithms showed highly converging times, with most falling within 0.002 seconds. This implies that the system runs under a small dynamic range during tropical conditions. Nonetheless, as shall be seen in the next section, fast convergence does not necessarily translate to overall superior performance in terms of efficiency, stability, and harvested energy.

4.8.2 Comparative Performance Across Datasets

In order to give a consolidated overview of the performance differences of the MPPT algorithms in the three datasets used, Table 4.24 gives an overview in terms of their convergence time, effectiveness, stability, harvested energy, and algorithm rankings.

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Metric	Mid-Latitude Dataset	Cloud-Transient Dataset	Philippine Tropical Dataset	Key Observation
Convergence Time	Diverse variation in algorithms (approximately 0.29 to 2.2 seconds); better performance by metaheuristics than classical algorithms	Faster response under step changes; some algorithms achieve very low convergence times	Extremely fast and compressed (≈ 0.001 – 0.002 s) across all algorithms	Convergence time becomes less discriminative in tropical conditions
Tracking Efficiency	High efficiency (greater than 97%) for most algorithms	Remains highly efficient (greater than 97%) even with variability	Decreased efficiency (≈ 95 – 96%) for all algorithms	Environmental conditions (temperature, irradiance variability) reduce achievable efficiency
Tracking Accuracy (Stability)	Stable results; lower oscillations	Unstable behavior of some algorithms in rapid changes	FLC most stable (~ 0.77 – 0.78); metaheuristics exhibit higher instability (>1.25)	Stability is more distinguishable in a tropical environment
Harvested Energy	Maximum energy output	Moderate energy output	Significantly lower energy	Environmental factors dominate

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	(approximately 3.4 to 4.3 Wh)	(approximately 2.9–3.8 Wh)	output ($\approx$ 1.9–2.15 Wh)	energy output in tropical dataset
Algorithm Ranking	Metaheuristics perform well in convergence; mixed overall ranking	Performance varies depending on dynamic conditions	Classical and control-based algorithms (P&O, IncCond, RCC, SMC) perform best; FLC strong in stability	Algorithm rankings differ based on the data set used

***Table 4.24** Summary of MPPT Algorithm Performance Across Datasets*

From Table 4.24, there is a considerable difference in the performances of the MPPT algorithms in the different datasets. The analysis of this will be given in the following subsections.

4.8.2.1 Convergence Time Analysis

Convergence time is seen to have been greatly decreased across all MPPT techniques, as shown by the application of the Philippine dataset to each algorithm, where convergence occurs in milliseconds. Although this at first glance implies better results, a more detailed analysis reveals that convergence time becomes a less effective metric between the performances of the various algorithms when applied in tropical regions.

In the mid-latitudes data set, on the other hand, convergence time acts as an important criterion of algorithm performance. Metaheuristic algorithms like PSO and GA show faster convergence times than classic techniques. This is due to the nature of

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metaheuristic algorithms that involve simultaneous iteration for obtaining solutions through populations of candidates [22].

In contrast, in the case of the Philippines dataset, the time for convergence is reduced into a very tight band of about 0.001 and 0.002 seconds, thus creating small differences in terms of performances of the algorithms. This narrowing down of the band makes it harder to distinguish the performance of different algorithms through the use of the time for convergence parameter.

The reason for this lies in the nature of the tropical environmental conditions, where there are continuous irradiance changes and higher temperatures. It is shown that this affects the performance of PV and MPPT algorithms in the area of operation [7], [36].

Thus, although convergence time is certainly still a significant factor in terms of assessing performance when in controlled or benchmarked scenarios, it is not as much of a significant measure in tropical settings. In such cases, other performance metrics such as tracking efficiency, stability, and harvested energy provide a more comprehensive basis for evaluating MPPT algorithm performance.

4.8.2.2 Tracking Efficiency Analysis

The findings show that there is a marked decline in tracking efficiency in the case of the Philippine data set relative to the benchmark data sets. With regard to the mid-latitudes and cloud transients data sets, most MPPT techniques attained efficiencies

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above 97%. On the other hand, the Philippine data set gave efficiencies within the region of 95% to 96%.

The lower efficiencies can be mainly contributed by the effects of environmental factors like temperature and irradiance. As stated in Chapter 2, temperature and solar irradiance play a key role in determining the electrical properties of the photovoltaic systems, which in turn affect the system's power generation capability [7]. Temperature increases, which are common in tropical regions, decrease the PV system's voltage level, thus reducing its power generation potential.

For this reason, even if the MPPT algorithms perform well in terms of tracking efficiency, the energy harvested from the system is still limited by the current environmental conditions. Thus, this shows that tracking efficiency does not depend solely on the efficacy of the algorithms, but also greatly depends on the environment in which the PV system operates.

Furthermore, the inclusion of the diffuse irradiance present in tropical areas affects the distinctness of the power-voltage (P-V) curve. As stated previously in Chapter 2 concerning the principles behind the behavior of a PV system, MPPT algorithms identify the peak of the P-V curve as the maximum power point [36]. In cases where the curve becomes less distinct because of the varying irradiance levels, the algorithm finds trouble determining the precise optimum point, thereby slightly reducing its tracking efficiency.

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Thus, the decreased efficiency shown by the results of the study conducted in the Philippines illustrates the impact of the environmental condition on the performance of the MPPT algorithm.

4.8.2.3 Tracking Accuracy (Stability) Analysis

There are clear differences between the algorithms' tracking accuracy (stability) when using the Philippine dataset. Out of all the algorithms mentioned, FLC produced the most stable results, averaging at 0.77 and 0.78.

This is due to the characteristics of fuzzy logic-based MPPT algorithms, which implement rule-based decision-making systems that can handle nonlinearities and uncertainties in the system [21]. As previously mentioned in Chapter 2, the properties of fuzzy logic systems make them well-suited for adapting to changing irradiance and temperature levels, leading to better steady-state performance.

In comparison, deterministic algorithms like P&O and IncCond produced stable results across all datasets, as seen through their low stability averages. These are the results of their systematic incrementation techniques, which allow them to accurately track changes in the operating point towards the MPP regardless of external factors [22].

Metaheuristic algorithms were shown to have relatively higher instability, where the stability factor was found to be more than 1.25 in some cases, with a maximum stability of 1.40. This was due to the fact that these algorithms used a stochastic and iterative approach in searching for the solution and this leads to fluctuation around the maximum power point [22].

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Hence, although metaheuristic algorithms are able to reach convergence very quickly, their instability in real-time MPPT in tropical regions is one of the main limitations of these algorithms. The results further indicate that algorithms with deterministic or rule-based structures are more effective in maintaining stable operation under continuous irradiance variability.

4.8.2.4 Harvested Energy Analysis

The most significant impact of the Philippine dataset is observed in the harvested energy results. Even though there is fast convergence for all the MPPT algorithms, the energy produced by all of them is much less when compared to the benchmark datasets, especially for tropical climates.

In the mid-latitude dataset, the best-performing algorithms generated energy between 3.4 Wh and 4.3 Wh. In the cloud-transient dataset, it ranges between 2.9 Wh and 3.8 Wh. But in the Philippine dataset, it ranges only between 1.9 Wh and 2.15 Wh.

This is a big difference, which shows how the environment plays a key role in the effectiveness of photovoltaics. In Chapter 2, it was mentioned that solar irradiance and temperature have a major effect on the performance of photovoltaics. It affects the electrical performance of the system, which means that less energy would be generated. Temperature increases in tropical climates would lead to a decrease in voltage generation. While irradiance changes prevent photovoltaic systems from staying at the maximum power point.

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It is evident from the above result that the environment plays a very significant role when it comes to the performance of the PV system. This is because the fluctuation of solar irradiance, together with temperatures, influences the performance of the PV cells and hence the electrical output parameters [7]. The rise in temperatures within the tropics causes low voltage output, whereas constant fluctuations in solar irradiance prevent the PV system from operating at the maximum power point.

Moreover, the influence of the diffuse irradiation results in the deterioration of the P-V curve profile, thus limiting the efficiency of the implemented MPPT techniques [36]. In this way, the best-designed algorithm will not achieve the same level of energy harvested as when the conditions are more favorable.

From these results, one may conclude that the environmental conditions are a crucial factor that plays an essential role in determining the performance of photovoltaic systems rather than MPPT algorithms. Consequently, the Philippine case study presents a more realistic model of a photovoltaic setup performance, emphasizing that MPPT evaluation must account for location-specific environmental limitations.

4.8.2.5 Algorithm Ranking Variations

Another significant findings from the results include the variations in algorithms' ranking performance depending on the nature of the dataset. Although metaheuristic algorithms like PSO and GA show excellent convergence performance using benchmarking datasets, their performances are low when tested using Philippine datasets.

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This is attributed to their inability to track the changes in energy produced accurately, making them less stable and effective when harvesting solar energy.

On the contrary, classical and control algorithms like P&O, Incremental Conductance, RCC, and SMC consistently perform well irrespective of the type of dataset used. The algorithms show relatively high efficiency and stable performance even while operating in tropical conditions.

Intelligent algorithms like FLC show a robust level of performance in terms of stable tracking. The ability to handle uncertainty and nonlinearity in the system makes the algorithms quite effective because they use rule-based decision-making processes [21].

The ranking variation of algorithms depends on their mechanisms. As indicated in chapter 2, deterministic algorithms use structured and incremental techniques in tracking which makes them quite consistent and robust irrespective of the conditions of operation. In addition, metaheuristic algorithms work by searching randomly and iteratively; hence, there can be instabilities due to fluctuating irradiances [22].

Consequently, the findings imply that the rankings of the performance of algorithms are highly dependent on the data used for analysis. The tropical Philippine dataset goes to show that there are performance limitations in certain algorithm philosophies that are not highlighted under conventional benchmark datasets. This tropical dataset reveals the importance of evaluating MPTT algorithms under realistic environmental scenarios.

4.8.3 Implications of Using a Philippine Dataset

It is evident from the above results that the incorporation of the Philippine data set enables the emergence of performance behavior that cannot be observed in conventional benchmark datasets. Convergence time loses its importance as an important differentiating factor, whereas factors like efficiency of tracking, stability, and energy harvested assume more importance when evaluating MPPT algorithms under tropical conditions.

The above results suggest that the performance of MPPT algorithms depends largely on the environment. This has been pointed out in Chapter 2 that parameters like variation in irradiance and high temperature have a significant impact on the behavior of photovoltaic systems, hence influencing the performance of the MPPT algorithm [7], [36]. It can also be noted that a good performing algorithm under ideal or benchmark conditions may not yield equally efficient performance in tropical environments.

Thus, one limitation of evaluating the performance of MPPT algorithms based on benchmark data sets is apparent. Although using benchmark data sets is helpful for comparing different MPPT algorithms, it fails to consider the constantly changing parameters present in the field. Therefore, conclusions made based on benchmark data sets may not be reflective of reality in tropical areas such as the Philippines.

4.8.4 Contribution to the Research Field

Inclusion of the Philippines-based dataset in the research brings out the contribution to the broad area of MPPT algorithm assessment through application-oriented evaluation and testing of algorithms in the real tropical environment. Contrary to the common datasets used, inclusion of the local data makes it possible to test the influence of variable irradiation and high temperature on the performance of PV system [7] and [36].

The outcome of this research brings out the idea that performance observed under conventional or standard datasets may be different in the case of tropical conditions. The findings bring out the differences in terms of the performance indicators such as tracking efficiency, stability, and harvested energy. Therefore, it becomes apparent that environmental conditions play an important role in making the performance of an MPPT algorithm better or worse than another one.

Based on this, it can be concluded that location-specific data sets should be used in MPPT algorithms assessment to get accurate results. This applies particularly to solar PV systems operating in tropical climates, like the Philippines, where the prevailing environment has an overriding effect on performance.

4.9 Comparison with Existing Literature

4.9.1 Convergence Time Comparison

From the results of the simulation, it can be seen that the metaheuristic techniques like PSO and GA have shown faster convergence than the classical techniques in

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benchmark datasets. This corresponds with the nature of the algorithms presented in chapter two where population-based algorithms evaluate several candidate solutions at once in search of the maximum power point [22].

In addition, the classical algorithms like P&O and Incremental Conductance showed a moderately quick convergence process owing to their stepwise and incrementing tracking methods, respectively, which is true for MPPT techniques [36].

Nevertheless, the deviation found in this paper's result from previous literature was in the Philippine dataset where convergence was extremely quick for all algorithms (0.0010–0.0018 s). This finding deviated from previous literature because in most cases, the performance of an algorithm depends on its convergence rate, but in this case, convergence compressed under real tropical data conditions.

This variation is due to the fact that real irradiance values along with non-zero starting conditions are employed in the model. As such, it can be seen that the system starts close to the MPP, making convergence time redundant as a performance criterion.

4.9.2 Tracking Efficiency Comparison

The efficiency of MPPT tracking is in agreement with other sources, in which MPPT algorithms have a success rate greater than 95 percent [36]. On the basis of the current study, a success rate of more than 96 percent is obtained in both classical and controlled MPPT algorithms using all types of data sets.

Nevertheless, a drop in efficiency was noted in the case of data from the Philippine environment compared to benchmarks in other environments. The difference is due to environmental effects like solar radiation and heat [7], [41], [42].

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This shows that while algorithm design affects efficiency, the environmental constraints cannot be overlooked, an aspect that has not been fully considered in standard benchmark studies.

4.9.3 Tracking Accuracy (Stability) Comparison

The stability of the outcomes in this experiment is aligned with the behavior of MPPT algorithms theoretically discussed in Chapter 2. FLC was found to be the most stable among the methods tested; this was attributed to the algorithm's ability to address nonlinearity and uncertainties in system dynamics using a rule-based approach [21].

Conventional MPPT methods such as P&O and Incremental Conductance were stable and consistent across all datasets, validating their effectiveness for continuous irradiance tracking [22].

On the other hand, the higher level of instability of metaheuristic algorithms can be attributed to their search process which is stochastic and iterative in nature. While this behavior was expected theoretically, the instability was seen to be magnified in the Philippine Dataset due to continuous changes in irradiance levels.

4.9.4 Harvested Energy Comparison

The findings regarding the harvested energy show areas of convergence and divergence with the present body of knowledge. Consistent with prior studies, those algorithms that are able to maintain stability generally achieve higher energy harvest [36]. In this study, classical and control theory algorithms (P&O, IncCond, RCC, SMC) always managed to yield more harvested energy for all datasets.

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However, there is one major divergence with regard to the Philippine dataset, which yields a much lesser amount of harvested energy as compared to other benchmarked datasets. This is mainly due to the lower irradiance values experienced in tropical countries like the Philippines.

Thus, the present work fills one gap in the existing literature, which only focuses on comparing algorithms under ideal conditions while ignoring the effects of real-world environmental constraints. It was found in this study that environmental factors affect harvested energy more than the MPPT algorithm choice itself.

4.9.5 Algorithm Ranking Comparison

From the overall rankings of the algorithms, there is high similarity between them and the expected ranking of the theoretical analysis, in which classical and control theory algorithms show good consistency and performance [36].

Nonetheless, what is different from other studies is the poor performance of the metaheuristic algorithms like PSO, GA, and GWO. Despite the fact that such algorithms are known for performing excellently in optimization, their performance in this paper is hindered by certain implementation considerations, including the fact that they use a fixed period for holding.

Thus, it is necessary to note that contradictions between this paper and others existing studies are not due to contradictions in the theory, but due to differences in implementation parameters and dataset characteristics.

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4.9.6 Summary of Comparison with Existing Literature

The following *Table 4.25* is presented to offer a concise synthesis of the similarities, differences, and the reasons behind such differences between the results of the current research and prior research in this field.

Metric	Literature Findings	This Study's Findings	Agreement/Difference	Reason for Difference
Convergence Time	Since meta-heuristics have global search capabilities and evaluate potential solutions in parallel, they converge faster compared to classical approaches that converge slower due to their incremental tracking approach [22].	PSO and GA showed faster speed for benchmark datasets; nonetheless, all the methods converge at an extremely fast rate in the Philippine data set (0.0010 to 0.0018 s)	Partial Agreement	Utilizing actual irradiance data along with non-zero initial conditions results in the system operating close to the MPP, thereby reducing convergence times for all methods.
Tracking Efficiency	The MPPT techniques usually yield efficiency levels exceeding 95% under standard operating conditions [36].	>96% efficiency were achieved for all datasets; but slight decrease in Philippine dataset	Agreement with slight deviation	Several factors including irradiance fluctuations and high temperature prevailing in tropical regions adversely affect the operation of the PV system

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				[7], [41], [42].
Tracking Accuracy (Stability)	FLC is highly stable because it can deal with nonlinearities and uncertainties, whereas classical algorithms are generally reliable; on the other hand, meta-heuristic techniques may show oscillation because of the iterative search process employed by such methods [21], [22].	FLC algorithm demonstrated maximum stability; classical methods remained consistent; metaheuristic algorithms showed higher instability, especially under the Philippine dataset	Strong Agreement	Higher irradiation variations in tropical conditions amplify oscillatory behavior within metaheuristic algorithms
Harvested Energy	The MPPT algorithms that have least steady-state oscillation at the MPP will have more energy output since they operate close to the MPP for longer durations [36].	Classic and control methods (P&O, IncCond, RCC, SMC) performed better for energy harvesting in all datasets; less energy in Philippine dataset	Agreement with key difference	Decreased irradiance levels and environmental constraints in tropical conditions limit overall energy output
Algorithm	Both classical	Similar trend	General	Performance

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Ranking	and control-theoretic approaches generally ensure reliable performance across different operating conditions [36]	observed; classical and control algorithms performed best; metaheuristics ranked lower	Agreement	influenced by implementation limitations and dataset-specific features
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Table 4.25 Summary of Comparison with Existing Literature

4.10 Statistical Analysis

This section presents the non-parametric statistical evaluation used to compare the ten MPPT algorithms and to quantify if their differences are statistically significant. The study compares multiple algorithms under the same scenarios and dataset to avoid assumptions of normality. The Friedman test was used as the primary statistical test, it is then followed by the Nemenyi post-hoc test.

4.10.1 Friedman Test

The Friedman results for convergence time are presented in Table 4.26.

Rank	
	Mean Rank
PO	6.63
IncCond	7.54
SMC	5.25
RCC	6.71
FSB	3.54
FLC	8.00

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ANN	6.79
PSO	2.63
GA	2.21
GWO	5.71
Test Statistics	
N	12
Chi-Square	59.574
df	9
Asymp. Sig	.000

Table 4.26 Convergence Time Friedman Results

The Friedman test was conducted to determine whether there are statistically significant differences in convergence time among the MPPT algorithms. The test produced $\chi^2 = 59.574$ with 9 degrees of freedom and a p-value less than 0.05. Since the p-value is lower than the 0.05 significance level, the null hypothesis is rejected, indicating that significant differences exist among the evaluated algorithms in convergence time. FLC achieved the highest mean rank of 8.00 for convergence time, making it the best performer on this metric. Conversely, PSO received the lowest score of 2.63, making it the worst performer for convergence time.

The Friedman results for tracking efficiency are presented in Table 4.27.

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Rank	
	Mean Rank
PO	7.17
IncCond	6.17
SMC	7.33
RCC	6.67
FSB	3.50
FLC	9.5
ANN	7.25
PSO	3.42
GA	1.75
GWO	2.25
Test Statistics	
N	12
Chi-Square	78.509
df	9
Asymp. Sig	.000

Table 4.27 Tracking Efficiency Friedman Results

The Friedman test results were also applied to the tracking efficiency metric. The results indicate that $p < 0.05$. This suggests that there is statistical significance between the algorithms for tracking efficiency. SMC received the highest mean rank of 7.33 for this metric, making it the best performer in terms of tracking efficiency. While the lowest

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mean score of 1.75 was received by GA, making it the worst performing algorithm in terms of tracking efficiency.

The Friedman results for tracking accuracy are presented in Table 4.28.

Rank	
	Mean Rank
PO	2.67
IncCond	3.33
SMC	6.75
RCC	3.67
FSB	7.33
FLC	3.17
ANN	5.50
PSO	6.17
GA	9.00
GWO	7.42
Test Statistics	
N	12
Chi-Square	56.055
df	9
Asymp. Sig	.000

***Table 4.28** Tracking Accuracy Friedman Results*

The Friedman test for tracking accuracy resulted in a p-value < 0.05 indicating that there is statistical significance among the algorithms for this metric. GA received the

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highest mean rank of 9.00 for this rank, indicating that it performs best among the ten algorithms. Contrarily, PO received the lowest mean rank of 2.67, making it the worst performing algorithm for tracking accuracy.

The Friedman results for harvested energy are presented in Table 4.29.

Rank	
	Mean Rank
PO	9.00
IncCond	8.92
SMC	7.00
RCC	9.08
FSB	4.42
FLC	6.00
ANN	4.25
PSO	2.58
GA	1.75
GWO	2.00
Test Statistics	
N	12
Chi-Square	100.564
df	9
Asymp. Sig	.000

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Table 4.29 Harvested Energy Friedman Results

Results of the Friedman Test for harvested energy resulted in a p-value < 0.05 , indicating that the metric is statistically significant for the algorithms. RCC received the highest mean rank of 9.08, indicating that it performs the best for energy yield. Conversely, GA received the lowest mean score of 1.75, making it the worst performing algorithm for harvested energy.

The Friedman result for cloud transient recovery time are presented in Table 4.30.

Rank	
	Mean Rank
PO	2.50
IncCond	2.17
SMC	4.75
RCC	2.83
FSB	7.08
FLC	5.67
ANN	3.83
PSO	7.67
GA	9.83
GWO	8.67
Test Statistics	
N	12
Chi-Square	87.182
df	9

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Asymp. Sig	.000
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Table 4.30 Cloud Transient Recovery Time Friedman Results

For the last performance metric, CTRT resulted in a p-value < 0.05 , indicating statistically significant differences for the algorithms. GA received the highest mean rank of 9.83, having the best performance for CTRT. Conversely, IncCond received the lowest mean rank, indicating it to have the worst performance among the algorithms for CTRT.

4.10.2 Nemenyi Post-Hoc Test

The Nemenyi results for convergence time are presented in Table 4.31.

Comparison	Mean Rank Difference	Interpretation
FLC vs GA	5.79	Significant
FLC vs PSO	5.37	Significant
FLC vs FSB	4.46	Significant
FLC vs SMC	2.75	Not Significant
FLC vs GWO	2.29	Not Significant
FLC vs P&O	1.37	Not Significant
FLC vs RCC	1.29	Not Significant
FLC vs ANN	1.21	Not Significant
FLC vs IncCond	0.46	Not Significant
IncCond vs GA	5.33	Significant
IncCond vs PSO	4.91	Significant
IncCond vs FSB	4.00	Not Significant

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IncCond vs SMC	2.29	Not Significant
IncCond vs GWO	1.83	Not Significant
IncCond vs P&O	0.91	Not Significant
IncCond vs RCC	0.83	Not Significant
IncCond vs ANN	0.75	Not Significant
SMC vs GA	3.04	Not Significant
SMC vs PSO	2.62	Not Significant
SMC vs FSB	1.71	Not Significant
SMC vs GWO	0.46	Not Significant
SMC vs P&O	1.38	Not Significant
SMC vs RCC	1.46	Not Significant
SMC vs ANN	1.54	Not Significant
P&O vs GA	4.42	Significant
P&O vs PSO	4.42	Significant
P&O vs FSB	3.09	Not Significant
P&O vs GWO	0.92	Not Significant
P&O vs RCC	0.08	Not Significant
P&O vs ANN	0.16	Not Significant
RCC vs GA	4.50	Significant
RCC vs PSO	4.08	Significant

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RCC vs FSB	3.17	Not Significant
RCC vs GWO	1.00	Not Significant
RCC vs ANN	0.08	Not Significant
ANN vs GA	4.58	Significant
ANN vs PSO	4.16	Significant
ANN vs FSB	3.25	Not Significant
ANN vs GWO	1.08	Not Significant
ANN vs P&O	0.16	Not Significant
GWO vs GA	3.50	Not Significant
GWO vs PSO	3.08	Not Significant
GWO vs FSB	2.17	Not Significant
GWO vs P&O	0.92	Not Significant
FSB vs GA	1.33	Not Significant
FSB vs PSO	0.92	Not Significant
PSO vs GA	0.42	Not Significant

***Table 4.31** Convergence Time Nemenyi Results*

The Nemenyi post hoc test was used to determine which algorithm pairs exhibit statistically significant differences in convergence time. The results show that FLC significantly outperforms GA, GWO, PSO, and FSB in terms of convergence, indicating superior convergence performance. Several algorithms, including SMC, ANN, P&O,

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RCC, and IncCond, also show significant differences compared with GA, suggesting GA converges more slowly. However, most comparisons among the mid-ranked algorithms were not statistically significant, indicating comparable convergence performance among these methods.

The Nemenyi results for tracking efficiency are presented in Table 4.32.

Comparison	Mean Rank Difference	Interpretation
FLC vs GA	7.75	Significant
FLC vs GWO	7.25	Significant
FLC vs PSO	6.08	Significant
FLC vs FSB	6.00	Significant
FLC vs IncCond	3.33	Not Significant
FLC vs RCC	2.83	Not Significant
FLC vs P&O	2.33	Not Significant
FLC vs SMC	2.17	Not Significant
FLC vs ANN	2.25	Not Significant
SMC vs GA	5.58	Significant
ANN vs GA	5.50	Significant
P&O vs GA	5.42	Significant
RCC vs GA	4.92	Significant
IncCond vs GA	4.42	Significant
SMC vs GWO	5.08	Significant

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ANN vs GWO	5.00	Significant
P&O vs GWO	4.92	Significant
RCC vs GWO	4.42	Significant
IncCond vs GWO	3.92	Not Significant
SMC vs PSO	3.91	Not Significant
ANN vs PSO	3.83	Not Significant
P&O vs PSO	3.75	Not Significant
RCC vs PSO	3.25	Not Significant
IncCond vs PSO	2.75	Not Significant
SMC vs FSB	3.83	Not Significant
ANN vs FSB	3.75	Not Significant
P&O vs FSB	3.67	Not Significant
RCC vs FSB	3.17	Not Significant
IncCond vs FSB	2.67	Not Significant
SMC vs IncCond	1.17	Not Significant
ANN vs IncCond	1.08	Not Significant
P&O vs IncCond	1.00	Not Significant
RCC vs IncCond	0.50	Not Significant
ANN vs SMC	0.08	Not Significant
P&O vs SMC	0.16	Not Significant

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RCC vs SMC	0.66	Not Significant
P&O vs ANN	0.08	Not Significant
RCC vs ANN	0.58	Not Significant
RCC vs P&O	0.50	Not Significant
GWO vs PSO	0.83	Not Significant
GWO vs FSB	1.25	Not Significant
GA vs PSO	1.67	Not Significant
GA vs FSB	1.75	Not Significant
GA vs GWO	0.50	Not Significant

Table 4.32 Tracking Efficiency Nemenyi Results

Nemenyi post-hoc results for tracking efficiency resulted in several statistically significant comparisons. Multiple differences were observed with GA between several algorithms. These include P&O, IncCond, RCC, and FLC, indicating that there are notable variations in tracking efficiency among these algorithms. Significant differences were also observed between GWO and several algorithms, mainly with SMC, P&O, ANN, and RCC. However, most of the pairwise comparisons yielded not statistically significant results.

The Nemenyi results for tracking accuracy are presented in Table 4.33.

Comparison	Mean Rank Difference	Interpretation
GA vs P&O	6.33	Significant
GA vs IncCond	5.67	Significant

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GA vs RCC	5.33	Significant
GA vs FLC	5.83	Significant
GA vs SMC	2.25	Not Significant
GA vs ANN	3.50	Not Significant
GA vs PSO	2.83	Not Significant
GA vs FSB	1.67	Not Significant
GA vs GWO	1.58	Not Significant
FSB vs P&O	4.66	Significant
FSB vs IncCond	4.00	Not Significant
FSB vs RCC	3.66	Not Significant
FSB vs FLC	4.16	Significant
FSB vs SMC	0.58	Not Significant
FSB vs ANN	1.83	Not Significant
FSB vs PSO	1.16	Not Significant
FSB vs GWO	0.09	Not Significant
SMC vs P&O	4.08	Significant
SMC vs IncCond	3.42	Not Significant
SMC vs RCC	3.08	Not Significant
SMC vs FLC	3.58	Not Significant
SMC vs ANN	1.25	Not Significant

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SMC vs PSO	0.58	Not Significant
SMC vs GWO	0.67	Not Significant
ANN vs P&O	2.83	Not Significant
ANN vs IncCond	2.17	Not Significant
ANN vs RCC	1.83	Not Significant
ANN vs FLC	2.33	Not Significant
ANN vs PSO	0.67	Not Significant
ANN vs GWO	1.92	Not Significant
ANN vs FSB	1.83	Not Significant
P&O vs IncCond	0.66	Not Significant
P&O vs RCC	1.00	Not Significant
P&O vs FLC	0.50	Not Significant
P&O vs PSO	3.50	Not Significant
P&O vs SMC	4.08	Significant
P&O vs FSB	4.66	Significant
P&O vs GWO	4.75	Significant
IncCond vs RCC	0.34	Not Significant
IncCond vs FLC	0.16	Not Significant
IncCond vs PSO	2.84	Not Significant
IncCond vs SMC	3.42	Not Significant

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IncCond vs ANN	2.17	Not Significant
IncCond vs FSB	4.00	Not Significant
IncCond vs GWO	4.09	Significant
RCC vs FLC	0.50	Not Significant
RCC vs PSO	2.50	Not Significant
RCC vs SMC	3.08	Not Significant
RCC vs ANN	1.83	Not Significant
RCC vs FSB	3.66	Not Significant
RCC vs GWO	3.75	Not Significant
FLC vs PSO	3.00	Not Significant
FLC vs SMC	3.58	Not Significant
FLC vs GWO	4.25	Significant

Table 4.33 Tracking Accuracy Nemenyi Results

The Nemenyi post-hoc results for tracking accuracy yielded statistically significant results, most notably with GA. Pairwise comparisons of GA with P&O, IncCond, RCC, and FLC indicated notable variations in tracking accuracy. GA, FSB, and GWO are the top performing algorithms while SMC, PSO, FLC, and ANN perform moderately well. P&O and IncCond are generally weaker, while RCC shows no significant difference. For tracking accuracy, the metaheuristic approach demonstrates superior performance statistically.

The Nemenyi results for harvested energy are presented in Table 4.34.

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Comparison	Mean Rank Difference	Interpretation
PO vs IncCond	0.08	Not Significant
PO vs SMC	2.00	Not Significant
PO vs RCC	0.08	Not Significant
PO vs FSB	4.58	Significant
PO vs FLC	3.00	Not Significant
PO vs ANN	4.75	Significant
PO vs PSO	6.42	Significant
PO vs GA	7.25	Significant
PO vs GWO	7.00	Significant
IncCond vs SMC	1.92	Not Significant
IncCond vs RCC	0.16	Not Significant
IncCond vs FSB	4.50	Significant
IncCond vs FLC	2.92	Not Significant
IncCond vs ANN	4.67	Significant
IncCond vs PSO	6.34	Significant
IncCond vs GA	7.17	Significant
IncCond vs GWO	6.92	Significant
SMC vs RCC	2.08	Not Significant
SMC vs FSB	2.58	Not Significant

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SMC vs FLC	1.00	Not Significant
SMC vs ANN	2.75	Not Significant
SMC vs PSO	4.42	Significant
SMC vs GA	5.25	Significant
SMC vs GWO	5.00	Significant
RCC vs FSB	4.66	Significant
RCC vs FLC	3.08	Not Significant
RCC vs ANN	4.83	Significant
RCC vs PSO	6.50	Significant
RCC vs GA	7.33	Significant
RCC vs GWO	7.08	Significant
FSB vs FLC	1.58	Not Significant
FSB vs ANN	0.17	Not Significant
FSB vs PSO	1.84	Not Significant
FSB vs GA	2.67	Not Significant
FSB vs GWO	2.42	Not Significant
FLC vs ANN	1.75	Not Significant
FLC vs PSO	3.42	Not Significant
FLC vs GA	4.25	Significant
FLC vs GWO	4.00	Not Significant

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ANN vs PSO	1.67	Not Significant
ANN vs GA	2.50	Not Significant
ANN vs GWO	2.25	Not Significant
PSO vs GA	0.83	Not Significant
PSO vs GWO	0.58	Not Significant
GA vs GWO	0.25	Not Significant

Table 4.34 Harvested Energy Nemenyi Results

The Nemenyi post-hoc results for harvested energy indicated traditional algorithms like P&O and IncCond are significantly outperformed by advanced algorithms such as FSB, ANN, PSO, GA, and GWO. SMC showed significant comparisons against PSO, GA, and GWO. On the other hand, RCC performed significantly worse when compared to FSB, ANN, PSO, GA, and GWO. The remaining algorithms showed no significant difference among their own performance.

Cloud Transient Recovery Time

Comparison	Mean Rank Difference	Interpretation
PO vs IncCond	0.33	Not Significant
PO vs SMC	2.25	Not Significant
PO vs RCC	0.33	Not Significant
PO vs FSB	4.58	Significant

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PO vs FLC	3.17	Not Significant
PO vs ANN	1.33	Not Significant
PO vs PSO	5.17	Significant
PO vs GA	7.33	Significant
PO vs GWO	6.17	Significant
IncCond vs SMC	2.58	Not Significant
IncCond vs RCC	0.66	Not Significant
IncCond vs FSB	4.91	Significant
IncCond vs FLC	3.50	Not Significant
IncCond vs ANN	1.66	Not Significant
IncCond vs PSO	5.50	Significant
IncCond vs GA	7.66	Significant
IncCond vs GWO	6.50	Significant
SMC vs RCC	1.92	Not Significant
SMC vs FSB	2.33	Not Significant
SMC vs FLC	0.92	Not Significant
SMC vs ANN	0.92	Not Significant
SMC vs PSO	2.92	Not Significant
SMC vs GA	5.08	Significant
SMC vs GWO	3.92	Not Significant

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RCC vs FSB	4.25	Significant
RCC vs FLC	2.84	Not Significant
RCC vs ANN	1.00	Not Significant
RCC vs PSO	4.84	Significant
RCC vs GA	7.00	Significant
RCC vs GWO	5.84	Significant
FSB vs FLC	1.41	Not Significant
FSB vs ANN	3.25	Not Significant
FSB vs PSO	0.59	Not Significant
FSB vs GA	2.75	Not Significant
FSB vs GWO	1.59	Not Significant
FLC vs ANN	1.84	Not Significant
FLC vs PSO	2.00	Not Significant
FLC vs GA	4.16	Not Significant
FLC vs GWO	3.00	Not Significant
ANN vs PSO	3.84	Not Significant
ANN vs GA	6.00	Significant
ANN vs GWO	4.84	Significant
PSO vs GA	2.16	Not Significant
PSO vs GWO	0.99	Not Significant

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GA vs GWO	1.16	Not Significant
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Table 4.35 Cloud Transient Recovery Time Nemenyi Results

The results indicate that GA, GWO, PSO, and FSB generally outperform traditional algorithms such as P&O, IncCond, RCC), and FLC, with significant differences observed in several pairwise comparisons. P&O, IncCond, SMC, and RCC exhibit no significant differences among themselves, suggesting comparable mid-tier performance. ANN demonstrates significant improvement over GA and GWO but remains similar to most other algorithms. In summary, metaheuristic algorithms tend to dominate, while traditional methods perform moderately, and no single method is universally superior across all comparisons.

4.11 Discussion of Results

This section connects sections 4.1 to 4.8; the quantitative findings of the individual metric analyses, the entropy weight derivation, the Algorithm Rank Score (ARS), the cross-dataset comparison, and the statistical analysis into a unified interpretation of algorithm performance. The discussion contextualizes the observed results compared to established literature and addresses the central research objective of identifying MPPT algorithms most suitable for Solar PV systems operating under Philippine tropical irradiance conditions.

Overall ranking of the algorithms is obtained using the entropy-weighted ARS evaluation method by considering the overall performance of each algorithm for all the

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evaluation metrics considered, instead of depending solely on a single indicator. For this analysis, harvested energy (0.4350) and convergence time (0.3043) have been found to carry the highest weights in distinguishing between algorithms owing to their high variance and discriminating capability, while tracking efficiency (0.0019) has the least effect on them since they obtain very high values in each algorithm. Therefore, algorithms like P&O, IncCond, RCC, and SMC perform best because of their superior performance in the most important metrics, especially in achieving maximum energy harvesting and having a fast dynamic response. Other algorithms may have better performance when compared with these in terms of low-importance metrics, like FLC for tracking efficiency and stability or ANN for prediction accuracy, yet their superiority in the latter metrics does not help them achieve a high overall rank as they are weak performers in energy-related metrics. This demonstrates that classical and control-based algorithms are superior not because of isolated metric dominance, but on consistent performance across high-weight performance metrics. This provides a realistic basis for the practical selection of MPPT algorithms.

The results show that the MPPT algorithms' performance varies across each metric. Metaheuristic algorithms such as PSO, GA, and GWO have relatively faster convergence in specific scenarios. However, classical and control-based algorithms such as P&O, IncCond, RCC, and SMC are consistent in higher harvested energy and tracking efficiency.

Due to this variation, the inherent tradeoff in MPPT design is highlighted, where faster convergence does not always translate to a better overall system performance.

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Although faster convergence allows an algorithm to approach the MPP faster, sustaining operation near the MPP with minimal oscillation is more important when considering the long-term harvested energy. This is especially true under highly variable tropical conditions.

Due to the multi-dimensional nature of the performance evaluation, this study employed the use of a multi-criteria decision-making framework using the Entropy Weight Method (EWM) in combination with the Algorithm Rank Stability (ARS). The use of the EWM assigns weights based on the importance of each metric by means of their variability and contribution to distinguishing algorithm performance, thus reducing subjective bias in the evaluation. Consequently, metrics such as harvested energy, convergence speed, and CTRT received greater influence due to their being directly connected to overall energy production. In contrast, metrics such as tracking efficiency and tracking accuracy received lower weights due to them producing similar results with small variations relative to the rest of the performance metrics.

Guided by this framework, classical and control-based algorithms achieved higher overall rankings because of their balanced and consistent performance across all evaluated metrics. Although it is observed that some metaheuristic algorithms have faster convergence time, they produce lower tracking accuracy and harvested energy, which diminishes their overall effectiveness and evaluation.

The use of ARS reinforced this conclusion by showing that classical and control-based algorithms maintained more consistent rankings across multiple datasets and scenarios. Due to this consistency, the use of these algorithms becomes a critical factor in real-world PV implementations due to the need for a reliable and predictable

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algorithm for variable weather conditions.

In conclusion, classical and control-based algorithms are optimal, not based on a single metric, but rather on their higher overall weighted performance, which is consistent across datasets. These algorithms also contribute to a better long-term harvested energy, which justifies their selection as the most suitable MPPT approach for PV systems under tropical conditions.

The validity of the cross-dataset results is influenced mainly by the inherent differences in each dataset's irradiance characteristics. For the Mid-Latitude dataset, it characteristically has smooth and gradual irradiance variation that results in low variability and a stable operation for MPPT algorithms. In contrast, the Cloud Transient dataset has controlled variability through a predefined ramp rate for irradiance, which enables structured evaluation of dynamic tracking performance under repeatable conditions.

The tropical dataset fundamentally has a different profile compared to the previously mentioned datasets. In comparison to the Mid-Latitude dataset of 479.2 W/m^2 , it has a lower mean irradiance of 254.1 W/m^2 . This is in combination with the short-term variability, irregular ramp rates, and frequent cloud transients. Analysis of the dataset shows that rapid irradiance fluctuations of up to 337 W/m^2 within a 5-minute interval, indicating conditions which are typical of tropical environments. Unlike the benchmark datasets, where irradiance changes are gradual, the tropical dataset profile exhibits non-uniform and unpredictable transitions that continually shift the MPP.

The characteristics of the tropical dataset show a more realistic and challenging

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operating environment. These directly impact the performance of the MPPTs across all evaluated metrics. The lower mean irradiance reduces the maximum achievable energy yield with increased variability and irregular transients. This imposes a greater demand on the responsiveness of the algorithm, tracking accuracy, and convergence. As a result, algorithms that would otherwise perform comparably in benchmark datasets start to show noticeable differences under the tropical dataset. This validates the necessity of including real measured irradiance data in MPPT benchmarking, as it reveals performance characteristics that are not observable under idealized or controlled test conditions.

The simulation results from the tropical dataset exhibited generally faster convergence time but had an overall lower performance in harvested energy, tracking efficiency, and tracking accuracy. The frequent and rapid fluctuations of irradiance, high humidity, and dynamic atmospheric conditions reinforce this kind of behavior for tropical irradiance profiles.

The sudden change in irradiance assists in achieving faster convergence by making the curve for power-voltage steeper. This results in the MPPT techniques responding more aggressively, resulting in faster convergence. Nevertheless, it does not mean that faster convergence necessarily means better tracking of the true MPP.

On the other hand, the same variability that facilitates quick adaptation also leads to constant perturbation in the system, making it unable to run smoothly around the MPP. As a result, algorithms experience frequent re-tracking, increased oscillations, and reduced tracking accuracy, which ultimately leads to lower harvested energy over time.

From the above, we conclude that convergence time alone cannot be taken to

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define the effectiveness of the algorithm. The importance of tracking efficiency, accuracy, and harvested energy becomes more important due to the highly dynamic environment. This reflects the algorithms' capability in maintaining optimal operation despite continuous disturbances.

Consequently, the tropical dataset provides a more accurate and realistic test environment for the effectiveness of an MPPT algorithm since it stresses the importance of reliability and flexibility compared to a perfect steady-state behavior. An algorithm that works efficiently with such parameters is more likely to provide stable results when implemented in real-world PV applications in the Philippines and other regions with similar climate conditions. This prioritizes the long-term energy capture and stability over isolated improvements in convergence speed by means of reinforcement of the multi-metric evaluation frameworks.

In this study, an algorithm is classified as 'underperformed' relative to the top-performing groups when (1) its ARS score falls below the lower quartile boundary of the distribution of all ten algorithm ARS scores, indicating it is a statistical outlier on the lower end of the overall performance distribution and (2) the Nemenyi post-hoc test identifies a statistically significant performance difference ($\alpha = 0.05$, critical difference $CD = 4.10$) between that algorithm and the top-ranked cluster in at least two of the five evaluated metrics. Applying these criteria, PSO (ARS: 0.6502), GA (ARS: 0.5085), and GWO (ARS: 0.4139) are classified as the underperforming group relative to other algorithms as they are statistically significantly outperformed in both harvested energy and CTRT.

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The most consistent finding across all three dataset and the five performance metrics calculated is the strong and stable performance of classical algorithms, mainly P&O, IncCond, SMC, and RCC. These algorithms maintained tracking efficiencies between 96.14% and 97.30%, harvested energy values in the range of 2.02-3.89 Wh, and convergence times between 1.646s and 1.657s under Mid-Latitude dataset, reflecting the deterministic and step-based nature of their control structures [66], [67], [71], [72]. This result is consistent with previous benchmarking studies that have also used P&O and IncCond as reliable baseline algorithms under both stable and moderately variable irradiance conditions [68], [70], [73]. Their near-identical performance across metrics such as differences of less than 0.04 Wh in harvested energy and less than 0.005s in convergence time was statistically confirmed by Friedman and Nemenyi tests: pairwise comparisons among P&O, IncCond, SMC, and RCC produced mean rank differences well below the critical difference of 4.10 across all metrics, indicating that these algorithms form a statistically indistinguishable performance cluster under the tested conditions. This is consistent with prior comparisons showing that fixed-step classical algorithms tend to converge toward similar operating points given comparable step sizes [69], [63].

Among all evaluated algorithms, SMC achieved the highest ARS of 0.8419, placing it first in the overall ranking. This result is primarily driven by its strong performance in the two highest-weighted metrics: harvested energy (entropy weight 0.4350) and convergence time (entropy weight 0.3043). SMC's sliding surface-based control law, which drives the system toward the MPP by minimizing the derivative of

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output power with respect to duty cycle, enables continuous and responsive tracking without requiring periodic perturbations [44], [74]. Under the harvested energy Friedman test, which produced the highest chi-square value of the study ($\chi^2 = 100.564$, $p = 0.000$), reflecting the highest algorithm differentiation; SMC's mean rank of 7.00 placed it clearly above the metaheuristic group, while Nemenyi comparisons confirmed that SMC significantly outperformed PSO ($\Delta = 4.42$), GA ($\Delta = 5.25$), and GWO ($\Delta = 5.00$). These findings align with literature that demonstrates SMC's robust MPPT performance due to its inherent insensitivity to system parameter variations and external disturbances [75], and the entropy-weighted ranking method amplifies SMC's consistent advantage in energy capture, elevating it to the top position overall.

The FLC achieved the highest tracking efficiency across all three datasets: 97.57%, 97.60%, and 96.47% for Mid-Latitude, Cloud Transient, and Tropical datasets respectively. It also achieved the lowest ripple ratio under the Tropical dataset (0.8178), indicating exceptional steady-state precision under real measured Philippine irradiance. The Friedman test for tracking efficiency ($\chi^2 = 78.509$, $p = 0.000$) confirmed that FLC's mean rank of 9.50 was the highest among all algorithms, and Nemenyi post-hoc analysis confirmed statistically significant advantages over GA ($\Delta = 7.75$), GWO ($\Delta = 7.25$), PSO ($\Delta = 6.08$), and FSB ($\Delta = 6.00$). These results are also consistent with literature showing that fuzzy rule-based controllers minimize oscillations around the MPP through adaptation of step size to the gradient of the P-V curve [61], [87], [88]. However, FLC ranked fifth in the ARS (0.7549) because tracking efficiency carries the lowest entropy weight in this study (0.0019), meaning the metric contributes minimally to the overall algorithm performance. FLC's harvested energy was also marginally low (3.5718,

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3.1488, and 1.8706 across Mid-Latitude, Cloud Transient, and Tropical datasets respectively) in comparison to the classical algorithms due to transient energy loss during adaptive step adjustments. This outcome highlights an important finding that tracking efficiency alone is not sufficient enough to base ranking of MPPT algorithms, as entropy weight reveals that energy yield under dynamic conditions bears more weight [59].

The ANN achieved a competitive ARS of 0.7238, ranking sixth overall and demonstrated tracking efficiency among the highest in both the Mid-Latitude (97.50%) and Cloud Transient (97.56%) datasets. This is consistent with literature that demonstrates the capability of ANN-based MPPT controllers to accurately approximate the MPP once the network has been sufficiently exposed to relevant input gradients [90], [91]. In the Friedman analysis for tracking efficiency, ANN's mean rank of 7.25 was significantly higher than GA ($\Delta = 5.50$) and GWO ($\Delta = 5.00$), but not significantly different from FLC, SMC, or P&O, which confirms comparable efficiency among the intelligent and classical algorithms. However, ANN shows a lower energy yield under Tropical dataset in comparison to classical algorithms (1.7314 Wh compared to 2.02-2.03 Wh), attributable to the online learning implementation used in this study. Because the ANN updates its weights through backpropagation rather than using pre-trained weights, it requires additional gradient observations before its updates fully converge. Under the tropical dataset, which begins at a non-zero irradiance ($\sim 433 \text{ W/m}^2$), this results in a higher energy loss relative to classical methods. This distinguishes the studied implementation from pre-trained ANN-MPPT architectures commonly reported in literature [90] and represents an implementation-specific characteristic that limits direct comparison with published results.

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The FSB algorithm ranked seventh in the ARS (0.6817), reflecting its deterministic search sequence which although was effective in locating the MPP during initial state of the simulation, does not re-mobilize efficiently following large irradiance changes. This is evident in its CTRT of 36.84s under the Mid-Latitude dataset and 45.96s under the Cloud Transient dataset, substantially longer than those of P&O IncCond at approximately 21s-30s. The Friedman test for CTRT ($\chi^2 = 87.182$, $p = 0.000$) placed FSB at a mean rank of 7.08, and Nemenyi comparisons confirmed that FSB was significantly slower than P&O ($\Delta = 4.58$), IncCond ($\Delta = 4.91$), and RCC ($\Delta = 4.25$). Literature on FSB-based MPPT acknowledges that the Fibonacci-step reduction sequence is well-suited to one-time MPP search but lacks the continuous perturbation mechanism needed to efficiently re-track after sudden irradiance drops [60], [84], [85]. The entropy weight of 0.2354 assigned to CTRT further penalizes FSB in the ARS, as this metric shows meaningful variability across algorithms and thus carries a greater discriminating power than tracking efficiency.

The performance of the three metaheuristic algorithms PSO, GA, and GWO was substantially lower than all of the other algorithms tested across most metrics, particularly in harvested energy and CTRT, yielding ARS scores of 0.6502, 0.5085, and 0.4139 and ranking eighth, ninth and tenth respectively. These results stand in contrast to literature reporting high MPPT performance for population-based methods under partial shading and multi-modal power curve conditions, where their global search capability provides a meaningful advantage over classical perturbation-based methods [92], [93], [96], [97], [100], [101]. The Friedman test for harvested energy ($\chi^2 = 100.564$, $p = 0.000$) placed GA, GWO, and PSO at mean ranks of 1.75, 2.00, and 2.58 which are the three

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lowest of all algorithms and Nemenyi post-hoc analysis confirmed statistically significant differences between this group and P&O, IncCond, RCC, and SMC, with mean rank differences ranging from 4.42 to 7.33 across these comparisons. For CTRT, GA (mean rank 9.83), GWO (mean rank 8.67), and PSO (mean rank 7.67) occupied the three highest ranks, indicating the slowest recovery, and were significantly outperformed by IncCond, P&O, and RCC in post-hoc comparisons. The divergence between these findings and literature expectations is attributable to the algorithm retaining its best solution for a fixed holding period of 2000 time steps between successive population updates, resulting in approximately 30 full optimization iterations over the 60-second simulation window. During each holding period, the converter operates at a fixed duty cycle, preventing the algorithm from tracking real-time irradiance variations. This design constraint limits the update rate of population-based metaheuristic algorithms to a small fraction of that of classical algorithms, which adjust their duty cycle at every simulation step. The low harvested energy values for PSO (0.6531 Wh Mid-Latitude, 2.4061 Wh Cloud Transient, 0.7754 Wh Tropical), GA (1.3095 Wh Mid-Latitude, 1.3842 Wh Cloud Transient, 0.6008 Wh Tropical), and GWO (1.2280 Wh Mid-Latitude, 1.2543 Wh Cloud Transient, 0.6545 Wh Tropical) therefore reflect the energy cost of extended holding periods rather than an intrinsic limitation of these algorithms under studied irradiance profiles [22], [98], [99], [102], [103]. These findings should be interpreted within the scope of the evaluated implementation parameters, and future work with reduced holding periods and larger population sizes may yield substantially different comparative outcomes.

The cross-dataset comparison revealed that algorithm performance rankings are largely preserved across the mid-latitude and cloud transient benchmark datasets, with the

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Tropical dataset introducing a consistent downward shift in absolute metric values, particularly in harvested energy, attributable to its lower mean irradiance in comparison the Mid-Latitude dataset (254.1 W/m² compared to 479.2 W/m²). This confirms that the relative ordering of algorithm performance is robust across irradiance environments, which is an important consideration for MPPT selection in the Philippine context where steady-state, high-irradiance, and cloud-transient dynamics are routinely encountered [7], [41]. Under the Tropical dataset specifically, FLC demonstrated the strongest stability performance (0.8178) while classical algorithms P&O, IncCond, RCC, and SMC maintained competitive performance under real measured Philippine irradiance profiles [42]. The Friedman tracking accuracy test ($\chi^2 = 56.055$, $p = 0.000$) placed P&O (mean rank 2.67), IncCond (mean rank 3.33), FLC (mean rank 3.17), and RCC (mean rank 3.67) among the most stable algorithms, while GA and GWO (9.00 and 7.42 respectively) ranked as the least stable, which is also a pattern seen across all three datasets. The absence of any performance advantage for metaheuristic algorithms under the tropical dataset further reinforces the interpretation that implementation constraints, rather than irradiance environment itself, drive the observed metaheuristic underperformance [116].

Taken together, the results of this study offer several contributions to the existing body of literature of MPPT benchmarking in tropical environments. First, the study confirms that conventional or classical algorithms such as P&O, IncCond, SMC, and RCC remain highly competitive under both standard benchmark and real measured tropical irradiance conditions, and that their performance equivalence is statistically verifiable through non-parametric analysis, consistent with calls in the literature for rigorous baseline comparisons before recommending more complex control strategies

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[114], [115]. Second, the application of the Entropy Weight Method as an objective weighting framework reveals that the harvested energy is the most influential differentiating metric among well-tuned algorithms, with the Friedman chi-square for harvested energy ($\chi^2 = 100.564$) being the largest of any metric, corroborating the entropy weight assignment. Third, the study documents specific implementation constraints that limited performance of metaheuristic algorithms under real-time simulation, providing a reproducible reference point for future work seeking to re-evaluate these algorithms under revised parameters. Finally, the inclusion of a real measured dataset from the Arayat-Mexico Solar Farm, Pampanga, as one of three evaluation environments provides a direct link between simulation-based benchmarking and the irradiance characteristics of an operational Philippine solar facility, addressing a gap identified in prior Philippine-context MPPT research [110], [41], [42].

Performance tradeoffs were identified among the MPPT algorithms across the different performance metrics. Algorithms with fast convergence, such as RCC, IncCond, and P&O, achieved low CTRT values. However, it is found that their fixed step mechanism adds oscillations that affect the tracking accuracy. In contrast, FLC is superior in tracking accuracy due to its rule-based control, but this comes at the cost of its harvested energy being reduced. This is because of the continuous adjustment that causes the FLC to deviate from the MPP. Similarly, ANN produced a high tracking efficiency under normal conditions, but it has reduced energy performance under tropical variability due to its slow adaptation during its real-time learning. For the metaheuristic algorithms, PSO, GA, and GWO, which are theoretically capable of global optimization, have

reduced convergence speed and harvested energy under rapidly changing irradiance conditions. These findings show that no single algorithm optimizes all performance metrics simultaneously. Selecting MPPT must consider the specific priorities of the PV system, mainly its responsiveness, stability, and maximizing energy yield.

4.11.1 Implications and Limitations of Simulation-Based Evaluation

Implications arise from the use of a purely simulation-based framework, despite having clear performance differences among the evaluated MPPT algorithms. Based on findings, a comparative evaluation is most reliable in finding the relative rankings and observed trends among the algorithms. The validity of classical and control-based algorithms, being strong performers, remains valid. However, convergence time, tracking efficiency, and harvested energy differ under real-world operating conditions when considering the metrics' absolute values.

A key implication is the inherently ideal behavior of the system in the simulated environment. In practical PV systems, system performance degrades when considering real-world factors such as switching losses, sensor noise, control delays, and parasitic resistances, which are excluded in the research. This results in the algorithms exhibiting fast convergence speed and high stability in the simulation. Algorithms such as SMC may experience slightly reduced performance when implemented in hardware due to these factors.

Furthermore, the MPPT algorithms' transient response may be affected by the resolution and accuracy of the weather datasets. Although the study utilized a localized tropical dataset, real-world irradiance variations can occur at finer time scales compared

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to the simulation. This influences metrics such as CTRT and tracking accuracy, especially for algorithms that are sensitive to fluctuations.

Implications also arise from the observed performances of the intelligent and metaheuristic algorithms. PSO, GA, and GWO ranked lower under selected parameter settings, which are highly dependent on tuning, computational constraints, and real-time implementation considerations. In practical systems, the behavior of these algorithms varies greatly when taking into account the stochastic environmental inputs and hardware limitations. The lower rankings of these algorithms in this study should not be indicative of a universally inferior performance.

Overall, the findings in this paper provide strong guidance for MPPT algorithm selection in tropical conditions. However, experimental validation is necessary to confirm their performance in real-world photovoltaic systems.

4.12 Bullet Summary of Key Findings

- SMC ranked overall first in ARS ($ARS = 0.8419$), which is mainly because of its consistently high performance in the two highest-weighted metrics: harvested energy (entropy weight 0.4350) and convergence time (0.3043) across all three datasets.
- P&O, IncCond, RCC, and SMC formed a statistically indistinguishable performance cluster in harvested energy and convergence time, with pairwise Nemenyi mean rank differences below the critical difference of 4.10, confirming functional equivalence among classical algorithms under the study's test conditions.

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- FLC achieved the highest tracking efficiency across all three datasets (97.57%, 97.60%, 96.47%) and the best stability under tropical conditions (ripple ratio 0.8178), yet ranked fifth in the ARS because tracking efficiency carries the lowest entropy weight (0.0019), indicating it is the least differentiating metric among well-tuned algorithms.
- Harvested energy was the most discriminating metric (entropy weight 0.4350, Friedman $\chi^2 = 100.564$), confirming that energy yield under dynamic conditions is a more reliable basis for algorithm ranking than tracking efficiency alone.
- The ANN online learning implementation showed competitive tracking efficiency (97.50% Mid-Latitude, 97.56% Cloud Transient) but lower energy yield under the Tropical dataset (1.7314 Wh) due to its weight update interval, distinguishing it from pre-trained ANN architectures in literature.
- PSO, GA, and GWO ranked eighth, ninth, and tenth respectively, with harvested energy values approximately 70% lower than IncCond under the Tropical dataset. This underperformance is attributable to a fixed holding period of 2000 time steps (~30 iterations per 60-second simulation) rather than an intrinsic algorithmic limitation, and should be interpreted within the scope of the evaluated implementation parameters.
- GA exhibited the worst cloud transient recovery time across all three datasets (~58.39s, 58.20s, 58.45s), with values nearly independent of the irradiance environment, indicating that its recovery time is dominated by fixed population generation overhead rather than irradiance conditions.

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- Algorithm performance rankings were largely preserved across all three datasets, with the Tropical dataset introducing a consistent downward shift in absolute energy values attributable to its lower mean irradiance (254.1 W/m² vs. 479.2 W/m² mid-latitude) rather than a change in relative algorithm ordering.
- The Friedman test confirmed statistically significant performance differences across all five metrics ($p = 0.000$), and Nemenyi post-hoc analysis identified P&O, IncCond, RCC, and SMC as a statistically superior group over GA, GWO, and PSO in harvested energy and cloud transient recovery time.
- Classical and control-theoretic algorithms are best suited for Philippine tropical PV deployment under the study's simulation conditions, maintaining tracking efficiencies above 96%, harvested energy above 2.0 Wh, and CTRT below 28s under real measured irradiance from the Arayat-Mexico Solar Farm, Pampanga.

CHAPTER 5.0 Conclusions and Recommendations

5.1 Conclusions

This chapter presents the conclusions drawn from the simulations of ten Maximum Power Point Tracking (MPPT) algorithms across three different irradiance datasets and four weather scenarios, ending with the results covering five performance metrics (convergence time, tracking efficiency, tracking accuracy, harvested energy, and cloud transient recovery time), Algorithm Ranking Stability (ARS), the Entropy Weight Method analysis, and the Friedman and Nemenyi statistical tests. Conclusions are organized according to the four research objectives of the study.

5.1.1 Effect of Irradiance and Temperature Variations on Algorithm Performance

Under Objective 1, the study demonstrated that rapid irradiance and temperature variations, which are a characteristic of the Philippine tropical climate, exerts a measurable and differentiating effect on the convergence speed, tracking efficiency, tracking accuracy, harvested energy, and cloud transient recovery time of MPPT algorithms. Under all three datasets (Mid-Latitude, Cloud Transient, and Tropical datasets), classical and control-theoretic algorithms (P&O, IncCond, SMC, and RCC) consistently achieved tracking efficiencies above 96%, convergence times within 1.646s-1.735s, and low ripple ratios, demonstrating that their step-based control logic are inherently well-suited to continuous irradiance tracking. The FLC shows the lowest ripple ratio across all datasets, with a 0.8178 ratio on the Tropical dataset, demonstrating superior steady-state stability under real Philippine irradiance conditions. In contrast, the metaheuristic algorithms PSO, GA, and GWO showed substantially reduced energy

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harvest and slower cloud transient recovery time under the study's implementation parameters, attributable to the fixed holding period between successive population updates, which limited the algorithm's real-time tracking responsiveness. The ANN, which is implemented via online backpropagation in this study, showed slightly reduced tropical performance relative to classical methods, consistent with the additional gradient observations required for its weight updates to converge under non-zero initial irradiance conditions.

5.1.2 Multi-Metric Performance Comparison and ARS Ranking

For the 2nd objective of the study, the evaluation of all five metrics which are convergence times, tracking efficiency, tracking accuracy, harvested energy, and cloud transient recovery time (CTRT) revealed clear and consistent performance tiers among ten algorithms. The Algorithm Ranking Stability (ARS), computed using entropy-derived objective weights, produced the following overall ranking: SMC (0.8419), P&O (0.8307), IncCond (0.8250), RCC (0.8188), FLC (0.7549), ANN (0.7238), FSB (0.6817), PSO (0.6502), GA (0.5085), and GWO (0.4139). The entropy weighting revealed that harvested energy (entropy weight 0.4350) and convergence time (entropy weight 0.3043) were the most discriminating metrics among evaluated algorithms, while tracking efficiency (entropy weight 0.0019) provided minimal differentiation. This reflects that most well-tuned algorithms achieve near-equivalent efficiency. SMC ranked first primarily due to its consistent advantage in the two highest weighted metrics calculated through entropy weighting. FLC, despite having the highest tracking efficiency of all algorithms across all datasets, only ranked fifth due to the lower energy yield during dynamic conditions. The FSB algorithms ranked seventh, which is penalized heavily by

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its high CTRT (up to 45.96s) compared to the 21-30 second range of classical algorithms. These results establish that harvested energy yield, not tracking efficiency, is the most practically relevant criterion in differentiating MPPT algorithm selections in the Philippine context.

5.1.3 Statistical Validation of Performance Differences

Objective 3 covered the Friedman non-parametric test, applied independently to each of the five performance metrics, confirmed statistically significant performance differences across the ten algorithms in all metrics ($p < 0.001$ for all five tests). The largest chi-square value was obtained for harvested energy ($\chi^2 = 100.564$), consistent with its highest entropy weight, which is followed by CTRT ($\chi^2 = 87.182$), tracking efficiency ($\chi^2 = 78.509$), convergence time ($\chi^2 = 59.574$), and tracking accuracy ($\chi^2 = 56.055$). The Nemenyi post-hoc analysis, conducted at a critical difference of $CD=4.10$ ($\alpha = 0.05$, $N = 12$, $k = 10$), identified the specific algorithm pairs with statistically significant performance separation. Classical algorithms P&O, IncCond, SMC, and RCC under the Nemenyi test formed a statistically indistinguishable cluster in harvested energy and convergence time which confirms their functional equivalence. FLC demonstrated statistically significant superiority in tracking efficiency over GA, GWO, PSO, and FSB. GA, GWO, and PSO were significantly outperformed by the top four classical algorithms in harvested energy, with mean rank differences ranging from 4.42 to 7.33, with both the entropy-weighted ARS and the Friedman and Nemenyi test analysis independently producing the same performance hierarchy, creating a mutual validation of the study's overall conclusion.

5.1.4 Cross-Dataset Comparisons of Tropical vs Mid-Latitude and Cloud Transient Datasets

Objective 4 refers to the cross-dataset analysis that confirmed that algorithm performance rankings are largely preserved across all three datasets, validating consistency and generalizability of the observed performance hierarchy. The Tropical dataset, which is based on real world irradiance data from Arayat-Mexico Solar Farm, introduced a consistent downward shift in metric values, particularly in harvested energy. This is attributable to its lower mean irradiance of 254.1 W/m^2 compared to 479.2 W/m^2 of the Mid-Latitude dataset. Despite this gap, relative rankings remained stable: the same classical and control-theoretic algorithms that performed best under the benchmark datasets continued to lead under the tropical dataset, with P&O, IncCond, RCC, and SMC sustaining tracking efficiencies above 96% and harvested energy above 2.0 Wh. The Tropical dataset distinctively showed near zero convergence times for all algorithms (0.0010s-0.0018s) mainly due to the non-zero initial irradiance condition at simulation start, putting all algorithms near the MPP at the beginning of simulation. This behaviour is specific to real measured irradiance profiles and does not occur in the benchmark datasets, highlighting an important difference in structure between synthetic and field-measured tropical data. FLC demonstrated the strongest stability performance under the Tropical dataset, while metaheuristic algorithms showed no improvement relative to benchmark conditions, confirming that their underperformance is leaning more towards implementation-driven rather than dataset-driven. These findings collectively demonstrate that standard benchmark datasets, while useful for general algorithm evaluation, do not fully replicate the irradiance dynamics of Philippine tropical

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conditions, and that real measured local data provides essential context for deployment-relevant algorithm selection.

5.1.5 Practicality of Results

Operationalizing the study's findings directly influences financial decision-making in PV system design by guiding engineers and adopters toward algorithms that maximize energy yield while minimizing implementation cost. Since classical and control-based methods (P&O, IncCond, RCC, and especially SMC) demonstrated the most consistent performance, they offer a strong balance between efficiency and practicality. These algorithms can be implemented on low- to mid-cost controllers without requiring high computational resources, reducing upfront hardware costs while still achieving high energy harvesting performance.

From an economic perspective, selecting high-performing yet efficient algorithms like SMC leads to faster convergence and better tracking accuracy, which translates to increased energy production over time. This improves return on investment through higher savings on electricity bills or greater revenue from net metering. Additionally, their stability under tropical conditions reduces performance losses and maintenance needs, lowering long-term operational costs. Thus, the research supports cost-effective PV system design decisions by identifying algorithms that optimize both technical performance and financial returns.

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5.1.6 Overall Conclusion

This study concludes that, under the evaluated simulation conditions, classical and control-theoretic MPPT algorithms, specifically P&O, IncCond, RCC, and SMC, provides the most competitive and consistent overall performance for solar PV systems operating under Philippine tropical irradiance conditions. SMC achieved the highest ARS ranking (0.8419), driven by its sustained advantage of the two highest entropy-weighted performance metrics: harvested energy and convergence time. These results are statistically verified and hold across synthetic benchmark and real measured tropical datasets. The population-based metaheuristic algorithms PSO, GA, and GWO ranked lowest under the tested implementation parameters. However, this outcome is specifically attributable to the $N_{\text{hold}}=2000$ step holding period used in the evaluated implementations, which constrained each algorithm to approximately 30 optimization iterations per 60-second simulation, and should not be interpreted as a general disqualification of these algorithms for MPPT applications. Future evaluations with revised implementation parameters may yield substantially different comparative outcomes of these algorithms.

5.2 Recommendations

5.2.1 Recommendations for Future Research

Based on the findings and limitations of this study, the following are recommended for future research implementations:

1. Re-evaluation of metaheuristic algorithms with reduced holding periods (N_{hold}).

Future work should re-simulate PSO, GA, and GWO with reduced N_{hold} values

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and increased population sizes to allow a greater number of optimization iterations within the simulation window. This improves the algorithm's responsiveness to real-time irradiance changes, finds true MPP faster, and potentially improves harvested energy and CTRT.

2. Hardware implementation and experimentation. As this study is limited to MATLAB/Simulink simulation, future work should validate the ranked algorithms through hardware testing or physical prototype implementation, particularly for the study's top-ranked algorithms (SMC, P&O, IncCond, RCC), to confirm that these simulation-based rankings also translate to physical system performance under real Philippine weather conditions/
3. Pre-trained ANN implementation. The ANN in this study was implemented with backpropagation learning, which introduces adaptation delays in the early simulation window. Future work should evaluate a pre-trained ANN-MPPT implementation using Philippine irradiance data as training input, which may substantially improve its tropical dataset performance and provide a more informative comparison against classical methods.
4. Expanded tropical dataset coverage. The tropical dataset used in this study represents a single site (Arayat-Mexico Solar Farm, Pampanga) under a specific period of time. Future studies should also incorporate irradiance profiles from multiple Philippine solar plant sites spanning different climate types, including typhoon-prone coastal zones and high-altitude inland regions, to assess if the observed algorithms generalize across broader Philippine climatic conditions.

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5. Hybrid algorithm architectures. The results of this study suggests that no single algorithm excels uniformly over all five performance metrics. Future research may explore hybrid algorithms that combine the fast convergence and energy efficiency of classical algorithms with the stability characteristics of intelligent control methods, such as using a SMC-FLC hybrid algorithm or a GWO initialized by IncCond algorithms, to investigate whether combined approaches can achieve superior multi-metric performance.
6. Consideration of non-ideal modeling of the system and high time resolution data. Through this investigation, this study used the MATLAB/Simulink simulation environment for controlled studies. Further research must improve simulation accuracy by considering non-ideal factors in system behavior, such as power electronics switching and conduction loss, resistor parasitics, measurement noise, and control latency. Moreover, increased temporal resolution irradiance and temperature data would be advantageous in representing rapid cloud motion that occurs in tropical settings. Such enhancements will result in improved accuracy of system models, thus allowing better validation of MPPT algorithms prior to practical application.

5.2.2 Recommendations for Practitioners and System Designers

Based on the simulation results and statistical findings of this study, the following practical recommendations are offered for system designers, engineers, and energy planners involved in solar PV system development in the Philippines:

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1. For general purpose MPPT implementation in Philippine solar farms, SMC is recommended as the top-ranked algorithm based on ARS analysis, offering the strongest balance of energy harvested and convergence speed across all datasets, including the tropical dataset based on Arayat-Mexico. Its robustness to parameter variations and continuous tracking behaviour makes it well suited for both stable and high variance high-irradiance environments.
2. For applications where low computational costs and simplicities are the primary constraints, P&O and IncCond remains highly competitive alternatives with ARS scores of 0.8307 and 0.8250 respectively, and statistically indistinguishable performance from SMC in harvested energy and convergence time based on the Nemenyi post-hoc analysis. These algorithms are recommended for low-cost embedded implementations where complex controls are impractical.
3. For applications where steady-state power quality and minimum ripple output are critical (such as sensitive load applications), the FLC is recommended, as it achieved the lowest ripple ratio across all datasets (0.8178 under tropical dataset) and the highest tracking efficiency of any evaluated algorithm (up to 97.60% under the cloud transient dataset)
4. Population-based metaheuristic algorithms (PSO, GA, GWO) are not recommended for continuous real-time single-peak MPPT applications under the implementation parameters evaluated in this study. However, their application should not be categorically excluded. Designers considering the implementation of these algorithms should validate its performance with adjusted holding periods before deployment.

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5. For large-scale Philippine solar farm operators selecting MPPT controllers, tracking efficiency alone is not a sufficient selection criterion. The Entropy Weight Method results demonstrate that harvested energy yield (entropy weight 0.4350) and CTRT (entropy weight 0.2354) are the most informative metrics for algorithm differentiation, and procurement decisions should prioritize these criteria over tracking efficiency.

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APPENDICES

APPENDIX A: Research Timetable

	APRIL				MAY			
	1	2	3	4	6	7	8	9

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DEPARTMENT**

1	D	D						
2		D	D					
3		D	D					
4		D	D	D				
5		D	D	D				
6					D	D	D	D
7					D	D	D	D
8								
9								
10								
11								
12							D	D
13a								
13b								
13c								
13d								
13e								
13f								
13g								
13h								
13i								
13j								
14								
15								
16								
17								
18								
19								
20								
21								

**UNIVERSITY OF SANTO TOMAS ELECTRICAL ENGINEERING
DEPARTMENT**

22								
23								

	AUGUST				SEPTEMBER			
	13	14	18	19	20	21	22	23
1								
2								
3								
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5								
6								
7								
8	D	D						
9			D	D	D	D		
10								
11								D
12								
13a								
13b								
13c								
13d								
13e								
13f								
13g								
13h								
13i								
13j								
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15								
16								

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DEPARTMENT**

17								
18							D	
19								
20								
21								
22								
23	D	D	D	D	D	D	D	D

	OCTOBER				NOVEMBER			
	24	25	26	27	28	29	30	32
1								
2								
3								
4								
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6								
7								
8					D	D	D	D
9								
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11								
12								
13a	D							
13b	D							
13c								
13d								
13e								
13f		D						
13g								

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13h								
13i								
13j								
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15								
16								
17								
18								
19					D	D	D	D
20								
21								
22				D				
23	D	D	D	D	D	D	D	D

	DECEMBER				JANUARY			
	33	34	35	36	37	38	39	40
1								
2								
3								
4								
5								
6								
7								
8	D	D	D	D				
9								
10								
11								
12								
13a					D			
13b					D			

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13c						D		
13d						D		
13e							D	
13f							D	
13g								D
13h								D
13i								
13j								
14					D	D	D	D
15								
16								
17								
18								
19	D	D	D	D				
20								
21								
22								
23	D	D	D	D	D	D	D	D

	FEBRUARY				MARCH			
	41	42	43	44	45	46	47	48
1								
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4								
5								
6								
7								
8								
9								

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10								
11								
12								
13a								
13b								
13c								
13d								
13e								
13f		D	D	D				
13g		D	D	D				
13h		D	D	D				
13i	D	D	D	D				
13j	D	D	D	D				
14	D	D	D	D				
15	D	D	D	D				
16								
17								
18								
19				D	D	D		
20								
21						D		
22								
23	D	D	D	D	D	D	D	D

Table A.1 Gantt Chart and Time Table

Legend:

D means the activity is 'DONE'	O means the activity is 'ON GOING'	F means this activity "WILL BE DONE IN THE FUTURE"
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List of Activities

- | | |
|------------|---|
| 1 | Select Project |
| 2 | Understanding of Problem |
| 3 | Checking of Related Research Papers in the Philippine Setting |
| 4 | List of Requirements/Objectives |
| 5 | Planning |
| 6 | Literature Review of Related Studies |
| 7 | Writing of Chapters 1, 2, 3 |
| 8 | Revise Chapters 1,2,3 to improve the paper |
| 9 | Get approval from a Research Adviser |
| 10 | Specifications on the PV Model |
| 11 | Collection of Weather Data |
| 12 | Selection of simulation tools and platform |
| 13a | Implementation (Perturb and Observe Algorithm) |
| 13b | Implementation (Incremental Conductance Algorithm) |
| 13c | Implementation (Sliding Mode Control Algorithm) |
| 13d | Implementation (Ripple Correlation Algorithm) |
| 13e | Implementation (Fibonacci Series Based Algorithm) |
| 13f | Implementation (Fuzzy Logic Control Algorithm) |
| 13g | Implementation (Artificial Neural Network Algorithm) |
| 13h | Implementation (Particle Swarm Optimization Algorithm) |
| 13i | Implementation (Genetic Algorithm) |
| 13j | Implementation (Grey Wolf Optimization Algorithm) |
| 14 | Testing and Validation under different weather profiles |
| 15 | Comparison of Results and Analysis of Results |
| 16 | Submit progress report to supervisor |
| 17 | Make corrections/adjustments based on feedback |
| 18 | Show logbook to supervisor |
| 19 | Write Thesis |
| 20 | Submit thesis draft to supervisor for feedback |
| 21 | Final Submission of thesis |

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22 Give Oral Presentation/Defense

23 Use Logbook

Figure A.1 List of Activities in Gannt Chart

APPENDIX B: Research Budget

1. Matlab Coder	Php 15,000.00
2. Statistician	Php 4,500.00
3. Research Ethics Committee Review	Php 3,500.00
Total	Php 23,000.0

Table B.1 List of Expenditures

APPENDIX C: Grammarly Report

Report: Chapter_1_Introduction

Chapter_1_Introduction

by ADRIAN ATIENZA

General metrics

26,132

characters

3,730

words

171

sentences

14 min 55 sec

reading time

28 min 41 sec

speaking time

Score

80

This text scores better than 80% of all texts checked by Grammarly

Writing Issues

132

Issues left

42

Critical

90

Advanced

Plagiarism

✓

This text seems 100% original. Grammarly found no matching text on the Internet or in ProQuest's databases.

Report was generated on Friday, May 8, 2026, 10:56 AM

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Grammarly Report for CHAPTER 1

Report: Chapter_2_Review_of_Related_Literature

Chapter_2_Review_of_Related_Literature

by ADRIAN ATIENZA

General metrics

75,679

characters

11,305

words

595

sentences

45 min 13 sec

reading time

1 hr 26 min

speaking time

Score

80

This text scores better than 80% of all texts checked by Grammarly

Writing Issues

390

Issues left

116

Critical

274

Advanced

Plagiarism

✓

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Grammarly Report for CHAPTER 2

UNIVERSITY OF SANTO TOMAS ELECTRICAL ENGINEERING DEPARTMENT

Report: Chapter_3_Methodology Chapter_3_Methodology by ADRIAN ATIENZA General metrics 76,428 characters 10,921 words 656 sentences 43 min 41 sec reading time 1 hr 24 min speaking time Score 80 This text scores better than 80% of all texts checked by Grammarly Writing Issues 349 Issues left 7 Critical 342 Advanced Plagiarism ✓ This text seems 100% original. Grammarly found no matching text on the Internet or in ProQuest's databases. Report was generated on Friday, May 8, 2026, 10:52 AM Page 1 of 97 Grammarly Report for CHAPTER 3	Report: Chapter_4_Part1_Performance_Metrics_and_Analysis Chapter_4_Part1_Performance_Metrics_and_Analysis by ADRIAN ATIENZA General metrics 73,204 characters 9,952 words 1162 sentences 39 min 48 sec reading time 1 hr 16 min speaking time Score 87 This text scores better than 87% of all texts checked by Grammarly Writing Issues 234 Issues left 31 Critical 203 Advanced Plagiarism ✓ This text seems 100% original. Grammarly found no matching text on the Internet or in ProQuest's databases. Report was generated on Friday, May 8, 2026, 10:42 AM Page 1 of 117 Grammarly Report for CHAPTER 4 Part 1
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Report: Chapter_4_Part2_Cross_Dataset_Statistical_Discussion

Chapter_4_Part2_Cross_Dataset_Statistical_Discussion

by ADRIAN ATIENZA

General metrics

87,295

characters

12,671

words

1587

sentences

50 min 41 sec

reading time

1 hr 37 min

speaking time

Score

80

This text scores better than 80% of all texts checked by Grammarly

Writing Issues

419

Issues left

61

Critical

358

Advanced

Plagiarism

✓

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Report: Chapter_5_References_Appendices_CV

Chapter_5_References_Appendices_CV

by ADRIAN ATIENZA

General metrics

64,071

characters

8,408

words

1096

sentences

33 min 37 sec

reading time

1 hr 4 min

speaking time

Score

94

This text scores better than 94% of all texts checked by Grammarly

Writing Issues

113

Issues left

63

Critical

50

Advanced

Plagiarism

✓

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Report was generated on Friday, May 8, 2026, 10:39 AM

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Table C.1 Grammarly Report

APPENDIX D: MATLAB/Simulink Public Repository

The complete MATLAB/Simulink implementation of all ten MPPT algorithms used in this study is publicly available at the following repository:

<https://github.com/Adrian-Atienza/10-mppt-algorithms-tested-under-3-different-datasets>

The repository includes: (1) the MATLAB/Simulink model file containing the unified PV system and MPPT controller architecture, (2) individual MATLAB function scripts for each of the ten MPPT algorithms, (3) the three irradiance datasets in CSV format as simulation inputs, (4) the scripts in building scenarios, calculating results, and entropy weight computation along with ARS calculations, and (5) a README file with

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step-by-step instruction for reproducing the simulation results of the study from a copy without results.

Sample Code (Code That Runs All Simulations)

```
clear; clc;
warning('off','Simulink:Engine:AccelBuildCleanup')
model = 'pv_environment';
load_system(model);
datasets = {'mid_latitude','cloud_transient','tropical'};
scenarios = {'clear','ramp','step','mixed'};
algorithms =
{'PO','IncCond','SMC','RCC','Fibonacci','FLC','ANN','PSO','GA','GWO'};
results_folder = 'results';
if ~exist(results_folder,'dir')
    mkdir(results_folder);
end
clc
sim_time = 60;
for d = 1:length(datasets)
    for s = 1:length(scenarios)

        dataset_name = [datasets{d} '_' scenarios{s}];
        load([dataset_name '.mat']);
        assignin('base','data',data);

        for a = 1:length(algorithms)

            filename = fullfile(results_folder, ...
                [dataset_name '_' algorithms{a} '.mat']);

            if exist(filename,'file')
                fprintf('Skipping (already done): %s\n', filename);
                continue;
            end

            fprintf('Running: %s | Algo: %s\n', ...
                dataset_name, algorithms{a});
```

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```
assignin('base','algo_id',a);

simOut = sim(model,'StopTime',num2str(sim_time));

Vpv = simOut.logouts.get('Vpv').Values.Data;
Ipv = simOut.logouts.get('Ipv').Values.Data;
Ppv = simOut.logouts.get('Ppv').Values.Data;
Duty = simOut.logouts.get('Duty').Values.Data;
Vout = simOut.logouts.get('Vout').Values.Data;
Time = simOut.logouts.get('Vpv').Values.Time;

result.time = single(Time);
result.Vpv = single(Vpv);
result.Ipv = single(Ipv);
result.Ppv = single(Ppv);
result.Duty = single(Duty);
result.Vout = single(Vout);

save(filename,'result','-v7');

clear simOut result Vpv Ipv Ppv Duty Vout Time

end
end
end
disp('ALL 120 SIMULATIONS COMPLETED SUCCESSFULLY.');
```

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Simulink PV System Block Diagram

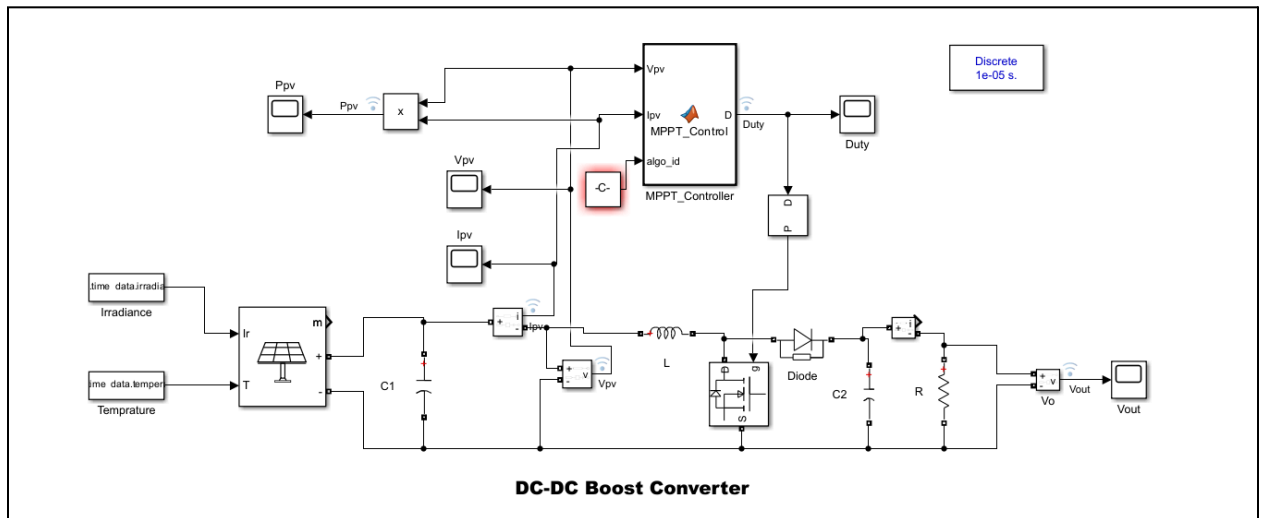


Figure D.1 Simulink Circuit

**UNIVERSITY OF SANTO TOMAS ELECTRICAL ENGINEERING
DEPARTMENT**

CURRICULUM VITAE

PERSONAL INFORMATION

Last Name: Atienza

First Name: Adrian

Middle Name: Sambajon

Student Number: 2022167630

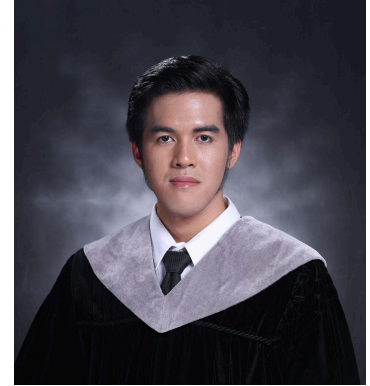
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City, 2101 Bataan
2010 - 2016

EXTRA-CURRICULAR INVOLVEMENT

Position: Class Officer - Class Deputy (2nd Semester)

Organization Name: University of Santo Tomas - Faculty of Engineering (UST - Manila)

Effectivity: 2025

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DEPARTMENT**

CURRICULUM VITAE

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EXTRA-CURRICULAR INVOLVEMENT

Position: Electrical Engineering Commissioner
Organization Name: UST Engineering Commission on Elections
Effectivity: 2023 - 2024

Position: Class Deputy
Organization Name: UST Engineering Commission on Elections
Effectivity: 2023 - 2024

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DEPARTMENT**

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Tertiary: University of Santo Tomas (UST-Manila)
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Secondary: Canossa Academy of Lipa City (CALC)
Calle Arzobispado Street, Barangay San Carlos - Lipa City
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Primary: Canossa Academy of Lipa City (CALC)
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2010 - 2016

EXTRA-CURRICULAR INVOLVEMENT

Position: Class Officer - Deputy (2nd Semester)

Organization Name: University of Santo Tomas - Faculty of Engineering (UST - Manila)

Effectivity: 2023 - 2024

Position: En Banc - EE Commissioner (1st and 2nd Semester)

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DEPARTMENT**

Organization Name: UST Engineering Commission on Elections - (UST - ENG
COMELEC)

Effectivity: 2024 - 2025

Position: Creatives - Executive Staff (1st and 2nd Semester)

Organization Name: UST Electrical Engineering Circuit - (UST - EEC)

Effectivity: 2024 - 2025

**UNIVERSITY OF SANTO TOMAS ELECTRICAL ENGINEERING
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Bachelor of Science in Electrical Engineering
2022 - present

Secondary: University of The East - Caloocan
2020-2022
Sto. Nino Academy, Binang 2nd, Bocaue, Bulacan
2016 - 2020

Primary: Sto. Nino Academy, Binang 2nd, Bocaue, Bulacan
2007 - 2016

EXTRA-CURRICULAR INVOLVEMENT

Position: Class Officer - Class Deputy (2nd Semester)

Organization Name: University of Santo Tomas - Faculty of Engineering (UST - Manila)

Effectivity: 2025

Position: Member

Organization Name: Institute of Integrated Electrical Engineers of the Philippines
(Student Chapter)

Effectivity: 2024 - 2025

**UNIVERSITY OF SANTO TOMAS ELECTRICAL ENGINEERING
DEPARTMENT**

Position: Member

Organization Name: Engineering Commission On Elections (Creatives Department)

Effectivity: 2024 - 2025

Position: Class Officer - Class Deputy (1st Semester)

Organization Name: University of Santo Tomas - Faculty of Engineering (UST - Manila)

Effectivity: 2024 - 2025